# Current Issues and Prospects of Vocational Education

# Current Issues and Prospects of Vocational Education

Dillon Reese

AMERICAN
ACADEMIC PUBLISHER
www.americanacademicpublisher.com

Published by
American Academic Publisher
1201 N. Orange St. Suit #600
Wilmington, De19801 USA.



ISBN: 978-1-6668-6996-5

**Current Issues and Prospects of Vocational Education**
*by Dillon Reese*





**Library of Congress Cataloging-in-Publication Data**
A catalogue record for this book is available from Library of Congress

For more information regarding American Academic Publisher and its products, please visit the publisher's website www.americanacademicpublisher.com

# TABLE OF CONTENTS

# Preface

Vocational education is training that equips students to work in various occupations, including those in trades, crafts, or as technicians. Technical education or career education are other names for vocational education. A sort of educational facility created expressly to offer vocational education is a vocational school. Vocational education is possible at the post-secondary, higher education, and further education levels. It can also work in conjunction with the apprenticeship system. At the post-secondary level, highly specialized trade, Technical schools, community colleges, institutions of further education UK, universities, and Institutes of Technology / Polytechnic Institutes frequently offer vocational education. Until recently, most vocational education took place in classrooms or on the job, where students were taught trade theory and skills by recognized academics or seasoned practitioners. However, the popularity of online vocational education has expanded, making it simpler for students to master different trade skills and soft skills from recognized authorities in the field. Vocational Education has been created and developed with the needs and requirements of target readers in mind. It briefly discusses the problems and difficulties faced by international, regional, and national educational systems and the growing demand for more vocational and technical courses and training programs.

***Author***

# 1
# CHAPTER

## Introduction

### THE MEANING OF VOCATION

At the present time the conflict of philosophic theories focuses in discussion of the proper place and function of vocational factors in education. The bald statement that significant differences in fundamental philosophical conceptions find their chief issue in connection with this point may arouse incredulity: there seems to be too great a gap between the remote and general terms in which philosophic ideas are formulated and the practical and concrete details of vocational education. But a mental review of the intellectual presuppositions underlying the oppositions in education of labour and leisure, theory and practice, body and mind, mental states and the world, will show that they culminate in the antithesis of vocational and cultural education. Traditionally, liberal culture has been linked to the notions of leisure, purely contemplative knowledge and a spiritual activity not involving the active use of bodily organs. Culture has also tended, latterly, to be associated with a purely private refinement, a cultivation of certain states and attitudes of consciousness, separate from either social direction or service. It has been an escape from the former, and a solace for the necessity of the latter.

So deeply entangled are these philosophic dualisms with the whole subject of vocational education, that it is necessary to define the meaning of vocation with some fullness in order to avoid the impression that an education which centers about it is narrowly practical, if not merely pecuniary. A vocation means nothing but such a direction of life activities as renders them perceptibly significant to a person, because of the consequences they accomplish, and also useful to his associates. The opposite of a career is neither leisure nor culture, but aimlessness, capriciousness, the absence of cumulative achievement in experience, on the personal side, and idle display, parasitic dependence upon the others, on the social side. Occupation is a concrete term for continuity. It includes the development of artistic capacity of any kind, of special scientific ability, of effective citizenship, as well as professional and business occupations, to say nothing of mechanical labour or engagement in gainful pursuits.

We must avoid not only limitation of conception of vocation to the occupations where immediately tangible commodities are produced, but also the notion that vocations are distributed in an exclusive way, one and only one to each person. Such restricted specialism is impossible; nothing could be more absurd than to try to educate individuals with an eye to only one line of activity. In the first place, each individual has of necessity a variety of callings, in each of which he should be intelligently effective; and in the second place any one occupation loses its meaning and becomes a routine keeping busy at something in the degree in which it is isolated from other interests. No one is just an artist and nothing else, and in so far as one approximates that condition, he is so much the less developed human being; he is a kind of monstrosity. He must, at some period of his life, be a member of a family; he must have friends and companions; he must either support himself or be supported by others, and thus he has a business career. He is a member of some organized political unit, and so on. We naturally name his vocation from that one of the callings which distinguishes him, rather than from those which he has in common with all others. But we should not allow ourselves to be so subject to words as to ignore and virtually deny his other callings when it comes to a consideration of the vocational phases of education.

As a man’s vocation as artist is but the emphatically specialized phase of his diverse and variegated vocational activities, so his efficiency in it, in the humane sense of efficiency, is determined by its association with other callings. A person must have experience, he must live, if his artistry is to be more than a technical accomplishment. He cannot find the subject matter of his artistic activity within his art; this must be an expression of what he suffers and enjoys in other relationships — a thing which depends in turn upon the alertness and sympathy of his interests. What is true of an artist is true of any other special calling. There is doubtless—in general accord with the principle of habit — a tendency for every distinctive vocation to become too dominant, too exclusive and absorbing in its specialized aspect. This means emphasis upon skill or technical method at the expense of meaning. Hence it is not the business of education to foster this tendency, but rather to safeguard against it, so that the scientific enquirer shall not be merely the scientist, the teacher merely the pedagogue, the clergyman merely one who wears the cloth, and so on.

## THE PLACE OF VOCATIONAL AIMS IN EDUCATION

Bearing in mind the varied and connected content of the vocation, and the broad background upon which a particular calling is projected, we shall now consider education for the more distinctive activity of an individual. An occupation is the only thing which balances the distinctive capacity of an individual with his social service.

To find out what one is fitted to do and to secure an opportunity to do it is the key to happiness. Nothing is more tragic than failure to discover one's true business in life, or to find that one has drifted or been forced by circumstance into an uncongenial calling. A right occupation means simply that the aptitudes of a person are in adequate play, working with the minimum of friction and the maximum of satisfaction. With reference to other members of a community, this adequacy of action signifies, of course, that they are getting the best service the person can render. It is generally believed, for example, that slave labour was ultimately wasteful even from the purely economic point of view — that there was not sufficient stimulus to direct the energies of slaves, and that there was consequent wastage.

Moreover, since slaves were confined to certain prescribed callings, much talent must have remained unavailable to the community, and hence there was a dead loss. Slavery only illustrates on an obvious scale what happens in some degree whenever an individual does not find himself in his work. And he cannot completely find himself when vocations are looked upon with contempt, and a conventional ideal of a culture which is essentially the same for all is maintained. Plato laid down the fundamental principle of a philosophy of education when he asserted that it was the business of education to discover what each person is good for, and to train him to mastery of that mode of excellence, because such development would also secure the fulfillment of social needs in the most harmonious way. His error was not in qualitative principle, but in his limited conception of the scope of vocations socially needed; a limitation of vision which reacted to obscure his perception of the infinite variety of capacities found in different individuals.

An occupation is a continuous activity having a purpose. Education through occupations consequently combines within itself more of the factors conducive to learning than any other method. It calls instincts and habits into play; it is a foe to passive receptivity. It has an end in view; results are to be accomplished. Hence it appeals to thought; it demands that an idea of an end be steadily maintained, so that activity cannot be either routine or capricious. Since the movement of activity must be progressive, leading from one stage to another, observation and ingenuity are required at each stage to overcome obstacles and to discover and readapt means of execution. In short, an occupation, pursued under conditions where the realization of the activity rather than merely the external product is the aim, fulfills the requirements which were laid down earlier in connection with the discussion of aims, interest, and thinking.

A calling is also of necessity an organizing principle for information and ideas; for knowledge and intellectual growth. It provides an axis which runs through an immense diversity of detail; it causes different experiences, facts, items of information to fall into order with one another. The lawyer, the physician, the laboratory investigator in some branch of chemistry, the parent, the citizen interested in his own locality, has a constant working stimulus to note and relate whatever has to do with his concern. He unconsciously, from the motivation of his occupation, reaches out for all relevant information, and holds to it. The vocation acts as both magnet to attract and as glue to hold. Such organization of knowledge is vital, because it has reference to needs; it is so expressed and readjusted in action that it never becomes stagnant. No classification, no selection and arrangement of facts, which is consciously worked out for purely abstract ends, can ever compare in solidity or effectiveness with that knit under the stress of an occupation; in comparison the former sort is formal, superficial, and cold.

The only adequate training for occupations is training through occupations. The principle stated early in this book that the educative process is its own end, and that the only sufficient preparation for later responsibilities comes by making the most of immediately present life, applies in full force to the vocational phases of education. The dominant vocation of all human beings at all times is living — intellectual and moral growth. In childhood and youth, with their relative freedom from economic stress, this fact is naked and unconcealed. To predetermine some future occupation for which education is to be a strict preparation is to injure the possibilities of present development and thereby to reduce the adequacy of preparation for a future right employment. To repeat the principle we have had occasion to appeal to so often, such training may develop a machine-like skill in routine lines but it will be at the expense of those qualities of alert observation and coherent and ingenious planning which make an occupation intellectually rewarding. In an autocratically managed society, it is often a conscious object to prevent the development of freedom and responsibility, a few do the planning and ordering, the others follow directions and are deliberately confined to narrow and prescribed channels of endeavor. However much such a scheme may inure to the prestige and profit of a class, it is evident that it limits the development of the subject class; hardens and confines the opportunities for learning through experience of the master class, and in both ways hampers the life of the society as a whole.

The only alternative is that all the earlier preparation for vocations be indirect rather than direct; namely, through engaging in those active occupations which are indicated by the needs and interests of the pupil at the time. Only in this way can there be on the part of the educator and of the one educated a genuine discovery of personal aptitudes so that the proper choice of a specialized pursuit in later life may be indicated. Moreover, the discovery of capacity and aptitude will be a constant process as long as growth continues. It is a conventional and arbitrary view which assumes that discovery of the work to be chosen for adult life is made once for all at some particular date. One has discovered in himself, say, an interest, intellectual and social, in the things which have to do with engineering and has decided to make that his calling. At most, this only blocks out in outline the field in which further growth is to be directed. It is a sort of rough sketch for use in direction of further activities. It is the discovery of a profession in the sense in which Columbus discovered America when he touched its shores. Future explorations of an indefinitely more detailed and extensive sort remain to be made.

When educators conceive vocational guidance as something which leads up to a definitive, irretrievable, and complete choice, both education and the chosen vocation are likely to be rigid, hampering further growth. In so far, the calling chosen will be such as to leave the person concerned in a permanently subordinate position, executing the intelligence of others who have a calling which permits more flexible play and readjustment. And while ordinary usages of language may not justify terming a flexible attitude of readjustment a choice of a new and further calling, it is such in effect. If even adults have to be on the lookout to see that their calling does not shut down on them and fossilize them, educators must certainly be careful that the vocational preparation of youth is such as to engage them in a continuous reorganization of aims and methods.

## PRESENT OPPORTUNITIES AND DANGERS

In the past, education has been much more vocational in fact than in name:

- The education of the masses was distinctly utilitarian. It was called apprenticeship rather than education, or else just learning from experience. The schools devoted themselves to the three R's in the degree in which ability to go through the forms of reading, writing, and figuring were common elements in all kinds of labour. Taking part in some special line of work, under the direction of others, was the out-of-school phase of this education. The two supplemented each other; the school work in its narrow and formal character was as much a part of apprenticeship to a calling as that explicitly so termed.
- To a considerable extent, the education of the dominant classes was essentially vocational — it only happened that their pursuits of ruling and of enjoying were not called professions. For only those things were named vocations or employments which involved manual labour, laboring for a reward in keep, or its commuted money equivalent, or the rendering of personal services to specific persons. For a long time, for example, the profession of the surgeon and physician ranked almost with that of the valet or barber — partly because it had so much to do with the body, and partly because it involved rendering direct service for pay to some definite person. But if we go behind words, the business of directing social concerns, whether politically or economically, whether in war or peace, is as much a calling as anything else; and where education has not been completely under the thumb of tradition, higher schools in the past have been upon the whole calculated to give preparation for this business. Moreover, display, the adornment of person, the kind of social companionship and entertainment which give prestige, and the spending of money, have been made into definite callings. Unconsciously to themselves the higher institutions of learning have been made to contribute to preparation for these employments. Even at present, what is called higher education is for a certain class mainly preparation for engaging effectively in these pursuits. In other respects, it is largely, especially in the most advanced work, training for

the calling of teaching and special research. By a peculiar superstition, education which has to do chiefly with preparation for the pursuit of conspicuous idleness, for teaching, and for literary callings, and for leadership, has been regarded as non-vocational and even as peculiarly cultural. The literary training which indirectly fits for authorship, whether of books, newspaper editorials, or magazine articles, is especially subject to this superstition: many a teacher and author writes and argues in behalf of a cultural and humane education against the encroachments of a specialized practical education, without recognizing that his own education, which he calls liberal, has been mainly training for his own particular calling. He has simply got into the habit of regarding his own business as essentially cultural and of overlooking the cultural possibilities of other employments. At the bottom of these distinctions is undoubtedly the tradition which recognizes as employment only those pursuits where one is responsible for his work to a specific employer, rather than to the ultimate employer, the community. There are, however, obvious causes for the present conscious emphasis upon vocational education — for the disposition to make explicit and deliberate vocational implications previously tacit. In the first place, there is an increased esteem, in democratic communities, of whatever has to do with manual labour, commercial occupations, and the rendering of tangible services to society. In theory, men and women are now expected to do something in return for their support — intellectual and economic — by society. Labour is extolled; service is a much-lauded moral ideal. While there is still much admiration and envy of those who can pursue lives of idle conspicuous display, better moral sentiment condemns such lives. Social responsibility for the use of time and personal capacity is more generally recognized than it used to be.

- In the second place, those vocations which are specifically industrial have gained tremendously in importance in the last century and a half. Manufacturing and commerce are no longer domestic and local, and consequently more or less incidental, but are world-wide. They engage the best energies of an increasingly large number of persons. The manufacturer, banker, and captain of industry have practically displaced a hereditary landed gentry as the immediate directors of social affairs. The problem of social readjustment is openly industrial, having to do with the relations of capital and labour. The great increase in the social importance of conspicuous industrial processes has inevitably brought to the front questions having to do with the relationship of schooling to industrial life. No such vast social readjustment could occur without offering a challenge to an education inherited from different social conditions, and without putting up to education new problems.
- In the third place, there is the fact already repeatedly mentioned: Industry has ceased to be essentially an empirical, rule-of-thumb procedure, handed down by custom. Its technique is now technological: that is to say, based upon

machinery resulting from discoveries in mathematics, physics, chemistry, bacteriology, etc. The economic revolution has stimulated science by setting problems for solution, by producing greater intellectual respect for mechanical appliances. And industry received back payment from science with compound interest. As a consequence, industrial occupations have infinitely greater intellectual content and infinitely larger cultural possibilities than they used to possess. The demand for such education as will acquaint workers with the scientific and social bases and bearings of their pursuits becomes imperative, since those who are without it inevitably sink to the role of appendages to the machines they operate. Under the old regime all workers in a craft were approximately equals in their knowledge and outlook. Personal knowledge and ingenuity were developed within at least a narrow range, because work was done with tools under the direct command of the worker. Now the operator has to adjust himself to his machine, instead of his tool to his own purposes. While the intellectual possibilities of industry have multiplied, industrial conditions tend to make industry, for great masses, less of an educative resource than it was in the days of hand production for local markets. The burden of realizing the intellectual possibilities inhering in work is thus thrown back on the school.

- In the fourth place, the pursuit of knowledge has become, in science, more experimental, less dependent upon literary tradition, and less associated with dialectical methods of reasoning, and with symbols. As a result, the subject matter of industrial occupation presents not only more of the content of science than it used to, but greater opportunity for familiarity with the method by which knowledge is made. The ordinary worker in the factory is of course under too immediate economic pressure to have a chance to produce a knowledge like that of the worker in the laboratory. But in schools, association with machines and industrial processes may be had under conditions where the chief conscious concern of the students is insight. The separation of shop and laboratory, where these conditions are fulfilled, is largely conventional, the laboratory having the advantage of permitting the following up of any intellectual interest a problem may suggest; the shop the advantage of emphasizing the social bearings of the scientific principle, as well as, with many pupils, of stimulating a livelier interest.
- Finally, the advances which have been made in the psychology of learning in general and of childhood in particular fall into line with the increased importance of industry in life. For modern psychology emphasizes the radical importance of primitive unlearned instincts of exploring, experimentation, and "trying on." It reveals that learning is not the work of something ready-made called mind, but that mind itself is an organization of original capacities into activities having significance. As we have already seen in older pupils work is to educative development of raw native activities what play is for younger pupils. Moreover, the passage from play to work should be gradual, not involving a radical change

> of attitude but carrying into work the elements of play, plus continuous reorganization in behalf of greater control. The reader will remark that these five points practically resume the main contentions of the previous part of the work. Both practically and philosophically, the key to the present educational situation lies in a gradual reconstruction of school materials and methods so as to utilize various forms of occupation typifying social callings, and to bring out their intellectual and moral content. This reconstruction must relegate purely literary methods — including textbooks—and dialectical methods to the position of necessary auxiliary tools in the intelligent development of consecutive and cumulative activities.

But our discussion has emphasized the fact that this educational reorganization cannot be accomplished by merely trying to give a technical preparation for industries and professions as they now operate, much less by merely reproducing existing industrial conditions in the school. The problem is not that of making the schools an adjunct to manufacture and commerce, but of utilizing the factors of industry to make school life more active, more full of immediate meaning, more connected with out-of-school experience.

The problem is not easy of solution. There is a standing danger that education will perpetuate the older traditions for a select few, and effect its adjustment to the newer economic conditions more or less on the basis of acquiescence in the untransformed, unrationalized, and unsocialized phases of our defective industrial regime. Put in concrete terms, there is danger that vocational education will be interpreted in theory and practice as trade education: as a means of securing technical efficiency in specialized future pursuits. Education would then become an instrument of perpetuating unchanged the existing industrial order of society, instead of operating as a means of its transformation. The desired transformation is not difficult to define in a formal way. It signifies a society in which every person shall be occupied in something which makes the lives of others better worth living, and which accordingly makes the ties which bind persons together more perceptible — which breaks down the barriers of distance between them. It denotes a state of affairs in which the interest of each in his work is uncoerced and intelligent: based upon its congeniality to his own aptitudes. It goes without saying that we are far from such a social state; in a literal and quantitative sense, we may never arrive at it. But in principle, the quality of social changes already accomplished lies in this direction. There are more ample resources for its achievement now than ever there have been before. No insuperable obstacles, given the intelligent will for its realization, stand in the way.

Success or failure in its realization depends more upon the adoption of educational methods calculated to effect the change than upon anything else. For the change is essentially a change in the quality of mental disposition — an educative change. This does not mean that we can change character and mind by direct instruction and exhortation, apart from a change in industrial and political conditions. Such a conception contradicts our basic idea that character and mind are attitudes of participative response in social affairs. But it does mean that we may produce in schools a projection in type of the society we should like to realize, and by forming minds in accord with it gradually modify the larger and more recalcitrant features of adult society. Sentimentally, it may seem harsh to say that the greatest evil of the present regime is not found in poverty and in the suffering which it entails, but in the fact that so many persons have callings which make no appeal to them, which are pursued simply for the money reward that accrues. For such callings constantly provoke one to aversion, ill will, and a desire to slight and evade. Neither men's hearts nor their minds are in their work. On the other hand, those who are not only much better off in worldly goods, but who are in excessive, if not monopolistic, control of the activities of the many are shut off from equality and generality of social intercourse. They are stimulated to pursuits of indulgence and display; they try to make up for the distance which separates them from others by the impression of force and superior possession and enjoyment which they can make upon others.

It would be quite possible for a narrowly conceived scheme of vocational education to perpetuate this division in a hardened form. Taking its stand upon a dogma of social predestination, it would assume that some are to continue to be wage earners under economic conditions like the present, and would aim simply to give them what is termed a trade education — that is, greater technical efficiency. Technical proficiency is often sadly lacking, and is surely desirable on all accounts — not merely for the sake of the production of better goods at less cost, but for the greater happiness found in work. For no one cares for what one cannot half do. But there is a great difference between a proficiency limited to immediate work, and a competency extended to insight into its social bearings; between efficiency in carrying out the plans of others and in one forming one's own. At present, intellectual and emotional limitation characterizes both the employing and the employed class.

While the latter often have no concern with their occupation beyond the money return it brings, the former's outlook may be confined to profit and power. The latter interest generally involves much greater intellectual initiation and larger survey of conditions. For it involves the direction and combination of a large number of diverse factors, while the interest in wages is restricted to certain direct muscular movements. But none the less there is a limitation of intelligence to technical and non- humane, non-liberal channels, so far as the work does not take in its social bearings. And when the animating motive is desire for private profit or personal power, this limitation is inevitable. In fact, the advantage in immediate social sympathy and humane disposition often lies with the economically unfortunate, who have not experienced the hardening effects of a one-sided control of the affairs of others.

Any scheme for vocational education which takes its point of departure from the industrial regime that now exists, is likely to assume and to perpetuate its divisions and weaknesses, and thus to become an instrument in accomplishing the feudal dogma of social predestination. Those who are in a position to make their wishes good, will demand a liberal, a cultural occupation, and one which fits for directive power the youth in whom they are directly interested. To split the system, and give to others, less fortunately situated, an education conceived mainly as specific trade preparation, is to treat the schools as an agency for transferring the older division of labour and leisure, culture and service, mind and body, directed and directive class, into a society nominally democratic. Such a vocational education inevitably discounts the scientific and historic human connections of the materials and processes dealt with. To include such things in narrow trade education would be to waste time; concern for them would not be "practical." They are reserved for those who have leisure at command—the leisure due to superior economic resources. Such things might even be dangerous to the interests of the controlling class, arousing discontent or ambitions "beyond the station" of those working under the direction of others. But an education which acknowledges the full intellectual and social meaning of a vocation would include instruction in the historic background of present conditions; training in science to give intelligence and initiative in dealing with material and agencies of production; and study of economics, civics, and politics, to bring the future worker into touch with the problems of the day and the various methods proposed for its improvement. Above all, it would train power of readaptation to changing conditions so that future workers would not become blindly subject to a fate imposed upon them. This ideal has to contend not only with the inertia of existing educational traditions, but also with the opposition of those who are entrenched in command of the industrial machinery, and who realize that such an educational system if made general would threaten their ability to use others for their own ends. But this very fact is the presage of a more equitable and enlightened social order, for it gives evidence of the dependence of social reorganization upon educational reconstruction. It is accordingly an encouragement to those believing in a better order to undertake the promotion of a vocational education which does not subject youth to the demands and standards of the present system, but which utilizes its scientific and social factors to develop a courageous intelligence, and to make intelligence practical and executive.

# VOCATIONAL GUIDANCE

## INTRODUCTION

We all will agree that in recent time there is variety of educational streams as new occupational opportunities are coming every other day. As there is variety of occupational opportunities available to the students, it is not always possible for students to meet the requirements of the available opportunities as all individual differ in the abilities, interest, attitude and skill from each other. Thus it is very necessary to know about ones own abilities, skills etc. To help in this process the role of vocational guidance becomes very important. In this unit we will discuss about vocational guidance, and other phenomenon related to the vocational guidance.

## NATURE AND SCOPE OF VOCATIONAL GUIDANCE

### Concept of Vocational Guidance

Vocational guidance according to DAW is that which have to be made well before the school leaving stage. Indeed with comprehensive education potentially offering a greater educational opportunity a good case for an increasingly important role for vocational guidance can be made.

It is worthwhile to look at some of the concepts or frames of references that are fundamental to a consideration of occupational choice.

Crites suggest that there are five sets of concepts that need to be considered. Firstly, how far is occupational choice the result of systematic behaviour as opposed to chance? Clearly the practice of vocational guidance is concerned with systematizing choice.

Thus, vocational guidance is not only adjustment but also achievement and development of the individuals to the maximum extent to be successful in their chosen occupations to be satisfied and useful to the society.

*Let us discuss the objectives of vocational guidance:*

- To assist the student to know about the characteristics, functions, duties of the occupations of his choice.
- To enable him to find out what are the specific criteria as skills, abilities, age etc. are required for the occupations of his choice.
- To assist the student to know about his skills, abilities, interests for making wise choices.
- To assist the student to acquire the technique of analysis before making any final choice.
- To provide opportunity for experiences in school and out of school this may provide an idea of the work environment to the student.
- To help the student realize that all honest labour is worthwhile.

The recommendations of Education Commission set up by Government of India in 1964 indicated the aims and scope of vocational guidance.

*These recommendations suggest the following:*

- The aim of guidance is not just to help student in making vocational choices but also be focused on the development of the child as good citizen.
- Guidance should be provided in solving learning difficulties, academic excellence, developing study habits and must be started at the primary stage itself.
- Guidance role is very crucial at the college level as at this stage student faces many problems related to educational vocational and personal. They should be make aware about the job market and how to make adjustment in it.
- Information may be provide about the training, competitive examinations and scholarships available.

**Need and Functions of Vocational Guidance**

- *Bringing individuals at par with the demands of the jobs:* As we all know that individual differences exist among all and no two individuals can have the same abilities, skills and personalities. We can understand it with an example as a person can be skilled in doing a desk job while the other can be skilled in the field job so the skill differs and thus the demand for the job also different from different individuals. Thus here vocational guidance is needed for matching the right job with right individual.
- *Assessing the right skill of an individual:* The capabilities and the limitations of an individual should be known to him *e.g.,* a person may have interest in painting but may not have the skill to take it as a career. So here the work of vocational guidance comes to assist the individual in assessing himself and then accordingly choosing the right kind of job.
- *Expansion of the world of work:* Nowadays jobs are available at national as well as international levels. And it gives rise to many queries as how one can choose the best from several alternatives? How to know about the details of the occupation? How to apply? And many more all these are dealt with the help of vocational guidance.

**Functions of Vocational Guidance**

- Assessment of the individual in terms of his academic achievement, skills attitude, interest, limitations etc.
- Assessment of individual's present environment such as financial resources, family background and family expectations etc.
- Assessing the suitability of the individual with that of the job demands.
- Helping the individual to come at par with the job requirements.
- Counseling for the adjustment in the job.
- Follow-up services in the form of feed back from the individuals for further modification in the guidance programme if needed.

**FACTORS EFFECTING VOCATIONAL CHOICE**

- Personal Factors Affecting Career Selection
  - Aptitudes - What natural abilities do you possess?
  - Interests inventories - what gives you satisfaction?
  - Your personality - Do you perform best in low-pressure or high-pressure working environments?
- Social Influences on Career Opportunities
  - Demographic trends
    a. Increase in working parents means more demand for food service and child care.
    b. More leisure time means more interest in health, and recreation products and services.

c. Increased demand for further employment training creates opportunities for teachers and trainers.

– Geographic trends

a. Where jobs are, salaries, and living costs. Economic Conditions affect career opportunities

• Trends in Industry and Technology affect career opportunities

– Automated production methods have decreased the need for many entry-level employees in factories.

**APPROACHES TO CAREER GUIDANCE**

Career guidance refers to services and activities which assist individuals, of any age and at any point throughout their lives, to make educational, training and occupational choices and to manage their careers. Such services may be found in schools, universities and colleges, in the workplace. It may take place individually or in group, and may be face-to-face or at a distance. It includes career information provision, assessment and self-assessment tools, counseling interviews, career education programmes. Guidance workers often encounter individuals having very little idea about the skills which they have and the career related to the skill. Sometimes, sufficient time devoted to the early stages of an interview, finding out about the individual. There are many approaches to career guidance as Computer-aided guidance. Others are facial expression, especially glints in the eyes, can be used to know levels of interest in particular types of work. Often, the client will be interested in more than one. Individual judgement is required to look at likely interactions between traits, and selecting the appropriate one.

**HOLLAND'S THEORY OF CAREER DEVELOPMENT**

The Holland Codes represents a set of personality types described in a theory of careers and vocational choice formulated by psychologist JOHN L. HOLLAND. Holland's theory argued that "the choice of a vocation is an expression of personality" and that the six factor typology he articulated could be used to describe both persons and work environments.

His model has been adopted by the U.S. Department of Labour for categorizing jobs relative to interests. The Holland Codes are usually referred to by their first letters: RIASEC. He presents his theory graphically as hexagon. The shorter the distance between their corners on the hexagon, the more closely they are related. Holland's theory of career guidance is based on four basic assumptions.

The first assumption is that most people can be characterized as one or a combination of six personality types. Second, the theory assumes that the work environment can be classified into the same six categories of personality. Third the theory assumes that people seek out environments compatible with their personality types. And, fourth, the theory holds that particular behavioural patterns emitted in any environment are determined by personality and environmental types.

*The six personality and work environment types described by Holland are as follows:*

- Realistic- Working with your hands, tools, machines, and things; practical, mechanically inclined, and physical. *viz.* agriculture, computer engineer, basket ball player, chef, Gardner, martial arts, pilot etc
- Investigative- Working with theory and information, analytical, intellectual, scientific. *Viz.* lawyer, statistician, surgeon.
- Artistic- Non-conforming, original, independent, chaotic, creative. *Viz.* actor, writer, dancer.
- Social- Cooperative environments, supporting, helping, healing/nurturing. *Viz.* psychologist, professor, social worker, physician.
- Enterprising- Competitive environments, leadership, persuading. Viz public relation, administration, journalism, marketing, management.
- Conventional- Detail-oriented, organizing, clerical. *Viz.* proof reader, copy editing, clerk, librarian.

## BURNOUT AND CAREER GUIDANCE

### Meaning of Burnout

First let us discuss about the concept of burnout. What is the meaning of the term burnout? Burnout is a psychological term for the experience of long-term exhaustion and diminished interest. It should not be looked as a disorder but as Problems related to life-management difficulty. The well-studied measurement of burnout in the literature is the Maslach Burnout Inventory. Maslach and her colleague Jackson first identified the construct "burnout" in the 1970s, and developed a measure that weighs the effects of emotional exhaustion and reduced sense of personal accomplishment. This indicator has become the standard tool for measuring burnout in research on the syndrome. The Maslach Burnout Inventory uses a three dimensional description of exhaustion, cynicism and inefficacy.

Many theories of burnout include negative outcomes related to burnout, including job function, health related outcomes, and mental health problems. The term burnout in psychology was coined by HERBERT FREUDENBERGER in 1974.

*Psychologists Herbert Freudenberger and Gail North gave 12 phases of burnout, which are not necessarily followed sequentially:*

- A compulsion to prove oneself
- Working harder
- Neglecting one's own needs
- Displacement of conflicts
- Revision of values
- Denial of emerging problems
- Withdrawal
- Behavioural changes become obvious to others

- Depersonalization
- Inner emptiness
- Depression
- Burnout syndrom.

The signs can vary from individual to individual, but the following are some universal indicators that one can use to determine if career burnout is occurring.

- Depression-Feelings of despair and sadness that last for weeks or months usually signal that something in your life is not working like it should and is cause for an investigation into the cause – potentially your job.
- Lack of energy-If individual experience constant fatigue throughout the day.
- Lack of desire-if an individual find that he just don‘t care if he is successful or not it‘s a warning that the individual may have become burned out.
- Decreased productivity- if the productivity of an individual is decreased and he is not coming up to the desired expected result.
- Increased absences and/or tardiness- if an individual finds every opportunity to skip out on work.
- Boredom-Occasional boredom in one‘s career is completely normal; however, pervasive feelings of weariness and dreariness are not and are an indicator of potential burnout.
- Anger/resentment in workplace-Frequently lashing out at coworkers and/or supervisors is unacceptable under any circumstance. This behaviour deserves immediate attention due to its potentially abusive nature.
- Sleep problems-Insomnia or occasional fatigue can happen to anyone but are a cause for concern if they become constant and a part of your everyday life. Sleep disturbances are your body‘s way of saying it is overworked.

**Coping with Burnout**

There are a variety of ways that both individuals and organizations can deal with burnout. In his book, *Managing stress: Emotion and power at work*, Newton argues that many of the remedies related to burnout are motivated not from an employee’s perspective, but from the organization’s perspective. Let us now discuss some of the common strategies for dealing with burnout.

***Organizational Aspects***

- *Employee assistance programmes (EAP):* Employee Assistance Programmes were designed to assist employees in dealing with the primary causes of stress. Some programmes included counseling and psychological services for employees. But now a days it is less utilized as compared to stress management training.
- *Stress management training (SMT):* Stress Management Training (SMT) is employed by many organizations today as a way to get employees to either

work through stress or to manage their stress levels; to maintain stress levels below that which might lead to higher instances of burnout.

***Individual Aspects***

- Problem-based coping-Individual can cope with the problems related to burnout and stress by focusing on the causes of the stress. This type of coping has successfully been linked to reductions in individual stress.
- Appraisal-based coping-Appraisal-based coping strategies deal with individual interpretations of what is and is not a stress inducing activity.

**LET US SUM UP**

In this unit we have discussed about nature and scope of vocational guidance, why vocational guidance is needed and how it helps an individual is discussed under the functions of vocational guidance. The factors which affect vocational choice as social, demographic, geographical, personal has also been talked about. An attempt has been made to acquaint you about the Holland‘s theory of career development, usually referred as RIASEC as it contains six factor typology, They are Realistic, Investigative, Artistic Social, Enterprising, Conventional. Meaning and clear understanding has been made by discussing about burnout and career guidance and also about the different strategies of coping with burnout.

# 2
# CHAPTER

## Curriculum Content

In education, a curriculum is broadly defined as the totality of student experiences that occur in the educational process. The term often refers specifically to a planned sequence of instruction, or to a view of the student's experiences in terms of the educator's or school's instructional goals. In a 2003 study Reys, Reys, Lapan, Holliday and Wasman refer to curriculum as a set of learning goals articulated across grades that outline the intended mathematics content and process goals at particular points in time throughout the K–12 school programme. Curriculum may incorporate the planned interaction of pupils with instructional content, materials, resources, and processes for evaluating the attainment of educational objectives. Curriculum is split into several categories, the explicit, the implicit (including the hidden), the excluded and the extra-curricular.

Curricula may be tightly standardized, or may include a high level of instructor or learner autonomy. Many countries have national curricula in primary and secondary education, such as the United Kingdom's National Curriculum. UNESCO's International Bureau of Education has the primary mission of studying curricula and their implementation worldwide.

**Definitions and Interpretations (sh)**

There is no generally agreed upon definition of curriculum.

*Some influential definitions combine various elements to describe curriculum as follows:*

- Kerr defines curriculum as, "All the learning which is planned and guided by the school, whether it is carried on in groups or individually, inside or outside of school."

- Braslavsky states that curriculum is an agreement among communities, educational professionals, and the State on what learners should take on during specific periods of their lives. Furthermore, the curriculum defines "why, what, when, where, how, and with whom to learn."
- Outlines the skills, performances, attitudes, and values pupils are expected to learn from schooling. It includes statements of desired pupil outcomes, descriptions of materials, and the planned sequence that will be used to help pupils attain the outcomes.
- The total learning experience provided by a school. It includes the content of courses (the syllabus), the methods employed (strategies), and other aspects, like norms and values, which relate to the way the school is organized.
- The aggregate of courses of study given in a learning environment. The courses are arranged in a sequence to make learning a subject easier. In schools, a curriculum spans several grades.
- Curriculum can refer to the entire programme provided by a classroom, school, district, state, or country. A classroom is assigned sections of the curriculum as defined by the school.
- Through the readings of Smith, Dewey, and Kelly, four curriculums could be defined as:
- Explicit curriculum: subjects that will be taught, the identified "mission" of the school, and the knowledge and skills that the school expects successful students to acquire.
- Implicit curriculum: lessons that arise from the culture of the school and the behaviours, attitudes, and expectations that characterize that culture, the unintended curriculum.
- Hidden curriculum: things which students learn, 'because of the way in which the work of the school is planned and organized but which are not in themselves overtly included in the planning or even in the consciousness of those responsible for the school arrangements (Kelly, 2009). The term itself is attributed to Philip W. Jackson and is not always meant to be a negative. Hidden curriculum, if its potential is realized, could benefit students and learners in all educational systems. Also, it does not just include the physical environment of the school, but the relationships formed or not formed between students and other students or even students and teachers (Jackson, 1986).
- Excluded curriculum: topics or perspectives that are specifically excluded from the curriculum.
- Extracurricular: May include school-sponsored programmes, which are intended to supplement the academic aspect of the school experience, or community-based programmes and activities. Examples of school-sponsored extracurricular programmes include sports, academic clubs, and performing arts. Community-based programmes and activities may take place at a school (after hours) but

are not linked directly to the school. Community-based programmes frequently expand on the curriculum that was introduced in the classroom. For instance, students may be introduced to environmental conservation in the classroom. This knowledge is further developed through a community-based programme. Participants then act on what they know with a conservation project. Community-based extracurricular activities may include "environmental clubs, 4-H, boy/girl scouts, and religious groups" (Hancock, Dyk, and Jones, 2012).

*Curriculum can be ordered into a procedure:*

- *Step 1:* Diagnosis of needs.
- *Step 2:* Formulation of objectives.
- *Step 3:* Selection of content.
- *Step 4:* Organization of content.
- *Step 5:* Selection of learning experiences.
- *Step 6:* Organization of learning experiences.
- *Step 7:* Determination of what to evaluate and of the ways and means of doing it.

Under some definitions, curriculum is prescriptive, and is based on a more general syllabus which merely specifies what topics must be understood and to what level to achieve a particular grade or standard. The word Syllabus originates from Greek. The Greek meaning of the word basically means a "concise statement or table of the heads of a discourse, the contents of a treatise, the subjects of series of lectures.

'Curriculum' has numerous definitions, which can be slightly confusing. In its broadest sense a curriculum may refer to all courses offered at a school, explicit. The intended curriculum, which the students learn through the culture of the school, implicit. The curriculum that is specifically excluded, like racism. Plus, the extra curricular activities like sports, and clubs. This is particularly true of schools at the university level, where the diversity of a curriculum might be an attractive point to a potential student.

A curriculum may also refer to a defined and prescribed course of studies, which students must fulfill in order to pass a certain level of education. For example, an elementary school might discuss how its curriculum, or its entire sum of lessons and teachings, is designed to improve national testing scores or help students learn the basics. An individual teacher might also refer to his or her curriculum, meaning all the subjects that will be taught during a school year.

On the other hand, a high school might refer to a curriculum as the courses required in order to receive one's diploma. They might also refer to curriculum in exactly the same way as the elementary school, and use curriculum to mean both individual courses needed to pass, and the overall offering of courses, which help prepare a student for life after high school.

Curriculum can be envisaged from different perspectives. What societies envisage as important teaching and learning constitutes the "intended" curriculum. Since it is usually presented in official documents, it may be also called the "written" and/or "official" curriculum. However, at classroom level this intended curriculum may be altered through a range of complex classroom interactions, and what is actually delivered can be considered the "implemented" curriculum. What learners really learn (*i.e.,* what can be assessed and can be demonstrated as learning outcomes/learner competencies) constitutes the "achieved" or "learned" curriculum. In addition, curriculum theory points to a "hidden" curriculum (*i.e.,* the unintended development of personal values and beliefs of learners, teachers and communities; unexpected impact of a curriculum; unforeseen aspects of a learning process). Those who develop the intended curriculum should have all these different dimensions of the curriculum in view. While the "written" curriculum does not exhaust the meaning of curriculum, it is important because it represents the vision of the society. The "written" curriculum is usually expressed in comprehensive and user-friendly documents, such as curriculum frameworks; subject curricula/ syllabuses, and in relevant and helpful learning materials, such as textbooks; teacher guides; assessment guides.

In some cases, people see the curriculum entirely in terms of the subjects that are taught, and as set out within the set of textbooks, and forget the wider goals of competencies and personal development. This is why a curriculum framework is important. It sets the subjects within this wider context, and shows how learning experiences within the subjects need to contribute to the attainment of the wider goals.

There are many common misconceptions of what curriculum is and one of the most common is that curriculum only entails a syllabus. Smith (1996,2000) says that, "A syllabus will not generally indicate the relative importance of its topics or the order in which they are to be studied. Where people still equate curriculum with a syllabus they are likely to limit their planning to a consideration of the content or the body of knowledge that they wish to transmit". Regardless of the definition of curriculum, one thing is certain. The quality of any educational experience will always depend to a large extent on the individual teacher responsible for it (Kelly, 2009).

Curriculum is almost always defined with relation to schooling. According to some, it is the major division between formal and informal education. However, under some circumstances it may also be applied to informal education or free-choice learning settings. For instance, a science museum may have a "curriculum" of what topics or exhibits it wishes to cover. Many after-school programmes in the US have tried to apply the concept; this typically has more success when not rigidly clinging to the definition of curriculum as a product or as a body of knowledge to be transferred. Rather, informal education and free-choice learning settings are more suited to the model of curriculum as practice or praxis.

## HISTORICAL CONCEPTION

In the early years of the 20th century, the traditional concepts held of the "curriculum is that it is a body of subjects or subject matter prepared by the teachers for the students to learn." It was synonymous to the "course of study" and "syllabus".

In *The Curriculum*, the first textbook published on the subject, in 1918, John Franklin Bobbitt said that curriculum, as an idea, has its roots in the Latin word for *race-course*, explaining the curriculum as the course of deeds and experiences through which children become the adults they should be, *for success in adult society*. Furthermore, the curriculum encompasses the entire scope of formative deed and experience occurring in and out of school, and not only experiences occurring in school; experiences that are unplanned and undirected, and experiences intentionally directed for the purposeful formation of adult members of society. (cf. image at right.)

To Bobbitt, the curriculum is a social engineering arena. Per his cultural presumptions and social definitions, his curricular formulation has two notable features:

- That scientific experts would best be qualified to and justified in designing curricula based upon their expert knowledge of what qualities are desirable in adult members of society, and which experiences would generate said qualities; and
- Curriculum defined as the deeds-experiences the student *ought to have* to become the adult he or she *ought to become*.

Hence, he defined the curriculum as an ideal, rather than as the concrete reality of the deeds and experiences that form people to who and what they are.

Contemporary views of curriculum reject these features of Bobbitt's postulates, but retain the basis of curriculum as the course of experience (s) that forms human beings into persons. Personal formation via curricula is studied at the personal level and at the group level, *i.e.,* cultures and societies (*e.g.,* professional formation, academic discipline via historical experience). The formation of a group is reciprocal, with the formation of its individual participants.

Although it formally appeared in Bobbitt's definition, curriculum as a course of formative experience also pervades John Dewey's work (who disagreed with Bobbitt on important matters). Although Bobbitt's and Dewey's idealistic understanding of "curriculum" is different from current, restricted uses of the word, curriculum writers and researchers generally share it as common, substantive understanding of curriculum. Development does not mean just getting something out of the mind. It is a development of experience and into experience that is really wanted.

Robert M. Hutchins, president of the University of Chicago, regarded curriculum as "permanent studies" where the rules of grammar, rhetoric and logic and mathematics for basic education are emphasized. Basic education should emphasize 3 Rs and college education should be grounded on liberal education. On the other hand, Arthur Bestor as an essentialist, believes that the mission of the school should be intellectual training, hence curriculum should focus on the fundamental intellectual disciplines of grammar, literature and writing. It should also include mathematics, science, history and foreign language.

This definition leads us to the view of Joseph Schwab that discipline is the sole source of curriculum. Thus in our education system, curriculum is divided into chunks of knowledge we call subject areas in basic education such as English, Mathematics, Science, Social Studies and others. In college, discipline may include humanities, sciences, languages and many more. Curriculum should consist entirely of knowledge which comes from various disciplines. To learn the lesson is more interesting than to take a scolding, be held up to general ridicule, stay after school, receive degrading low marks, or fail to be promoted.

Thus, curriculum can be viewed as a field of study. It is made up of its foundations (philosophical, historical, psychological, and social foundations); domains of knowledge as well as its research theories and principles. Curriculum is taken as scholarly and theoretical. It is concerned with broad historical, philosophical and social issues and academics. Under a starting definition offered by John Kerr and taken up by Vic Kelly in his standard work on the curriculum, curriculum is "all the learning which is planned and guided by the school, whether it is carried on in groups or individually, inside or outside the school.

*Four ways of approaching curriculum theory and practice:*

1. Curriculum as a body of knowledge to be transmitted.
2. Curriculum as an attempt to achieve certain ends in students – products.
3. Curriculum as a process.
4. Curriculum as praxis.

In recent years the field of education, and therefore curriculum, has expanded outside the walls of the classroom and into other settings such as museums. Within these settings curriculum is an even broader topic, including various teachers such as other visitors, inanimate objects such as audio tour devices, and even the learners themselves. As with the traditional idea of curriculum, curriculum in a free choice learning environment can consist of the explicit stated curriculum and the hidden curriculum, both of which contribute to the learner's experience and lessons from the experience. These elements are further compounded by the setting, cultural influences, and the state of mind of the learner. Museums and other similar settings are most commonly leveraged within traditional classroom settings as enhancements to the curriculum when educators develop curriculum that encompasses visits to museums, zoos, and aquarium.

## PROGRESSIVIST VIEWS

On the other hand, to a progressivist, a listing of school subjects, syllabi, course of study, and list of courses of specific discipline do not make a curriculum. These can only be called curriculum if the written materials are actualized by the learner. Broadly speaking, curriculum is defined as the total learning experiences of the individual. This definition is anchored on John Dewey's definition of experience and education. He believed that reflective thinking is a means that unifies curricular elements. Thought is not derived from action but tested by application.

Caswell and Campbell viewed curriculum as "all experiences children have under the guidance of teachers." This definition is shared by Smith, Stanley and shores when they defined “curriculum as a sequence of potential experiences set up in schools for the purpose of disciplining children and youth in group ways of thinking and acting.”

Curriculum as a process is when a teacher enters a particular schooling and situation with: an ability to think critically, in-action; an understanding of their role and the expectations others have of them; and a proposal for action which sets out essential principles and features of the educational encounter. Guided by these, they encourage conversations between, and with, people in the situation out of which may comes a course of thinking and action. Plus, the teacher continually evaluates the process and what they can see of outcomes.

Marsh and Willis on the other hand view curriculum as all the “experiences in the classroom which are planned and enacted by teacher, and also learned by the students.

Any definition of curriculum, if it is to be practically effective and productive, must offer much more than a statement about knowledge-content or merely the subjects which schooling is to teach or transmit or deliver. Some would argue of the course that the values implicit in the arrangements made by schools for their pupils are quite clearly in the consciousness of teachers and planners, again especially when the planners are politicians, and are equally clearly accepted by them as part of what pupils should learn in school, even though they are not overtly recognized by the pupils themselves. In other words, those who design curricula deliberately plan the schools’ ‘expressive culture’. If this is the case, then, the curriculum is ‘hidden’ only to or from the pupils, and the values to be learnt clearly from a part of what is planned for pupils. They must, therefore, be accepted as fully a part of the curriculum, and most especially as an important focus for the kind of study of curriculum with which we are concerned here, not least because important questions must be asked concerning the legitimacy of such practices.

Currently, a spiral curriculum is promoted as allowing students to revisit a subject matter’s content at the different levels of development of the subject matter being studied. The constructivist approach proposes that children learn best via pro-active engagement with the educational environment, *i.e.,* learning through discovery.

**Primary and Secondary Education**

A curriculum may be partly or entirely determined by an external, authoritative body (*e.g.,* the National Curriculum for England in English schools).

Crucial to the curriculum is the definition of the course objectives that usually are expressed as learning outcomes and normally include the program’s assessment strategy. These outcomes and assessments are grouped as units (or modules), and, therefore, the curriculum comprises a collection of such units, each, in turn, comprising a specialised, specific part of the curriculum. So, a typical curriculum includes communications, numeracy, information technology, and social skills units, with specific, specialized teaching of each.

Core curricula are often instituted, at the primary and secondary levels, by school boards, Departments of Education, or other administrative agencies charged with overseeing education. A core curriculum is a curriculum, or course of study, which is deemed central and usually made mandatory for all students of a school or school system. However, even when core requirements exist, they do not necessarily involve a requirement for students to engage in one particular class or activity. For example, a school might mandate a music appreciation class, but students may opt out if they take a performing musical class, such as orchestra, band, chorus, etc.

**AUSTRALIA**

In Australia, the Australian Curriculum took effect nationwide in 2014, after a curriculum development process that began in 2010. Previously, each state's Education Department had traditionally established curricula. The Australian Curriculum consists of one curriculum covering eight subject areas through year 10, and another covering fifteen subjects for the senior secondary years.

**CANADA**

In Canada each province and territory has the authority to create its own curriculum. However, the Northwest Territories and Nunavut both choose to use the Alberta Curriculum for select parts of their curriculum. The territories also use Alberta's standardized tests in some subjects.

**SOUTH AFRICA**

In South Africa the Caps curriculum is used in Public schools. Private schools use IEB, Cambridge, etc.

**SOUTH KOREA**

The National Curriculum of Korea covers kindergarten, primary, and secondary education, as well as special education. The version currently in place is the 7th National Curriculum, which has been revised in 2007 and 2009. The curriculum provides a framework for a common set of subjects through 9th grade, and elective subjects in grades 10 through 12.

**JAPAN**

The Japanese educational system is based off traditional values from their heritage with curriculum ideas borrowed from England, Germany, France and the United States. The Japanese curriculum is world-famous. Their math and science standards are among the most demanding in the developed countries. Students in Japan are expected to know more about another country's history, economics, and geography than their own country. Japanese students cannot skip grades and are not held back. They are expected to master the curriculum at every level. Due to their meritocratic nature, all students are funded equitably and follow exactly the same curriculum with the same expectations. Students that are ahead in class are expected to help those that are not. Beyond the academics, students are expected to clean the classrooms and the hallways to teach respect and responsibility.

**THE NETHERLANDS**

The Dutch system is based on directives coming from the Ministry of Education, Culture and Science (OCW). Primary and secondary education use key objectives to create curricula. For primary education the total number of objectives has been reduced from 122 (back in 1993) to 58 in 2006. Starting in 2009/2010 all key objectives are obligatory for primary education. The key objectives are oriented towards subject areas such as language, mathematics/arithmetic, orientation towards self and the world, art, and physical education. All of the objectives have accompanying concrete activities. Also final exams are determined by the OCW and required. Parts of those exams are taken in a national setting, created by a committee: Centrale examencommissie vaststelling opgaven. Furthermore, OCW will determine the number of hours to be spent per subject. Apart from these directives every school can determine its own curriculum.

**NIGERIA**

In 2005, the Nigerian government adopted a national Basic Education Curriculum for grades 1 through 9. The policy was an outgrowth of the Universal Basic Education programme announced in 1999, to provide free, compulsory, continuous public education for these years. In 2014, the government implemented a revised version of the national curriculum, reducing the number of subjects covered from 20 to 10.

**SWEDEN**

In Sweden since 2011, the primary school curriculum is Lgr 11, while secondary schools use Lgy 11.

**SCOTLAND**

In Scotland, the Curriculum for Excellence (CfE) was introduced in August 2010 in all schools. The national qualifications were introduced in 2013 by the Scottish Qualifications Authority (SQA). The national qualifications include: Life Skills Coursework (SFL), National 3 (NAT3), National 4 (NAT4), National 5 (NAT5), Higher, Advanced Higher.

**UNITED KINGDOM**

The National Curriculum was introduced into England, Wales and Northern Ireland as a nationwide curriculum for primary and secondary state schools following the

Education Reform Act 1988. Notwithstanding its name, it does not apply to independent schools, which may set their own curricula, but it ensures that state schools of all local education authorities have a common curriculum. Academies, while publicly funded, have a significant degree of autonomy in deviating from the National Curriculum.

The purpose of the National Curriculum was to standardise the content taught across schools to enable assessment, which in turn enabled the compilation of league tables detailing the assessment statistics for each school. These league tables, together with the provision to parents of some degree of choice in assignment of the school for their child (also legislated in the same act) were intended to encourage a ‘free market’ by allowing parents to choose schools based on their measured ability to teach the National Curriculum.

**UNITED STATES**

In the U.S., each state, with the individual school districts, establishes the curricula taught. Each state, however, builds its curriculum with great participation of national academic subject groups selected by the United States Department of Education, *e.g.,* National Council of Teachers of Mathematics (NCTM) nctm.org for mathematical instruction.

The Common Core State Standards Initiative promulgates a core curriculum for states to adopt and optionally expand upon. This coordination is intended to make it possible to use more of the same textbooks across states, and to move towards a more uniform minimum level of educational attainment.

**Higher Education**

Many educational institutions are currently trying to balance two opposing forces. On the one hand, some believe students should have a common knowledge foundation, often in the form of a core curriculum; on the other hand, others want students to be able to pursue their own educational interests, often through early specialty in a major, however, other times through the free choice of courses. This tension has received a large amount of coverage due to Harvard University’s reorganization of its core requirements.

Many labour economics studies report that employment and earnings vary by college major and this appears to be *caused* by differences in the labour market value of the skills taught in different majors. Majors also have different labour market value even after students complete graduate degrees such as law degrees or business degrees.

An essential feature of curriculum design, seen in every college catalogue and at every other level of schooling, is the identification of prerequisites for each course. These prerequisites can be satisfied by taking particular courses, and in some cases by examination, or by other means, such as work experience. In general, more advanced courses in any subject require some foundation in basic courses, but some coursework requires study in other departments, as in the sequence of math classes required for a physics major, or the language requirements for students preparing in literature, music, or scientific research. A more detailed curriculum design must deal with prerequisites within a course for each topic taken up. This in turn leads to the problems of course organization and scheduling once the dependencies between topics are known.

## RUSSIA

Core curriculum has typically been highly emphasized in Soviet and Russian universities and technical institutes.

## UNITED STATES

### Core Curriculum

At the undergraduate level, individual college and university administrations and faculties sometimes mandate core curricula, especially in the liberal arts. But because of increasing specialization and depth in the student's major field of study, a typical core curriculum in higher education mandates a far smaller proportion of a student's course work than a high school or elementary school core curriculum prescribes.

Amongst the best known and most expansive core curricula programmes at leading American colleges and universities are that of Columbia University, as well as the University of Chicago's. Both can take up to two years to complete without advanced standing, and are designed to foster critical skills in a broad range of academic disciplines, including: the social sciences, humanities, physical and biological sciences, mathematics, writing and foreign languages.

In 1999, the University of Chicago announced plans to reduce and modify the content of its core curriculum, including lowering the number of required courses from 21 to 15 and offering a wider range of content. When *The New York Times*, *The Economist*, and other major news outlets picked up this story, the University became the focal point of a national debate on education. The National Association of Scholars released a statement saying, "It is truly depressing to observe a steady abandonment of the University of Chicago's once imposing undergraduate core curriculum, which for so long stood as the benchmark of content and rigour among American academic institutions." Simultaneously, however, a set of university administrators, notably then-President Hugo Sonnenschein, argued that reducing the core curriculum had become both a financial and educational imperative, as the university was struggling to attract a commensurate volume of applicants to its undergraduate division compared to peer schools as a result of what was perceived by the pro-change camp as a reaction by "the average eighteen-year-old" to the expanse of the collegiate core.

As core curricula began to diminish over the course of the twentieth century at many American schools, some smaller institutions became famous for embracing a core curriculum that covers nearly the student's entire undergraduate education, often utilizing classic texts of the western canon to teach all subjects including science. Four Great

Books colleges in the United States follow this approach: St. John’s, Shimer, Thomas Aquinas, Gutenberg College and Thomas More.

**Distribution Requirements**

Some colleges opt for the middle ground of the continuum between specified and unspecified curricula by using a system of distribution requirements. In such a system, students are required to take courses in particular fields of learning, but are free to choose specific courses within those fields.

**Open Curriculum**

Other institutions have largely done away with core requirements in their entirety. Brown University offers the “New Curriculum,” implemented after a student-led reform movement in 1969, which allows students to take courses without concern for any requirements except those in their chosen concentrations (majors), plus two writing courses. In this vein it is certainly possible for students to graduate without taking college-level science or math courses, or to take only science or math courses. Amherst College requires that students take one of a list of first-year seminars, but has no required classes or distribution requirements. Similarly, Grinnell College requires students to take a First-Year Tutorial in their first semester, and has no other class or distribution requirements. Others include Evergreen State College, Hamilton College, and Smith College.

Wesleyan University is another school that has not and does not require any set distribution of courses. However, Wesleyan does make clear “General Education Expectations” such that if a student does not meet these expectations, he/she would not be eligible for academic honours upon graduation.

**Gender Inequality in Curricula**

Gender inequality in curricula shows how men and women are not treated equally in several types of curricula. More precisely, gender inequality is visible in the curriculum of both schools and Teacher Education Institutes (TEIs). Physical education (PE) is an example where gender equality issues are highlighted because of preconceived stereotyping of boys and girls. The general belief is that boys are better at physical activities than girls, and that girls are better at ‘home’ activities such as sewing and cooking. This is the case in many cultures around the world and is not specific to one culture only.

# CURRICULUM SOURCES

## FACTORS AFFECTING CURRICULUM

**Sources of Curriculum Materials**

Curriculum materials centers are essential to the instructional and research needs of students and faculty in programmes preparing educators for P-12* schools. These guidelines describe essential elements of administration, services, and collections for curriculum materials centers in all university and college settings.

These guidelines are intended for administrators at all levels of post-secondary education, particularly education deans or department chairs; library deans or directors; librarians responsible for curriculum materials centers; and accrediting and licensure agencies.

***Terms Used***

- Curriculum materials are educational resources that provide curriculum and instructional experiences for P-12* students. These materials are used by educators to develop curricula and lesson plans and may also be used in actual instructional situations with P-12* students. These materials also provide information for those doing research.
- Curriculum Materials Center (CMC) refers to a physical location of a curriculum materials collection. Curriculum materials centers are often housed in a main campus library, a branch library building, or in an academic building housing the campus education academic programmes.
- CMC users are education students and faculty, and may also include P-12* educators, other students, and community members as defined by the CMC's mission.
- Director refers to the librarian who has primary responsibility for the CMC, including its facilities, administration, collection, personnel, and services.

***Administration***

*Mission/Goals*

The CMC should have a written mission statement with articulated goals that reflect these guidelines.

- Collaboration—The mission statement, goal setting, and planning should be jointly developed by the CMC director, an administrator from the unit to which the CMC administratively reports, and faculty representatives from the college or department of education.
- Review—The mission statement and goals should be regularly reviewed and updated as needed.
- Compliance—Goal setting should be in compliance with this document of CMC guidelines and appropriate accreditation standards.

*Budget*

The CMC should have a budget that adequately addresses its needs.

- Funding Responsibility—The CMC director and the administrator (s) responsible for budgeting for the unit to which the CMC administratively reports should jointly plan the CMC budget.

- Funding Level—The CMC budget should be adequate to ensure compliance with state department of education and other accrediting bodies' standards, college/department of education programme needs, as well as particular guidelines in this document in the areas of collection, facilities, services, and personnel. It should be reflective of the college of education or department of education enrollment.
- Funding Source—The CMC budget should be funded as part of the unit under which the CMC is administered. This budget does not preclude additional funding from other units or sources.
- Administraton—The CMC budget should be administered by the CMC director.

*Personnel*

The CMC staff should include a director and support staff sufficient to maintain the CMC and all services.

- Director—The CMC director should have a master's degree from an ALA-accredited programme or equivalent and have preparation in curriculum development, teaching methodology, media, and technology. The director should be assigned no less than half time to the management of the CMC.
- Support Staff—The CMC should have sufficient support staff to maintain the CMC and all its functions. The support staff may consist of paraprofessionals or clerical aides, with at least one being a permanent staff member, and graduate assistants and student assistants. Support staff should have sufficient training to provide a basic level of assistance to CMC users, or refer users as appropriate.
- Continuing Education—The CMC director and staff should have regular opportunities for continuing education so that the CMC reflects current trends in curriculum materials and technology. Continuing education opportunities should extend to support staff as needed.

*Facilities*

The CMC should be a distinct facility that provides for effective use of its resources.

- Location—The curriculum materials center should be located in proximity to the education holdings of the college or university library, or alternatively it should be in the building that houses the college/department of education. The location should be completely accessible as detailed in the Americans with Disabilities Act (ADA).
- Hours—The CMC should, if housed in the college or university library, be open the hours of that facility's operation. If housed separately, or with the college/department of education, it should be open enough hours to meet the needs of its users. Evening and weekend hours should be included if needed.
- Size—The size of the public area of the facility should be adequate to comfortably hold all materials, associated equipment, user study areas, and

workstations. Room for collection growth should be available. Staff workspace should be adequate to complete work activities efficiently and effectively, including technical library functions, when necessary.

- Seating—There should be enough seating in the CMC to allow users to work individually or collaboratively. Sufficient seating should be available to accommodate the students in an average-sized class in the teacher education programme. A variety of seating types may be available, including, but not limited to study tables, carrels, and lounge seating. If the CMC will be used by small children, appropriately sized seating for them may also be available.
- Maintenance—The facility should be maintained in such a way as to ensure the security and safety of materials, staff, and users. There should be an adequate number of electrical connections and computer ports to meet user and staff needs.
- Classroom—The CMC should have its own classroom, or have a convenient space available for formal instruction. This classroom, or its equivalent, should have adequate seating for the average-sized class in the teacher education programme. It should be equipped with technology appropriate for demonstration (and if possible, hands-on practice) of electronic and media resources.

*Promotion*

The CMC should have a plan for promotion of the CMC, its services, and its collection. Promotion should be directed towards all CMC user groups and stakeholders and should include both formal and informal means.

- Web Presence—A Web presence should be used to promote the CMC, and should be linked to and from the library site and the education site. The Web site should include, but not be limited to, the resources and services of the CMC and links to appropriate curriculum materials sites, such as teaching activities, standards, children's literature, publishers, etc.
- Printed Brochures/Guides—CMC brochures/guides should be available in the library publicity area, the college/department of education office area, and appropriate distance locations.
- Informal Campus Contacts—The CMC director should make use of faculty liaison activities mentioned in this document to informally promote the CMC.
- School Contacts—The CMC should be promoted to appropriate personnel in local schools/districts.

**Services**

*Reference*

The CMC staff should provide reference service to its users.

- Delivery Of Service—Reference service should be available during all hours

the CMC is open, and may include face-to-face, telephone, electronic, or other appropriate methods of delivery.
- Staff—CMC staff should be trained to conduct an effective reference interview. They should also have knowledge of the CMC's collection and of external resources in order to provide both ready-reference and in-depth research assistance. CMC student assistants should be knowledgeable about the CMC collection and be trained to provide basic assistance. A professional librarian located in an adjacent area may be called on if the CMC is not otherwise staffed.

*Instruction*

The CMC should have a programme for instruction in the use of curriculum-related resources.
- Collaboration—The instruction programme should be developed in collaboration with education faculty, librarians, and others as appropriate.
- Setting—Instruction may take place within the CMC, in the classroom, or in a virtual environment.
- Delivery—Instruction, both in person and virtual, should include all appropriate techniques such as guides, lectures, Web pages, tutorials, bibliographies, workshops, orientations, tours, and point of need instruction.
- Content—Instruction should include research strategies and the selection and evaluation of resources, as well as the use of the CMC collection and services, and instructional technology.

*Faculty Liaison*

The CMC staff should seek out and maintain professional contact with teacher education instructional units and with individual faculty members.
- Faculty Contact—Faculty contact should be maintained through both formal and informal means, including, but not limited to, telephone, electronic communication, attendance at faculty meetings, instruction sessions, and specialized programming.
- Accrediting Bodies—CMC staff should prepare documentation for visiting accrediting organizations as needed and requested.
- Collection Development—In collaboration with faculty, CMC staff should develop the CMC collection to meet instructional and curriculum materials research needs of both faculty and students.

*Distance Learning*

The CMC should have a programme for serving distance learning users.
- Collaboration—The distance learning programme should be developed in collaboration with teaching faculty, librarians, and others as appropriate.

- Users—Distance learning users should include distance learning faculty and students, whether courses are offered in an off-campus classroom, through teleconferencing, online, or by other means.
- Services—Services offered should be equivalent to services at the main campus and should include reference, instruction, and access to CMC materials.
- Delivery—Distance learning services should be provided by various means as appropriate. Electronic means are particularly well suited to off-campus situations and should be used to their best advantage. Electronic delivery options include, but are not limited to: web pages, CMC online catalogue, online CMC instruction, e-mail/mailing lists, online discussion groups, and access/subscriptions to online databases. Other means should be used as appropriate and may include librarian visits to off-campus classrooms, interlibrary loan, document delivery, and agreements with other libraries/CMCs.

*Outreach*

The CMC may offer outreach services.

- Users—Users may include students from other universities, P-12 educators, those who home-school, and other community members.
- Services—Services may include tours, exhibits, speakers, continuing education opportunities, or other activities that meet the needs of the users.
- Delivery—Policies regarding delivery of services and access to resources should be in agreement with the appropriate governing unit (s).

*Production*

The CMC may provide modern, high quality equipment and supplies to meet user needs for production of instructional materials.

- Equipment—The equipment provided for production should allow users to create instructional materials similar to those currently being used in schools, utilizing both traditional and emerging technologies. The equipment should be kept updated, well maintained, and in sufficient quantity to meet typical demand levels.
- Supplies—Supplies necessary for production of instructional materials should be provided to users, either for free or on a cost-recovery basis and in sufficient quantity to meet demands.
- Assistance—CMC staff should provide ideas and basic assistance to users, although the responsibility for creating the materials remains with the users.

***Collection***

*General Characteristics*

The CMC collection supports the institution's teacher education curriculum with an organized collection of current and high quality educational materials created for use with children from preschool through grade twelve, and adult education materials, when appropriate.

- Selection—The selection of curriculum materials should be the responsibility of a professional librarian specifically charged with building the curriculum materials collection.
- Collection Development Policy—The CMC should have a written collection development policy, as described in the policy section of these guidelines.
- Organization—The CMC collection should be organized in accordance with current national standards and practices, as described in the access section of these guidelines.
- Location—All of the collection should be available in the CMC.
- Size—The size of the CMC collection should be sufficient to meet the needs of its users, as well as to ensure compliance with state department of education standards.
- Format—These resources should represent a variety of formats including print, non-print, and electronic.
- Funding Level—Funding level for collection materials should reflect the enrollment of education majors and pre-service teachers in comparison to other majors within the institution.

*Collection Categories*

The CMC should collect materials in both print and electronic formats, including, but not limited to, textbooks, curriculum guides, children's literature, professional literature, reference materials, education periodicals, media materials, educational tests and measures, and digital content.

- Textbooks—Current textbooks in all major P-12 curricular subjects should be collected. Several publishers should be represented for each grade level in major curriculum areas. This collection may reflect the texts used in the public schools in the region, and schools in which the teacher education students receive field placements. The scope and depth of each subject area should depend upon each institution's needs.
- Curriculum Guides/Courses Of Study—Curriculum guides, preschool through grade twelve, should be collected annually on the local, state, and national levels. All major curriculum areas should be represented, with emphasis on the certification programmes of the college/department of education of the institution.
- Children's And Young Adult Literature—This collection should include fiction, non-fiction, picture books, folk and fairy tales, plays, poetry, and graphic literature appropriate for preschool through grade twelve. The collection should be consistent with the recommendations of standard reviewing tools and include annual acquisition of award books and books from various notable books lists.

- Teaching Activity Materials—Professional teaching materials that provide ideas and activities for lesson planning and curricular development should be collected. All major curriculum areas and grade levels should be represented in accordance with the needs of the college/department of education.
- Reference Materials—Current reference materials, in print and electronic formats, should be acquired. These include materials related to other resources in the CMC (children's literature indexes and bibliographies, educational software directories, etc), as well as reference works intended for use by children and young adults.
- P-12 Magazines—Magazines intended for use by children and young adults should be included. Professional education periodicals that provide teaching ideas and review curriculum materials, educational media, and children's and young adult literature may also be represented.
- Media Materials—A variety of formats, in both traditional and emerging technologies, should be acquired annually. A range of curriculum concepts, skills, topics, and trends in P-12 curricula should be represented. Materials collected may include instructional games, posters, kits, transparencies, models, flat pictures, video recordings, sound recordings, computer-based instructional materials, and miscellaneous instructional materials such as puppets, manipulatives, rock collections, etc.
- Tests—Educational tests and measures that support education courses may be collected.
- Web Resources—The CMC Web site should include links to the vast array of online resources available to teaching professionals for lesson planning and curricular development. These resources should reflect both current and emerging technologies.

*Collection Development Policy*

The CMC should provide a written collection development policy that guides the selection and acquisition of materials.

- Mission Statement—The policy should reflect and support the mission of the curriculum materials center.
- Users—The policy should include a statement concerning those served by the curriculum materials center and the extent of that service.
- Collaboraton—The policy should be developed in collaboration with the education faculty.
- Objectives—The policy should identify the scope and objectives of the collection.
- Format—The policy should identify the formats in which materials are to be collected.
- Tools And Criteria—The policy should identify selection tools, criteria, and processes to be used in choosing materials.

- Categories And Balance—The policy should set forth the categories in which materials will be collected, such as textbooks, media materials, periodicals, etc., and give guidance for allocating budget resources among the categories.
- Compliance—The policy should address compliance with state standards and appropriate treatment of gender, racial, ethnic, and cultural issues. The policy may address maintenance of a collection of less appropriate materials for research and teaching purposes.
- Maintenance And Weeding—The policy should address regular maintenance of the collection and weeding as appropriate.

***Access–Physical***

*Organization*

The CMC collection should be displayed in an organized manner that makes it easily accessible to users.

- Arrangement—The collection should be arranged in the CMC in a systematic pattern with some materials inter-shelved while others are shelved as distinct collections within the CMC.
- Access—The collection should be organized in such a way as to make it physically and easily accessible and ADA compliant. All collection materials, except reserve or historic, should be openly available rather than remotely stored.
- Storage—The shelving should be appropriate for the various types, sizes and shapes of materials and sufficient to accommodate all items.
- Signage—Adequate and appropriate signage should be clearly posted and visible to direct CMC users to the various areas of the collection.

*Processing*

The CMC collection should be processed to promote easy access.

- Preservation—The collection items should be processed with appropriate reinforcement so that the items are preserved for multiple circulation transactions, yet convenient enough for easy access.
- Integrity Of Unit—Packaging of multiple-piece units should be sturdy and easily maintained to keep the various pieces intact and should be packaged with user access in mind; multiple-piece containers should be labeled with numbers and types of items contained within; when appropriate, individual pieces should be marked with identifying call numbers so that individual pieces can be readily returned to their appropriate container when separated.
- Item Labeling—Collection items should be clearly and consistently labeled to promote easy retrieval from shelving areas.
- Security—Theft detection devices should be used whenever possible.

*Circulation Policy*

The CMC should provide a written circulation policy.

- User Groups—The policy should identify the various user groups served, noting restrictions and privileges for each group.
- Circulation Periods—The policy should identify circulation periods and restrictions for each type of material.
- Penalties—The policy should state the penalties, if any, that are imposed.
- Other Policies—The policy should state other regulations, including but not limited to those concerning holds, recalls, interlibrary loan policies, and distance learning students.
- Automation—The policy should support or encourage use of an automated circulation system.

*Equipment*

The CMC should provide updated, appropriate equipment, in close proximity to the CMC non-print materials and in sufficient numbers to meet the needs of users to access all of the various non-print materials available in the collection.

- Appropriateness—Appropriate equipment should be provided so all types of non-print media in the CMC collection can be accessed.
- Quantity—A sufficient quantity of equipment should be maintained to meet typical demand levels.
- Location—The equipment should be in close proximity to the CMC non-print media collection so that access is convenient.
- Maintenance—The equipment should be regularly maintained and kept in good working condition, with a budget and technical support to ensure this.
- Updating—The equipment should be regularly updated to meet the needs of new technologies.
- Computer Hardware And Software—Appropriate computer hardware and software should be provided to support access to hypermedia applications. Hypermedia is defined as the amalgamation and/or use of graphics, audio, video, plain text, and hyperlinks to present information in a non-linear format.

***Access–Bibliographic***

*Cataloging*

The CMC collection should be cataloged in accordance with current national standards, including full subject access.

- Description And Subjects—The physical description of items should follow currently accepted models, including uniform information (title, author, etc.) and subject headings.
- Documentation—Local policies ands procedures as they relate to CMC materials should be documented in writing.

- Classification—The call numbers on items should follow the latest, most current edition of a nationally accepted classification scheme (*e.g.,* Dewey, LC). The choice of scheme and call numbers can be tailored to fit the CMC's needs. The CMC collection should be cataloged in a timely fashion, with sufficient levels of support.
- Cataloger Support—Because CMC materials often require longer cataloging time, a model timetable and dedicated time/librarian should be provided for cataloging.
- Equipment/Supplies—Sufficient equipment and supplies for cataloging and processing should be maintained.

*Indexing*

Bibliographic and holdings information about the CMC collection should exist on the same retrieval mechanism as other library materials.

- Electronic and Remote Access—The CMC (and its parent institution) should have, or strive to have, electronic and remote access to the collection, with sufficient terminals in the CMC.
- Indexes for Uncataloged Items—The CMC shall have indexes, preferably electronic, to access non-cataloged items (*e.g.,* curriculum guides on microfiche, etc.).

***Assessment***

The CMC should have a plan in place for evaluating the achievement of its mission and goals.

- Plan—The plan should focus on how well the CMC is meeting its goals and objectives relative to its collection, administration, facilities, and service.
- Frequency—The evaluation should take place on a periodic basis.
- Methodology—The method used could be accomplished through focus groups, surveys, questionnaires, or other evaluation strategies and should include participation by all user groups.
- Resources—A variety of published materials related to the management of CMCs are available and should be consulted regularly.
- Results—The results of the evaluation should be recorded and used in reviewing the viability of the current goals and objectives with changes being made where appropriate.

*Appendix I*

Adequate and appropriate documentation is vitally important to evaluation of the CMC. Following are examples of types of documentation that may be gathered to show compliance with the guidelines.

- Budget reports

- Calendars
- Collection development policy
- Floor plans
- Inventories
- Policies and procedures
- Publication examples (handouts/bibliographies/pathfinders)
- Publicity materials
- Schedules
- Statistics
-
  a. Reference statistics
  b. User statistics
  c. Usage statistics
- Web sites

*Appendix II*

Bibliography of resources that are recommended for consultation by CMC directors.

Carr, J. (Ed.). (2001). *A guide to the management of curriculum materials centers for the 21st century: The promise and the challenge*. Chicago: IL: Association of College and Research Libraries, American Library Association.

Curriculum Materials Committee of the Education and Behavioural Sciences Section. (2007). *A guide to writing CMC collection development policies*. Chicago, IL: Association of College and Research Libraries, American Library Association. Retrieved from: http://www.ala.org/ala/mgrps/divs/acrl/acrlpubs/downloadables/guidetowritingcmc.pdf.

Lare, G. (2004). *Acquiring and organizing curriculum materials: A guide and directory of resources*. (2nd Ed.). Lanham, MD: Scarecrow Press.

Olive, F. (Ed.) (2001). *Directory of curriculum materials centers*. Association of College and Research Libraries, American Library Association. Retrieved from: http://acrl.telusys.com/cmc/index (New edition forthcoming.)

**Textbooks**

A textbook or coursebook (UK English) is a manual of instruction in any branch of study. Textbooks are produced according to the demands of educational institutions. Schoolbooks are textbooks and other books used in schools. Although most textbooks aren't only published in printed format, many are now available as online electronic books.

The ancient Greeks wrote *texts* intended for education. The modern textbook has its roots in the standardization made possible by the printing press. Johannes Gutenberg himself may have printed editions of Ars Minor, a schoolbook on Latin grammar by Aelius Donatus. Early textbooks were used by tutors and teachers, who used the books as instructional aids (*e.g.,* alphabet books), as well as individuals who taught themselves.

The Greek philosopher Plato lamented the loss of knowledge because the media of transmission were changing. Before the invention of the Greek alphabet 2,500 years ago, knowledge and stories were recited aloud, much like Homer's epic poems. The new technology of writing meant stories no longer needed to be memorized, a development Socrates feared would weaken the Greeks' mental capacities for memorizing and retelling. (Ironically, we know about Socrates' concerns only because they were written down by his student Plato in his famous Dialogues.)

The next revolution for books came with the 15th-century invention of printing with changeable type. The invention is attributed to German metalsmith Johannes Gutenberg, who cast type in molds using a melted metal alloy and constructed a wooden-screw printing press to transfer the image onto paper.

Gutenberg's first and only large-scale printing effort was the now iconic Gutenberg Bible in the 1450s — a Latin translation from the Hebrew Old Testament and the Greek New Testament, copies of which can be viewed on the British Library web site. Gutenberg's invention made mass production of texts possible for the first time. Although the Gutenberg Bible itself was expensive, printed books began to spread widely over European trade routes during the next 50 years, and by the 16th century, printed books had become more widely accessible and less costly. Compulsory education and the subsequent growth of schooling in Europe led to the printing of many standardized texts for children. Textbooks have become the primary teaching instrument for most children since the 19th century. Two textbooks of historical significance in United States schooling were the 18th century New England Primer and the 19th century McGuffey Readers.

Technological advances change the way people interact with textbooks. Online and digital materials are making it increasingly easy for students to access materials other than the traditional print textbook. Students now have access to electronic and PDF books, online tutoring systems and video lectures. An example of an electronically published book, or e-book, is *Principles of Biology* from Nature Publishing.

Most notably, an increasing number of authors are foregoing commercial publishers and offering their textbooks under a creative commons or other open license.

**Journals**

A journal (through French from Latin *diurnalis*, daily) has several related meanings:

- A record of events or business; a private journal is usually referred to as a diary
- A newspaper or other periodical, in the literal sense of one published each day
- Many publications issued at stated intervals, such as academic journals (including scientific journals), or the record of the transactions of a society, are often called journals. In academic use, a journal refers to a serious, scholarly publication that is peer-reviewed. A non-scholarly magazine written for an educated audience about an industry or an area of professional activity is usually called a trade magazine.

The word “journalist”, for one whose business is writing for the public press and nowadays also other media, has been in use since the end of the 17th century.

***Public Journal***

A public journal is a record of day-by-day events in a parliament or congress. It is also called minutes or records.

***Business and Accounting***

The term “journal” is also used in business:

- A journal is a book or computer file in which monetary transactions are entered the first time they are processed. This journal lists transactions in chronological sequence by date prior to a transfer of the same transactions to a ledger in the process of bookkeeping
- Narrations or equivalent to a ship’s log, as a record of the daily run, such as observations, weather changes, or other events of daily importance

**Dictionaries**

A dictionary is a collection of words in one or more specific languages, often arranged alphabetically (or by radical and stroke for ideographic languages), which may include information on definitions, usage, etymologies, phonetics, pronunciations, translation, etc. or a book of words in one language with their equivalents in another, sometimes known as a lexicon. It is a lexicographical product which shows inter-relationships among the data.

A broad distinction is made between general and specialized dictionaries. Specialized dictionaries include words in specialist fields, rather than a complete range of words in the language. Lexical items that describe concepts in specific fields are usually called terms instead of words, although there is no consensus whether lexicology and terminology are two different fields of study. In theory, general dictionaries are supposed to be semasiological, mapping word to definition, while specialized dictionaries are supposed to be onomasiological, first identifying concepts and then establishing the terms used to designate them. In practice, the two approaches are used for both types. There are other types of dictionaries that do not fit neatly into the above distinction, for instance bilingual (translation) dictionaries, dictionaries of synonyms (thesauri), and rhyming dictionaries. The word dictionary (unqualified) is usually understood to refer to a general purpose monolingual dictionary.

There is also a contrast between *prescriptive* or *descriptive* dictionaries; the former reflect what is seen as correct use of the language while the latter reflect recorded actual use. Stylistic indications (*e.g.,* “informal” or “vulgar”) in many modern dictionaries are also considered by some to be less than objectively descriptive.

Although the first recorded dictionaries date back to Sumerian times (these were bilingual dictionaries), the systematic study of dictionaries as objects of scientific interest themselves is a 20th-century enterprise, called lexicography, and largely initiated by Ladislav Zgusta. The birth of the new discipline was not without controversy, the practical dictionary-makers being sometimes accused by others of "astonishing" lack of method and critical-self reflection.

***History***

The oldest known dictionaries were Akkadian Empire cuneiform tablets with bilingual Sumerian–Akkadian wordlists, discovered in Ebla (modern Syria) and dated roughly 2300 BCE. The early 2nd millennium BCE *Urra=hubullu* glossary is the canonical Babylonian version of such bilingual Sumerian wordlists. A Chinese dictionary, the c. 3rd century BCE *Erya*, was the earliest surviving monolingual dictionary; although some sources cite the c. 800 BCE Shizhoupian as a "dictionary", modern scholarship considers it a calligraphic compendium of Chinese characters from Zhou dynasty bronzes. Philitas of Cos (fl. 4th century BCE) wrote a pioneering vocabulary *Disorderly Words* (ôáêôïé ãëóóáé, *Átaktoi glôssai*) which explained the meanings of rare Homeric and other literary words, words from local dialects, and technical terms. Apollonius the Sophist (fl. 1st century CE) wrote the oldest surviving Homeric lexicon. The first Sanskrit dictionary, the Amarakoœa, was written by Amara Sinha c. 4th century CE. Written in verse, it listed around 10,000 words. According to the *Nihon Shoki*, the first Japanese dictionary was the long-lost 682 CE *Niina* glossary of Chinese characters. The oldest existing Japanese dictionary, the c. 835 CE *Tenrei Banshô Meigi*, was also a glossary of written Chinese. A 9th-century CE Irish dictionary, Sanas Cormaic, contained etymologies and explanations of over 1,400 Irish words. In India around 1320, Amir Khusro compiled the Khaliq-e-bari which mainly dealt with Hindavi and Persian words.

Arabic dictionaries were compiled between the 8th and 14th centuries CE, organizing words in rhyme order (by the last syllable), by alphabetical order of the radicals, or according to the alphabetical order of the first letter (the system used in modern European language dictionaries). The modern system was mainly used in specialist dictionaries, such as those of terms from the Qur'an and hadith, while most general use dictionaries, such as the *Lisan al-'Arab* (13th century, still the best-known large-scale dictionary of Arabic) and *al-Qamus al-Muhit* (14th century) listed words in the alphabetical order of the radicals. The *Qamus al-Muhit* is the first handy dictionary in Arabic, which includes only words and their definitions, eliminating the supporting examples used in such dictionaries as the *Lisan* and the *Oxford English Dictionary*.

In medieval Europe, glossaries with equivalents for Latin words in vernacular or simpler Latin were in use (*e.g.,* the Leiden Glossary). The *Catholicon* (1287) by Johannes Balbus, a large grammatical work with an alphabetical lexicon, was widely adopted. It served as the basis for several bilingual dictionaries and was one of the earliest books (in 1460) to be printed. In 1502 Ambrogio Calepino's *Dictionarium* was published, originally a monolingual Latin dictionary, which over the course of the 16th century was enlarged to become a multilingual glossary. In 1532 Robert Estienne published the

*Thesaurus linguae latinae* and in 1572 his son Henri Estienne published the *Thesaurus linguae graecae*, which served up to the 19th century as the basis of Greek lexicography. The first monolingual dictionary written in Europe was the Spanish, written by Sebastián Covarrubias' *Tesoro de la lengua castellana o española*, published in 1611 in Madrid, Spain. In 1612 the first edition of the *Vocabolario dell'Accademia della Crusca*, for Italian, was published. It served as the model for similar works in French and English. In 1690 in Rotterdam was published, posthumously, the *Dictionnaire Universel* by Antoine Furetière for French. In 1694 appeared the first edition of the *Dictionnaire de l'Académie française*. Between 1712 and 1721 was published the *Vocabulario portughez e latino* written by Raphael Bluteau. The Real Academia Española published the first edition of the *Diccionario de la lengua española* in 1780, but their *Diccionario de Autoridades*, which included quotes taken from literary works, was published in 1726. The *Totius Latinitatis lexicon* by Egidio Forcellini was firstly published in 1777; it has formed the basis of all similar works that have since been published.

The first edition of *A Greek-English Lexicon* by Henry George Liddell and Robert Scott appeared in 1843; this work remained the basic dictionary of Greek until the end of the 20th century. And in 1858 was published the first volume of the Deutsches Wörterbuch by the Brothers Grimm; the work was completed in 1961. Between 1861 and 1874 was published the *Dizionario della lingua italiana* by Niccolò Tommaseo. Between 1862 and 1874 was published the six volumes of *A magyar nyelv szótára* (Dictionary of Hungarian Language) by Gergely Czuczor and János Fogarasi. Émile Littré published the Dictionnaire de la langue française between 1863 and 1872. In the same year 1863 appeared the first volume of the *Woordenboek der Nederlandsche Taal* which was completed in 1998. Also in 1863 Vladimir Ivanovich Dahl published the *Explanatory Dictionary of the Living Great Russian Language*. The Duden dictionary dates back to 1880, and is currently the prescriptive source for the spelling of German. The decision to start work on the *Svenska Akademiens ordbok* was taken in 1787.

*English Dictionaries in Britain*

The earliest dictionaries in the English language were glossaries of French, Spanish or Latin words along with their definitions in English. The word "dictionary" was invented by an Englishman called John of Garland in 1220 — he had written a book *Dictionarius* to help with Latin "diction". An early non-alphabetical list of 8000 English words was the *Elementarie*, created by Richard Mulcaster in 1582.

The first purely English alphabetical dictionary was *A Table Alphabeticall*, written by English schoolteacher Robert Cawdrey in 1604. The only surviving copy is found at the Bodleian Library in Oxford. This edictionary, and the many imitators which followed it, was seen as unreliable and nowhere near definitive. Philip Stanhope, 4th Earl of Chesterfield was still lamenting in 1754, 150 years after Cawdrey's publication, that it is "a sort of disgrace to our nation, that hitherto we have had no… standard of our language; our dictionaries at present being more properly what our neighbours the Dutch and the Germans call theirs, word-books, than dictionaries in the superior sense of that title." In 1616, John Bullokar described the history of the dictionary with his "English Expositor". *Glossographia* by Thomas Blount, published in 1656, contains more than 10,000 words along with their etymologies or histories. Edward Phillips wrote another dictionary in 1658, entitled "The New World of English Words: Or a General Dictionary" which boldly plagiarized Blount's work, and the two renounced each other. This created more interest in the dictionaries. John Wilkins' 1668 essay on philosophical language contains a list of 11,500 words with careful distinctions, compiled by William Lloyd. Elisha Coles published his "English Dictionary" in 1676.

It was not until Samuel Johnson's *A Dictionary of the English Language* (1755) that a more reliable English dictionary was produced. Many people today mistakenly believe that Johnson wrote the first English dictionary: a testimony to this legacy. By this stage, dictionaries had evolved to contain textual references for most words, and were arranged alphabetically, rather than by topic (a previously popular form of arrangement, which meant all animals would be grouped together, etc.). Johnson's masterwork could be judged as the first to bring all these elements together, creating the first "modern" dictionary.

Johnson's dictionary remained the English-language standard for over 150 years, until the Oxford University Press began writing and releasing the *Oxford English Dictionary* in short fascicles from 1884 onwards. It took nearly 50 years to complete this huge work, and they finally released the complete *OED* in twelve volumes in 1928. It remains the most comprehensive and trusted English language dictionary to this day, with revisions and updates added by a dedicated team every three months. One of the main contributors to this modern dictionary was an ex-army surgeon, William Chester Minor, a convicted murderer who was confined to an asylum for the criminally insane.

*American English Dictionaries*

In 1806, American Noah Webster published his first dictionary, *A Compendious Dictionary of the English Language*. In 1807 Webster began compiling an expanded and fully comprehensive dictionary, *An American Dictionary of the English Language;* it took twenty-seven years to complete. To evaluate the etymology of words, Webster learned twenty-six languages, including Old English (Anglo-Saxon), German, Greek, Latin, Italian, Spanish, French, Hebrew, Arabic, and Sanskrit.

Webster completed his dictionary during his year abroad in 1825 in Paris, France, and at the University of Cambridge. His book contained seventy thousand words, of which twelve thousand had never appeared in a published dictionary before. As a spelling reformer, Webster believed that English spelling rules were unnecessarily complex, so his dictionary introduced American English spellings, replacing "colour" with "colour", substituting "wagon" for "waggon", and printing "center" instead of "centre". He also added American words, like "skunk" and "squash", that did not appear in British dictionaries. At the age of seventy, Webster published his dictionary in 1828; it sold

2500 copies. In 1840, the second edition was published in two volumes. Austin (2005) explores the intersection of lexicographical and poetic practices in American literature, and attempts to map out a “lexical poetics” using Webster’s definitions as his base. He explores how American poets used Webster’s dictionaries, often drawing upon his lexicography in order to express their word play. Austin explicates key definitions from both the *Compendious* (1806) and *American* (1828) dictionaries, and brings into its discourse a range of concerns, including the politics of American English, the question of national identity and culture in the early moments of American independence, and the poetics of citation and of definition. Austin concludes that Webster’s dictionaries helped redefine Americanism in an era of an emergent and unstable American political and cultural identity. Webster himself saw the dictionaries as a nationalizing device to separate America from Britain, calling his project a “federal language”, with competing forces towards regularity on the one hand and innovation on the other. Austin suggests that the contradictions of Webster’s lexicography were part of a larger play between liberty and order within American intellectual discourse, with some pulled towards Europe and the past, and others pulled towards America and the new future.

***Types***

In a general dictionary, each word may have multiple meanings. Some dictionaries include each separate meaning in the order of most common usage while others list definitions in historical order, with the oldest usage first.

In many languages, words can appear in many different forms, but only the undeclined or unconjugated form appears as the headword in most dictionaries. Dictionaries are most commonly found in the form of a book, but some newer dictionaries, like StarDict and the *New Oxford American Dictionary* are dictionary software running on PDAs or computers. There are also many online dictionaries accessible via the Internet.

*Specialized Dictionaries*

According to the *Manual of Specialized Lexicographies* a specialized dictionary (also referred to as a technical dictionary) is a dictionary that focuses upon a specific subject field. Following the description in *The Bilingual LSP Dictionary*, lexicographers categorize specialized dictionaries into three types. A multi-field dictionary broadly covers several subject fields (*e.g.,* a business dictionary), a single-field dictionary narrowly covers one particular subject field (*e.g.,* law), and a sub-field dictionary covers a more specialised field (*e.g.,* constitutional law). For example, the 23-language Inter-Active Terminology for Europe is a multi-field dictionary, the American National Biography is a single-field, and the African American National Biography Project is a sub-field dictionary. In terms of the above coverage distinction between “minimizing dictionaries” and “maximizing dictionaries”, multi-field dictionaries tend to minimize coverage across subject fields (for instance, *Oxford Dictionary of World Religions* and *Yadgar Dictionary of Computer and Internet Terms*) whereas single-field and sub-field dictionaries tend to maximize coverage within a limited subject field *(The Oxford Dictionary of English Etymology*). A 'phonetic dictionary is one] that allows the reader to locate words by the "way they sound", *i.e.,* a dictionary that matches common or phonetic misspellings with the correct spelling of a word; such a dictionary uses pronunciation respelling to aid the search for or recognition of a word, and differs from a pronouncing dictionary, such as the CMU Pronouncing Dictionary, which maps from a written word to its pronunciation in a chosen dialect.

Another variant is the glossary, an alphabetical list of defined terms in a specialised field, such as medicine (medical dictionary).

*Defining Dictionaries*

The simplest dictionary, a defining dictionary, provides a core glossary of the simplest meanings of the simplest concepts. From these, other concepts can be explained and defined, in particular for those who are first learning a language. In English, the commercial defining dictionaries typically include only one or two meanings of under 2000 words. With these, the rest of English, and even the 4000 most common English idioms and metaphors, can be defined.

*Prescriptive vs. Descriptive*

Lexicographers apply two basic philosophies to the defining of words: *prescriptive* or *descriptive*. Noah Webster, intent on forging a distinct identity for the American language, altered spellings and accentuated differences in meaning and pronunciation of some words. This is why American English now uses the spelling *colour* while the rest of the English-speaking world prefers *colour*. (Similarly, British English subsequently underwent a few spelling changes that did not affect American English; see further at American and British English spelling differences.)

Large 20th-century dictionaries such as the *Oxford English Dictionary* (OED) and *Webster's Third* are descriptive, and attempt to describe the actual use of words. Most dictionaries of English now apply the descriptive method to a word's definition, and then, outside of the definition itself, add information alerting readers to attitudes which may influence their choices on words often considered vulgar, offensive, erroneous, or easily confused. *Merriam-Webster* is subtle, only adding italicized notations such as, *sometimes offensive* or *non-stand* (non-standard). *American Heritage* goes further, discussing issues separately in numerous "usage notes." *Encarta* provides similar notes, but is more prescriptive, offering warnings and admonitions against the use of certain words considered by many to be offensive or illiterate, such as, "an offensive term for..." or "a taboo term meaning...".

Because of the widespread use of dictionaries in schools, and their acceptance by many as language authorities, their treatment of the language does affect usage to some degree, with even the most descriptive dictionaries providing conservative continuity. In the long run, however, the meanings of words in English are primarily determined by usage, and the language is being changed and created every day. As Jorge Luis Borges says in the prologue to "El otro, el mismo": "*It is often forgotten that (dictionaries) are artificial repositories, put together well after the languages they define. The roots of language are irrational and of a magical nature.*"

Sometimes the same dictionary can be descriptive in some domains and prescriptive in others. For example, according to Ghil'ad Zuckermann, the *Oxford English-Hebrew Dictionary* is "at war with itself": whereas its coverage (lexical items) and glosses (definitions) are descriptive and colloquial, its vocalization is prescriptive. This internal conflict results in absurd sentences such as *hi taharóg otí kshetiré me asíti lamkhonít* (she'll tear me apart when she sees what I've done to the car). Whereas *hi taharóg otí*, literally 'she will kill me', is colloquial, me (a variant of ma 'what') is archaic, resulting in a combination that is unutterable in real life.

*Dictionaries for Natural Language Processing*

In contrast to traditional dictionaries, which are designed to be used by human beings, dictionaries for natural language processing (NLP) are built to be used by computer programmes. The final user is a human being but the direct user is a programme. Such a dictionary does not need to be able to be printed on paper. The structure of the content is not linear, ordered entry by entry but has the form of a complex network. Because most of these dictionaries are used to control machine translations or cross-lingual information retrieval (CLIR) the content is usually multilingual and usually of huge size. In order to allow formalized exchange and merging of dictionaries, an ISO standard called Lexical Markup Framework (LMF) has been defined and used among the industrial and academic community.

*Other Types*

- Bilingual dictionary
- Electronic dictionary
- Encyclopedic dictionary
- Monolingual learner's dictionary
  - Advanced learner's dictionary
- By sound
  - Phonetic dictionary
  - Rhyming dictionary
- Reverse dictionary
- Visual dictionary
- Satirical dictionary

**Thesis**

A thesis or dissertation is a document submitted in support of candidature for an academic degree or professional qualification presenting the author's research and findings. In some contexts, the word “thesis” or a cognate is used for part of a bachelor’s or master’s course, while “dissertation” is normally applied to a doctorate, while in other contexts, the reverse is true. The term *graduate thesis* is sometimes used to refer to both master’s theses and doctoral dissertations.

The required complexity or quality of research of a thesis or dissertation can vary by country, university, or programme, and the required minimum study period may thus vary significantly in duration.

The word dissertation can at times be used to describe a treatise without relation to obtaining an academic degree. The term thesis is also used to refer to the general claim of an essay or similar work.

### *Structure and Presentation Style*

#### *Structure*

A thesis (or dissertation) may be arranged as a thesis by publication or a monograph, with or without appended papers, respectively. An ordinary monograph has a title page, an abstract, a table of contents, comprising the various chapters (introduction, literature review, findings, etc.), and a bibliography or (more usually) a references section. They differ in their structure in accordance with the many different areas of study (arts, humanities, social sciences, technology, sciences, etc.) and the differences between them. In a thesis by publication, the chapters constitute an introductory and comprehensive review of the appended published and unpublished article documents. Dissertations normally report on a research project or study, or an extended analysis of a topic.

The structure of the thesis or dissertation explains the purpose, the previous research literature which impinges on the topic of the study, the methods used and the findings of the project. Most world universities use a multiple chapter format: a) an introduction, which introduces the research topic, the methodology, as well as its scope and significance; b) a literature review, reviewing relevant literature and showing how this has informed the research issue; c) a methodology chapter, explaining how the research has been designed and why the research methods/population/data collection and analysis being used have been chosen; d) a findings chapter, outlining the findings of the research itself; e) an analysis and discussion chapter, analysing the findings and discussing them in the context of the literature review (this chapter is often divided into two—analysis and discussion); f) a conclusion.

#### *Style*

Degree-awarding institutions often define their own house style that candidates have to follow when preparing a thesis document. In addition to institution-specific house styles, there exist a number of field-specific, national, and international standards and recommendations for the presentation of theses, for instance ISO 7144. Other applicable international standards include ISO 2145 on section numbers, ISO 690 on bibliographic references, and ISO 31 on quantities or units.

Some older house styles specify that front matter (title page, abstract, table of content, etc.) uses a separate page-number sequence from the main text, using Roman numerals. The relevant international standard and many newer style guides recognize that this book design practice can cause confusion where electronic document viewers number all pages of a document continuously from the first page, independent of any printed page numbers. They therefore avoid the traditional separate number sequence for front matter and require a single sequence of Arabic numerals starting with 1 for the first printed page (the recto of the title page).

Presentation requirements, including pagination, layout, type and colour of paper, use of acid-free paper (where a copy of the dissertation will become a permanent part of the library collection), paper size, order of components, and citation style, will be checked page by page by the accepting officer before the thesis is accepted and a receipt is issued.

However, strict standards are not always required. Most Italian universities, for example, have only general requirements on the character size and the page formatting, and leave much freedom on the actual typographic details.

### *Thesis Committee*

A thesis or dissertation committee is a committee that supervises a student's dissertation. These committees, at least in the US model, usually consist of a primary supervisor or advisor and two or more committee members, who supervise the progress of the dissertation and may also act as the examining committee, or jury, at the oral examination of the thesis.

At most universities, the committee is chosen by the student in conjunction with his or her primary adviser, usually after completion of the comprehensive examinations or prospectus meeting, and may consist of members of the comps committee. The committee members are doctors in their field (whether a PhD or other designation) and have the task of reading the dissertation, making suggestions for changes and improvements, and sitting in on the defence. Sometimes, at least one member of the committee must be a professor in a department that is different from that of the student.

### *Regional and Degree-specific Practices and Terminologies*

#### *Argentina*

In the *Latin American docta*, the academic dissertation can be referred to as different stages inside the academic programme that the student is seeking to achieve into a recognized Argentine University, in all the cases the students must develop original contribution in the chosen fields by means of several paper work and essays that comprehend the body of the *thesis*. Correspondingly to the academic degree, the last phase of an academic thesis is called in Spanish a *defensa de grado*, *defensa magistral* or *defensa doctoral* in cases in which the university candidate is finalizing his or her licentiate, master's, or PhD programme. According to a committee resolution, the dissertation can be approved or rejected by an academic committee consisting of the thesis director, the thesis coordinator, and at least one evaluator from another recognized university in which the student is pursuing his or her academic programme. All the dissertation referees must already have achieved at least the academic degree that the candidate is trying to reach.

*Canada*

At English-speaking Canadian universities, writings presented in fulfillment of undergraduate coursework requirements are normally called *papers*, *term papers* or *essays*. A longer paper or essay presented for completion of a 4-year bachelor's degree is sometimes called a *major paper*. High-quality research papers presented as the empirical study of a "postgraduate" consecutive bachelor *with Honours* or Baccalaureatus Cum Honore degree are called *thesis* (Honours Seminar Thesis). Major papers presented as the final project for a master's degree are normally called *thesis*; and major papers presenting the student's research towards a doctoral degree are called *theses* or *dissertations*.

At Canadian universities under the French influenced system, students may have a choice between presenting a "*mémoire*"', which is a shorter synthetic work (roughly 75 pages) and a *thèse* which is one hundred pages or more. A synthetic monograph associated with doctoral work is referred to as a "*thèse*". Either work can be awarded a "*mention d'honneur*" (excellence) as a result of the decision by the examination committee, although these are rare.

A typical undergraduate paper or essay might be forty pages. Master's theses are approximately one hundred pages. PhD theses are usually over two hundred pages. This may vary greatly by discipline, programme, college, or university. However, normally the required minimum study period is primarily depending on the complexity or quality of research requirements. Theses Canada acquires and preserves a comprehensive collection of Canadian theses at Library and Archives Canada' (LAC) through partnership with Canadian universities who participate in the programme.

*Croatia*

At most university faculties in Croatia, a degree is obtained by defending a thesis after having passed all the classes specified in the degree programme. In the Bologna system, the bachelor's thesis, called *završni rad* (literally "final work" or "concluding work") is usually defended after 3 years of study and is about 30 pages long. Most students with bachelor's degrees continue onto master's programmes which end with a thesis called *diplomski rad*. The term dissertation is used for a doctoral degree paper *(doktorska disertacija*).

*Czech Republic*

In the Czech Republic, higher education is completed by passing all classes remaining to the educational compendium for given degree and defending a thesis. For bachelors programme the thesis is called *bakaláøská práce* (bachelors thesis), for master's degrees and also doctor of medicine or dentistry degrees it is the *diplomová práce* (master's thesis), and for Philosophiae doctor (PhD.) degree it is dissertation *dizertaèní práce*. Thesis for so called Higher-Professional School (Vyšší odborná škola, VOŠ) is called *absolventská práce*.

*Finland*

The following types of thesis are used in Finland (names in Finnish/Swedish):

- *Kandidaatintutkielma/kandidatavhandling* is the dissertation associated with lower-level academic degrees (bachelor's degree), and at universities of applied science.
- *Pro gradu (-tutkielma)/ (avhandling) pro gradu*, colloquially referred to simply as 'gradu', is the dissertation for master's degrees, which make up the majority of degrees conferred in Finland, and this is therefore the most common type of thesis submitted in the country. The equivalent for engineering and architecture students is *diplomityö/diplomarbete*.
- The highest-level theses are called *lisensiaatintutkielma/licentiatavhandling* and *(tohtorin) väitöskirja/doktorsavhandling*, for licentiate and doctoral degrees, respectively.

*France*

In France, the academic dissertation or thesis is called a *thèse* and it is reserved for the final work of doctoral candidates. The minimum page length is generally (and not formally) 100 pages (or about 400,000 characters), but is usually several times longer (except for technical theses and for "exact sciences" such as physics and maths).

To complete a master's degree in research, a student is required to write a *mémoire*, the French equivalent of a master's thesis in other higher education systems.

The word *dissertation* in French is reserved for shorter (1,000–2,000 words), more generic academic treatises.

The defence is called a *soutenance*.

*Germany*

In Germany, an academic thesis is called *Abschlussarbeit* or, more specifically, the basic name of the degree complemented by *-arbeit* (*e.g., Diplomarbeit*, *Masterarbeit*, *Doktorarbeit*). For bachelor's and master's degrees, the name can alternatively be complemented by *-thesis* instead (*e.g., Bachelorthesis*).

Length is often given in page count and depends upon departments, faculties, and fields of study. A bachelor's thesis is often 40–60 pages long, a diploma thesis and a master's thesis usually 60–100. The required submission for a doctorate is called a *Dissertation* or *Doktorarbeit*. The submission for a Habilitation, which is an academic qualification, not an academic degree, is called *Habilitationsschrift*, not *Habilitationsarbeit*. PhD by publication is becoming increasingly common in many fields of study.

A doctoral degree is often earned with multiple levels of a Latin honours remark for the thesis ranging from *summa cum laude* (best) to *rite* (duly). A thesis can also be rejected with a Latin remark *(non-rite*, *non-sufficit* or worst as *sub omni canone*). Bachelor's and master's theses receive numerical grades from 1.0 (best) to 5.0 (failed).

*India*

In India the thesis defence is called a *viva voce* (Latin for "by live voice") examination *(viva* in short). Involved in the *viva* are two examiners and the candidate. One examiner is an academic from the candidate's own university department (but not one of the candidate's supervisors) and the other is an external examiner from a different university.

In India, PG Qualifications such as MSc Physics accompanies submission of dissertation in Part I and submission of a Project (a working model of an innovation) in Part II. Engineering qualifications such as Diploma, BTech or B.E., MTech or M.Des also involves submission of dissertation. In all the cases, the dissertation can be extended for summer internship at certain research and development organizations or also as PhD synopsis.

*Indonesia*

In Indonesia, the term thesis is used specifically to refer to master's theses. The undergraduate thesis is called *skripsi*, while the doctoral dissertation is called *disertasi*. In general, those three terms are usually called as *tugas akhir* (final assignment), which is mandatory for the completion of a degree. Undergraduate students usually begin to write their final assignment in their third, fourth or fifth enrollment year, depends on the requirements of their respective disciplines and universities. In some universities, students are required to write a *proposal skripsi*, *proposal thesis* or thesis proposal before they could write their final assignment. If the thesis proposal is considered to fulfill the qualification by the academic examiners, students then may proceed to write their final assignment.

*Italy*

In Italy there are normally three types of thesis. In order of complexity: one for the Laurea (equivalent to the UK Bachelor's Degree), another one for the Laurea Magistrale (equivalent to the UK Master's Degree) and then a thesis to complete the Dottorato di Ricerca (PhD). Thesis requirements vary greatly between degrees and disciplines, ranging from as low as 3–4 ECTS credits to more than 30. Thesis work is mandatory for the completion of a degree.

*Malaysia*

Malaysian universities often follow the British model for dissertations and degrees. However, a few universities follow the United States model for theses and dissertations. Some public universities have both British and US style PhD programmes. Branch campuses of British, Australian and Middle East universities in Malaysia use the respective models of the home campuses.

*Pakistan*

In Pakistan, at undergraduate level the thesis is usually called final year project, as it is completed in the senior year of the degree, the name project usually implies that the work carried out is less extensive than a thesis and bears lesser credit hours too. The undergraduate level project is presented through an elaborate written report and a presentation to the advisor, a board of faculty members and students. At graduate level however, *i.e.,* in MS, some universities allow students to accomplish a project of 6 credits or a thesis of 9 credits, at least one publication is normally considered enough for the awarding of the degree with project and is considered mandatory for the awarding of a degree with thesis. A written report and a public thesis defence is mandatory, in the presence of a board of senior researchers, consisting of members from an outside organization or a university. A PhD candidate is supposed to accomplish extensive research work to fulfill the dissertation requirements with international publications being a mandatory requirement. The defence of the research work is done publicly.

*Philippines*

In the Philippines, an academic thesis is named by the degree, such as bachelor/ undergraduate thesis or masteral thesis. However, in Philippine English, the term *doctorate* is typically replaced with *doctoral* (as in the case of “doctoral dissertation”), though in official documentation the former is still used. The terms *thesis* and *dissertation* are commonly used interchangeably in everyday language yet it generally understood that a *thesis* refers to bachelor/undergraduate and master academic work while a *dissertation* is named for doctorate work.

The Philippine system is influenced by American collegiate system, in that it requires a *research project* to be submitted before being allowed to write a thesis. This is mostly given as a prerequisite writing course to the actual thesis and is accomplished in the term period before; supervision is provided by one professor assigned to a class. This is later to be presented in front of an academic panel, often the entire faculty of an academic department, with their recommendations contributing to the acceptance, revision, or rejection of the initial topic. In addition, the presentation of the research project will help the candidate choose their primary thesis adviser.

An undergraduate thesis is completed in the final year of the degree alongside existing seminar (lecture) or laboratory courses, and is often divided into two presentations: *proposal* and *thesis* presentations (though this varies across universities), whereas a master thesis or doctorate dissertation is accomplished in the last term alone and is defended once. In most universities, a thesis is required for the bestowment of a degree to a candidate alongside a number of units earned throughout their academic period of stay, though for practice and skills-based degrees a *practicum* and a written report can be achieved instead. The examination board often consists of 3 to 5 examiners, often professors in a university (with a Masters or PhD degree) depending on the university's examination rules. Required word length, complexity, and contribution to scholarship varies widely across universities in the country.

*Poland*

In Poland, a bachelor's degree usually requires a *praca licencjacka* (bachelor's thesis) or the similar level degree in engineering requires a *praca inzynierska* (engineer's thesis/ bachelor's thesis), the master's degree requires a *praca magisterska* (master's thesis). The academic dissertation for a PhD is called a *dysertacja* or *praca doktorska*. The submission for the Habilitation is called *praca habilitacyjna" or* dysertacja habilitacyjna". Thus the term *dysertacja* is reserved for PhD and Habilitation degrees. All the theses need to be "defended" by the author during a special examination for the given degree. Examinations for PhD and Habilitation degrees are public.

*Portugal and Brazil*

In Portugal and Brazil, a dissertation *(dissertação*) is required for completion of a master or PhD degree. The defence is done in a public presentation in which teachers, students, and the general public can participate. For the PhD a thesis *(tese*) is presented for defence in a public exam. The exam typically extends over 3 hours. The examination board typically involves 5 to 6 scholars (including the advisor) or other experts with a PhD degree (generally at least half of them must be external to the university where the candidate defends the thesis, but may depend on the University). Each university/ faculty defines the length of these documents, but typical numbers of pages are around 60–80 for MSc and 150–250 for PhD.

*Russia, Kazakhstan, Belarus, Ukraine*

In Russia, Kazakhstan, Belarus, and Ukraine an academic dissertation or thesis is called what can be literally translated as a "master's degree work" (thesis), whereas the word *dissertation* is reserved for doctoral theses (Candidate of Sciences). To complete a master's degree, a student is required to write a thesis and to then defend the work publicly. Length of this manuscript usually is given in page count and depends upon educational institution, its departments, faculties, and fields of study.

*Slovenia*

At universities in Slovenia, an academic thesis called *diploma thesis* is a prerequisite for completing undergraduate studies. The thesis used to be 40–60 pages long, but has been reduced to 20–30 pages in new Bologna process programmes. To complete Master's studies, a candidate must write *magistrsko delo* (Master's thesis) that is longer and more detailed than the undergraduate thesis. The required submission for the doctorate is called *doktorska disertacija* (doctoral dissertation). In pre Bologna programmes students were able to skip the preparation and presentation of a Master's thesis and continue straightforward towards doctorate.

*Slovakia*

In Slovakia, higher education is completed by defending a thesis, which is called bachelors thesis "bakalárska práca" for bachelors programme, master's thesis or "diplomová práca" for master's degrees and also doctor of medicine or dentistry degrees and dissertation "dizertaèná práca" for Philosophiae doctor (PhD.) degree.

*Sweden*

In Sweden, there are different types of theses. Practices and definitions vary between fields but commonly include the *C thesis*/Bachelor thesis, which corresponds to 15 HP or 10 weeks of independent studies, *D thesis*/' /Magister/one year master's thesis, which corresponds to 15 HP or 10 weeks of independent studies and *E Thesis*/two-year master's thesis, which corresponds to 30 HP or 20 weeks of independent studies.

After that there are two types of post graduate degrees, Licentiate dissertation and PhD dissertation. A licentiate degree is approximately "half a PhD" in terms of size and scope of the thesis. Swedish PhD studies should in theory last for four years, including course work and thesis work, but as many PhD students also teach, the PhD often takes longer to complete.

*United Kingdom*

Outside the academic community, the terms *thesis* and *dissertation* are interchangeable. At universities in the United Kingdom, the term *thesis* is usually associated with PhD/EngD (doctoral) and research master's degrees, while *dissertation* is the more common term for a substantial project submitted as part of a taught master's degree or an undergraduate degree (*e.g.,* BA, BSc, BMus, BEd, BEng etc.).

Thesis word lengths may differ by faculty/department and are set by individual universities.

A wide range of supervisory arrangements can be found in the British academy, from single supervisors (more usual for undergraduate and Masters level work) to supervisory teams of up to three supervisors. In teams, there will often be a Director of Studies, usually someone with broader experience (perhaps having passed some threshold of successful supervisions).

The Director may be involved with regular supervision along with the other supervisors, or may have more of an oversight role, with the other supervisors taking on the more day-to-day responsibilities of supervision.

*United States*

In some U.S. doctoral programmes, the “dissertation” can take up the major part of the student’s total time spent (along with two or three years of classes), and may take years of full-time work to complete. At most universities, *dissertation* is the term for the required submission for the doctorate, and *thesis* refers only to the master’s degree requirement.

*Thesis* is also used to describe a cumulative project for a bachelor’s degree, and is more common at selective colleges and universities, or for those seeking admittance to graduate school or to obtain an honours academic designation. These are called “senior projects” or “senior theses;” they are generally done in the senior year near graduation after having completed other courses, the independent study period, and the internship or student teaching period (the completion of most of the requirements before the writing of the paper ensures adequate knowledge and aptitude for the challenge). Unlike a dissertation or master’s thesis, they are not as long, they do not require a novel contribution to knowledge, or even a very narrow focus on a set subtopic. Like them, they can be lengthy and require months of work, they require supervision by at least one professor adviser, they must be focused on a certain area of knowledge, and they must use an appreciable amount of scholarly citations. They may or may not be defended before a committee, but usually are not; there is generally no preceding examination before the writing of the paper, except for at a very few colleges. Because of the nature of the graduate thesis or dissertation having to be more narrow and more novel, the result of original research, these usually have a smaller proportion of the work that is cited from other sources, though the fact that they are lengthier may mean they still have more total citations.

Specific undergraduate courses, especially writing-intensive courses or courses taken by upperclassmen, may also require one or more extensive written assignments referred to variously as theses, essays, or papers. Increasingly, high schools are requiring students to complete a senior project or senior thesis on a chosen topic during the final year as a prerequisite for graduation. The extended essay component of the International Baccalaureate Diploma Programme, offered in a growing number of American high schools, is another example of this trend.

Generally speaking, a dissertation is judged as to whether or not it makes an original and unique contribution to scholarship. Lesser projects (a master’s thesis, for example) are judged by whether or not they demonstrate mastery of available scholarship in the presentation of an idea.

The required complexity or quality of research of a thesis may vary significantly among universities or programmes.

### *Thesis Examinations*

One of the requirements for certain advanced degrees is often an oral examination (a.k.a. *viva voce* examination or just *viva*). This examination normally occurs after the dissertation is finished but before it is submitted to the university, and may comprise a presentation by the student and questions posed by an examining committee or jury. In North America, an initial oral examination in the field of specialization may take place just before the student settles down to work on the dissertation. An additional oral exam may take place after the dissertation is completed and is known as a thesis or dissertation "defence," which at some universities may be a mere formality and at others may result in the student being required to make significant revisions. In the UK and certain other English-speaking countries, an oral examination is called a *viva voce*.

*Examination Results*

The result of the examination may be given immediately following deliberation by the examiners (in which case the candidate may immediately be considered to have received his or her degree), or at a later date, in which case the examiners may prepare a defence report that is forwarded to a Board or Committee of Postgraduate Studies, which then officially recommends the candidate for the degree.

Potential decisions (or "verdicts") include:

- Accepted/pass with no corrections.

The thesis is accepted as presented. A grade may be awarded, though in many countries PhDs are not graded at all, and in others only one of the theoretically possible grades (the highest) is ever used in practice.

- The thesis must be revised.

Revisions (for example, correction of numerous grammatical or spelling errors; clarification of concepts or methodology; addition of sections) are required. One or more members of the jury or the thesis supervisor will make the decision on the acceptability of revisions and provide written confirmation that they have been satisfactorily completed. If, as is often the case, the needed revisions are relatively modest, the examiners may all sign the thesis with the verbal understanding that the candidate will review the revised thesis with his or her supervisor before submitting the completed version.

- Extensive revision required.

The thesis must be revised extensively and undergo the evaluation and defence process again from the beginning with the same examiners. Problems may include theoretical or methodological issues. A candidate who is not recommended for the degree after the second defence must normally withdraw from the programme.

- Unacceptable.

The thesis is unacceptable and the candidate must withdraw from the programme.

This verdict is given only when the thesis requires major revisions and when the examination makes it clear that the candidate is incapable of making such revisions.

At most North American institutions the latter two verdicts are extremely rare, for two reasons. First, to obtain the status of doctoral candidates, graduate students typically write a qualifying examination or comprehensive examination, which often includes an oral defence. Students who pass the qualifying examination are deemed capable of completing scholarly work independently and are allowed to proceed with working on a dissertation. Second, since the thesis supervisor (and the other members of the advisory committee) will normally have reviewed the thesis extensively before recommending the student proceed to the defence, such an outcome would be regarded as a major failure not only on the part of the candidate but also by the candidate's supervisor (who should have recognized the substandard quality of the dissertation long before the defence was allowed to take place). It is also fairly rare for a thesis to be accepted without any revisions; the most common outcome of a defence is for the examiners to specify minor revisions (which the candidate typically completes in a few days or weeks).

On the other hand, at universities on the British pattern it is not uncommon for theses to be failed at the *viva* stage, in which case either a major rewrite is required, followed by a new *viva*, or the thesis may be awarded the lesser degree of M.Phil (Master of Philosophy) instead, preventing the candidate from resubmitting the thesis.

*Australia*

In Australia, doctoral theses are usually examined by three examiners although some, like the Australian Catholic University and the University of New South Wales, have shifted to using only two examiners; without a live defence except in extremely rare exceptions. In the case of a master's degree by research the thesis is usually examined by only two examiners. Typically one of these examiners will be from within the candidate's own department; the other (s) will usually be from other universities and often from overseas. Following submission of the thesis, copies are sent by mail to examiners and then reports sent back to the institution.

Similar to a master's degree by research thesis, a thesis for the research component of a master's degree by coursework is also usually examined by two examiners, one from the candidate's department and one from another university. For an Honours year, which is a fourth year in addition to the usual three-year bachelor's degree, the thesis is also examined by two examiners, though both are usually from the candidate's own department. Honours and Master's by coursework theses do not require an oral defence before they are accepted.

*Germany*

In Germany, a thesis is usually examined with an oral examination. This applies to almost all Diplom, Magister, master's and doctoral degrees as well as to most bachelor's degrees. However, a process that allows for revisions of the thesis is usually only implemented for doctoral degrees.

There are several different kinds of oral examinations used in practice. The *Disputation*, also called *Verteidigung* ("defence"), is usually public (at least to members of the university) and is focused on the topic of the thesis. In contrast, the *Rigorosum* is not held in public and also encompasses fields in addition to the topic of the thesis. The

*Rigorosum* is only common for doctoral degrees. Another term for an oral examination is *Kolloquium*, which generally refers to a usually public scientific discussion and is often used synonymously with *Verteidigung*.

In each case, what exactly is expected differs between universities and between faculties. Some universities also demand a combination of several of these forms.

*Malaysia*

Like the British model, the PHD or MPhil student is required to submit their theses or dissertation for examination by two or three examiners. The first examiner is from the university concerned, the second examiner is from another local university and the third examiner is from a suitable foreign university (usually from Commonwealth countries). The choice of examiners must be approved by the university senate. In some public universities, a PhD or MPhil candidate may also have to show a number publications in peer reviewed academic journals as part of the requirement. An oral viva is conducted after the examiners have submitted their reports to the university. The oral viva session is attended by the Oral Viva chairman, a rapporteur with a PhD qualification, the first examiner, the second examiner and sometimes the third examiner.

Branch campuses of British, Australian and Middle East universities in Malaysia use the respective models of the home campuses to examine their PhD or MPhil candidates.

*Philippines*

In the Philippines, a thesis is followed by an oral defence. In most universities, this applies to all bachelor, master, and doctorate degrees. However, the oral defence is held in once per semester (usually in the middle or by the end) with a presentation of revisions (so-called "plenary presentation") at the end of each semester. The oral defence is typically not held in public for bachelor and master oral defences, however a colloquium is held for doctorate degrees.

*Portugal*

In Portugal, a thesis is examined with an oral defence, which includes an initial presentation by the candidate followed by an extensive questioning/answering period. Typical duration for the total exam is 1 hour 30 minutes for the MSc and 3 hours for the PhD.

*North America*

In North America, the *thesis defence* or *oral defence* is the final examination for doctoral candidates, and sometimes for master's candidates.

The examining committee normally consists of the thesis committee, usually a given number of professors mainly from the student's university plus his or her primary supervisor, an external examiner (someone not otherwise connected to the university), and a chair person. Each committee member will have been given a completed copy of the dissertation prior to the defence, and will come prepared to ask questions about the thesis itself and the subject matter. In many schools, master's thesis defences are restricted to the examinee and the examiners, but doctoral defences are open to the public.

The typical format will see the candidate giving a short (20–40-minute) presentation of his or her research, followed by one to two hours of questions.

At some U.S. institutions, a longer public lecture (known as a "thesis talk" or "thesis seminar") by the candidate will accompany the defence itself, in which case only the candidate, the examiners, and other members of the faculty may attend the actual defence.

*Russia and Ukraine*

A student in Ukraine or Russia has to complete a thesis and then defend it in front of their department. Sometimes the defence meeting is made up of the learning institute's professionals, and sometimes the students peers are allowed to view or join in. After the presentation and defence of the thesis, the final conclusion of the department should be that none of them have reservations on the content and quality of the thesis.

A conclusion on the thesis has to be approved by the rector of the educational institute. This conclusion (final grade so to speak) of the thesis can be defended/argued not only in the thesis council at, but also in any other thesis council of Russia or Ukraine.

*Spain*

The Diploma de estudios avanzados (DEA) can last two years and candidates must complete coursework and demonstrate their ability to research the specific topics they have studied. After completing this part of the PhD, students begin a dissertation on a set topic. The dissertation must reach a minimum length depending on the subject and it is valued more highly if it contains field research. Once candidates have finished their written dissertations, they must present them before a committee. Following this presentation, the examiners will ask questions.

*United Kingdom, Ireland and Hong Kong*

In Hong Kong, Ireland and the United Kingdom, the thesis defence is called a *viva voce* (Latin for "by live voice") examination (*viva* for short). A typical *viva* lasts for approximately 3 hours, though there is no formal time limit. Involved in the *viva* are two examiners and the candidate. Usually, one examiner is an academic from the candidate's own university department (but not one of the candidate's supervisors) and the other is an external examiner from a different university. Increasingly, the examination may involve a third academic, the 'chair'; this person, from the candidate's institution, acts as an impartial observer with oversight of the examination process to ensure that the examination is fair. The 'chair' does not ask academic questions of the candidate.

In the United Kingdom, there are only two or at most three examiners, and in many universities the examination is held in private. The candidate's primary supervisor is not permitted to ask or answer questions during the viva, and their presence is not necessary. However, some universities permit members of the faculty or the university to attend. At the University of Oxford, for instance, any member of the University may attend a DPhil viva (the University's regulations require that details of the examination and its time and place be published formally in advance) provided he or she attends in full academic dress.

***Submission of the Thesis***

A submission of the thesis is the last formal requirement for most students after the defence. By the final deadline, the student must submit a complete copy of the thesis to the appropriate body within the accepting institution, along with the appropriate forms, bearing the signatures of the primary supervisor, the examiners, and, in some cases, the head of the student's department. Other required forms may include library authorizations (giving the university library permission to make the thesis available as part of its collection) and copyright permissions (in the event that the student has incorporated copyrighted materials in the thesis). Many large scientific publishing houses (*e.g.,* Taylor and Francis, Elsevier) use copyright agreements that allow the authors to incorporate their published articles into dissertations without separate authorization.

Failure to submit the thesis by the deadline may result in graduation (and granting of the degree) being delayed. At most U.S. institutions, there will also be various fees (for binding, microfilming, copyright registration, and the like), which must be paid before the degree will be granted.

Once all the paperwork is in order, copies of the thesis may be made available in one or more university libraries. Specialist abstracting services exist to publicize the content of these beyond the institutions in which they are produced. Many institutions now insist on submission of digitized as well as printed copies of theses; the digitized versions of successful theses are often made available online.

**Encyclopaedia**

An encyclopaedia or encyclopaedia is a type of reference work or compendium holding a comprehensive summary of information from either all branches of knowledge or a particular branch of knowledge. Encyclopaedias are divided into articles or entries, which are usually accessed alphabetically by article name. Encyclopaedia entries are longer and more detailed than those in most dictionaries. Generally speaking, unlike dictionary entries which focus on linguistic information about words, encyclopaedia articles focus on factual information concerning the subject for which the article is named.

Encyclopaedias have existed for around 2,000 years; the oldest still in existence, *Naturalis Historia*, was written starting in ca. AD 77 by Pliny the Elder and was not fully revised at the time of his death in AD 79. The modern encyclopaedia evolved out of dictionaries around the 17th century. Historically, some encyclopaedias were contained in one volume, whereas others, such as the *Encyclopædia Britannica*, the *Enciclopedia Italiana* (62 volumes, 56,000 pages) or the world's largest, *Enciclopedia universal ilustrada europeo-americana* (118 volumes, 105,000 pages), became huge multi-volume works. Some modern encyclopaedias, such as Wikipedia, are often electronic and freely available.

***Characteristics***

The modern encyclopaedia was developed from the dictionary in the 18th century. Historically, both encyclopaedias and dictionaries have been researched and written by well-educated, well-informed content experts, but they are significantly different in structure. A dictionary is a linguistic work which primarily focuses on alphabetical listing of words and their definitions. Synonymous words and those related by the subject matter are to be found scattered around the dictionary, giving no obvious place for in-depth treatment. Thus, a dictionary typically provides limited information, analysis or background for the word defined. While it may offer a definition, it may leave the reader lacking in understanding the meaning, significance or limitations of a term, and how the term relates to a broader field of knowledge. An encyclopaedia is, allegedly, not written in order to convince, although one of its goals is indeed to convince its reader about its own veracity. In the terms of Aristotle's Modes of persuasion, a dictionary should persuade the reader through *logos* (conveying only appropriate emotions); it will be expected to have a lack of pathos (it should not stir up irrelevant emotions), and to have little ethos except that of the dictionary itself.

To address those needs, an encyclopaedia article is typically not limited to simple definitions, and is not limited to defining an individual word, but provides a more extensive meaning for a *subject or discipline*. In addition to defining and listing synonymous terms for the topic, the article is able to treat the topic's more extensive meaning in more depth and convey the most relevant accumulated knowledge on that subject. An encyclopaedia article also often includes many maps and illustrations, as well as bibliography and statistics.

*Four major elements define an encyclopaedia: its subject matter, its scope, its method of organization, and its method of production:*

- Encyclopaedias can be general, containing articles on topics in every field (the English-language *Encyclopædia Britannica* and German *Brockhaus* are well-known examples). General encyclopaedias often contain guides on how to do a variety of things, as well as embedded dictionaries and gazetteers. There are also encyclopaedias that cover a wide variety of topics but from a particular cultural, ethnic, or national perspective, such as the *Great Soviet Encyclopaedia* or *Encyclopaedia Judaica.*
- Works of encyclopedic scope aim to convey the important accumulated knowledge for their subject domain, such as an encyclopaedia of medicine, philosophy, or law. Works vary in the breadth of material and the depth of

discussion, depending on the target audience. (For example, the Medical encyclopaedia produced by A.D.A.M., Inc. for the U.S. National Institutes of Health.)

- Some systematic method of organization is essential to making an encyclopaedia usable as a work of reference. There have historically been two main methods of organizing printed encyclopaedias: the alphabetical method (consisting of a number of separate articles, organized in alphabetical order), or organization by hierarchical categories. The former method is today the most common by far, especially for general works. The fluidity of electronic media, however, allows new possibilities for multiple methods of organization of the same content. Further, electronic media offer previously unimaginable capabilities for search, indexing and cross reference. The epigraph from Horace on the title page of the 18th century *Encyclopédie* suggests the importance of the structure of an encyclopaedia: "What grace may be added to commonplace matters by the power of order and connection."
- As modern multimedia and the information age have evolved, they have had an ever-increasing effect on the collection, verification, summation, and presentation of information of all kinds. Projects such as Everything2, Encarta, h2g2, and Wikipedia are examples of new forms of the encyclopaedia as information retrieval becomes simpler. The method of production for an encyclopaedia historically has been supported in both for-profit and non-profit contexts. The *Great Soviet Encyclopaedia* mentioned above was entirely state sponsored, while the *Britannica* was supported as a for-profit institution. By comparison, Wikipedia is supported by volunteers contributing in a non-profit environment under the organization of the Wikimedia Foundation.

Some works entitled "dictionaries" are actually similar to encyclopaedias, especially those concerned with a particular field (such as the *Dictionary of the Middle Ages*, the *Dictionary of American Naval Fighting Ships*, and *Black's Law Dictionary*). The *Macquarie Dictionary,* Australia's national dictionary, became an encyclopedic dictionary after its first edition in recognition of the use of proper nouns in common communication, and the words derived from such proper nouns.

There are some broad differences between encyclopaedias and dictionaries. Most noticeably, encyclopaedia articles are longer, fuller and more thorough than entries in most general-purpose dictionaries. There are differences in content as well. Generally speaking, dictionaries provide linguistic information about words themselves, while encyclopaedias focus more on the thing for which those words stand. Thus, while dictionary entries are inextricably fixed to the word described, encyclopaedia articles can be given a different entry name. As such, dictionary entries are not fully translatable into other languages, but encyclopaedia articles can be.

In practice, however, the distinction is not concrete, as there is no clear-cut difference between factual, "encyclopedic" information and linguistic information such as appears in dictionaries. Thus encyclopaedias may contain material that is also found in dictionaries, and vice versa. In particular, dictionary entries often contain factual information about the thing named by the word.

### *History*

Encyclopaedias have progressed from the beginning of history in written form, through medieval and modern times in print, and most recently, displayed on computer and distributed via computer networks.

#### *Ancient times*

One of the earliest encyclopedic works to have survived to modern times is the *Naturalis Historia* of Pliny the Elder, a Roman statesman living in the 1st century AD. He compiled a work of 37 chapters covering natural history, architecture, medicine, geography, geology, and all aspects of the world around him. He stated in the preface that he had compiled 20,000 facts from 2000 works by over 200 authors, and added many others from his own experience. The work was published around AD 77-79, although he probably never finished proofing the work before his death in the eruption of Vesuvius in AD 79.

#### *Middle Ages*

Isidore of Seville, one of the greatest scholars of the early Middle Ages, is widely recognized for writing the first encyclopaedia of the Middle Ages, the *Etymologiae (The Etymologies*) or *Origines* (around 630), in which he compiled a sizable portion of the learning available at his time, both ancient and contemporary. The work has 448 chapters in 20 volumes, and is valuable because of the quotes and fragments of texts by other authors that would have been lost had he not collected them.

The most popular encyclopaedia of the Carolingian Age was the *De universo* or *De rerum naturis* by Rabanus Maurus, written about 830; it was based on *Etymologiae*.

The encyclopaedia of Suda, a massive 10th-century Byzantine encyclopaedia, had 30 000 entries, many drawing from ancient sources that have since been lost, and often derived from medieval Christian compilers. The text was arranged alphabetically with some slight deviations from common vowel order and place in the Greek alphabet.

The early Muslim compilations of knowledge in the Middle Ages included many comprehensive works. Around year 960, the Brethren of Purity of Basra were engaged in their Encyclopaedia of the Brethren of Purity. Notable works include Abu Bakr al-Razi's encyclopaedia of science, the Mutazilite Al-Kindi's prolific output of 270 books, and Ibn Sina's medical encyclopaedia, which was a standard reference work for centuries. Also notable are works of universal history (or sociology) from Asharites, al-Tabri, al-Masudi, Tabari's *History of the Prophets and Kings*, Ibn Rustah, al-Athir, and Ibn Khaldun, whose Muqadimmah contains cautions regarding trust in written records that remain wholly applicable today.

The enormous encyclopedic work in China of the *Four Great Books of Song*, compiled by the 11th century AD during the early Song dynasty (960–1279), was a massive literary undertaking for the time. The last encyclopaedia of the four, the *Prime Tortoise of the Record Bureau*, amounted to 9.4 million Chinese characters in 1000 written volumes.

In the late Middle Ages, several authors had the ambition of compiling the sum of human knowledge in a certain field or overall, for example Bartholomew of England, Vincent of Beauvais, Radulfus Ardens, Sydrac, Brunetto Latini, Giovanni da Sangiminiano, Pierre Bersuire. Some were women, like Hildegard of Bingen and Herrad of Landsberg. The most successful of those publications were the *Speculum maius (Great Mirror)* of Vincent of Beauvais and the *De proprietatibus rerum (On the Properties of Things)* by Bartholomew of England. The latter was translated (or adapted) into French, Provençal, Italian, English, Flemish, Anglo-Norman, Spanish, and German during the Middle Ages. Both were written in the middle of the 13th century. No medieval encyclopaedia bore the title *Encyclopaedia* – they were often called *On nature (De natura, De naturis rerum), Mirror (Speculum maius, Speculum universale), Treasure (Trésor).*

*Renaissance*

These works were all hand copied and thus rarely available, beyond wealthy patrons or monastic men of learning: they were expensive, and usually written for those extending knowledge rather than those using it.

During the Renaissance, the creation of printing allowed a wider diffusion of encyclopaedias and every scholar could have his or her own copy. The *De expetendis et fugiendis rebus* by Giorgio Valla was posthumously printed in 1501 by Aldo Manuzio in Venice. This work followed the traditional scheme of liberal arts. However, Valla added the translation of ancient Greek works on mathematics (firstly by Archimedes), newly discovered and translated. The *Margarita Philosophica* by Gregor Reisch, printed in 1503, was a complete encyclopaedia explaining the seven liberal arts.

The term encyclopaedia was coined by 16th century humanists who misread copies of their texts of Pliny and Quintilian, and combined the two Greek words "*enkyklios paideia*" into one word, εγκυκλοπαιδεια. The phrase *enkyklios paideia* (εγκυκλιοζ παιδεια) was used by Plutarch and the Latin word Encyclopaedia came from him.

The first work titled in this way was the *Encyclopaedia orbisque doctrinarum, hoc est omnium artium, scientiarum, ipsius philosophiae index ac divisio* written by Johannes Aventinus in 1517.

The English physician and philosopher, Sir Thomas Browne used the word 'encyclopaedia' in 1646 in the preface to the reader to define his *Pseudodoxia Epidemica*, a major work of the 17th-century scientific revolution. Browne structured his encyclopaedia upon the time-honoured schemata of the Renaissance, the so-called 'scale of creation' which ascends through the mineral, vegetable, animal, human, planetary, and cosmological worlds. *Pseudodoxia Epidemica* was a European best-seller, translated into French, Dutch, and German as well as Latin it went through no fewer than five editions, each revised and augmented, the last edition appearing in 1672.

Factors that changed the size of encyclopaedias were financial, commercial, legal, and intellectual. During the Renaissance, middle classes were having more time to read and encyclopaedias helped them to learn more. Publishers were wanting to increase their output so some countries like Germany start selling books that were missing alphabetical sections to publish faster. Also, publishers could not afford all the resources by themselves so they multiple publishers would come together with their resources to create better encyclopaedias. Soon it was difficult financially to continue to produce the amount they wanted to produce so then subscriptions and serial publications were made. This was risky for publishers because they had to find people that would pay all upfront or make payments. When this did work out capital would rise and there would be a steady income for encyclopaedias. Later rivalry grew causing copyright to occur due to weak underdeveloped laws. Some publishers would copy another publisher's work to produce an encyclopaedia faster and cheaper so consumers did not have to pay a lot and they would sell more. Encyclopaedias made it to where middle class citizens could basically have a small library in their own house. Europeans were becoming more curious about their society around them causing them to revolt against their government.

*18th–19th Centuries*

The beginnings of the modern idea of the general-purpose, widely distributed printed encyclopaedia precede the 18th century encyclopedists. However, Chambers' *Cyclopaedia, or Universal Dictionary of Arts and Sciences* (1728), and the *Encyclopédie* of Denis Diderot and Jean le Rond d'Alembert (1751 onwards), as well as *Encyclopædia Britannica* and the *Conversations-Lexikon*, were the first to realize the form we would recognize today, with a comprehensive scope of topics, discussed in depth and organized in an accessible, systematic method. Chambers, in 1728, followed the earlier lead of John Harris's *Lexicon Technicum* of 1704 and later editions; this work was by its title and content "A Universal English Dictionary of Arts and Sciences: Explaining not only the Terms of Art, but the Arts Themselves".

During the 19th and early 20th century, many smaller or less developed languages saw their first encyclopaedias, using French, German, and English role models. While encyclopaedias in larger languages, having large markets that could support a large editorial staff, churned out new 20-volume works in a few years and new editions with brief intervals, such publication plans often spanned a decade or more in smaller languages.

*20th Century*

Popular and affordable encyclopaedias such as Harmsworth's Universal Encyclopaedia and the Children's Encyclopaedia appeared in the early 1920s.

In the United States, the 1950s and 1960s saw the introduction of several large popular encyclopaedias, often sold on installment plans. The best known of these were *World Book* and *Funk and Wagnalls*.

The second half of the 20th century also saw the proliferation of specialized encyclopaedias that compiled topics in specific fields. This trend has continued. Encyclopaedias of at least one volume in size now exist for most if not all academic disciplines, including such narrow topics such as bioethics.

By the late 20th century, encyclopaedias were being published on CD-ROMs for use with personal computers. Microsoft's *Encarta*, launched in 1993, was a landmark example as it had no printed equivalent. Articles were supplemented with both video and audio files as well as numerous high-quality images. After sixteen years, Microsoft discontinued the Encarta line of products in 2009.

*21st Century*

In 2001, Jimmy Wales and Larry Sanger launched Wikipedia, a collaboratively edited, multilingual, open-source, free Internet encyclopaedia supported by the non-profit Wikimedia Foundation. As of 23 April 2017, there are 5,391,001 articles in the English Wikipedia. There are 287 different editions of Wikipedia. As of February 2014, it had 18 billion page views and nearly 500 million unique visitors each month. Wikipedia has more than 25 million accounts, out of which there were over 118,000 active editors globally, as of August 2015. Wikipedia's accuracy was found by a *Nature* study to be close to that of *Encyclopædia Britannica*, with Wikipedia being much larger.

However, critics argue Wikipedia exhibits systemic bias, and its group dynamics hinder its goals. Many academics, historians, teachers, and journalists reject Wikipedia as a reliable source of information, primarily for being a mixture of truths, half truths, and some falsehoods, and that as a resource about many controversial topics, is notoriously subject to manipulation and spin.

While Wikipedia is by far the largest web-based encyclopaedia, it is not the only one in existence. There are several much smaller, usually more specialized, encyclopaedias on various themes, sometimes dedicated to a specific geographic region or time period.

**Magazines**

A magazine is a publication, usually a periodical publication, which is printed or electronically published (sometimes referred to as an online magazine). Magazines are generally published on a regular schedule and contain a variety of content. They are generally financed by advertising, by a purchase price, by prepaid subscriptions, or a combination of the three. At its root, the word "magazine" refers to a collection or storage location. In the case of written publication, it is a collection of written articles. This explains why magazine publications share the word root with gunpowder magazines, artillery magazines, firearms magazines, and, in French, retail stores such as department stores.

***Definition***

By definition, a *magazine* paginates with each issue starting at page three, with the standard sizing being 8 3/8 × 10 7/8 inches. However, in the technical sense a *journal* has continuous pagination throughout a volume. Thus *Business Week*, which starts each issue anew with page one, is a magazine, but the *Journal of Business Communication*, which starts each volume with the winter issue and continues the same sequence of pagination throughout the coterminous year, is a journal. Some professional or trade publications are also peer-reviewed, an example being the *Journal of Accountancy*. Academic or professional publications that are not peer-reviewed are generally *professional magazines*. That a publication calls itself a *journal* does not make it a journal in the technical sense; *The Wall Street Journal* is actually a newspaper.

***Distribution***

Magazines can be distributed through the mail, through sales by newsstands, bookstores, or other vendors, or through free distribution at selected pick-up locations. The subscription business models for distribution fall into three main categories.

*Paid Circulation*

In this model, the magazine is sold to readers for a price, either on a per-issue basis or by subscription, where an annual fee or monthly price is paid and issues are sent by post to readers. Paid circulation allows for defined readership statistics.

*Non-paid Circulation*

This means that there is no cover price and issues are given away, for example in street dispensers, airline, or included with other products or publications. Because this model involves giving issues away to unspecific populations, the statistics only entail the number of issues distributed, and not who reads them.

*Controlled Circulation*

This is the model used by many trade magazines (industry-based periodicals) distributed only to qualifying readers, often for free and determined by some form of survey. Because of costs (*e.g.,* printing and postage) associated with the medium of print, publishers may not distribute free copies to everyone who requests one (unqualified leads); instead, they operate under controlled circulation, deciding who may receive free subscriptions based on each person's qualification as a member of the trade (and likelihood of buying, for example, likelihood of having corporate purchasing authority, as determined from job title). This allows a high level of certainty that advertisements will be received by the advertiser's target audience, and it avoids wasted printing and distribution expenses. This latter model was widely used before the rise of the World Wide Web and is still employed by some titles. For example, in the United Kingdom, a number of computer-industry magazines use this model, including *Computer Weekly* and *Computing*, and in finance, *Waters Magazine*. For the global media industry, an example would be *VideoAge International.*

***History***

The earliest example of magazines was *Erbauliche Monaths Unterredungen*, a literary and philosophy magazine, which was launched in 1663 in Germany. *The Gentleman's Magazine*, first published in 1731, in London was the first general-interest magazine. Edward Cave, who edited *The Gentleman's Magazine* under the pen name "Sylvanus Urban", was the first to use the term "magazine," on the analogy of a military storehouse. Founded by Herbert Ingram in 1842, *The Illustrated London News* was the first illustrated magazine.

*Britain*

The oldest consumer magazine still in print is *The Scots Magazine*, which was first published in 1739, though multiple changes in ownership and gaps in publication totalling over 90 years weaken that claim. *Lloyd's List* was founded in Edward Lloyd's England coffee shop in 1734; it is still published as a daily business newspaper. Despite being among the first mass media outlets to venture from the bible, periodicals still remained rooted in the naturalized class and gender system held by European and American society.

Manufacturing of the early magazines were done via an archaic form of the printing press, using large hand engraved wood blocks for printing. When production of magazines increased, entire production lines were created to manufacture these wooden blocks.

*France*

Under the ancien regime, the most prominent magazines were *Mercure de France*, *Journal des sçavans*, founded in 1665 for scientists, and *Gazette de France*, founded in 1631. Jean Loret was one of France's first journalists. He disseminated the weekly news of music, dance and Parisian society from 1650 until 1665 in verse, in what he called a *gazette burlesque*, assembled in three volumes of *La Muse historique* (1650, 1660, 1665). The French press lagged a generation behind the British, for they catered to the needs the aristocracy, while the newer British counterparts were oriented towards the middle and working classes.

Periodicals were censored by the central government in Paris. They were not totally quiescent politically—often they criticized Church abuses and bureaucratic ineptitude. They supported the monarchy and they played at most a small role in stimulating the revolution. During the Revolution, new periodicals played central roles as propaganda organs for various factions. Jean-Paul Marat (1743–1793) was the most prominent editor. His *L'Ami du peuple* advocated vigorously for the rights of the lower classes against the enemies of the people Marat hated; it closed when he was assassinated. After 1800

Napoleon reimposed strict censorship. Magazines flourished after Napoleon left in 1815. Most were based in Paris and most emphasized literature, poetry and stories. They served religious, cultural and political communities. In times of political crisis they expressed and helped shape the views of their readership and thereby were major elements in the changing political culture. For example, there were eight Catholic periodicals in 1830 in Paris. None were officially owned or sponsored by the Church and they reflected a range of opinion among educated Catholics about current issues, such as the 1830 July Revolution that overthrew the Bourbon monarchy. Several were strong supporters of the Bourbon kings, but all eight ultimately urged support for the new government, putting their appeals in terms of preserving civil order. They often discussed the relationship between church and state. Generally, they urged priests to focus on spiritual matters and not engage in politics. Historian M. Patricia Dougherty says this process created a distance between the Church and the new monarch and enabled Catholics to develop a new understanding of church-state relationships and the source of political authority.

*United States*

Late 19th century:

- In the mid-1800s, monthly magazines gained popularity. They were general interest to begin, containing some news, vignettes, poems, history, political events, and social discussion. Unlike newspapers, they were more of a monthly record of current events along with entertaining stories, poems, and pictures. The first periodicals to branch out from news were *Harper's* and *The Atlantic*, which focused on fostering the arts. Both *Harper's* and *The Atlantic* persist to this day, with Harper's being a cultural magazine and The Atlantic focusing mainly on world events. Early publications of Harper's even held famous works such as early publications of *Moby Dick* or famous events such as the laying of the world's first transatlantic telegraph cable however the majority of early content was trickle down from British events.
- The development of the magazines stimulated an increase in literary criticism and political debate, moving towards more opinionated pieces from the objective newspapers. The increased time between prints and the greater amount of space to write provided a forum for public arguments by scholars and critical observers.
- The early periodical predecessors to magazines started to evolve to modern definition in the late 1800s. Works slowly became more specialized and the general discussion or cultural periodicals were forced to adapt to a consumer market which yearned for more localization of issues and events.

Progressive Era: 1890s-1920s:

- Mass circulation magazines became much more common after 1900, some with circulations in the hundreds of thousands of subscribers. Some passed the million-mark in the 1920s. It was an age of Mass media. Thanks to the rapid expansion of national advertising, the cover price fell sharply to about 10 cents. One cause was the heavy coverage of corruption in politics, local government and big business, especially by *Muckrakers.* They were journalists who wrote for popular magazines to expose social and political sins and shortcomings. They relied on their own investigative journalism reporting; muckrakers often worked to expose social ills and corporate and political corruption. Muckraking magazines–notably *McClure's*–took on corporate monopolies and crooked political machines while raising public awareness of chronic urban poverty, unsafe working conditions, and social issues like child labour.

- The journalists who specialized in exposing waste, corruption, and scandal operated at the state and local level, like Ray Stannard Baker, George Creel, and Brand Whitlock. Other like Lincoln Steffens exposed political corruption in many large cities; Ida Tarbell went after John D. Rockefeller's Standard Oil Company. Samuel Hopkins Adams in 1905 showed the fraud involved in many patent medicines, Upton Sinclair's 1906 novel *The Jungle* gave a horrid portrayal of how meat was packed, and, also in 1906, David Graham Phillips unleashed a blistering indictment of the U.S. Senate. Roosevelt gave these journalists their nickname when he complained they were not being helpful by raking up all the muck.

21st century:

- In 2011, 152 magazines ceased operations and in 2012, 82 magazines were closed down. Between the years of 2008 to 2015, Oxbridge communications announced that 227 magazines launched and 82 magazines closed in 2012 in North America. Furthermore, according to MediaFinder.com, 93 new magazines launched between the first six months of 2014 and just 30 closed. The category that produced new publications was "Regional interest", six new magazines were launched, including 12th and Broad and Craft Beer and Brewing. However, two magazines had to change their print schedules. Johnson Publishing's Jet stopped printing regular issues making the transition to digital format, however still print an annual print edition. Ladies Home Journal, stopped their monthly schedule and home delivery for subscribers to become a quarterly newsstand-only special interest publication. According to statistics from the end of 2013, subscription levels for 22 of the top 25 magazines declined from 2012 to 2013, with just *Time*, *Glamour* and *ESPN The Magazine* gaining numbers.

### *Women's Magazines*

#### *Fashion*

Immortalized in movies and magazines, young women's fashions of the 1920s set both a trend and social statement, a breaking-off from the rigid Victorian way of life. Their glamorous life style was celebrated in the feature pages and in the advertisements, where they learned the brands that best exemplified the look they sought. These young, rebellious, middle-class women, labeled “flappers” by older generations, did away with the corset and donned slinky knee-length dresses, which exposed their legs and arms. The hairstyle of the decade was a chin-length bob, which had several popular variations. Cosmetics, which until the 1920s were not typically accepted in American society because of their association with prostitution, became, for the first time, extremely popular.

In the 1920s new magazines appealed to young German women with a sensuous image and advertisements for the appropriate clothes and accessories they would want to purchase. The glossy pages of *Die Dame* and *Das Blatt der Hausfrau* displayed the “Neue Frauen,” “New Girl” - what Americans called the flapper. She was young and fashionable, financially independent, and was an eager consumer of the latest fashions. The magazines kept her up to date on fashion, arts, sports, and modern technology such as automobiles and telephones.

**Internet**

The Internet is the global system of interconnected computer networks that use the Internet protocol suite (TCP/IP) to link devices worldwide. It is a *network of networks* that consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies. The Internet carries an extensive range of information resources and services, such as the inter-linked hypertext documents and applications of the World Wide Web (WWW), electronic mail, telephony, and peer-to-peer networks for file sharing.

The origins of the Internet date back to research commissioned by the United States federal government in the 1960s to build robust, fault-tolerant communication via computer networks. The primary precursor network, the ARPANET, initially served as a backbone for interconnection of regional academic and military networks in the 1980s. The funding of the National Science Foundation Network as a new backbone in the 1980s, as well as private funding for other commercial extensions, led to worldwide participation in the development of new networking technologies, and the merger of many networks. The linking of commercial networks and enterprises by the early 1990s marks the beginning of the transition to the modern Internet, and generated a sustained exponential growth as generations of institutional, personal, and mobile computers were connected to the network. Although the Internet was widely used by academia since the 1980s, the commercialization incorporated its services and technologies into virtually every aspect of modern life.

Internet use grew rapidly in the West from the mid-1990s and from the late 1990s in the developing world. In the two decades since then, Internet use has grown 100-times, measured for the period of one year, to over one third of the world population. Most traditional communications media, including telephony, radio, television, paper mail and newspapers are being reshaped or redefined by the Internet, giving birth to new services such as e-mail, Internet telephony, Internet television music, digital newspapers, and video streaming web sites. Newspaper, book, and other print publishing are adapting to web site technology, or are reshaped into blogging, web feeds and online news aggregators. The entertainment industry was initially the fastest growing segment on the Internet. The Internet has enabled and accelerated new forms of personal interactions through instant messaging, Internet forums, and social networking. Online shopping has grown exponentially both for major retailers and small businesses and entrepreneurs, as it enables firms to extend their "bricks and mortar" presence to serve a larger market or even sell goods and services entirely online. Business-to-business and financial services on the Internet affect supply chains across entire industries.

The Internet has no centralized governance in either technological implementation or policies for access and usage; each constituent network sets its own policies. Only the overreaching definitions of the two principal name spaces in the Internet, the Internet Protocol address space and the Domain Name System (DNS), are directed by a maintainer organization, the Internet Corporation for Assigned Names and Numbers (ICANN). The technical underpinning and standardization of the core protocols is an activity of the Internet Engineering Task Force (IETF), a non-profit organization of loosely affiliated international participants that anyone may associate with by contributing technical expertise.

***Terminology***

The term *Internet*, when used to refer to the specific global system of interconnected Internet Protocol (IP) networks, is a proper noun and may be written with an initial capital letter. In common use and the media, it is often not capitalized, *viz. the internet.* Some guides specify that the word should be capitalized when used as a noun, but not capitalized when used as an adjective.

The Internet is also often referred to as *the Net*, as a short form of *network*. Historically, as early as 1849, the word *internetted* was used uncapitalized as an adjective, meaning *interconnected* or *interwoven*. The designers of early computer networks used *internet* both as a noun and as a verb in shorthand form of internetwork or internetworking, meaning interconnecting computer networks.

The terms *Internet* and *World Wide Web* are often used interchangeably in everyday speech; it is common to speak of "*going on the Internet*" when invoking a web browser to view web pages. However, the World Wide Web or *the Web* is only one of a large number of Internet services. The Web is a collection of interconnected documents (web pages) and other web resources, linked by hyperlinks and URLs. As another point of comparison, Hypertext Transfer Protocol, or HTTP, is the language used on the Web for information transfer, yet it is just one of many languages or protocols that can be used for communication on the Internet. The term *Interweb* is a portmanteau of *Internet* and *World Wide Web* typically used sarcastically to parody a technically unsavvy user.

***History***

Research into packet switching started in the early 1960s, and packet switched networks such as the ARPANET, CYCLADES, the Merit Network, NPL network, Tymnet, and Telenet, were developed in the late 1960s and 1970s using a variety of protocols. The ARPANET project led to the development of protocols for internetworking, by which multiple separate networks could be joined into a single network of networks. ARPANET development began with two network nodes which were interconnected between the Network Measurement Center at the University of California, Los Angeles (UCLA) Henry Samueli School of Engineering and Applied Science directed by Leonard Kleinrock, and the NLS system at SRI International (SRI) by Douglas Engelbart in Menlo Park, California, on 29 October 1969. The third site was the Culler-Fried Interactive Mathematics Center at the University of California, Santa Barbara, followed by the University of Utah Graphics Department. In an early sign of future growth, fifteen sites were connected to the young ARPANET by the end of 1971. These early years were documented in the 1972 film *Computer Networks: The Heralds of Resource Sharing*.

Early international collaborations on the ARPANET were rare. European developers were concerned with developing the X.25 networks. Notable exceptions were the Norwegian Seismic Array (NORSAR) in June 1973, followed in 1973 by Sweden with satellite links to the Tanum Earth Station and Peter T. Kirstein's research group in the United Kingdom, initially at the Institute of Computer Science, University of London and later at University College London. In December 1974, RFC 675 (*Specification of Internet Transmission Control Programme*), by Vinton Cerf, Yogen Dalal, and Carl Sunshine, used the term *internet* as a shorthand for *internetworking* and later RFCs repeated this use. Access to the ARPANET was expanded in 1981 when the National Science Foundation (NSF) funded the Computer Science Network (CSNET). In 1982, the Internet Protocol Suite (TCP/IP) was standardized, which permitted worldwide proliferation of interconnected networks.

TCP/IP network access expanded again in 1986 when the National Science Foundation Network (NSFNet) provided access to supercomputer sites in the United States for researchers, first at speeds of 56 kbit/s and later at 1.5 Mbit/s and 45 Mbit/s. Commercial Internet service providers (ISPs) emerged in the late 1980s and early 1990s. The ARPANET was decommissioned in 1990. By 1995, the Internet was fully commercialized in the U.S. when the NSFNet was decommissioned, removing the last restrictions on use of the Internet to carry commercial traffic. The Internet rapidly expanded in Europe and Australia in the mid to late 1980s and to Asia in the late 1980s and early 1990s. The beginning of dedicated transatlantic communication between the NSFNET and networks in Europe was established with a low-speed satellite relay between Princeton University and Stockholm, Sweden in December 1988. Although other network protocols such as UUCP had global reach well before this time, this marked the beginning of the Internet as an intercontinental network.

Public commercial use of the Internet began in mid-1989 with the connection of MCI Mail and Compuserve's e-mail capabilities to the 500,000 users of the Internet. Just months later on January 1, 1990, PSInet launched an alternate Internet backbone for commercial use; one of the networks that would grow into the commercial Internet we know today. In March 1990, the first high-speed T1 (1.5 Mbit/s) link between the NSFNET and Europe was installed between Cornell University and CERN, allowing much more robust communications than were capable with satellites. Six months later Tim Berners-Lee would begin writing WorldWideWeb, the first web browser after two years of lobbying CERN management. By Christmas 1990, Berners-Lee had built all the tools necessary for a working Web: the HyperText Transfer Protocol (HTTP) 0.9, the HyperText Markup Language (HTML), the first Web browser (which was also a HTML editor and could access Usenet newsgroups and FTP files), the first HTTP server software (later known as CERN httpd), the first web server, and the first Web pages that described the project itself. In 1991 the Commercial Internet eXchange was founded, allowing PSInet to communicate with the other commercial networks CERFnet and Alternet. Since 1995 the Internet has tremendously impacted culture and commerce, including the rise of near instant communication by e-mail, instant messaging, telephony (Voice over Internet Protocol or VoIP), two-way interactive video calls, and the World Wide Web with its discussion forums, blogs, social networking, and online shopping sites. Increasing amounts of data are transmitted at higher and higher speeds over fibre optic networks operating at 1-Gbit/s, 10-Gbit/s, or more.

The Internet continues to grow, driven by ever greater amounts of online information and knowledge, commerce, entertainment and social networking. During the late 1990s, it was estimated that traffic on the public Internet grew by 100 percent per year, while the mean annual growth in the number of Internet users was thought to be between 20% and 50%. This growth is often attributed to the lack of central administration, which allows organic growth of the network, as well as the non-proprietary nature of the Internet protocols, which encourages vendor interoperability and prevents any one company from exerting too much control over the network. As of 31 March 2011, the estimated total number of Internet users was 2.095 billion (30.2% of world population). It is estimated that in 1993 the Internet carried only 1% of the information flowing through two-way telecommunication, by 2000 this figure had grown to 51%, and by 2007 more than 97% of all telecommunicated information was carried over the Internet.

***Governance***

The Internet is a global network comprising many voluntarily interconnected autonomous networks. It operates without a central governing body. The technical underpinning and standardization of the core protocols (IPv4 and IPv6) is an activity of the Internet Engineering Task Force (IETF), a non-profit organization of loosely affiliated international participants that anyone may associate with by contributing technical expertise. To maintain interoperability, the principal name spaces of the Internet are administered by the Internet Corporation for Assigned Names and Numbers (ICANN). ICANN is governed by an international board of directors drawn from across the Internet technical, business, academic, and other non-commercial communities. ICANN coordinates the assignment of unique identifiers for use on the Internet, including domain names, Internet Protocol (IP) addresses, application port numbers in the transport protocols, and many other parameters. Globally unified name spaces are essential for maintaining the global reach of the Internet. This role of ICANN distinguishes it as perhaps the only central coordinating body for the global Internet.

*Regional Internet Registries (RIRs) allocate IP addresses:*

- African Network Information Center (AfriNIC) for Africa
- American Registry for Internet Numbers (ARIN) for North America
- Asia-Pacific Network Information Centre (APNIC) for Asia and the Pacific region
- Latin American and Caribbean Internet Addresses Registry (LACNIC) for Latin America and the Caribbean region
- Réseaux IP Européens – Network Coordination Centre (RIPE NCC) for Europe, the Middle East, and Central Asia

The National Telecommunications and Information Administration, an agency of the United States Department of Commerce, continues to have final approval over changes to the DNS root zone. The Internet Society (ISOC) was founded in 1992 with a mission to *"assure the open development, evolution and use of the Internet for the benefit of all people throughout the world"*. Its members include individuals (anyone may join) as well as corporations, organizations, governments, and universities. Among other activities ISOC provides an administrative home for a number of less formally organized groups that are involved in developing and managing the Internet, including: the Internet Engineering Task Force (IETF), Internet Architecture Board (IAB), Internet Engineering Steering Group (IESG), Internet Research Task Force (IRTF), and Internet Research Steering Group (IRSG). On 16 November 2005, the United Nations-sponsored World Summit on the Information Society in Tunis established the Internet Governance Forum (IGF) to discuss Internet-related issues.

### *Infrastructure*

The communications infrastructure of the Internet consists of its hardware components and a system of software layers that control various aspects of the architecture.

#### *Routing and Service Tiers*

Internet service providers establish the worldwide connectivity between individual networks at various levels of scope. End-users who only access the Internet when needed to perform a function or obtain information, represent the bottom of the routing hierarchy. At the top of the routing hierarchy are the tier 1 networks, large telecommunication companies that exchange traffic directly with each other via peering agreements. Tier 2 and lower level networks buy Internet transit from other providers to reach at least some parties on the global Internet, though they may also engage in peering. An ISP may use a single upstream provider for connectivity, or implement multihoming to achieve redundancy and load balancing. Internet exchange points are major traffic exchanges with physical connections to multiple ISPs. Large organizations, such as academic institutions, large enterprises, and governments, may perform the same function as ISPs, engaging in peering and purchasing transit on behalf of their internal networks. Research networks tend to interconnect with large subnetworks such as GEANT, GLORIAD, Internet2, and the UK's national research and education network, JANET. Both the Internet IP routing structure and hypertext links of the World Wide Web are examples of scale-free networks. Computers and routers use routing tables in their operating system to direct IP packets to the next-hop router or destination. Routing tables are maintained by manual configuration or automatically by routing protocols. End-nodes typically use a default route that points towards an ISP providing transit, while ISP routers use the Border Gateway Protocol to establish the most efficient routing across the complex connections of the global Internet.

*Access*

Common methods of Internet access by users include dial-up with a computer modem via telephone circuits, broadband over coaxial cable, fibre optics or copper wires, Wi-Fi, satellite and cellular telephone technology (3G, 4G). The Internet may often be accessed from computers in libraries and Internet cafes. Internet access points exist in many public places such as airport halls and coffee shops. Various terms are used, such as *public Internet kiosk*, *public access terminal*, and *Web payphone*. Many hotels also have public terminals, though these are usually fee-based. These terminals are widely accessed for various usages, such as ticket booking, bank deposit, or online payment. Wi-Fi provides wireless access to the Internet via local computer networks. Hotspots providing such access include Wi-Fi cafes, where users need to bring their own wireless devices such as a laptop or PDA. These services may be free to all, free to customers only, or fee-based.

Grassroots efforts have led to wireless community networks. Commercial Wi-Fi services covering large city areas are in place in New York, London, Vienna, Toronto, San Francisco, Philadelphia, Chicago and Pittsburgh. The Internet can then be accessed from such places as a park bench. Apart from Wi-Fi, there have been experiments with proprietary mobile wireless networks like Ricochet, various high-speed data services over cellular phone networks, and fixed wireless services. High-end mobile phones such as smartphones in general come with Internet access through the phone network. Web browsers such as Opera are available on these advanced handsets, which can also run a wide variety of other Internet software. More mobile phones have Internet access than PCs, though this is not as widely used. An Internet access provider and protocol matrix differentiates the methods used to get online.

*Structure*

Many computer scientists describe the Internet as a "prime example of a large-scale, highly engineered, yet highly complex system". The structure was found to be highly robust to random failures, yet, very vulnerable to intentional attacks. The Internet structure and its usage characteristics have been studied extensively and the possibility of developing alternative structures has been investigated.

***Protocols***

While the hardware components in the Internet infrastructure can often be used to support other software systems, it is the design and the standardization process of the software that characterizes the Internet and provides the foundation for its scalability and success. The responsibility for the architectural design of the Internet software systems has been assumed by the Internet Engineering Task Force (IETF). The IETF conducts standard-setting work groups, open to any individual, about the various aspects of Internet architecture. Resulting contributions and standards are published as *Request for Comments* (RFC) documents on the IETF web site. The principal methods of networking that enable the Internet are contained in specially designated RFCs that constitute the Internet Standards. Other less rigorous documents are simply informative, experimental, or historical, or document the best current practices (BCP) when implementing Internet technologies.

The Internet standards describe a framework known as the Internet protocol suite. This is a model architecture that divides methods into a layered system of protocols, originally documented in RFC 1122 and RFC 1123. The layers correspond to the environment or scope in which their services operate. At the top is the application layer, space for the application-specific networking methods used in software applications.

For example, a web browser programme uses the client-server application model and a specific protocol of interaction between servers and clients, while many file-sharing systems use a peer-to-peer paradigm. Below this top layer, the transport layer connects applications on different hosts with a logical channel through the network with appropriate data exchange methods.

Underlying these layers are the networking technologies that interconnect networks at their borders and exchange traffic across them. The Internet layer enables computers to identify and locate each other via Internet Protocol (IP) addresses, and routes their traffic via intermediate (transit) networks. Last, at the bottom of the architecture is the link layer, which provides logical connectivity between hosts on the same network link, such as a local area network (LAN) or a dial-up connection. The model, also known as TCP/IP, is designed to be independent of the underlying hardware used for the physical connections, which the model does not concern itself with in any detail. Other models have been developed, such as the OSI model, that attempt to be comprehensive in every aspect of communications. While many similarities exist between the models, they are not compatible in the details of description or implementation. Yet, TCP/IP protocols are usually included in the discussion of OSI networking.

The most prominent component of the Internet model is the Internet Protocol (IP), which provides addressing systems, including IP addresses, for computers on the network. IP enables internetworking and, in essence, establishes the Internet itself. Internet Protocol Version 4 (IPv4) is the initial version used on the first generation of the Internet and is still in dominant use. It was designed to address up to ~4.3 billion (10) hosts. However, the explosive growth of the Internet has led to IPv4 address exhaustion, which entered its final stage in 2011, when the global address allocation pool was exhausted. A new protocol version, IPv6, was developed in the mid-1990s, which provides vastly larger addressing capabilities and more efficient routing of Internet traffic. IPv6 is currently in growing deployment around the world, since Internet address registries (RIRs) began to urge all resource managers to plan rapid adoption and conversion.

IPv6 is not directly interoperable by design with IPv4. In essence, it establishes a parallel version of the Internet not directly accessible with IPv4 software. Thus, translation facilities must exist for internetworking or nodes must have duplicate networking software for both networks. Essentially all modern computer operating systems support both versions of the Internet Protocol. Network infrastructure, however, has been lagging in this development. Aside from the complex array of physical connections that make up its infrastructure, the Internet is facilitated by bi- or multi-lateral commercial contracts, *e.g.,* peering agreements, and by technical specifications or protocols that describe the exchange of data over the network. Indeed, the Internet is defined by its interconnections and routing policies.

***Services***

The Internet carries many network services, most prominently mobile apps such as social media apps, the World Wide Web, electronic mail, multiplayer online games, Internet telephony, and file sharing services.

*World Wide Web*

Many people use the terms *Internet* and *World Wide Web*, or just the *Web*, interchangeably, but the two terms are not synonymous. The World Wide Web is the primary application that billions of people use on the Internet, and it has changed their lives immeasurably. However, the Internet provides many other services. The Web is a global set of documents, images and other resources, logically interrelated by hyperlinks and referenced with Uniform Resource Identifiers (URIs). URIs symbolically identify services, servers, and other databases, and the documents and resources that they can provide. Hypertext Transfer Protocol (HTTP) is the main access protocol of the World Wide Web. Web services also use HTTP to allow software systems to communicate in order to share and exchange business logic and data.

World Wide Web browser software, such as Microsoft's Internet Explorer, Mozilla Firefox, Opera, Apple's Safari, and Google Chrome, lets users navigate from one web page to another via hyperlinks embedded in the documents. These documents may also contain any combination of computer data, including graphics, sounds, text, video, multimedia and interactive content that runs while the user is interacting with the page. Client-side software can include animations, games, office applications and scientific demonstrations. Through keyword-driven Internet research using search engines like Yahoo! and Google, users worldwide have easy, instant access to a vast and diverse amount of online information. Compared to printed media, books, encyclopaedias and traditional libraries, the World Wide Web has enabled the decentralization of information on a large scale.

The Web has also enabled individuals and organizations to publish ideas and information to a potentially large audience online at greatly reduced expense and time delay. Publishing a web page, a blog, or building a web site involves little initial cost and many cost-free services are available. However, publishing and maintaining large, professional web sites with attractive, diverse and up-to-date information is still a difficult and expensive proposition. Many individuals and some companies and groups use *web logs* or blogs, which are largely used as easily updatable online diaries. Some commercial organizations encourage staff to communicate advice in their areas of specialization in the hope that visitors will be impressed by the expert knowledge and free information, and be attracted to the corporation as a result.

One example of this practice is Microsoft, whose product developers publish their personal blogs in order to pique the public's interest in their work. Collections of personal web pages published by large service providers remain popular and have become increasingly sophisticated. Whereas operations such as Angelfire and GeoCities have existed since the early days of the Web, newer offerings from, for example, Facebook and Twitter currently have large followings. These operations often brand themselves as social network services rather than simply as web page hosts.

Advertising on popular web pages can be lucrative, and e-commerce which is the sale of products and services directly via the Web, continues to grow. Online advertising is a form of marketing and advertising which uses the Internet to deliver promotional marketing messages to consumers. It includes e-mail marketing, search engine marketing (SEM), social media marketing, many types of display advertising (including web banner advertising), and mobile advertising. In 2011, Internet advertising revenues in the United States surpassed those of cable television and nearly exceeded those of broadcast television. Many common online advertising practices are controversial and increasingly subject to regulation.

When the Web developed in the 1990s, a typical web page was stored in completed form on a web server, formatted in HTML, complete for transmission to a web browser in response to a request. Over time, the process of creating and serving web pages has become dynamic, creating a flexible design, layout, and content. Web sites are often created using content management software with, initially, very little content. Contributors to these systems, who may be paid staff, members of an organization or the public, fill underlying databases with content using editing pages designed for that purpose while casual visitors view and read this content in HTML form. There may or may not be editorial, approval and security systems built into the process of taking newly entered content and making it available to the target visitors.

*Communication*

E-mail is an important communications service available on the Internet. The concept of sending electronic text messages between parties in a way analogous to mailing letters or memos predates the creation of the Internet. Pictures, documents, and other files are sent as e-mail attachments. E-mails can be cc-ed to multiple e-mail addresses.

Internet telephony is another common communications service made possible by the creation of the Internet. VoIP stands for Voice-over-Internet Protocol, referring to the protocol that underlies all Internet communication. The idea began in the early 1990s with walkie-talkie-like voice applications for personal computers. In recent years many VoIP systems have become as easy to use and as convenient as a normal telephone. The benefit is that, as the Internet carries the voice traffic, VoIP can be free or cost much less than a traditional telephone call, especially over long distances and especially for those with always-on Internet connections such as cable or ADSL. VoIP is maturing into a competitive alternative to traditional telephone service. Interoperability between different providers has improved and the ability to call or receive a call from a traditional telephone is available. Simple, inexpensive VoIP network adapters are available that eliminate the need for a personal computer.

Voice quality can still vary from call to call, but is often equal to and can even exceed that of traditional calls. Remaining problems for VoIP include emergency telephone number dialing and reliability. Currently, a few VoIP providers provide an emergency service, but it is not universally available. Older traditional phones with no "extra features" may be line-powered only and operate during a power failure; VoIP can never do so without a backup power source for the phone equipment and the Internet access devices. VoIP has also become increasingly popular for gaming applications, as a form of communication between players. Popular VoIP clients for gaming include Ventrilo and Teamspeak. Modern video game consoles also offer VoIP chat features.

*Data transfer*

File sharing is an example of transferring large amounts of data across the Internet. A computer file can be e-mailed to customers, colleagues and friends as an attachment. It can be uploaded to a web site or File Transfer Protocol (FTP) server for easy download by others. It can be put into a "shared location" or onto a file server for instant use by colleagues. The load of bulk downloads to many users can be eased by the use of "mirror" servers or peer-to-peer networks. In any of these cases, access to the file may be controlled by user authentication, the transit of the file over the Internet may be obscured by encryption, and money may change hands for access to the file. The price can be paid by the remote charging of funds from, for example, a credit card whose details are also passed – usually fully encrypted – across the Internet. The origin and authenticity of the file received may be checked by digital signatures or by MD5 or other message digests. These simple features of the Internet, over a worldwide basis, are changing the production, sale, and distribution of anything that can be reduced to a computer file for transmission. This includes all manner of print publications, software products, news, music, film, video, photography, graphics and the other arts. This in turn has caused seismic shifts in each of the existing industries that previously controlled the production and distribution of these products.

Streaming media is the real-time delivery of digital media for the immediate consumption or enjoyment by end users. Many radio and television broadcasters provide Internet feeds of their live audio and video productions. They may also allow time-shift viewing or listening such as Preview, Classic Clips and Listen Again features. These providers have been joined by a range of pure Internet "broadcasters" who never had on-air licenses.

This means that an Internet-connected device, such as a computer or something more specific, can be used to access on-line media in much the same way as was previously possible only with a television or radio receiver. The range of available types of content is much wider, from specialized technical webcasts to on-demand popular multimedia services. Podcasting is a variation on this theme, where – usually audio – material is downloaded and played back on a computer or shifted to a portable media player to be listened to on the move. These techniques using simple equipment allow anybody, with little censorship or licensing control, to broadcast audio-visual material worldwide.

Digital media streaming increases the demand for network bandwidth. For example, standard image quality needs 1 Mbit/s link speed for SD 480p, HD 720p quality requires 2.5 Mbit/s, and the top-of-the-line HDX quality needs 4.5 Mbit/s for 1080p.

Webcams are a low-cost extension of this phenomenon. While some webcams can give full-frame-rate video, the picture either is usually small or updates slowly. Internet users can watch animals around an African waterhole, ships in the Panama Canal, traffic at a local roundabout or monitor their own premises, live and in real time. Video chat rooms and video conferencing are also popular with many uses being found for personal webcams, with and without two-way sound.

YouTube was founded on 15 February 2005 and is now the leading web site for free streaming video with a vast number of users. It uses a flash-based web player to stream and show video files. Registered users may upload an unlimited amount of video and build their own personal profile. YouTube claims that its users watch hundreds of millions, and upload hundreds of thousands of videos daily. Currently, YouTube also uses an HTML5 player.

***Social Impact***

The Internet has enabled new forms of social interaction, activities, and social associations. This phenomenon has given rise to the scholarly study of the sociology of the Internet.

*Users*

Internet usage has seen tremendous growth. From 2000 to 2009, the number of Internet users globally rose from 394 million to 1.858 billion. By 2010, 22 percent of the world's population had access to computers with 1 billion Google searches every day, 300 million Internet users reading blogs, and 2 billion videos viewed daily on YouTube. In 2014 the world's Internet users surpassed 3 billion or 43.6 percent of world population, but two-thirds of the users came from richest countries, with 78.0 percent of Europe countries population using the Internet, followed by 57.4 percent of the Americas.

The prevalent language for communication on the Internet has been English. This may be a result of the origin of the Internet, as well as the language's role as a lingua franca. Early computer systems were limited to the characters in the American Standard Code for Information Interchange (ASCII), a subset of the Latin alphabet.

After English (27%), the most requested languages on the World Wide Web are Chinese (25%), Spanish (8%), Japanese (5%), Portuguese and German (4% each), Arabic, French and Russian (3% each), and Korean (2%). By region, 42% of the world's Internet users are based in Asia, 24% in Europe, 14% in North America, 10% in Latin America and the Caribbean taken together, 6% in Africa, 3% in the Middle East and 1% in Australia/Oceania. The Internet's technologies have developed enough in recent years, especially in the use of Unicode, that good facilities are available for development and communication in the world's widely used languages. However, some glitches such as *mojibake* (incorrect display of some languages' characters) still remain.

In an American study in 2005, the percentage of men using the Internet was very slightly ahead of the percentage of women, although this difference reversed in those under 30. Men logged on more often, spent more time online, and were more likely to be broadband users, whereas women tended to make more use of opportunities to communicate (such as e-mail). Men were more likely to use the Internet to pay bills, participate in auctions, and for recreation such as downloading music and videos. Men and women were equally likely to use the Internet for shopping and banking. More recent studies indicate that in 2008, women significantly outnumbered men on most social networking sites, such as Facebook and Myspace, although the ratios varied with age. In addition, women watched more streaming content, whereas men downloaded more. In terms of blogs, men were more likely to blog in the first place; among those who blog, men were more likely to have a professional blog, whereas women were more likely to have a personal blog.

According to forecasts by Euromonitor International, 44% of the world's population will be users of the Internet by 2020. Splitting by country, in 2012 Iceland, Norway, Sweden, the Netherlands, and Denmark had the highest Internet penetration by the number of users, with 93% or more of the population with access.

Several neo-logisms exist that refer to Internet users: Netizen (as in as in "citizen of the net") refers to those actively involved in improving online communities, the Internet in general or surrounding political affairs and rights such as free speech, Internaut refers to operators or technically highly capable users of the Internet, digital citizen refers to a person using the Internet in order to engage in society, politics, and government participation.

*Usage*

The Internet allows greater flexibility in working hours and location, especially with the spread of unmetered high-speed connections. The Internet can be accessed almost anywhere by numerous means, including through mobile Internet devices. Mobile phones, datacards, handheld game consoles and cellular routers allow users to connect to the Internet wirelessly. Within the limitations imposed by small screens and other limited facilities of such pocket-sized devices, the services of the Internet, including e-mail and the web, may be available. Service providers may restrict the services offered and mobile data charges may be significantly higher than other access methods.

Educational material at all levels from pre-school to post-doctoral is available from web sites. Examples range from CBeebies, through school and high-school revision guides and virtual universities, to access to top-end scholarly literature through the likes of Google Scholar. For distance education, help with homework and other assignments, self-guided learning, whiling away spare time, or just looking up more detail on an interesting fact, it has never been easier for people to access educational information at any level from anywhere. The Internet in general and the World Wide Web in particular are important enablers of both formal and informal education. Further, the Internet allows universities, in particular, researchers from the social and behavioural sciences, to conduct research remotely via virtual laboratories, with profound changes in reach and generalizability of findings as well as in communication between scientists and in the publication of results.

The low cost and nearly instantaneous sharing of ideas, knowledge, and skills have made collaborative work dramatically easier, with the help of collaborative software. Not only can a group cheaply communicate and share ideas but the wide reach of the Internet allows such groups more easily to form. An example of this is the free software movement, which has produced, among other things, Linux, Mozilla Firefox, and OpenOffice.org. Internet chat, whether using an IRC chat room, an instant messaging system, or a social networking web site, allows colleagues to stay in touch in a very convenient way while working at their computers during the day. Messages can be exchanged even more quickly and conveniently than via e-mail. These systems may allow files to be exchanged, drawings and images to be shared, or voice and video contact between team members.

Content management systems allow collaborating teams to work on shared sets of documents simultaneously without accidentally destroying each other's work. Business and project teams can share calendars as well as documents and other information. Such collaboration occurs in a wide variety of areas including scientific research, software development, conference planning, political activism and creative writing. Social and political collaboration is also becoming more widespread as both Internet access and computer literacy spread.

The Internet allows computer users to remotely access other computers and information stores easily from any access point. Access may be with computer security, *i.e.,* authentication and encryption technologies, depending on the requirements. This is encouraging new ways of working from home, collaboration and information sharing in many industries. An accountant sitting at home can audit the books of a company based in another country, on a server situated in a third country that is remotely maintained by IT specialists in a fourth. These accounts could have been created by home-working bookkeepers, in other remote locations, based on information e-mailed to them from offices all over the world. Some of these things were possible before the widespread use of the Internet, but the cost of private leased lines would have made many of them infeasible in practice. An office worker away from their desk, perhaps on the other side of the world on a business trip or a holiday, can access their e-mails, access their data using cloud computing, or open a remote desktop session into their office PC using a secure virtual private network (VPN) connection on the Internet. This can give the worker complete access to all of their normal files and data, including e-mail and other applications, while away from the office. It has been referred to among system administrators as the Virtual Private Nightmare, because it extends the secure perimeter of a corporate network into remote locations and its employees' homes.

*Social Networking and Entertainment*

Many people use the World Wide Web to access news, weather and sports reports, to plan and book vacations and to pursue their personal interests. People use chat, messaging and e-mail to make and stay in touch with friends worldwide, sometimes in the same way as some previously had pen pals. Social networking web sites such as Facebook, Twitter, and Myspace have created new ways to socialize and interact. Users of these sites are able to add a wide variety of information to pages, to pursue common interests, and to connect with others. It is also possible to find existing acquaintances, to allow communication among existing groups of people. Sites like LinkedIn foster commercial and business connections. YouTube and Flickr specialize in users' videos and photographs. While social networking sites were initially for individuals only, today they are widely used by businesses and other organizations to promote their brands, to market to their customers and to encourage posts to "go viral". "Black hat" social media techniques are also employed by some organizations, such as spam accounts and astroturfing.

A risk for both individuals and organizations writing posts (especially public posts) on social networking web sites, is that especially foolish or controversial posts occasionally lead to an unexpected and possibly large-scale backlash on social media from other Internet users. This is also a risk in relation to controversial *offline* behaviour, if it is widely made known. The nature of this backlash can range widely from counter-arguments and public mockery, through insults and hate speech, to, in extreme cases, rape and death threats. The online disinhibition effect describes the tendency of many individuals to behave more stridently or offensively online than they would in person. A significant number of feminist women have been the target of various forms of harassment in response to posts they have made on social media, and Twitter in particular has been criticised in the past for not doing enough to aid victims of online abuse.

For organizations, such a backlash can cause overall brand damage, especially if reported by the media. However, this is not always the case, as any brand damage in the eyes of people with an opposing opinion to that presented by the organization could sometimes be outweighed by strengthening the brand in the eyes of others. Furthermore, if an organization or individual gives in to demands that others perceive as wrong-headed, that can then provoke a counter-backlash.

Some web sites, such as Reddit, have rules forbidding the posting of personal information of individuals (also known as doxxing), due to concerns about such postings leading to mobs of large numbers of Internet users directing harassment at the specific individuals thereby identified. In particular, the Reddit rule forbidding the posting of personal information is widely understood to imply that all identifying photos and names must be censored in Facebook screenshots posted to Reddit. However, the interpretation of this rule in relation to public Twitter posts is less clear, and in any case, like-minded people online have many other ways they can use to direct each other's attention to public social media posts they disagree with.

Children also face dangers online such as cyberbullying and approaches by sexual predators, who sometimes pose as children themselves. Children may also encounter material which they may find upsetting, or material which their parents consider to be not age-appropriate. Due to naivety, they may also post personal information about themselves online, which could put them or their families at risk unless warned not to do so. Many parents choose to enable Internet filtering, and/or supervise their children's online activities, in an attempt to protect their children from inappropriate material on the Internet. The most popular social networking web sites, such as Facebook and Twitter, commonly forbid users under the age of 13. However, these policies are typically trivial to circumvent by registering an account with a false birth date, and a significant number of children aged under 13 join such sites anyway. Social networking sites for younger children, which claim to provide better levels of protection for children, also exist.

The Internet has been a major outlet for leisure activity since its inception, with entertaining social experiments such as MUDs and MOOs being conducted on university servers, and humor-related Usenet groups receiving much traffic. Many Internet forums have sections devoted to games and funny videos. The Internet pornography and online gambling industries have taken advantage of the World Wide Web, and often provide a significant source of advertising revenue for other web sites. Although many governments have attempted to restrict both industries' use of the Internet, in general, this has failed to stop their widespread popularity.

Another area of leisure activity on the Internet is multiplayer gaming. This form of recreation creates communities, where people of all ages and origins enjoy the fast-paced world of multiplayer games. These range from MMORPG to first-person shooters, from role-playing video games to online gambling. While online gaming has been around since the 1970s, modern modes of online gaming began with subscription services such as GameSpy and MPlayer. Non-subscribers were limited to certain types of game play or certain games.

Many people use the Internet to access and download music, movies and other works for their enjoyment and relaxation. Free and fee-based services exist for all of these activities, using centralized servers and distributed peer-to-peer technologies. Some of these sources exercise more care with respect to the original artists' copyrights than others.

Internet usage has been correlated to users' loneliness. Lonely people tend to use the Internet as an outlet for their feelings and to share their stories with others, such as in the "I am lonely will anyone speak to me" thread.

Cybersectarianism is a new organizational form which involves: "highly dispersed small groups of practitioners that may remain largely anonymous within the larger social context and operate in relative secrecy, while still linked remotely to a larger network of believers who share a set of practices and texts, and often a common devotion to a particular leader.

Overseas supporters provide funding and support; domestic practitioners distribute tracts, participate in acts of resistance, and share information on the internal situation with outsiders. Collectively, members and practitioners of such sects construct viable virtual communities of faith, exchanging personal testimonies and engaging in the collective study via e-mail, on-line chat rooms, and web-based message boards." In particular, the British government has raised concerns about the prospect of young British Muslims being indoctrinated into Islamic extremism by material on the Internet, being persuaded to join terrorist groups such as the so-called "Islamic State", and then potentially committing acts of terrorism on returning to Britain after fighting in Syria or Iraq.

Cyberslacking can become a drain on corporate resources; the average UK employee spent 57 minutes a day surfing the Web while at work, according to a 2003 study by Peninsula Business Services. Internet addiction disorder is excessive computer use that interferes with daily life. Nicholas G. Carr believes that Internet use has other effects on individuals, for instance improving skills of scan-reading and interfering with the deep thinking that leads to true creativity.

*Electronic Business*

Electronic business *(e-business)* encompasses business processes spanning the entire value chain: purchasing, supply chain management, marketing, sales, customer service, and business relationship. E-commerce seeks to add revenue streams using the Internet to build and enhance relationships with clients and partners. According to International Data Corporation, the size of worldwide e-commerce, when global business-to-business and -consumer transactions are combined, equate to $16 trillion for 2013. A report by Oxford Economics adds those two together to estimate the total size of the digital economy at $20.4 trillion, equivalent to roughly 13.8% of global sales. While much has been written of the economic advantages of Internet-enabled commerce, there is also evidence that some aspects of the Internet such as maps and location-aware services may serve to reinforce economic inequality and the digital divide. Electronic commerce may be responsible for consolidation and the decline of mom-and-pop, brick and mortar businesses resulting in increases in income inequality. Author Andrew Keen, a long-time critic of the social transformations caused by the Internet, has recently focused on the economic effects of consolidation from Internet businesses. Keen cites a 2013 Institute for Local Self-Reliance report saying brick-and-mortar retailers employ 47 people for every $10 million in sales while Amazon employs only 14. Similarly, the 700-employee room rental start-up Airbnb was valued at $10 billion in 2014, about half as much as Hilton Hotels, which employs 152,000 people. And car-sharing Internet startup Uber employs 1,000 full-time employees and is valued at $18.2 billion, about the same valuation as Avis and Hertz combined, which together employ almost 60,000 people.

*Telecommuting*

Telecommuting is the performance within a traditional worker and employer relationship when it is facilitated by tools such as groupware, virtual private networks, conference calling, videoconferencing, and voice over IP (VOIP) so that work may be performed from any location, most conveniently the worker's home. It can be efficient and useful for companies as it allows workers to communicate over long distances, saving significant amounts of travel time and cost. As broadband Internet connections become commonplace, more workers have adequate bandwidth at home to use these tools to link their home to their corporate intranet and internal communication networks.

*Crowdsourcing*

The Internet provides a particularly good venue for crowdsourcing, because individuals tend to be more open in web-based projects where they are not being physically judged or scrutinized and thus can feel more comfortable sharing.

*Collaborative Publishing*

Wikis have also been used in the academic community for sharing and dissemination of information across institutional and international boundaries. In those settings, they have been found useful for collaboration on grant writing, strategic planning, departmental documentation, and committee work. The United States Patent and Trademark Office uses a wiki to allow the public to collaborate on finding prior art relevant to examination of pending patent applications. Queens, New York has used a wiki to allow citizens to collaborate on the design and planning of a local park. The English Wikipedia has the largest user base among wikis on the World Wide Web and ranks in the top 10 among all Web sites in terms of traffic.

*Politics and Political Revolutions*

The Internet has achieved new relevance as a political tool. The presidential campaign of Howard Dean in 2004 in the United States was notable for its success in soliciting donation via the Internet. Many political groups use the Internet to achieve a new method of organizing for carrying out their mission, having given rise to Internet activism, most notably practiced by rebels in the Arab Spring. *The New York Times* suggested that social media web sites, such as Facebook and Twitter, helped people organize the political revolutions in Egypt, by helping activists organize protests, communicate grievances, and disseminate information.

The potential of the Internet as a civic tool of communicative power was explored by Simon R. B. Berdal in his 2004 thesis:

- As the globally evolving Internet provides ever new access points to virtual discourse forums, it also promotes new civic relations and associations within which communicative power may flow and accumulate. Thus, traditionally … national-embedded peripheries get entangled into greater, international peripheries, with stronger combined powers... The Internet, as a consequence, changes the topology of the "centre-periphery" model, by stimulating conventional peripheries to interlink into "super-periphery" structures, which enclose and "besiege" several centres at once.

Berdal, therefore, extends the Habermasian notion of the *public sphere* to the Internet, and underlines the inherent global and civic nature that interwoven Internet technologies provide. To limit the growing civic potential of the Internet, Berdal also notes how "self-protective measures" are put in place by those threatened by it:

- If we consider China's attempts to filter "unsuitable material" from the Internet, most of us would agree that this resembles a self-protective measure by the system against the growing civic potentials of the Internet. Nevertheless, both types represent limitations to "peripheral capacities". Thus, the Chinese government tries to prevent communicative power to build up and unleash (as the 1989 Tiananmen Square uprising suggests, the government may find it wise to install "upstream measures"). Even though limited, the Internet is proving to be an empowering tool also to the Chinese periphery: Analysts believe that Internet petitions have influenced policy implementation in favour of the public's online-articulated will …

Incidents of politically motivated Internet censorship have now been recorded in many countries, including western democracies.

*Philanthropy*

The spread of low-cost Internet access in developing countries has opened up new possibilities for peer-to-peer charities, which allow individuals to contribute small amounts to charitable projects for other individuals. Web sites, such as DonorsChoose and GlobalGiving, allow small-scale donors to direct funds to individual projects of their choice. A popular twist on Internet-based philanthropy is the use of peer-to-peer lending for charitable purposes. Kiva pioneered this concept in 2005, offering the first web-based service to publish individual loan profiles for funding. Kiva raises funds for local intermediary microfinance organizations which post stories and updates on behalf of the borrowers. Lenders can contribute as little as $25 to loans of their choice, and receive their money back as borrowers repay. Kiva falls short of being a pure peer-to-peer charity, in that loans are disbursed before being funded by lenders and borrowers do not communicate with lenders themselves.

However, the recent spread of low-cost Internet access in developing countries has made genuine international person-to-person philanthropy increasingly feasible. In 2009, the US-based non-profit Zidisha tapped into this trend to offer the first person-to-person microfinance platform to link lenders and borrowers across international borders without intermediaries. Members can fund loans for as little as a dollar, which the borrowers then use to develop business activities that improve their families' incomes while repaying loans to the members with interest. Borrowers access the Internet via public cybercafes, donated laptops in village schools, and even smart phones, then create their own profile pages through which they share photos and information about themselves and their businesses. As they repay their loans, borrowers continue to share updates and dialogue with lenders via their profile pages. This direct web-based connection allows members themselves to take on many of the communication and recording tasks traditionally performed by local organizations, bypassing geographic barriers and dramatically reducing the cost of microfinance services to the entrepreneurs.

***Security***

Internet resources, hardware, and software components are the target of criminal or malicious attempts to gain unauthorized control to cause interruptions, commit fraud, engage in blackmail or access private information.

*Malware*

Malicious software used and spread on the Internet includes computer viruses which copy with the help of humans, computer worms which copy themselves automatically, software for denial of service attacks, ransomware, botnets, and spyware that reports on the activity and typing of users. Usually, these activities constitute cybercrime.

Defence theorists have also speculated about the possibilities of cyber warfare using similar methods on a large scale.

*Surveillance*

The vast majority of computer surveillance involves the monitoring of data and traffic on the Internet. In the United States for example, under the Communications Assistance For Law Enforcement Act, all phone calls and broadband Internet traffic (e-mails, web traffic, instant messaging, etc.) are required to be available for unimpeded real-time monitoring by Federal law enforcement agencies. Packet capture is the monitoring of data traffic on a computer network. Computers communicate over the Internet by breaking up messages (e-mails, images, videos, web pages, files, etc.) into small chunks called “packets”, which are routed through a network of computers, until they reach their destination, where they are assembled back into a complete “message” again. Packet Capture Appliance intercepts these packets as they are traveling through the network, in order to examine their contents using other programmes. A packet capture is an information *gathering* tool, but not an *analysis* tool. That is it gathers “messages” but it does not analyze them and figure out what they mean. Other programmes are needed to perform traffic analysis and sift through intercepted data looking for important/ useful information. Under the Communications Assistance For Law Enforcement Act all U.S. telecommunications providers are required to install packet sniffing technology to allow Federal law enforcement and intelligence agencies to intercept all of their customers’ broadband Internet and voice over Internet protocol (VoIP) traffic.

The large amount of data gathered from packet capturing requires surveillance software that filters and reports relevant information, such as the use of certain words or phrases, the access of certain types of web sites, or communicating via e-mail or chat with certain parties. Agencies, such as the Information Awareness Office, NSA, GCHQ and the FBI, spend billions of dollars per year to develop, purchase, implement, and operate systems for interception and analysis of data. Similar systems are operated by Iranian secret police to identify and suppress dissidents. The required hardware and software was allegedly installed by German Siemens AG and Finnish Nokia.

*Censorship*

Some governments, such as those of Burma, Iran, North Korea, the Mainland China, Saudi Arabia and the United Arab Emirates restrict access to content on the Internet within their territories, especially to political and religious content, with domain name and keyword filters.

In Norway, Denmark, Finland, and Sweden, major Internet service providers have voluntarily agreed to restrict access to sites listed by authorities. While this list of forbidden resources is supposed to contain only known child pornography sites, the content of the list is secret. Many countries, including the United States, have enacted laws against the possession or distribution of certain material, such as child pornography, via the Internet, but do not mandate filter software. Many free or commercially available software programmes, called content-control software are available to users to block offensive web sites on individual computers or networks, in order to limit access by children to pornographic material or depiction of violence.

***Performance***

As the Internet is a heterogeneous network, the physical characteristics, including for example the data transfer rates of connections, vary widely. It exhibits emergent phenomena that depend on its large-scale organization.

*Outages*

An Internet blackout or outage can be caused by local signalling interruptions. Disruptions of submarine communications cables may cause blackouts or slowdowns to large areas, such as in the 2008 submarine cable disruption. Less-developed countries are more vulnerable due to a small number of high-capacity links. Land cables are also vulnerable, as in 2011 when a woman digging for scrap metal severed most connectivity for the nation of Armenia. Internet blackouts affecting almost entire countries can be achieved by governments as a form of Internet censorship, as in the blockage of the Internet in Egypt, whereby approximately 93% of networks were without access in 2011 in an attempt to stop mobilization for anti-government protests.

*Energy use*

In 2011, researchers estimated the energy used by the Internet to be between 170 and 307 GW, less than two percent of the energy used by humanity. This estimate included the energy needed to build, operate, and periodically replace the estimated 750 million laptops, a billion smart phones and 100 million servers worldwide as well as the energy that routers, cell towers, optical switches, Wi-Fi transmitters and cloud storage devices use when transmitting Internet traffic.

## INTEGRATION OF PHYSICAL EDUCATION WITH OTHER SPORTS SCIENCES

The real world is neither fragmented nor boxed into separate compartments that are totally independent of each other. Life is a mix of experiences, each relating to another. Similarly, the school curriculum should be integrated to actively engage children in a learning process consisting of complementary curricular areas.

### What is Integrated/ Interdisciplinary Learning?

- Several subject areas being integrated with the goal of enhancing learning in each subject area.
- Not a new concept but in the past it has not involved the integration of physical education with other subject areas.

- There is no one specific model that can be followed to integrate subjects.
- Integration can occur when there is a “focal point” which can be targeted across curricular areas.
- It is important to implement integration when a learning problem arises which requires involvement of more than one subject area to solve.

The SACSA Framework believes integration of the physical education curriculum is vital. It states;

Learning in health and physical education promotes the integration of physical, social, emotional, environmental and spiritual dimensions of living, and includes areas such as health education, physical education, home economics, outdoor education and sport education…it also includes work education, community studies, integrated studies, nutrition and personal development studies, and cross-disciplinary studies such as women’s issues.

With the current focus in Australia on the numeracy and literacy deficiencies of our children, it is also important to note that the framework also states;

Within the studies of health and physical education there are significant opportunities provided for learners to use a range of literacy, numeracy and information and communication technologies skills. This is evident in the use of appropriate terminology and Information Communications Technology (ICTs) to communicate ideas and explore concepts of space, time, shape and measurement. Learners select and use a range of ways and modes to structure and report their learning, enhancing the process by the use and application of ICTs in planning, drafting, editing and presenting.

In texts published prior to the 1990’s it is impossible to find reference to integration of physical education with other subjects. This is an accurate reflection of how physical education has been neglected until recent years as a curricular area which can be integrated with maths, english, science, languages and many other subjects.

Physical education provides social, physical, mental and emotional learning that can be applied not only in the physical education curriculum but all learning areas, and indeed throughout a lifetime. (Singer, R. and Dick, W.,1974, p 16). Ancient Greeks recognised these widespread benefits by placing great emphasis on physical activity and development. (Halsey, E., 1964, p 11) Nowadays, however, physical education is being increasingly squeezed out of school programmes for a number of reasons. One line of thought is that subjects such as maths, literacy and science are far more important to students because these are what are required to proceed through further education towards employment. This argument falls apart when one becomes aware of how physical education can be integrated with these subject areas to maximise the learning in both of them

There are many ways physical education can be integrated with other subject areas. The key to this being achieved successfully is that there is a central theme of “focal point” which relates to the subjects being integrated. This “focal point” will then be learnt more effectively when integration takes place.

**Examples of how Physical Education can be Integrated with Other Subjects**

- An athletics unit in physical education can involve students recording their achievements in long jump, sprints, javelin etc. This data can then be compiled and used in a maths class where students can obtain a class average, an average for boys/girls. Following on from this, in an IT lesson, students could use this data to compile graphs, tables, spreadsheets, even a PowerPoint presentation.
- An exercise circuit in which students take their pulse can be followed by a discussion of how exercise effects the cardio-vascular system. This can be complemented in a science classroom where a cow's heart may be dissected to demonstrate how our hearts function and are affected by exercise.
- A dance unit in the physical education curriculum which emphasises movement as a form of expression can be integrated with a creative writing exercise where students have to write what they feel a dance represents.
- A physical education unit involving foreign sports such as bocce or hurling can be incorporated with a society and environment unit looking at countries and cultures around the world.
- Any physical education lesson can be integrated with the concept of biomechanics and the subject area of physics to explain how our muscles and bones operate. This could then lead to a unit in maths looking at angles such as those formed at the knees and elbows during specific physical activities.

**Advantages of Integration**

- Can help students recognise the links between their learning and real life/other contexts.
- Can alleviate the pressure on teachers to try and fit every subject area into their planning.
- Connects ideas in the curriculum.
- Enhances learning by introducing content from new perspectives.
- Breaks disciplinary boundaries, thereby providing teacher more flexibility.
- Students can often draw more meaning and relevance from learning.
- Acts as a reinforcement tool.

**Disadvantages of Integration**

- Requires additional programme planning to coordinate different subject areas to run concurrently.
- The scope and purity of some subjects can be lost/confused.
- The teachers need expertise to be able to integrate effectively so as to maximise learning
- Needs to be taught at appropriate time/context or can become confusing and detrimental to learning.

Integrating of physical education can go beyond simply integrating with other subjects. Physical education can be integrated into many aspects of the school environment. The Active For Life, Active Schools and Jump Rope for Heart programmes are just some of the initiatives a school can get involved in to integrate physical activity throughout the school.

The physical education curriculum can also incorporate skills that can be applied in many subjects. A project which requires research in the library, on the internet or through conducting interviews as well as any form of resource-based learning requires students to practice skills that can be used in any other subject. (Harris, J. p 67)

**CURRICULUM RESEARCH**

Curriculum Studies researchers ranging across the spectrum of paradigms — from conservative structural-functionalists, to neo-Marxists to narrative- and arts-based researchers — have examined formal curricula, experienced curricula, and hidden curricula. Progressive researchers like Paul Willis (1977, *Learning to Labour: How Working Class Kids Get Working Class Jobs*), Jean Anyon (1980, "Social Class and the Hidden Curriculum of Work." *Journal of Education* 162), and Annette Laureau (1989. *Home Advantage: Social Class, and Parental Intervention in Elementary Education*) have examined the ways that hidden and overt curricula reproduce social class position. Narrative and arts-based researchers like Thomas Barone (2001, *Touching Eternity: The Enduring Outcomes of Teaching*) have enquired about the long-term effects of curricula on student lives.

Critical theorists like Henry Giroux (1983 "Theories of Reproduction and Resistance in the New Sociology of Education: A critical analysis." *Harvard Educational Review* 53) began to examine the roles of students and teachers in resisting curricula both official and hidden. So-called "resistance theorists" conceptualized students and teachers as active agents working to subvert, reject, or change curricula. They noted that "curriculum" was not a unified structure but incoherent conflicting and contradictory messages. Other researchers have examined the interactions between racial and ethnic cultures and the dominant curricula of the school. For instance the Anthropologist John Ogbu examined curricula established by African American students (Signithia Fordham and John Ogbu 1986 "Black Students' School Success: Coping with the Burden of 'Acting White.' *The Urban Review*18. Critical Race Theorists Critical race theory like Daniel Solórzano examined how racial attitudes constitute another "hidden" curriculum in teacher education programmes (1997, Images and Words that Wound: Critical Race Theory and Racial Stereotyping, and Teacher Education, *Teacher Education Quarterly*, 24). Additionally, Judith Stacey proposed in the 1960s schools conveyed a hidden curriculum that perpetuated the "sexist beliefs, attitudes, and values" of the time period (1974 "And Jill Came Tumbling After Sexism in American Education").

The interest in Curriculum Studies is thus cross disciplinary and of increasing importance to educational research and to the philosophy of education.

## OBJECTIVES OF CURRICULUM RESEARCH

### Curriculum Objectives

*The objective of the professional master programme is to prepare graduates who can:*

- Practice independently in a range of settings and in a variety of roles; for example, in
- Interdisciplinary programmes, in private practice, in primary health care settings, and as consultants and case managers
- Supervise rehabilitation assistants, OT aides or other support workers
- Use principles of evidence-based practice to guide and evaluate service delivery
- Contribute to research that will advance the knowledge base of the discipline
- Assume management roles
- Take leadership roles in the profession
- Take leadership roles in health care and other sectors including social services, education and labour
- Fill academic-practitioner positions
- Pursue a PhD and careers in academia or clinical research, following some additional preparation

### Research Objectives

*The Department's research is focused on occupation. In broad terms, the faculty's research interests are:*

- Culture, transcultural competence
- International Development
- Return to Work
- Aging in Place
- Kids' Skills
- Technology (Hardware, Instruments and Measurement Tools, Rx Protocols, Orthotics)
- Theory and Model Development
- Effectiveness and Accountability (Programme Evaluation, Practice Processes, Clinical Decision-making)
- Social and Policy Contexts of Practice.

## IMPORTANCE OF CURRICULUM RESEARCH

In most countries the ministry of education establishes curricular standards. In the United States, however, control of education is a state's right, and in many states, for example, Montana, state constitutions give control of education to local districts. The federal government influences education through funding initiatives, such as the No Child Left Behind Act in 2001. The 2010 Common Core State Standards (CCSS) initiative is not a federal programme but has been adopted and is being implemented by 45 of the 50 states and the District of Columbia. China also does not have a mandated national curriculum. China Mainland, including Shanghai, has common standards; Hong Kong, Taiwan and Macau create their own standards/curriculum goals.

In many countries, standards/curricular goals are set by historical tradition or cultural norms. For example, Namibia used the Cambridge curriculum when they became independent in 1990 and only recently has begun to develop their its own standards. Brazil ‘s standards are attributed to the history of the discipline, the prescribed curricula, and the comparative analysis among national documents from different historical periods and national and international documents. Some countries base their standards and guidelines on those of countries with high achievement scores on recent international exams. For example, both England and the United States cite countries such those from the Pacific Rim and Finland as resources for their new standards. Peru noted that an analysis of documents from other countries in South American and from TIMSS, Programme for International Student Assessment (PISA), and National Council of Teachers of Mathematics (NCTM) contributed to the development of their Diseño Curricular Nacional (CND) (National Curricular Design) (2009).

***Why Standards?***

Over time, many countries have changed from local standards to national standards. For example, Brazil found that the lack of national standards contributed to unequal opportunity for education. For much the same reason, the documented difference in the rigour and quality of individual state standards, the state governors in the United States supported the development and adoption of the CCSS. The new US standards are intended to be substantially more focused and coherent.

Standards are viewed as political: *i.e.,* Brazil suggests that mathematics curricular goals depend more on political timing, election campaigns and government administrations, where “the logic of an education agenda that transcends governments and politicians’ mandates, set as a goal for a democratic and developed society, is not the rule” (Response to ICME 12 Curriculum Survey 2011, p. 6). In the United States the two major political parties have different views on education, its funding and its goals. This has recently given rise to the creation of publicly funded schools governed by a group or organization with a legislative contract or charter from a state or jurisdiction that exempts the school from selected state or local regulations in keeping with its charter. Hong Kong also reported that writing standards seems to be more politically based than research based. Many of the changes in England’s National Curriculum (NC) are the result of criticism from the current government that the NC is over-prescriptive, includes non-essential material, and specifies teaching method rather than content. In Peru each new curricular proposal is viewed as an adjustment to the prior curriculum. In this process, radical changes do occur, such as changing the curriculum by capabilities (CND 2005) to the curriculum by competencies (CND 2008) in the secondary education level. These decisions are often the result of a policy change with each new government.

In most countries surveyed, a diverse team, including mathematics education researchers, ministry of education staff, curriculum supervisors, and representatives of boards of education are responsible for developing the standards/goals. In some countries (Japan, Australia) teachers are involved, but in others the design teams are primarily experts from universities, teaching universities or the ministry of education (Indonesia, Egypt). The design of the framework for the National Curriculum in England is carried out by a panel of four, not necessarily mathematics educators, charged to reflect the view of the broader mathematics education community including teachers.

***What is the Role of Research?***

Research has different interpretations and meanings in relation to the development and implementation of standards or curricula guidelines. One common response in the surveys was to cite as research the resources used in preparing standards (for example, other countries' standards). In addition, the degree to which research is used in compiling the standards often depends on the vision, perspectives and beliefs of the team responsible for the development.

The use of research related to student learning in developing standards/curricular goals is not common among the countries surveyed. A typical description of the process was given by Hong Kong, where the development team might do a literature review and refer to documents of other countries, but the process is not necessarily well structured and often depends on the expertise of the team members. England, however, noted that the first version of their National Curriculum (NC) was largely based on the Concepts in Secondary Mathematics and Science project, (Hart 1981) that sought to formulate hierarchies of understanding in 10 mathematical topics normally taught in British secondary schools based on the results of testing 10,000 children in 1976 and 1977.

The NC was also based on the ILEA Checkpoints (1979) and the Graded Assessment in Mathematics (1988, 1989, 1990) projects. The original research-based design of the NC had many unintended consequences. Although the attainment targets were intended to measure learning outcomes on particular tasks, the levels were used to define the order in which topics should be taught, rather than paying attention to the development of concepts over time. The processes of mathematics, originally called "Using and applying mathematics" were defined in a general way related to progressions and levels that made interpretation difficult. As a consequence, the NC was revised several times and as of summer 2012 was again in the process of revision.

After a 1996 survey showed that social segmentation in Brazil seemed to be an obstacle to access to a quality education, research led to the development of the National Curricular Parameters in Brazil (1997). The Board of National Standards for Education (*Badan Standar Nasional Pendidikan*) in Indonesia examined the national needs for education, the vision of the country, societal demands, challenges for the future, and used their findings in developing the curriculum (Ministry of National Education 2006).

***What is the Nature of Standards?***

In Brazil, Indonesia, Namibia and Peru, the standards/curricular frameworks are general and provide overarching guidelines for the development of discipline specific content. In the United States, Australia, and Japan, the mathematical standards essentially stand alone, although supporting documents may illustrate how the maths standards fit into the larger national education philosophy and perspective. Some standards include process goals. For example, Australia includes standards for four proficiencies (understanding, fluency, problem solving and reasoning) based on those described in *Adding It Up* (Kilpatrick et al. 2001). The new Australian standards want students to see that mathematics is about creating connections, developing strategies, and effective communication, as well as following rules and procedures. The United States CCSS has mathematical practice standards specifying eight "habits of mind" students should have when doing mathematics. In Brazil ideas such as "learn to learn", "promote independence", "learn to solve problems" are being incorporated into new curricula. In Peru and Indonesia the emphasis is primarily on the processes of problem solving, reasoning and proof, and mathematical communication.

In some cases standards reinforce the role of education in responding to the needs of the country. For example, the Curriculum for Basic Education (1st–9th grade) in Honduras (Department of Education 2003) was developed under three axes: personal, national and cultural identity, and democracy and work. The four pillars of lifelong learning defined by Delors (1996) (personal fulfilment, active citizenship, social inclusion and employability/adaptability) were used to define the mathematical content and methodological guides with problem solving as the central umbrella. Namibia's National Curriculum for a Basic Education outlines the aims of a basic education for the society of the future and specifies a few very general learning outcomes for each educational level (Namibia MoE 2008).

Standards span different sets of school grades or levels and differ in generality. Some countries have grade specific standards for what students should know throughout their primary and secondary schooling (*i.e.,* US, Japan). Australia specifies a common curriculum for grades 1–10 and course options for students in upper secondary. Egypt and Honduras have curricular goals for students in grades 1–9 (age 14). At the high school level, Honduras focuses on post high school preparation with more than 53 career- focused schools for students.

The development of fractions in Australia by the Australian Curriculum and Assessment Reporting Authority (ACARA 2011), the Japanese Ministry of Education, Culture, Sports, Science and Technology (MEXT 2008), the Ministry of Education in Namibia (MoE 2005, 2006), and the US (CCSS 2010) illustrates the difference in standards across countries In grade 1, the standards/goals in the US, Namibia and

Australia introduce words such as half, quarter and whole; this happens in grade 2 in Japan. Both US and Japan treat fraction as a number on the number line beginning in grade 3, emphasize equal partitioning of a unit and consider a fraction as composed of unit fractions: 4/3 = 4 units of 1/3. Australia suggests relating fractions to a number line only for unit fractions in grade 3, while Namibia does not mention fractions in relation to the number line. Equivalent fractions are taught in grade 4 in US, Japan, and Australia and in grade 6 in Namibia. Addition and subtraction of fractions with like denominators occurs in grade 4 in Japan, with unlike denominators in grade 5 in the US and Japan, and grade 7 in Namibia and Australia. Australia and Namibia have fractions as parts of collections in grade 2 and again in grade 4 in Namibia, but fractions as subsets of a collection are not mentioned in the standards/goals in the US and Japan. Students are expected to multiply and divide fractions in grade 5 in the US (with the exception of division of a fraction by a fraction, which happens in grade 6), in grade 6 in Japan, and in grade 7 in Australia and Namibia.

## EVALUATION OF CURRICULUM

The term "evaluation" generally applies to the process of making a value judgement. In education, the term "evaluation" is used in reference to operations associated with curricula, programmes, interventions, methods of teaching and organizational factors. Curriculum evaluation aims to examine the impact of implemented curriculum on student (learning) achievement so that the official curriculum can be revised if necessary and to review teaching and learning processes in the classroom. Curriculum evaluation establishes:

- Specific strengths and weaknesses of a curriculum and its implementation;
- Critical information for strategic changes and policy decisions;
- Inputs needed for improved learning and teaching;
- Indicators for monitoring.

Curriculum evaluation may be an internal activity and process conducted by the various units within the education system for their own respective purposes. These units may include national Ministries of Education, regional education authorities, institutional supervision and reporting systems, departments of education, schools and communities.

Curriculum evaluation may also be external or commissioned review processes. These may be undertaken regularly by special committees or task forces on the curriculum, or they may be research-based studies on the state and effectiveness of various aspects of the curriculum and its implementation. These processes might examine, for example, the effectiveness of curriculum content, existing pedagogies and instructional approaches, teacher training and textbooks and instructional materials.

## METHODS OF EVALUATION

An evaluation can use quantitative or qualitative data, and often includes both. Both methods provide important information for evaluation, and both can improve community engagement. These methods are rarely used alone; combined, they generally provide the best overview of the project. This section describes both quantitative and qualitative methods, and Table 7.1 shows examples of quantitative and qualitative questions according to stage of evaluation.

**Quantitative Methods**

Quantitative data provide information that can be counted to answer such questions as “How many?”, “Who was involved?”, “What were the outcomes?”, and “How much did it cost?” Quantitative data can be collected by surveys or questionnaires, pretests and posttests, observation, or review of existing documents and databases or by gathering clinical data. Surveys may be self- or interviewer-administered and conducted face-to-face or by telephone, by mail, or online. Analysis of quantitative data involves statistical analysis, from basic descriptive statistics to complex analyses.

Quantitative data measure the depth and breadth of an implementation (*e.g.,* the number of people who participated, the number of people who completed the programme). Quantitative data collected before and after an intervention can show its outcomes and impact. The strengths of quantitative data for evaluation purposes include their generalizability (if the sample represents the population), the ease of analysis, and their consistency and precision (if collected reliably). The limitations of using quantitative data for evaluation can include poor response rates from surveys, difficulty obtaining documents, and difficulties in valid measurement. In addition, quantitative data do not provide an understanding of the program’s context and may not be robust enough to explain complex issues or interactions (Holland et al., 2005; Garbarino et al., 2009).

**Qualitative Methods**

Qualitative data answer such questions as “What is the value added?”, “Who was responsible?”, and “When did something happen?” Qualitative data are collected through direct or participant observation, interviews, focus groups, and case studies and from written documents. Analyses of qualitative data include examining, comparing and contrasting, and interpreting patterns. Analysis will likely include the identification of themes, coding, clustering similar data, and reducing data to meaningful and important points, such as in grounded theory-building or other approaches to qualitative analysis (Patton, 2002).

Observations may help explain behaviours as well as social context and meanings because the evaluator sees what is actually happening. Observations can include watching a participant or programme, videotaping an intervention, or even recording people who have been asked to “think aloud” while they work (Ericsson et al., 1993).

Interviews may be conducted with individuals alone or with groups of people and are especially useful for exploring complex issues. Interviews may be structured and conducted under controlled conditions, or they may be conducted with a loose set of questions asked in an open-ended manner. It may be helpful to tape-record interviews, with appropriate permissions, to facilitate the analysis of themes or content. Some interviews have a specific focus, such as a critical incident that an individual recalls and describes in detail. Another type of interview focuses on a person's perceptions and motivations.

Focus groups are run by a facilitator who leads a discussion among a group of people who have been chosen because they have specific characteristics (*e.g.,* were clients of the programme being evaluated). Focus group participants discuss their ideas and insights in response to open-ended questions from the facilitator. The strength of this method is that group discussion can provide ideas and stimulate memories with topics cascading as discussion occurs (Krueger et al., 2000; Morgan, 1997).

The strengths of qualitative data include providing contextual data to explain complex issues and complementing quantitative data by explaining the "why" and "how" behind the "what." The limitations of qualitative data for evaluation may include lack of generalizability, the time-consuming and costly nature of data collection, and the difficulty and complexity of data analysis and interpretation (Patton, 2002).

**Mixed Methods**

The evaluation of community engagement may need both qualitative and quantitative methods because of the diversity of issues addressed (*e.g.,* population, type of project, and goals). The choice of methods should fit the need for the evaluation, its timeline, and available resources (Holland et al., 2005; Steckler et al., 1992).

# 3
# CHAPTER

## Vocational School: Trade and Industrial

A school is an institution designed to provide learning spaces and learning environments for the teaching of students (or "pupils") under the direction of teachers. Most countries have systems of formal education, which is commonly compulsory. In these systems, students progress through a series of schools. The names for these schools vary by country (discussed in the *Regional* section below) but generally include primary school for young children and secondary school for teenagers who have completed primary education. An institution where higher education is taught, is commonly called a university college or university.

In addition to these core schools, students in a given country may also attend schools before and after primary and secondary education. Kindergarten or pre-school provide some schooling to very young children (typically ages 3–5). University, vocational school, college or seminary may be available after secondary school. A school may be dedicated to one particular field, such as a school of economics or a school of dance. Alternative schools may provide non-traditional curriculum and methods.

There are also non-government schools, called private schools. Private schools may be required when the government does not supply adequate, or special education. Other private schools can also be religious, such as Christian schools, madrasa, hawzas (Shi'a schools), yeshivas (Jewish schools), and others; or schools that have a higher standard of education or seek to foster other personal achievements. Schools for adults include institutions of corporate training, military education and training and business schools.

In home schooling and online schools, teaching and learning take place outside a traditional school building. Schools are commonly organized in several different organizational models, including departmental, small learning communities, academies, integrated, and schools-within-a-school.

## HISTORY AND DEVELOPMENT

The concept of grouping students together in a centralized location for learning has existed since Classical antiquity. Formal schools have existed at least since ancient Greece, ancient Rome ancient India, and ancient China. The Byzantine Empire had an established schooling system beginning at the primary level. According to *Traditions and Encounters*, the founding of the primary education system began in 425 AD and "... military personnel usually had at least a primary education...". The sometimes efficient and often large government of the Empire meant that educated citizens were a must. Although Byzantium lost much of the grandeur of Roman culture and extravagance in the process of surviving, the Empire emphasized efficiency in its war manuals. The Byzantine education system continued until the empire's collapse in 1453 AD.

In Western Europe a considerable number of cathedral schools were founded during the Early Middle Ages in order to teach future clergy and administrators, with the oldest still existing, and continuously operated, cathedral schools being The King's School, Canterbury (established 597 CE), King's School, Rochester (established 604 CE), St Peter's School, York (established 627 CE) and Thetford Grammar School (established 631 CE). Beginning in the 5th century CE monastic schools were also established throughout Western Europe, teaching both religious and secular subjects.

Islam was another culture that developed a school system in the modern sense of the word. Emphasis was put on knowledge, which required a systematic way of teaching and spreading knowledge, and purpose-built structures. At first, mosques combined both religious performance and learning activities, but by the 9th century, the madrassa was introduced, a school that was built independently from the mosque, such as al-Qarawiyyin, founded in 859 CE. They were also the first to make the *Madrassa* system a public domain under the control of the Caliph.

Under the Ottomans, the towns of Bursa and Edirne became the main centers of learning. The Ottoman system of Külliye, a building complex containing a mosque, a hospital, madrassa, and public kitchen and dining areas, revolutionized the education system, making learning accessible to a wider public through its free meals, health care and sometimes free accommodation.

In Europe, universities emerged during the 12th century; here, scholasticism was an important tool, and the academicians were called *schoolmen*. During the Middle Ages and much of the Early Modern period, the main purpose of schools (as opposed to universities) was to teach the Latin language. This led to the term grammar school, which in the United States informally refers to a primary school, but in the United Kingdom means a school that selects entrants based on ability or aptitude. Following this, the school curriculum has gradually broadened to include literacy in the vernacular language as well as technical, artistic, scientific and practical subjects. Obligatory school attendance became common in parts of Europe during the 18th century. In Denmark-Norway, this was introduced as early as in 1739-1741, the primary end being to increase the literacy of the *almue*, *i.e.,* the "regular people". Many of the earlier public schools in the United States and elsewhere were one-room schools where a single teacher taught seven grades of boys and girls in the same classroom. Beginning in the 1920s, one-room schools were consolidated into multiple classroom facilities with transportation increasingly provided by kid hacks and school buses.

## REGIONAL TERMS

The use of the term *school* varies by country, as do the names of the various levels of education within the country.

### UNITED KINGDOM AND COMMONWEALTH OF NATIONS

In the United Kingdom, the term *school* refers primarily to pre-university institutions, and these can, for the most part, be divided into pre-schools or nursery schools, primary schools (sometimes further divided into infant school and junior school), and secondary schools. Various types of secondary schools in England and Wales include grammar schools, comprehensives, secondary moderns, and city academies. In Scotland, while they may have different names, all Secondary schools are the same, except in that they may be funded by the state, or independently funded. It is unclear if "Academies", which are a hybrid between state and independently funded/controlled schools and have been introduced to England in recent years, will ever be introduced to Scotland. School performance in Scotland is monitored by Her Majesty's Inspectorate of Education. Ofsted reports on performance in England and Estyn reports on performance in Wales.

In the United Kingdom, most schools are publicly funded and known as state schools or maintained schools in which tuition is provided free. There are also private schools or independent schools that charge fees. Some of the most selective and expensive private schools are known as public schools, a usage that can be confusing to speakers of North American English. In North American usage, a public school is one that is publicly funded or run.

In much of the Commonwealth of Nations, including Australia, New Zealand, India, Pakistan, Bangladesh, Sri Lanka, South Africa, Kenya, and Tanzania, the term *school* refers primarily to pre-university institutions.

### INDIA

In ancient India, schools were in the form of Gurukuls. Gurukuls were traditional Hindu residential schools of learning; typically the teacher's house or a monastery. During the Mughal rule, Madrasahs were introduced in India to educate the children of Muslim parents. British records show that indigenous education was widespread in the

18th century, with a school for every temple, mosque or village in most regions of the country. The subjects taught included Reading, Writing, Arithmetic, Theology, Law, Astronomy, Metaphysics, Ethics, Medical Science and Religion.

Under the British rule in India, Christian missionaries from England, USA and other countries established missionary and boarding schools throughout the country. Later as these schools gained in popularity, more were started and some gained prestige. These schools marked the beginning of modern schooling in India and the syllabus and calendar they followed became the benchmark for schools in modern India. Today most of the schools follow the missionary school model in terms of tutoring, subject/ syllabus, governance etc. with minor changes. Schools in India range from schools with large campuses with thousands of students and hefty fees to schools where children are taught under a tree with a small/ no campus and are totally free of cost. There are various boards of schools in India, namely Central Board for Secondary Education (CBSE), Council for the Indian School Certificate Examinations (CISCE), Madrasa Boards of various states, Matriculation Boards of various states, State Boards of various boards, Anglo Indian Board, and so on. The typical syllabus today includes Language (s), Mathematics, Science — Physics, Chemistry, Biology, Geography, History, General Knowledge, Information Technology/ Computer Science etc.. Extra curricular activities include physical education/ sports and cultural activities like music, choreography, painting, theater/ drama etc.

**EUROPE**

In much of continental Europe, the term *school* usually applies to primary education, with primary schools that last between four and nine years, depending on the country. It also applies to secondary education, with secondary schools often divided between *Gymnasiums* and vocational schools, which again depending on country and type of school educate students for between three and six years. In Germany students graduating from Grundschule are not allowed to directly progress into a vocational school, but are supposed to proceed to one of Germany's general education schools such as Gesamtschule, Hauptschule, Realschule or Gymnasium. When they leave that school, which usually happens at age 15-19 they are allowed to proceed to a vocational school. The term school is rarely used for tertiary education, except for some *upper* or *high* schools (German: Hochschule), which describe colleges and universities.

In Eastern Europe modern schools (after World War II), of both primary and secondary educations, often are combined, while secondary education might be split into accomplished or not. The schools are classified as middle schools of general education and for the technical purposes include "degrees" of the education they provide out of three available: the first — primary, the second — unaccomplished secondary, and the third — accomplished secondary. Usually the first two degrees of education (eight years) are always included, while the last one (two years) gives option for the students to pursue vocational or specialized educations.

### NORTH AMERICA AND THE UNITED STATES

In North America, the term *school* can refer to any educational institution at any level, and covers all of the following: preschool (for toddlers), kindergarten, elementary school, middle school (also called intermediate school or junior high school, depending on specific age groups and geographic region), high school (or in some cases senior high school), college, university, and graduate school.

In the United States, school performance through high school is monitored by each state's department of education. Charter schools are publicly funded elementary or secondary schools that have been freed from some of the rules, regulations, and statutes that apply to other public schools. The terms grammar school and *grade school* are sometimes used to refer to a primary school.

## OWNERSHIP AND OPERATION

Many schools are owned or funded by states. Private schools operate independently from the government. Private schools usually rely on fees from families whose children attend the school for funding; however, sometimes such schools also receive government support (for example, through School vouchers). Many private schools are affiliated with a particular religion; these are known as parochial schools.

### STARTING A SCHOOL

The Toronto District School Board is an example of a school board that allows parents to design and propose new schools.

*When designing a school, factors that need to be decided include:*

- *Goals:* What is the purpose of education, and what is the school's role?
- *Governance:* Who will make which decisions?
- *Parent involvement:* In which ways are parents welcome at the school?
- *Student body:* Will it be, for example, a neighbourhood school or a specialty school?
- *Student conduct:* What behaviour is acceptable, and what happens when behaviour is inappropriate?
- *Curriculum:* What will be the curriculum model, and who will decide on curricula?

## COMPONENTS OF MOST SCHOOLS

Schools are organized spaces purposed for teaching and learning. The classrooms, where teachers teach and students learn, are of central importance. Classrooms may be specialized for certain subjects, such as laboratory classrooms for science education and workshops for industrial arts education.

Typical schools have many other rooms and areas, which may include:

- Cafeteria (Commons), dining hall or canteen where students eat lunch and often breakfast and snacks.

- Athletic field, playground, gym, and/or track place where students participating in sports or physical education practice
- School yards, that is, all-purpose playfields typically in elementary schools, often made of concrete, although some are being transformed into environmentally friendly teaching gardens by landscape artists such as Sharon Gamson Danks.
- Auditorium or hall where student theatrical and musical productions can be staged and where all-school events such as assemblies are held
- Office where the administrative work of the school is done
- Library where students ask librarians reference questions, check out books and magazines, and often use computers
- Computer labs where computer-based work is done and the internet accessed

## SECURITY

The safety of staff and students is increasingly becoming an issue for school communities, an issue most schools are addressing through improved security. Some have also taken measures such as installing metal detectors or video surveillance. Others have even taken measures such as having the children swipe identification cards as they board the school bus. For some schools, these plans have included the use of door numbering to aid public safety response. Other security concerns faced by schools include bomb threats, gangs, vandalism, and bullying.

## HEALTH SERVICES

School health services are services from medical, teaching and other professionals applied in or out of school to improve the health and well-being of children and in some cases whole families. These services have been developed in different ways around the globe but the fundamentals are constant: the early detection, correction, prevention or amelioration of disease, disability and abuse from which school-aged children can suffer.

## ONLINE SCHOOLS AND CLASSES

Some schools offer remote access to their classes over the Internet. Online schools also can provide support to traditional schools, as in the case of the School Net Namibia. Some online classes also provide experience in a class, so that when people take them, they have already been introduced to the subject and know what to expect, and even more classes provide High School/College credit allowing people to take the classes at their own pace. Many online classes cost money to take but some are offered free.

Internet-based distance learning programmes are offered widely through many universities. Instructors teach through online activities and assignments. Online classes are taught the same as physically being in class with the same curriculum. The instructor offers the syllabus with their fixed requirements like any other class. Students can virtually turn their assignments in to their instructors according to deadlines. This being through via e-mail or in the course webpage. This allowing students to work at their own pace, yet meeting the correct deadline. Students taking an online class have more flexibility in their schedules to take their classes at a time that works best for them. Conflicts with taking an online class may include not being face to face with the instructor when learning or being in an environment with other students. Online classes can also make understanding the content difficult, especially when not able to get in quick contact with the instructor. Online students do have the advantage of using other online sources with assignments or exams for that specific class. Online classes also have the advantage of students not needing to leave their house for a morning class or worrying about their attendance for that class. Students can work at their own pace to learn and achieve within that curriculum.

The convenience of learning at home has been a major attractive point for enrolling online. Students can attend class anywhere a computer can go—at home, a library or while traveling internationally. Online school classes are designed to fit your needs, while allowing you to continue working and tending to your other obligations. Online school education is divided into three subcategories: Online Elementary School, Online Middle School, Online High school.

## STRESS

As a profession, teaching has levels of work-related stress (WRS) that are among the highest of any profession in some countries, such as the United Kingdom and the United States. The degree of this problem is becoming increasingly recognized and support systems are being put into place. Teacher education increasingly recognizes the need to train those new to the profession to be aware of and overcome mental health challenges they may face.

Stress sometimes affects students more severely than teachers, up to the point where the students are prescribed stress medication. This stress is claimed to be related to standardized testing, and the pressure on students to score above average.

According to a 2008 mental health study by the Associated Press and mtvU, eight in 10 college students said they had sometimes or frequently experienced stress in their daily lives. This was an increase of 20% from a survey five years previously. 34 percent had felt depressed at some point in the past three months, 13 percent had been diagnosed with a mental health condition such as an anxiety disorder or depression, and 9 percent had seriously considered suicide.

## DISCIPLINE TOWARDS STUDENTS

Schools and their teachers have always been under pressure — for instance, pressure to cover the curriculum, to perform well in comparison to other schools, and to avoid the stigma of being “soft” or “spoiling” towards students. Forms of discipline, such as control over when students may speak, and normalized behaviour, such as raising a hand to speak, are imposed in the name of greater efficiency. Practitioners of critical pedagogy maintain that such disciplinary measures have no positive effect on student learning. Indeed, some argue that disciplinary practices detract from learning, saying that they undermine students' individual dignity and sense of self-worth—the latter occupying a more primary role in students' hierarchy of needs.

# VOCATIONAL SCHOOL

A vocational school, sometimes also called a trade school, career center, or vocational college, is a type of educational institution, which, depending on country, may refer to secondary or post-secondary education designed to provide vocational education, or technical skills required to perform the tasks of a particular and specific job. In the case of secondary education, these schools differ from academic high schools which usually prepare students who aim to pursue tertiary education, rather than enter directly into the workforce. With regard to post-secondary education, vocational schools are traditionally distinguished from four-year colleges by their focus on job-specific training to students who are typically bound for one of the skilled trades, rather than providing academic training for students pursuing careers in a professional discipline. While many schools have largely adhered to this convention, the purely vocational focus of other trade schools began to shift in the 1990s "towards a broader preparation that develops the academic" as well as technical skills of their students.

## AUSTRALIA

Vocational schools were called "technical colleges" in Australia, and there were more than 20 schools specializing in vocational educational training (VET). Only four technical colleges remain, and these are now referred to as "trade colleges". At these colleges, students complete a modified year 12 certificate and commence a school-based apprenticeship in a trade of their choice. There are two trade colleges in Queensland; Brisbane, the Gold Coast, Australian Industry Trade College and one in Adelaide, Vergalargona rico, Technical College, and another in Perth, Australian Trades College.

In Queensland, students can also undertake VET at private and public high schools instead of studying for their overall position (OP), which is a tertiary entrance score. However these students usually undertake more limited vocational education of one day per week whereas in the trade colleges the training is longer.

## CANADA

Vocational schools are sometimes called "colleges" in Canada. However, a college may also refer to an institution that offers part of a university degree, or credits that may be transferred to a university.

In Ontario, secondary schools were separated into three streams: technical schools, commercial or business schools, and collegiates (the academic schools). Those schools still exist; however, the curriculum has changed that no matter which type of school one attends, they can still attend any post-secondary institution and still study a variety of subjects and others (either academic or practical).

In Ontario, the Ministry of Training, Colleges and Universities have divided postsecondary education into universities, community colleges and private career colleges.

In the Province of Quebec, there are some vocational programmes offered at institutions called CEGEPs *(collège d'enseignement général et professionnel*), but these too may function as an introduction to university. Generally, students complete two years at a CEGEP directly out of high school, and then complete three years at a university (rather than the usual four), to earn an undergraduate degree. Alternatively, some CEGEPs offer vocational training, but it is more likely that vocational training will be found at institutions separate from the academic institutions, though they may still be called colleges.

## CENTRAL AND EASTERN EUROPE

In Central and Eastern Europe, a vocational education is represented in forms of (professional) vocational technical schools often abbreviated as PTU and technical colleges (technikum).

### Vocational School (College)

Vocational school or Vocational college ([vyshche] uchylyshche - study school) is considered a post-secondary education type school, but combines coursework of a high school and junior college stretching for six years. In Ukraine, the term is used mostly for sports schools sometimes interchangeably with the term college. Such college could be a separate entity or a branch of bigger university. Successful graduates receive a specialist degree.

### PTU

PTUs are usually a preparatory vocational education and are equivalent to the general education of the third degree in the former Soviet education, providing a lower level of vocational education (apprenticeship).

It could be compared to a trade high school. In the 1920-30s, such PTUs were called schools of factory and plant apprenticeship, and later 1940s - vocational schools.

Sometime after 1959, the name PTU was established, however, with the reorganization of the Soviet educational system these vocational schools renamed into lyceums. There were several types of PTUs such as middle city PTU and rural PTU.

**Technicum**

Technical college (technicum) is becoming an obsolete term for a college in different parts of Central and Eastern Europe. Technicums provided a middle level of vocational education. Aside of technicums and PTU there also were vocational schools that also provided a middle level of vocational education. In 1920-30s Ukraine, technicums were a (technical) vocational institutes, however, during the 1930-32s Soviet educational reform they were degraded in their accreditation.

**Institute**

Institutes were considered a higher level of education; however, unlike universities, they were more oriented to a particular trade. With the reorganization of the Soviet education system, most institutes have been renamed as technical universities.

**FINLAND**

The Finnish system is divided between vocational and academic paths. Currently about 47 percent of Finnish students at age 16 go to vocational school. The vocational school is a secondary school for ages 16–21, and prepares the students for entering the workforce. The curriculum includes little academic general education, while the practical skills of each trade are stressed. The education is divided into eight main categories with a total of about 50 trades. The basic categories of education are

- Humanist and educational branch (typical trade: youth- and free-time director)
- Cultural branch (typical trade: artisan, media-assistant)
- The branch of social sciences, business and merchandise (typical trade: vocational qualification in business and administration (Finnish: *merkonomi*))
- Natural science (typical trade: IT worker (Finnish: *datanomi*))
- Technology and traffic (typical trades: machinist, electrician, process worker)
- The branch of natural resources and environment (typical trade: rural entrepreneur, forest worker)
- The branch of social work, health care and physical exercise (typical trade: practical nurse (Finnish: *lähihoitaja*))
- The branch of travel, catering and domestic economics (typical trade: institutional catering worker)

In addition to these categories administered by the Ministry of Education, the Ministry of Interior provides vocational education in the security and rescue branch for policemen, prison guards and firefighters.

The vocational schools are usually owned by the municipalities, but in special cases, private or state vocational schools exist. The state grants aid to all vocational schools on the same basis, regardless of the owner. On the other hand, the vocational schools are not allowed to operate for profit. The Ministry of Education issues licences to provide vocational education. In the licence, the municipality or a private entity is given permission to train a yearly quota of students for specific trades. The licence also specifies the area where the school must be located and the languages used in the education. The vocational school students are selected by the schools on the basis of criteria set by the Ministry of Education.

The basic qualification for the study is completed nine-year comprehensive school. Anyone may seek admission in any vocational school regardless of their domicile. In certain trades, bad health or invalidity may be acceptable grounds for refusing admission. The students do not pay tuition and they must be provided with health care and a free daily school lunch. However, the students must pay for the books, although the tools and practice material are provided to the students for free.

In tertiary education, there are higher vocational schools *(ammattikorkeakoulu* which is translated to “polytechnic” or “university of applied sciences”), which give three- to four-year degrees in more involved fields, like engineering and nursing.

In contrast to the vocational school, an academically orientated upper secondary school, or senior high school (Finnish: *lukio*) teaches no vocational skills. It prepares students for entering the university or a higher vocational school.

**IRELAND**

A vocational school in Ireland is a type of secondary education school which places a large emphasis on vocational and technical education; this led to some conflict in the 1960s when the Regional Technical College system was in development.

Since 2013 the schools have been managed by Education and Training Boards, which replaced Vocational Education Committees which were largely based on city or county boundaries. Establishment of the schools is largely provided by the state; funding is through block grant system providing about 90% of necessary funding requirements.

Vocational schools typically have further education courses in addition to the traditional courses at secondary level. For instance, post leaving certificate courses which are intended for school leavers and pre-third level education students.

Until the 1970s the vocational schools were seen as inferior to the other schools then available in Ireland. This was mainly because traditional courses such as the leaving certificate were not available at the schools, however this changed with the *Investment in Education* (1962) report which resulted in an upgrade in their status. Currently about 25% of secondary education students attend these schools.

**JAPAN**

In Japan vocational schools are known as *senmon gakkô*. They are a part of Japan’s higher education system.

There are two-year schools that many students study at after finishing high school (although it is not always required that students graduate from high school). Some have a wide range of majors, others only a few majors. Some examples are computer technology, fashion and English.

## NETHERLANDS

In the Middle Ages boys learned a vocation through an apprenticeship. They were usually 10 years old when they entered service, and were first called *leerling* (apprentice), then *gezel* (journeyman) and after an exam - sometimes with an example of workmanship called a *meesterproef* (masterpiece) - they were called *meester* (master craftsman). In 1795, all of the guilds in the Netherlands were disbanded by Napoleon, and with them the guild vocational schooling system. After the French occupation, in the 1820s, the need for quality education caused more and more cities to form day and evening schools for various trades. In 1854, the society *Maatschappij tot verbetering van den werkenden stand* (society to improve the working class) was founded in Amsterdam, that changed its name in 1861 to the *Maatschappij voor de Werkende Stand* (Society for the working class). This society started the first public vocational school *(De Ambachtsschool*) in Amsterdam, and many cities followed. At first only for boys, later the *Huishoudschool* (housekeeping) was introduced as vocational schooling for girls. Housekeeping education began in 1888 with the Haagsche Kookschool in The Hague.

In 1968 the law called the *Mammoetwet* changed all of this, effectively dissolving the Ambachtsschool and the Huishoudschool. The name was changed to LTS *(lagere technische school*, lower technical school), where mainly boys went because of its technical nature, and the other option, where most girls went, was LBO *(lager beroepsonderwijs*, lower vocational education). In 1992 both LTS and LBO changed to VBO *(voorbereidend beroepsonderwijs*, preparatory vocational education) and since 1999 VBO together with MAVO *(middelbaar algemeen voortgezet onderwijs*, intermediate general secondary education) changed to the current VMBO *(voorbereidend middelbaar beroepsonderwijs*, preparatory intermediate vocational education).

## UNITED STATES

In the United States, there is a very large difference between career college and vocational college. The term *career college* is generally reserved for post-secondary for-profit institutions. Conversely, vocational schools are government-owned or at least government-supported institutions, occupy two full years of study, and their credits are by and large accepted elsewhere in the academic world and in some instances such as charter academies or magnet schools may take the place of the final years of high school.

Career colleges on the other hand are generally not government supported in any capacity, occupy periods of study less than a year, and their training and certifications are rarely recognized by the larger academic world. In addition, as most career colleges are private schools, this group may be further subdivided into non-profit schools and proprietary schools, operated for the sole economic benefit of their owners.

As a result of this emphasis on the commercialization of education, a widespread poor reputation for quality was retained by a great number of career colleges for over promising what the job prospects for their graduates would actually be in their field of study upon completion of their programme, and for emphasizing the number of careers from which a student could choose.

Even though the popularity of career colleges has exploded in recent years, the number of government-sponsored vocational schools in the United States has decreased significantly..

The Association for Career and Technical Education (ACTE) is the largest American national education association dedicated to the advancement of career and technical education or vocational education that prepares youth and adults for careers. Earlier vocational schools such as California Institute of Technology and Carnegie Mellon University have gone on to become full degree-granting institutions.

# NIGHT SCHOOL

A night school is an adult learning school that holds classes in the evening or at night to accommodate people who work during the day. It can also apply to a community college or university that hold classes for the aforementioned purpose, but also admits undergraduates. Certain high schools, almost always private or charter do exist in fact, but are usually hard to find.

# INDUSTRIAL SCHOOL

In England, the 1857 Industrial Schools Act was intended to solve problems of juvenile vagrancy, by removing poor and neglected children from their home environment to a boarding school. The Act allowed magistrates to send disorderly children to a residential industrial school. An 1876 Act led to non-residential day schools of a similar kind.

There were similar arrangements in Scotland, where the Industrial Schools Act came into force in 1866. The schools cared for neglected children and taught them a trade, with an emphasis on preventing crime. Glasgow Industrial School for Girls is an example formed in 1882.

They were distinct from reformatories set up under the 1854 Youthful Offenders Act (the Reformatory Schools Act) which included an element of punishment. Both agreed in 1927 to call themselves approved schools.

In Ireland, the Industrial Schools Act of 1868 established industrial schools (Irish: *scoileanna saothair*) to care for “neglected, orphaned and abandoned children”. By 1884 there were 5,049 children in such institutions.

## CONTEXT

An Industrial Feeding School was opened in Aberdeen in 1846. Industrial schools, like the contemporaneous ragged schools, were set up by volunteers to help destitute children. Their philosophy differed in that they believed that an education was not enough: these children needed to be removed from the harmful environment of the street, trained to be industrious, and given a trade they could practise.

Some schools were residential but other children were ‘day-boys’. The regime was severe, with a tightly timetabled daily routine that stretched from waking at 6:00am to bedtime at 7:00pm.

During the day there were set times for religion and moral guidance, formal schooling, doing housework, eating, and learning trades, with three invervals for play. The boys’ trades were gardening, tailoring, and shoemaking; the girls learned housework and washing, knitting, and sewing.

## INDUSTRIAL SCHOOLS ACT 1857

The Industrial Schools Act 1857 gave magistrates the power to sentence homeless children between the ages of 7 and 14 years, who were brought before the courts for vagrancy to a spell in an industrial school.

In 1861 a further act strengthened the powers of the magistrate “to include:

- Any child apparently under the age of fourteen found begging or receiving alms (money or goods given as charity to the poor).
- Any child apparently under the age of fourteen found wandering and not having any home or visible means of support, or in company of reputed thieves.
- Any child apparently under the age of twelve who, having committed an offence punishable by imprisonment or less.
- Any child under the age of fourteen whose parents declare him to be beyond their control”.

It was not until 1875 that it became compulsory to register births and vagrants genuinely didn’t know their age. Also children would say any age that was convenient to them at the time. Homeless children who had not committed any more serious crime were sent to industrial schools, while children who had been arrested for other crimes were sent to juvenile prisons known as reformatories.

## FINANCIAL ACCOUNTABILITY

The intention was that the cost of attending such a school would be met by the parents, which was an interesting concept for a child who was sent there because he was already homeless.

The state had to provide. The system expanded rapidly, and magistrates preferred to send miscreants to a school rather than a reformatory. There was public unrest about the mounting cost. From 1870, responsibility for the Industrial schools passed to the local Committee of Education.

Best estimates suggest that, at the peak, there were 224 certified industrial schools in England and 50 in Scotland. There were others operating without certification or when certification had been removed. Name changing was common. Directly after World War I, one hundred closed and in 1933 there were only 55. They agreed to change their names to approved schools in 1927 and the Approved Schools Act of 1933 merged them to form a single system.

# PART-TIME LEARNER IN HIGHER EDUCATION

A part-time learner is a non-traditional student who pursues higher education, typically after reaching physical maturity, while living off-campus, and possessing responsibilities related to family and/or employment. Part-time student status is based on taking fewer course credits in a semester than full-time students.

In the United States, the number of part-time students rose 16 percent between 2004 and 2014. In 2015, 23 percent of undergraduate students at 4-year institutions attended part-time, compared to 61 percent of students at 2-year institutions.

In Canada, the course load that constitutes part-time student status varies between institutions. The University of British Columbia, for example, defines a part-time undergraduate student as one enrolled in less than 80 percent of the standard 30 credit-hour course load. The University of Manitoba defines the part-time undergraduate student as an individual enrolled in less than 60 percent of the standard full 30 credit hour course load. The Government of Canada national student loans programme defines a part-time student as one who is enrolled in 20-59 percent of a full course load.

In Canada part-time undergraduate enrolment grew by 25 percent from 1980 to 1992. From 2000 to 2010, part-time enrolment grew by one percent a year compared to four percent for full-time enrolment. A high number of part-time students are adult students. In 2010, approximately 24 percent of undergraduate students in Canada were studying part-time, and 60 percent of part-time students were 25 years old or older.

In the United Kingdom, while full-time students have been increasing, part-time student enrolment has been steady decreasing since 2009-2010. In 2011-2012, 31 percent of all enrolments were part-time, while in 2015-2016 part-time students consisted of 24 percent of all enrolments. Between 2011-2012 and 2015-2016 there was an overall 30 percent decrease of part-time students.

In Australia, 31.2 percent of students in 2008 were enrolled part-time. Between 2003 and 2008, while the number of students attending full time increased by 21.1 percent, the number attending part-time enrollments increased by only 2.5 percent.

# VOCATIONAL EDUCATION COMMITTEE

A Vocational Education Committee (VEC) (Irish: *Coiste Gairmoideachais*) was a statutory local education body in the Republic of Ireland that administered some secondary education, most adult education and a very small amount of primary education in the state. Before 1992 VECs had authority over the Dublin Institute of Technology and the Regional Technical Colleges. They existed from 1930 to 2013, when they were replaced by Education and Training Boards.

## ESTABLISHMENT

VECs were originally created by the Vocational Education Act 1930, as successors to the Technical Instruction Committees established by the Agriculture and Technical

Instruction (Ireland) Act 1899. The original purpose of the committees was to administer continuation and technical education for 14- to 16-year-olds. Continuation education was defined as *"general and practical training in preparation for employment in trades"*, while technical education was described as *"pertaining to trades, manufacturers, commerce and other industrial pursuits"*. To this end the VECs were charged with the duty of setting up and maintaining vocational schools.

**DUTIES**

Over time the duties of VECs increased, in particular in the area of adult education. These included:

*Post-primary (secondary) education*

- Vocational Schools
- Community Colleges

*Further and adult education*

- Post-Leaving Certificate courses (PLCs)
- Youthreach Services
- Vocational Training Opportunities Scheme (VTOS)
- Community Education
- Adult Literacy and Basic Education
- Back To Education Initiative (BTEI)
- Adult Refugee Programme (ARP)

VECs also administered maintenance grants and bursaries for third-level education until 2012.

In September 2008, County Dublin VEC opened the first Community National School, in Porterstown, Dublin 15. This marked the first time a VEC had become involved in primary school education.

**ORGANISATION**

VECs were originally established in each administrative county and county borough in the then Irish Free State.

In addition, a VEC was formed in those municipal boroughs and urban districts which had a separate Technical Instruction Committee under the 1899 legislation (namely Bray, Drogheda, Sligo, Tralee and Wexford) and in the newly created Borough of Dún Laoghaire.

The number of VECs was reduced to the final number of thirty-three when five town committees were amalgamated with the adjacent county committees, leaving Dún Laoghaire as the only VEC area not consisting of a city or county.

**Committees**

Each Vocational Education Committee was elected by the county, borough or urban district council and consisted partly of councillors and partly of persons with an *"interest* *and experience in education"* and who could be recommended by bodies *"interested in manufacture or trades"*. The Vocational Education (Amendment) Act 2001 changed the composition of the committees, and parents of students under 18 and members of the staff were also entitled to elect committee members. Members could also be appointed to represent the interests of students, voluntary organisations, community organisations, Irish language interests and business.

List of Vocational Education Committees 1997–2013

| VEC | Notes |
|---|---|
| County Carlow | |
| County Cavan | |
| County Clare | |
| City of Cork | |
| County Cork | |
| County Donegal | |
| City of Dublin | |
| County Dublin | Covered Fingal, South Dublin and part of Dún Laoghaire–Rathdown |
| Dún Laoghaire | Part of Dún Laoghaire–Rathdown (the former borough of Dún Laoghaire) |
| City of Galway | |
| County Galway | |
| Kerry Education Service (County Kerry VEC) | Absorbed Town of Tralee VEC 1997 |
| County Kildare | |
| County Kilkenny | |
| County Laois | |
| County Leitrim | |
| City of Limerick | |
| County Limerick | |
| County Longford | |
| County Louth | Absorbed Town of Drogheda VEC 1997 |
| County Mayo | |
| County Meath | |
| County Monaghan | |
| County Offaly | |
| County Roscommon | |
| County Sligo | Absorbed Town of Sligo VEC 1997 |
| North Tipperary | |
| South Tipperary | |
| City of Waterford | |
| County Waterford | |
| County Westmeath | |
| County Wexford | Absorbed Town of Wexford VEC 1997 |
| County Wicklow | Absorbed Town of Bray VEC 1997 |

**AMALGAMATIONS AND ABOLITION**

In October 2010 the Department of Education and Skills announced that the number of VECs was to be reduced from 33 to 16 by amalgamation In 2011 the new government confirmed that scale of reduction but revised the scheme of amalgamations It subsequently announced the establishment of SOLAS, a new steering and funding agency to cover the further education provision of the VECs, which also absorbed the training activities formerly carried out by FÁS.

Under the terms of the Education and Training Boards Act 2013 the VECS were dissolved, being replaced by 16 Education and Training Boards (ETBs) on 1 July 2013:

1. County Dublin VEC and Dún Laoghaire VEC replaced by Dublin and Dún Laoghaire ETB
2. City of Dublin VEC replaced by City of Dublin ETB
3. City of Galway VEC, County Galway VEC and County Roscommon VEC replaced by Galway and Roscommmon ETB
4. County Cork VEC and City of Cork VEC replaced by Cork ETB
5. County Kerry VEC replaced by Kerry ETB
6. City of Limerick VEC, County Limerick VEC and County Clare VEC replaced by Limerick and Clare ETB
7. City of Waterford VEC, County Waterford VEC and County Wexford VEC replaced by Waterford and Wexford ETB
8. North Tipperary VEC and South Tipperary VEC replaced by Tipperary ETB
9. County Donegal VEC replaced by Donegal ETB
10. County Kildare VEC and County Wicklow VEC replaced by Kildare and Wicklow ETB
11. County Mayo VEC, County Sligo VEC and County Leitrim VEC replaced by Mayo, Sligo and Leitrim ETB
12. County Cavan VEC and County Monaghan VEC replaced by Cavan and Monaghan ETB
13. County Laois VEC and County Offaly VEC replaced by Laois and Offaly ETB
14. County Carlow VEC and County Kilkenny VEC replaced by Kilkenny and Carlow ETB
15. County Longford VEC and County Westmeath VEC replaced by Longford and Westmeath ETB
16. County Meath VEC and County Louth VEC replaced by Louth and Meath ETB

# TECHNICAL SCHOOL

In the United States, a technical school is a two-year college that provides mostly employment-preparation skills for trained labour, such as welding, culinary arts and office management. The Association for Career and Technical Education is the largest U.S. education association dedicated to promoting career and technical education for youth and adults. Technical school is also the term used in the United States Armed Forces for the job specific training given immediately after recruit training. Though similar to the training provided by a two-year college, the training is much more concise, eschewing any coursework outside the minimum necessary to begin working in the chosen career field; additionally, the training is more time intensive, often including more than nine in-class hours per day. Military technical school is typically one to three months in duration, though some schools are as short as two weeks or as long as two years.

Upon graduation, military technical school recruits are qualified only as apprentices and must work under supervision until they have completed a more extensive on-the-job training programme. Many times the military training can be converted to standard university credits, leaving the graduating recruit with only a few general education requirement courses (such as speech or composition) to complete in order to receive the more traditional two-year technical school diploma referenced above.

## NORTH AMERICAN TRADE SCHOOLS

North American Trade Schools (NATS) is a private vocational school located in Baltimore, Maryland. In 1970, the Diesel Institute of America (DIA) was founded by Sheldon Monsein, then president of Central GMC/Kenworth inc. in Landover, Maryland. His intent in starting the DIA was born from his company's need for additional qualified mechanics to service the diesel engine vehicles in the Central GMC/Kenworth fleet. Not too long afterwards, DIA offered its first class for diesel engine technicians in a 500-square-foot (46 m) garage in Landover, making the DIA the first trade school in the state of Maryland to offer training in the repair and maintenance of diesel engines. In September 1983, the Diesel Institute of America moved its operation to Grantsville, Maryland and, three years later, began offering training in driving commercial trucks. In 2002, the DIA was purchased by EFC Trade, a Canada based business that specializes in providing career training through specialty schools. It was in 2004 that the Diesel Institute of America relocated from Grantsville to its current location in Baltimore and its name changed to North American Trade Schools. The school's campus is located in the unincorporated community of Woodlawn, Baltimore County, Maryland near Security Square Mall. Its current enrollment consists of 529 full-time students.

# 4

# CHAPTER

# Information Technology Education

## ACADEMY OF INFORMATION TECHNOLOGY (AUSTRALIA)

AIT, Creative Technology Educators (also known as the *Academy of Information Technology*) is a specialist higher education and vocational education institution based in Sydney, Australia, with a second campus opening in Melbourne, Australia in 2015. AIT was established in 1999, and specialises in three disciplines: digital media, information technology, and business. AIT is a member of the RedHill Education group, alongside Greenwich English College, Go Study Australia, and the International School of Colour and Design (ISCD). AIT was the first in Australia to offer courses in Motion capture technology, and has the latest markerless Motion Capture system installed in its Sydney campus, in 2013, AIT was still the first Mobile Applications Development course provided. AIT holds and participates in many events, including the AIT Oscar Night, the AIT Games Night, and the Vivid Light Festival. AIT is recognised as a quality tertiary education provider by the Australian Government's Tertiary Education Quality and Standards Agency (TEQSA) and the Australian Skills Quality Authority (ASQA).

### STUDENT COUNCIL AND SOCIAL ACTIVITIES

#### Student Representative Council

The Student Representative Council (SRC) is a small group of students selected by the student body. Their job is to help improve the quality of life on the AIT campus, and do so by communicating with teachers and staff members on the student bodies behalf. The current president of the SRC is Mikhaila Katte.

**Clubs**

As of 2015 AIT currently has five clubs running: the 3D Club, Film Club, Drawing Club, Table Top Club and Fight Club. These clubs are overseen and assisted by the SRC.

**Events**

AIT holds regular events throughout the year, held by a mixture of staff members, teachers and the SRC. Smaller events include barbecues, Christmas parties and Trivia Nights.

***AIT Oscar Night***

The AIT Oscar Night is an annual event held at AIT. This event allows student works to be recognised and rewarded by their teacher and peers in the fields of Animation and Film. The event is run by Film Coordinator, Patrick Huang.

***AIT Games Night***

The AIT Games Night is an annual event held at AIT. This event features student games alongside other activities like cosplay competitions. Prizes are awarded to the best student games. This event is run by both the teachers and SRC, allowing multiple activities to run in one night.

***Vivid***

In 2015 AIT participated in the Vivid Light Festival. Their entry, titled "Lightwell", was a collaborative task between teachers and students. The structure itself was created by Helen Goritsas, Patrick Huang, Nik Sutila, Adam Katz, Sharon Sanders, Carlton Zhu, Kwan Chemsripong and Kriss Mahatumaratana. The interactive structure displayed a collection of student artworks onto a wall along Walsh Bay, Sydney.

**FACILITIES**

**Motion Capture**

The Motion Capture (or Organic Motion) is a markerless motion capture system that was designed and manufactured by Organic Motion of New York. This type of technology was the first of its kind implemented in Australia. The Motion Capture, aka MoCap, allows users to motion track and animate their 3D characters easily.

**Green Screen Room**

The Green Screen Room is a state of the art filming environment located on the Sydney campus. The space was launched on 28 May 2014 and allows students to extend their editing skills in Adobe Premiere and Adobe After Effects for Green Screen compositing.

**COURSES**

2D and 3D Animation:

- Bachelor of Interactive Media (Specialisation: Animation)
- Diploma of Digital Media Technologies (Specialisation: Animation)

Design

- Bachelor of Interactive Media (Specialisation: Design)
- Diploma of Digital Media Technologies (Specialisation: Design)

Film Making

- Bachelor of Interactive Media (Specialisation: Film Making)
- Diploma of Digital Media Technologies (Specialisation: Film Making)

Game Design

- Bachelor of Interactive Media (Specialisation: Game Development)
- Diploma of Digital Media Technologies (Specialisation: Game Development)

Digital Design

- Bachelor of Digital Design
- Diploma of Design

IT

- Bachelor of IT (Mobile Applications Development)
- Diploma of Software Development

# ASSOCIATION OF INFORMATION TECHNOLOGY PROFESSIONALS

The Association of Information Technology Professionals (AITP) is a professional association that focuses on information technology education for business professionals. The group is a non-profit US-oriented group, but its activities are performed by 62 local chapters organized on a geographic basis, and 286 student chapters at college and universities.

**HISTORY**

The group was first organized in Chicago, United States, in 1949, as the National Machine Accountants Association (NMAA). The State of Illinois granted the NMAA a charter on December 26, 1951. It soon had over 20 chapters. Harvey W. Protzel, systems manager for Protzel's Company, was elected the first International President at the 1952 First Annual Convention in Minneapolis. In 1962, it adopted the more inclusive name, Data Processing Management Association (DPMA), and in 1996, the present name. The Association began its certification programme with the Certificate in Data Processing (CDP) professional examination, first held in 1962 in New York City. In

1970, it added the Registered Business Programmer (RBP) examination. In 1974, the Association joined in establishing the Institute for the Certification of Computer Professionals (ICCP) to administer the examination programme and stimulate industry acceptance of the examinations. The certification is now called Certified Computing Professional (CCP).

In 1969, the Association created the annual DPMA Man of the Year Award, awarding it (incongruously, given its name) to Dr. Grace Murray Hopper. After Dr. Ruth Davis of the US National Bureau of Standards won the 1979 award, it was renamed the Distinguished Information Sciences Award.

### ETHICS

The Association requires its members to abide by a code of ethics that has been in place for decades, predating both the HIPAA and Sarbanes-Oxley legislation. The Association also requires members to operate by a *"Standards of Conduct"* for IT Professionals, which adds specifics.

## BECTA

Becta, originally known as the British Educational Communications and Technology Agency, was a non-departmental public body (popularly known as a Quango) funded by the Department for Education and its predecessor departments, in the United Kingdom. It was a charity and a company limited by guarantee. The abolition of Becta was announced in the May 2010 post-election spending review. Government funding was discontinued in March 2011. Becta went into liquidation in April 2011.

### ROLE

Becta was the lead agency in the United Kingdom for promotion and integration of information and communications technology (ICT) in education. Becta was a company limited by guarantee with charitable status. It was established in 1998 through the reconstitution of the National Council for Educational Technology (NCET), which oversaw the procurement of all ICT equipment and e-learning strategy for schools.

### POLICY

Foremost among the 2005–2008 Becta strategic objectives were "to influence strategic direction and development of national education policy to best take advantage of technology" and "to develop a national digital infrastructure and resources strategy leading to greater national coherence."

### NATIONAL GRID FOR LEARNING

The National Grid for Learning (NGfL) was managed by Becta and was set up as a gateway to educational resources on to support schools and colleges across the UK.

The NGfL portal was launched in November 1998, as one of several new programmes initiated by the new Labour government which took office in May 1997 and had a linked budget of earmarked funds to be spent on schools' internet connections and ICT.

### PURCHASING FRAMEWORKS

Purchasing Frameworks

Becta awarded certain vendors placement on approved "purchasing frameworks":

- The frameworks are awarded in accordance with EU procurement legislation... against a range of criteria based around quality of provision and service, and against the extent to which they meet the requirements of the functional and technical specifications – specifications that have been developed in conjunction with all stakeholders, including members of the open source community...

The purchasing frameworks were criticised as being outdated, and for effectively denying schools the option of benefiting from both free and open source and the value and experience of small and medium ICT companies. Participating companies had to have a net worth of at least £700,000 to qualify and had to satisfy a list of functional requirements. A concern was raised about the "over-comfortable relationship the government has with some of the bigger players."

In January 2007, Crispin Weston, who had helped Becta draw up the criteria used to select suppliers, asked the EC Competition Commission to investigate his allegation that a significant number of the successful tenders had failed to implement the mandatory functional requirements, including particular aspects of inter-operability. He also added in his letter to the Commission that they should take action on the further issue of:

- [T]he insistence that many different categories of software within a particular school or Local Authority should all be supplied by a single supplier [which] has serious anti-competitive implications."

## BETT

Bett or The Bett Show (formerly known as the British Educational Training and Technology Show) is an annual trade show in the United Kingdom marketing information technology in education organised by Ascential. The first Bett show was held in London, England in 1985, and since 2013 it has been held annually in January at the ExCeL London.

### HISTORY

The first Bett was held in January 1985 as the "Hi Technology and Computers in Education Exhibition" at the Barbican Centre, central London, in association with the British Educational Suppliers Association.

- The 1993 Bett show moved locations from the Barbican Centre to the National and Grand Halls at the Olympia exhibition centre.

- The 2012 Bett show had keynote delivered by Education Secretary Michael Gove.
- The 2013 Bett show moved locations from the Olympia exhibition centre to the ExCeL London.
- The 2014 Bett show celebrated its 30th anniversary, which attracted 35,044 visitors from 113 countries.
- The 2015 Bett show introduced the "Bett Futures" feature, aimed at selecting and promoting 30 educational technology start-ups, selected by a panel of expert educators. Keynote presenters also included Education Secretary, Nicky Morgan.
- The 2017 Bett show saw the return of Sir Ken Robinson who had previously delivered keynotes in 2015.

**EXHIBITORS**

A full and updated list is available at the Bett exhibitor list web site.

- Weeras Author (Barcelona)
- 3P Learning
- Adobe Systems
- BBC
- BioStore Ltd
- Business I.T. Systems Ltd
- Capita Group
- Cappelen Damm
- Channel 4 Learning
- Classroom Professor
- Clevertouch
- CORE ECS
- Dell
- The Dutch School
- EducationCity
- emap
- European Electronique
- EuroTalk
- Exa Networks Ltd
- Frog Education
- Genisys Enterprise
- Gojimo
- Groupcall
- iBoardTouch
- Impero Software
- itslearning
- Jigsaw School Apps

- Kudlian Software - Animation, Green Screen and App Creation
- LJ Create
- Microsoft
- Mozaik Education
- Nationwide Retail Systems Ltd
- NetSupport
- Norman ASA
- $O_2$
- Pearson PLC
- Promethean Ltd
- Rapid Electronics
- Redstor
- RM
- SAM Learning
- Sammy Herring Enterprises
- Schools Broadband
- senso.cloud
- Sherston Software
- Smart Systems
- Smart Technologies
- Spire Education
- Studyzone.tv
- Synergy School Radio
- Twig World
- VS School MIS
- Zu3D Animation Software

**AWARDS**

The Bett Awards are an annual celebration for the highest levels of achievement by companies that exhibit at the Bett show and supply Information and Communications Technology (ICT) for education.

# CANADIAN IT BODY OF KNOWLEDGE

Canadian Information Technology Body of Knowledge (CITBOK) is the project sponsored and undertaken by Canadian Information Processing Society (CIPS) to define and outline the body of knowledge that defines a Canadian Information Systems Professional (ISP).

CIPS recognizes that in order to strengthen the criteria of the ISP designation successfully while still maintaining high standards, the association needs to develop a Canadian Information Technology Body of Knowledge (CITBOK) that will set standards for knowledge. The CITBOK is an outline of the knowledge bases that form the intellectual basis for the IT profession. CIPS has identified a core CITBOK that all professionals would be expected to master and specialty bodies of knowledge that would depend on your area of practice. When setting this Canadian standard for Information Technology (IT) knowledge, CIPS looked to other organizations and internationally for standards and bodies of knowledge that CIPS could adopt or adapt, especially for the specialty areas. For example, in 2004, CIPS adopted the British Computer Society (BCS) Professional Examination Study Guide Syllabus Diploma level (Core and 11 specialist modules) as the initial Body of Knowledge for CIPS.

The CITBOK aims to:

- Be an industry structure model that can be used to define the set of performance, training and development standards for all IT practitioners (CIPS members and non-members) in Canada;
- Create alternate paths to certification based on concepts of CITBOK;
- Establish guidelines for accreditation criteria based on concepts of CITBOK; and
- Create a professional development model that sets the standards criteria for the knowledge base including knowledge, skills, and professional activities for IT practitioners.

## EMPOWER UP

Empower Up, formerly known as CREAM (Computer Reuse Education And Marketing), a 501 (c) (3), was a computer recycling center in SW Washington. Started in mid-2002, Empower Up took old electronics in for donation, and recycled/distributed them.

Empower Up's history can be traced back to a collection event started by the City of Vancouver and Clark County in June 2001. The success of this collection prompted the city to start another collection in January 2002, to collect only reusable computers and monitors for the community. Due to the success of the second event, comments by sponsors, and the general public showed an interest in these collections.

In mid-2002, after both an event for re-usage and recycling of computers, departments of the City of Vancouver created the programme Computer REuse and Marketing, or CREAM. This new organization would have a permanent location, permanent collection sites, and mobile collection units to travel around the area. CREAM would also make sure of the proper disposal of materials in electronics not suitable for redistribution. By 2008, CREAM had distributed 250 refurbish computers to needy families (with the help of Salvation Army Family Services), and recycled 3.5 million pounds of electronic waste.

Near the beginning of 2009, due to new laws in Washington, it was recommended that the organization become an independent non-profit organization. With the help of

Habitat for Humanity, and Clark County ReStore, the state created a 3-year plan to make CREAM an independent non-profit. In February 2009, they established their first permanent building in Vancouver.

In April 2010, CREAM's name was changed to Empower Up, with the motto "*Technology Reuse In Action*", and changed to their current logo.

In October 2013 Empower Up moved to 3206 N E 52nd Street Vancouver Washington 98663 on St Johns Road. They expanded the Reuse store and added an eBay store.

On 27 August 2016 Empower Up was shut down permanently, citing "economic pressure".

Empower Up has recently received attention from many local and state news sources. They have been featured in places such as KATU local news, *The Vancouver Voice*, and *The Columbian*.

# FREE GEEK

Free Geek is a non-profit organization started in Portland, Oregon in 2000. Free Geek has two central goals: to reuse or recycle used computer equipment that might otherwise become hazardous waste, and to make computer technology more accessible to those who lack financial means or technical knowledge.

Free Geek's refurbished computers are either granted to schools, churches, non-profit or community change organizations, given to volunteers, or sold in Free Geek's thrift store.

## ACTIVITIES

Free Geek offers a wide range of free classes to its volunteers and to the general public. These classes generally fall into a few major categories including basic computer use, advanced computer courses, digital arts creation, digital nativity/online safety, and workplace readiness. Free Geek also offers phone and drop-in technical support on specific days of the week for the computers it produces.

### Programmes

People who wish to volunteer at Free Geek's community technology center usually choose between two programmes: the Build Programme and the Adoption Programme. The Adoption Programme allows volunteers to get to work immediately on relatively simple tasks, such as keeping incoming equipment organized or sorting metals and plastics. In the Build Programme, volunteers learn while they work, and are trained to build refurbished computers. Volunteers completing the Adoption Programme are given a computer after 24 hours of volunteering (one free computer per year) while the Build Programme volunteers get to keep the sixth computer they build.

Free Geek has numerous other programmes, which are generally run by interns, longer-serving volunteers and staff members. A few examples: The Hardware Grants programme reviews requests for computer equipment from schools, churches, non-profit and community change organizations. The Reuse programme works to ensure that reuse is prioritized over recycling, and finds new ways to get equipment into the hands of people who will put it to use. The Technocrats oversee the network infrastructure of the organization.

### FREE SOFTWARE

Free Geek's refurbished computers run Linux Mint, Ubuntu or other free and open-source software. The use of free software has several major benefits to the organization, and to the recipients of equipment: Free Geek operates without needing to devote resources to managing software licenses, and may install software where it is needed with minimal complications from legal considerations; and computer recipients get a wide range of software, which they may easily expand without paying money or entering into restrictive contracts.

Free Geek was the joint winner of the first Chris Nicol FOSS Prize awarded by the Association for Progressive Communications (APC) in 2007.

### LOCATIONS

In addition to Portland, a number of other cities have started their own Free Geek organizations.

- Portland, OR ("the Mothership")
- Fayetteville, AR
- Chicago, IL
- Vancouver, BC
- Seattle, WA
- Minneapolis-Saint Paul, MN
- Toronto, ON
- Providence, RI
- Ephrata, PA (South East Pennsylvania)
- Athens, GA (Free I.T. Athens)
- Oslo, Norway
- Montreal, QC (opening Dec 2017)

## GEEKCORPS

Geekcorps is a non-profit organization that sends people with technical skills to developing countries to assist in computer infrastructure development.

The non-profit was created in 2000 by Ethan Zuckerman and Elisa Korentayer in North Adams, Massachusetts.

In 2001 Geekcorps became a division of the International Executive Service Corps located in Washington, D.C.

### CREATION

After a visit to a Ghana library by Zuckerman in 1993, the lack of up-to-date resources available prompted him to create Geekcorps years later. Humanitarian and banker Elisa Korentayer became co-founder of Geekcorps due to the organization's need of financial wisdom. In effort to increase access to current information and bridge the digital divide in developing nations Zuckerman, and associates from his now bought out internet company tripod, funded most of the $350,000 budget for Geekcorps' first year.

### MAJOR PROJECTS

#### Ghana

Starting September 2000, with 6 volunteers selected from over 200 applicants, Geekcorps first mission was in Accra, Ghana. Co-founder Zuckerman was already familiar with the infrastructure of Ghana. Zuckerman stated, "The government has relatively liberal telecommunications and investment policies, making it possible for IT businesses to be built there." Geekcorps initial focus in Ghana was assisting Accran companies with it's IT expertise. Geekcorps had an understanding with local businesses, after receiving help, the businesses involved were to help the locals with their newfound resources. Initial challenges for Geekcorps were communication and teaching skills needed by volunteers, and reliance on outdated programming languages for local businesses. Geekcorps involvement lead to innovations such as a new java based payment system for local businesses in Ghana. Geekcorps was also instrumental in the creation of Ghana's internet exchange in 2005.

#### Mali

Initial assistance in Mali came from the CMRT (Community Mobilization through Radio Technology) programme sponsored by USAID. Under CMRT, Geekcorps installed 5 radio stations to enable local communication through the area. Later under another programme Radio for Peace Building, Geekcorps installed another 11 stations, and renovations were done to older existing stations.

Geekcorps set up ICT stations in less populated areas of Mali in 2006. These stations were updated by a memory stick delivered from a computer center with internet access in Ouelessebougou. Although this allowed many rural locations access to specific web resources, such as web pages and digital media, due to lack of interest the programme was modified after a year. Yearly updates to more desired information such as Moulin, a french version of Wikipedia, became the focus.

## NETCORPS

NetCorps was a volunteer-organizing coalition consisting of nine Canadian non-governmental organizations (NGOs), funded by the Canadian International Development

Agency (CIDA) and managed by the NetCorps coordination unit. Through the programme, the organizations created international information and communication technologies (ICT) internships in developing countries around the world. Interns typically participated in six-month programmes, leaving between August and November for host organizations in the placement countries. Positions were limited to 19–30-year-old Canadian citizens or landed immigrants who had "appropriate information and communication technologies skills". Typical duties included creating webpages, developing databases, computer networking, setting up hardware, preparing manuals and other documentation, and general-to-advanced computer instruction.

## PROGRAMME

### Objectives

- To strengthen the ICT capacity of civil society and public-sector organizations to contribute to development goals in governance, education, health, gender equality and environmental sustainability
- Increase the professional skills and capacities of young Canadians aged 19 to 30 to contribute positively to Canadian international development efforts and secure meaningful employment
- To increase the awareness and promote greater engagement of young Canadians with respect to international development issues

### Budget

During the 2007-2008 season, the programme placed 240 interns and was funded by the Canadian International Development Agency (CIDA) and Human Resources and Social Development Canada (HRSDC) at a cost of C$4.6 million. The programme covered all travel and accommodation expenses for the interns, provided a modest living allowance and a completion bonus.

## HISTORY

### NetCorps Creation

NetCorps Canada International was conceived in 1996 by Dr. David Johnston (Governor General of Canada as of 2011 and former President of the University of Waterloo), a prominent lawyer and adviser to the federal government on information-highway issues.

### Phase I

In 1997 Industry Canada, in partnership with the Special Initiatives Division of Canada's International Development Research Centre (IDRC), funded a NetCorps-style test project that sent two residents of Cape Breton to Angola for Internet and Geographic

Information Systems projects. This NetCorps conceptualization and placement was done through the Centre for Community and Enterprise Networking (C/CEN) and Dr. Michael Gurstein, Director of the Centre and (at the time) Chair in the Management of Technology Change at the University College of Cape Breton. An initial NetCorps training programme and manual were produced as a result of this programme. Notably, the two placements in Angola were of previously unemployed young adults from Cape Breton; they were provided with training in Linux through C/CEN as part of a demonstration programme in the economic benefits that could be achieved through local investment in Information and Communications Technologies in economically marginal regions.

**Phase II**

In 1998 the efforts of Industry Canada, the Canadian International Development Agency (CIDA) and a consortium of Canada's largest international development volunteer-sending agencies secured 14 internships in Africa, Asia and Latin America.

**Phase III**

Further collaboration with Industry Canada and the consortium of international development agencies in 1998 saw the Canadian International Development Agency offer 30 internships in the Americas.

This phase was announced by the Prime Minister at the 2nd Summit of the Americas in Santiago, Chile in April 1998.

**Phase IV - NetCorps Canada International**

From 1999 to 2006, the NetCorps consortium sent more than 1700 NetCorps volunteers overseas, due to the collaboration of Industry Canada and the Youth Employment Strategy financed by Human Resources and Skills Development Canada (HRSDC).

**2006–2007 – Transition Year**

In 2006–2007, the consortium successfully organized the placement of 236 volunteers in collaboration with Industry Canada and the Canadian International Development Agency (CIDA). 2006-2007 was the last year that Industry Canada would be involved in the NetCorps programme.

**Phase VI – 2007–2008**

For 2007–2008, (CIDA) replaced Industry Canada as funders of the NetCorps programme. The 240 placements offered that year were once again part of the Government of Canada's youth-employment strategy.

The programme was discontinued indefinitely as of the 2008–2009 season.

**MEMBER ORGANIZATIONS**

- WUSC
- CWY
- Alternatives
- CUSO
- Oxfam Québec
- VSO International
- Canadian Crossroads International
- Canadian Society for International Health
- International Institute for Sustainable Development
- Human Rights Internet.

# NETDAY

NetDay (1995–2004) was an event established in 1995 that "called on high-tech companies to commit resources to schools, libraries, and clinics worldwide so that they could connect to the Internet". It was developed by John Gage (then-chief science officer at Sun Microsystems) and activist Michael Kaufman. The first official NetDay was held in 1996. In 2005, NetDay merged with *Project Tomorrow (tomorrow.org)*, a California non-profit involved with math and science education. The organization is continuing to work with schools to improve the use of technology in education.

NetDay was established to take place over the course of one Saturday, whereby designated schools would receive full connection to the Internet. Activities were coordinated at the web site *netday.org*. The HTML Writers Guild (quoting the NetDay FAQ) defined the day as an:

- Historic grassroots effort in the classic American barn-raising tradition. Using volunteer labour, our goal is to install all the basic wiring needed to make five classrooms and a library or a computer lab in every school Internet-ready. If the same work were financed by taxpayers, it would cost more than $1,000 per classroom. Volunteers from businesses, education, and the community will acquire all of the equipment and will install and test it at every school site. Your support and participation is crucial to our success. In addition, by bringing together these diverse elements, NetDay establishes a framework for lasting partnerships among business, government, educational institutions, and local communities to provide ongoing support for our schools.

Some argued that access to the Internet should not be a priority when schools lack even basic resources like library books (although in many cases the project added needed materials and efforts to computing projects already underway).

**NETDAY '96**

The first NetDay was held on March 9, 1996. NetDay '96 created considerable excitement amongst participating schools. The day was organized via the web site *netday96.com*. 20,000 volunteers helped to wire 20 percent of California schools to the Internet. 2,500 wiring kits were donated by telephone companies. Of the event, John Gage commented, “NetDay96 is a demonstration of what can happen when people coalesce around a community project [...] In one day, we can begin to reverse California’s abysmal record of putting technology into its classrooms.”

President Bill Clinton and Vice President Al Gore were also involved with NetDay ’96, spending the day at Ygnacio Valley High School, as part of the drive to connect California public schools to the Internet. In a speech given at YVH, Clinton stated that he was excited to see that his challenge the previous September to “Californians to connect at least 20 percent of your schools to the Information Superhighway by the end of this school year” was met. Clinton also described this event as part of a time of “absolutely astonishing transformation; a moment of great possibility. All of you know that the information and technology explosion will offer to you and to the young people of the future more opportunities and challenges than any generation of Americans has ever seen”. Clinton acknowledged the support of the State Superintendent of Public Instruction Delaine Eastin, Lieutenant Governor Gray Davis, Senator Barbara Boxer and Representative George Miller. In a prepared statement, Gore added that NetDay was part of one of the major goals of the Clinton administration, which was “to give every child in America access to high quality educational technology by the dawn of the new century.” Gore also stated that the administration planned “to connect every classroom to the Internet by the year 2000”.

### NETDAY ACTIVITIES

On April 28, 1998, Gore honoured numerous volunteers who had been involved with NetDay and “who helped connect students to the Internet in 700 of the poorest schools in the country” via “an interactive online session with children across the country.” Volunteers continued to play a central role in NetDay school wiring activities, and by the end of 2001, NetDay events were held in 40 states and engaged more than 500,000 volunteers to wire more than 75,000 classrooms across the USA. Archived versions of the NetDay Web site, netday.org, say that NetDay events were planned for March 20, 1999, April 8, 2000, October 28, 2000, March 31, 2001 and October 27, 2001.

In its last four years of operation, NetDay changed its focus from one or two days a year to wire schools to the Internet. Its new events and activities included collaborating with Join Hands Day in 2002, and three new initiatives: NetDay TechDay, NetDay Speakup Day and TESS - Technology Enhancing Student Success. In February 2006, NetDay merged with Project Tomorrow, a national education non-profit organization.

## WORLD COMPUTER EXCHANGE

World Computer Exchange (WCE) is a United States Canada based charity organization whose mission is “to reduce the digital divide for youth in developing countries, to use our global network of partnerships to enhance communities in these countries, and to promote the reuse of electronic equipment and its ultimate disposal in an environmentally responsible manner." According to UNESCO, it is North America's largest non-profit supplier of tested used computers to schools and community organizations in developing countries.

## HISTORY

The organization was founded in 1999 by Timothy Anderson.

The organization is a non-profit organization, and tax-exempt under section 501 (c) (3) in the US. WCE headquarters is in Hull, Massachusetts, and there are 15 chapters in the USA and five in Canada. In 2015, WCE opened a chapter in Puerto Rico.

By November 2002, the organisation shipped 4,000 computers to 585 schools in many developing countries.

By October, 2011, along with partner organizations, WCE has shipped 30,000 computers, established 2,675 computer labs. In February 2012, Boston chapter (also headquarters chapter) in Hull, Massachusetts sent out their 68th shipment bringing their total to 13,503 computers.

## ACTIVITIES

WCE provides computers and technology, and the support to make them useful in the developing communities. WCE delivers educational content and curriculum on agriculture, health entrepreneurship, and even water and energy. The programme also ensures that teachers will know how to use the technology and content by providing staff and teacher training, as well as ongoing tech team support.

Each chapter of WCE collects donated computers, refurbishes and prepares the shipment for the projects. While preparing shipments, they also raise funds to ship the computers. Volunteers of WCE chapters inspect each computer, repair if necessary, install operating system as well as educational material to each computer.

WCE calls recipients of its computers "partners." The requests of computer donations originate from the partners. Once the refurbished computers and the funds to ship the computers are fulfilled, WCE initiates shipment. When possible, WCE coordinates the shipments of computers with other non-profit organizations, such as University of the People, Peace Corps, Computers4Africa.org, ADEA (Assoc. for the development of Education in Africa) and others.

In November 2015, WCE sent two 2 Spanish speakers to visit Honduras for 2 weeks in 2015 to pilot an employable tech skills training for youth under a WCE contract with World Vision.

The WCE Computers for Girls (C4G) initiative is field testing of eight tools to provide technological training and STEM education for interested teachers helping their girl students in four West African countries (Ghana, Liberia, Mali, and Zambia) and Pakistan. In September 2016, World Computer Exchange-Puerto Rico and 4GCommunity.org, two not-for-profit corporations, have announced their alliance to improve public school and family access to technology where needed throughout Puerto Rico.

**ECORPS**

To install the computers at a partner site without access to experts otherwise, WCE recruits and supports volunteers from the USA under its "eCorp" initiative. To be eligible, volunteers must be 21 years of age, have necessary tech skills, and be prepared to self-fund their travel and accommodation expenses. 18 eCorps training teams have worked in Dominican Republic (2), Ethiopia, Georgia, Ghana (2), Honduras, Kenya, Liberia, Mali, Nepal, Nicaragua, Nigeria, Philippines, Puerto Rico, Tanzania (4), and Zimbabwe.

The eCorps "Travelers" programme is geared towards those already planning to go to one of the countries in the WCE network, to provide tech support during their trip. 79 eCorps "Travelers" have visited the following 41 developing countries including: Armenia, Bolivia (2), Cambodia, Cameroon, Democratic Republic of Congo, Dominican Republic (2), Ecuador, Ethiopia, Haiti, Honduras, India (2), Indonesia, Jordan, Kenya (2), Liberia, Malawi, Mexico (2), Namibia, Nepal, Pakistan, Palestine, Panama, Peru (2), Puerto Rico, Qatar, Senegal, Sierra Leone, South Africa (2), Swaziland, Tanzania (2), Togo, and Uganda. In FY'15, "Travelers" visited: Cambodia, Haiti, Honduras, Mexico, Puerto Rico, and South Africa.

**COMPUTERS**

WCE uses Ubuntu operating system to prepare computers, citing the cost of license and less prone computer viruses while providing sufficient computing environment such as word processor and printer drivers. Unlike One Laptop per Child, the prepared computers do not contain specialized software. Each computer is loaded with educational materials which allows users to learn materials without internet connection.

# CHARTERED IT PROFESSIONAL

Chartered IT Professional (in full, Chartered Information Technology Professional) denoted by CITP is a professional qualification awarded under Royal Charter to IT professionals who satisfy strict criteria set by the British Computer Society (BCS), which is a professional body for IT in the United Kingdom. In order to qualify for this award a person normally needs to have at least 8 to 10 years professional experience in IT, with evidence of experience at a senior level (5) in the Skills Framework for the Information Age (SFIA), have passed a professional competency examination and successfully completed a skills assessment interview with two BCS assessors.

The title Chartered IT Professional is aligned with The Skills Framework for the Information Age (SFIA), the UK Government backed competency framework, CITP is the benchmark of IT excellence and is terminal (final/top most) qualification in IT.

CITP status is gaining increasing national and international recognition as the benchmark of excellence for IT professionals. It provides evidence of an individual's commitment to their profession and endorsement of their experience and knowledge. A fast track scheme exists for holders of Certified Architect and Certified IT Specialist certifications from The Open Group, which exempts applicants from the initial review and interview elements of the application process.

In order to maintain a certificate of current competence, demonstrable CPD must be undertaken.

Criteria and requirements for chartered status in the UK have to be approved by the Privy Council and as such the CITP designation is on par with other chartered qualifications in other fields (such as the Chartered Accountant qualification awarded by the ICAEW).

### DESIGNATORY LETTERING

Chartered IT Professionals are entitled to use the suffix CITP after their names. This is written after honours, decorations and university degrees and after letters denoting membership of professional engineering institutions - for example: BSc (Hons), MBCS CITP. The BCS maintains an online register of members with CITP status.

### CHARTERED I.T PROFESSIONAL IRELAND

The Irish Computer Society is the awarding body of CITP status in Ireland. The standard has been developed in consultation with BCS and applicants undergo the same assessment process.

## COUNCIL FOR EDUCATIONAL TECHNOLOGY

The Council for Educational Technology (originally called the National Council for Educational Technology (NCET) but reformed as the Council for Educational Technology (CET) in 1972) was set up in 1967 by the Department of Education and Science in the UK. Initially it consisted of a large council of experts with a small administrative team whose purpose was to "advise educational services and industrial training organisations on the use of audio visual aids and media" but it quickly became more than this, developing projects, producing an academic journal BJET and advising government on setting up major computer aided learning programmes (NDPCAL and MEP). It was amalgamated with the Microelectronics Education Support Unit (MESU) in 1989 to form the National Council for Educational Technology (NCET which later was renamed the British Educational and Communication Technology Agency (Becta) in 1997).

### FORMATION

The original Council (NCET) consisted of a Chairman appointed by the Secretary of State for Education and Science after consultation with the Secretary of State for Scotland, 31 members appointed by the former and 4 members appointed by the latter. In addition, assessors from eight government departments and educational bodies attended meetings. In 1973, as a result of the recommendations of the Hudson Working Party the Council was a representative body, consisting of 59 people.

**DIRECTORS**

- Professor Tony Becher 1967-1969: Tony Becher was the first Director appointed from the Nuffield Foundation
- Geoffrey Hubbard 1969-1986: Geoffrey Hubbard was appointed as Director in June 1969. He was previously an engineer and then a civil servant at the Ministry of Technology. He successfully steered the Council through its sometimes difficult relationship with government. He retired in 1986 but continued his role as Chairman of the National Extension College.
- Richard Fothergill 1987-1988: Richard Fothergill was appointed Director following his role as Director of the Microelectronics Education Programme.

**THE BRITISH JOURNAL OF EDUCATIONAL TECHNOLOGY**

The British Journal of Educational Technology (BJET) was sponsored and funded by the Council.

It published its first issue in January 1970 and Professor Norman Mackenzie was its first editor and the prime mover behind its creation. Although sponsored by the Council it always kept a strong, peer-reviewed, academic approach to its work - as it said in its "Auspices" at the front of each volume.

*Whilst the British Journal of Educational Technology is supported by the Council for Educational Technology for the United Kingdom, it nevertheless reflects an independent, and not official view, of developments or opinions on educational technology.*

BJET continued through the decades and is now published by Blackwell and continues to publish academic articles on educational technology. Importantly its back numbers chronicle much of the history of educational technology in the UK and elsewhere.

**DEVELOPING NDPCAL**

During the late 1960s computers were beginning to make an impact on education and John Duke, the Council's newly appointed assistant Director proposed a major initiative in computer-based learning. The Council set up a Working Party to investigate the potential role of the computer and to outline a programme of research and development. Following a feasibility study the Council set out the case for a 5-year programme in 'computer-based learning' in 1969. The Government, following much discussion amongst the interested departments and an intervening general election, announced the approval of Mrs Thatcher, Secretary of State for Education and Science to a 'National Development Programme in Computer Aided Learning' in a DES press release dated 23 May 1972.

### DEVELOPING MEP

During the late 1970s, with the rapid rise in the use of microelectronics, the Prime Minister Jim Callaghan, is reported to have asked each government department to draw up an action plan to meet the challenge of new technologies and the DES asked CET to create plans for a new programme - the Microelectronics Education Programme. The Programme was aimed at primary and secondary schools in England, Northern Ireland and Wales.

Although it was delayed by the change of government in 1979, Keith Joseph as Education Secretary finally approved it in 1980 and in March a four-year programme for schools, costing £9 million.

### PUBLICATIONS

As well as BJET CET published a range of publications, many the result of projects it set up and funded. The following is a selection of these publications to give a flavour of the breadth and range of the Council’s activity:

- 1972 Educational Technology: The Development, Application and Evaluation of Systems, Techniques and Aids to Improve the Process of Human Learning
- 1974 Copyright and education: a guide to the use of copyright material in educational institutions
- 1978 Deciding to Individualize Learning: A Study of the Process, Malcolm L. MacKenzie
- 1980 How to write a distance learning course, Roger Lewis and Glyn Jones
- 1981 The Kingdom of Sand: Essays to Salute a World in Process of Being Born, William Gosling
- 1984 Open Learning in Action: Case Studies, Roger Lewis
- 1986 The Magic of the Micro: a resource for children with learning difficulties, Mary Hope

## DE LA SALLE UNIVERSITY COLLEGE OF COMPUTER STUDIES

The College of Computer Studies (CCS) is one of the seven colleges of De La Salle University. It was established in 1981 as the Center for Planning, Information, and Computer Science offering only a Bachelor of Science degree in Computer Science.

The department was formally declared as a college in 1984. In 1990, the college was transferred to its new building, the INTELLECT (Information Technology Lecture) Building, which was eventually renamed as the Gokongwei Building. In 1996 the college was granted semi-autonomous status along with the Graduate School of Business which led into the establishment of De La Salle-Professional Schools, Inc.. The college became a part of De La Salle Professional Schools but later transferred back to the university.

## ACCREDITATION

The BS Computer Science programme of the college was the first Computer Science programme in the country to be given accreditation by the Philippine Accrediting Association of Schools, Colleges, and Universities in 1989. It was granted a Level II accreditation in 1993, and was re-accredited to Level II in 1998. In 2000, the college was named as one of the Commission on Higher Education's Center of Development and Excellence in Information Technology.

In 2005, the Computer Science programme of the college received a Level III accreditation by PAASCU, becoming the first Level III Accredited Computer Science Programme in the Philippines. It is the highest accreditation granted in the Philippines to this date for Computer Science programmes. In March 2007, the Commission on Higher Education recognized the college as Center of Excellence in Information Technology.

## ACADEMIC CENTERS

### Advanced Research Institute for Informatics, Computing and Networking (AdRIC)

Formerly known as Advanced Research Institute for Computing (AdRIC), the Advanced Research Institute for Informatics, Computing and Networking is the research center of the college.

The aim of the center is to produce both local and international researches through faculty researchers. The center was established in 1994 to facilitate research directives of the college. It also has its research laboratories on the fourth floor of the Gokongwei Hall.

*The following are the activities done by the center:*

- Symposium on Semantic Web and Ontology
- 2nd Natural Language Processing Research Symposium
- Constraint-based Action Planning
- De La Salle University Science and Technology Congress 2005
- DOST Research Proposal Preparation
- Machine Translation Project.

### Consulting and Education Center

Consulting and Education Center is a continuing non-degree computer training center of the college currently headed by Engr. Jesus Gonzalez. The center provides quality education for both professionals and students. Included in its services are corporate trainings, tutorials, and summer classes catering children, secondary students, and professionals.

As an academic center of the college, it also holds seminars and major conferences such as the Emerging Technologies for Philippines 2020 (ETFP) held last April 17–18, 2006 at the New World Renaissance Hotel.

**Cisco Academy Training Center**

The college is the Cisco Academy Training Center for Expanded Courses (Sponsored Curriculum) or CATC-SC of the country. The center is responsible for conducting trainings on sponsored curriculum courses such as Fundamentals of Unix, Fundamentals of Java, IT Essentials, and Voice and Data Cabling. Under DLSU, trainings for sponsored curriculum courses are categorized into two: instructor and professional trainings.

**Center for Empathic Human-Computer Interactions (CEHCI)**

The Center for Empathic Human-Computer Interactions laboratory investigates the problem of multimodal emotion modeling and empathic response modeling to build human-centered design systems.

By multimodal, we mean we recognize a human's emotion based on facial expressions, speech, and movement (such as posture and gait). Because we extend a computing system from a software to a physical space, novel approaches to providing empathic responses are required.

To achieve this, we use emotion-based interactions together with sensor-rich, ubiquitous computing and ambient intelligence.

Currently, we are focused on the development of a self-improving, ambient intelligent empathic space. It is a three-year project funded by the Philippine Government's Department of Science and Technology - Philippine Council for Advanced Science and Technology Research and Development. It is on its first year with an initial grant of PhP 3,300,000.

**STUDENT ORGANIZATIONS**

- Computer Studies Government (formerly Computer Studies Assembly)
  - Batch Assemblies, or Computer Batches (CATCH)
- Peer Tutors Society (PTS)
- Society of Proactive Role Models Inspiring Total Development (SPRINT)
- Student Research and Development Programme (SRDP)
- DLSU Game Development Laboratory (DGDL)
- CCS Programmers' League (CPL) - formerly known as the InterCollegiate Programming Contest Training Pool (ICPC TraP).
- MooMedia - a brainchild organization of the Instructional Systems Technology programme of the college, but was later accredited as a Special Interest university-wide organization by the Council of Student Organizations of DLSU.
- La Salle Computer Society (LSCS) - the only professional organization in the College of Computer Studies accredited by the Council of Student Organizations of DLSU.

## ACADEMIC DEPARTMENTS

### Computer Technology

Computer Technology emphasizes the fundamental concepts of electronics, the theory and use of analog and digital components, the analysis and design of combinational and sequential digital circuits and the techniques required in order to interface a microprocessor to memories, input-output ports and other devices or to other microprocessors.

### Information Technology

Information Technology focuses on the study, design, implementation, and evaluation of information systems for organizations. This programme aims to make the student capable of recognizing problems that are amenable to computer solutions. It also covers the study of management and organizations in order to bridge the gap between specialists and decision-makers in the preparation of support systems development.

### Software Technology

Software Technology features courses designed to provide students with a deeper understanding of the design and analysis of computer algorithms, thus strengthening their ability to write correct and efficient programmes. It integrates software engineering into major courses in a way that students are able to apply and discover different methodologies appropriate for different software projects.

## DEGREE OFFERINGS

Undergraduate Degree Programmes:

- BS in Computer Science
  - Major in Computer Systems Engineering (tracks: Robotics and Automation, Digital Signal Processing, Computer Architecture)
  - Specialization in Network Engineering (research areas: Network Security, Network Management, Converged Networks)
  - Specialization in Software Technology (research areas: Natural Language Processing, Game Development, Artificial Intelligence)
- BS in Information Systems
  - Information Management Track
    a. Organizational Development and Information Technology
    b. Supply Chain Management
    c. Enterprise Resource Planning
    d. Financial Management Systems
    e. e-Commerce and Mobile computing
    f. Systems Quality, Testing and Assurance

- Interactive Technologies Track
  a. Digital Media Design and Construction
  b. Interactive Multimedia Applications Development
  c. Cognition and Technology
  d. Game Development
  e. Mobile and Wireless computing.
- BS in Information Technology

*Graduate Degree Programmes:*

- Doctor of Philosophy in Computer Science
- Doctor in Information Technology
- Master of Science in Information Technology
- Master of Science in Computer Science

*Diploma Programme:*

- Post-Graduate Diploma in Computer Science

# 5
# CHAPTER

# Teaching: Vocational Teaching and Learning in Practice

## TEACHING AND PUBLIC SOCIOLOGY

### PRESENTING SOCIOLOGY'S FOUR FACES

The clearest trend in American sociology over the past several years has been towards pursuing what Michael Burawoy coined "public sociology." Just as clear is that the idea of giving sociology a more public face has moved American sociologists, albeit in two opposite directions. Indeed, public sociology's proponents are matched in their enthusiasm only by the vehemence of their opponents. Regardless of their respective positions, sociologists brave enough to enter the debate over the virtues and vices of public sociology have largely ignored a crucial way in which to take the discipline public; namely, through teaching. In all that has been written about public sociology, very little has been said about students, perhaps sociologists' largest and most obvious public audience. And the few published essays that have discussed the teaching of sociology have focused solely on undergraduate and graduate students.

To this point, every aspiring public sociologist, as well as each opponent of public sociology, has completely ignored what may very well be a much more important public: high school students. This chapter begins with a brief outline of the intellectual lineage of public sociology and a summary of the little that has been written about the connection between public sociology and the teaching of sociology. The bulk of the chapter is devoted to discussing how Burawoy's four "faces" of sociology have been, are, and could be applied in high school classrooms. In the process, it brings to light the absolute necessity of a solid foundation of professional sociology. I ultimately argue that without the strong presence of professional sociology, the other three types cannot and will not be developed in high school sociology courses.

**Teaching Sociology**

Sociologists have always been the most introspective of social scientists. Since Robert Lynd famously asked Knowledge for What? 70 years ago, each succeeding generation of sociologists has struggled to define the discipline's purpose, mark its boundaries, and confront its theoretical and methodological weaknesses. Michael Burawoy's relatively recent call for a more "public sociology," and the lively debates it has stirred up, is only the most recent case in point. Like previous sociologists who have been courageous enough to offer a vision of what the discipline could and should be, Burawoy has said relatively little about how his notion might be realized by teaching sociology. In fact, most sociologists—critics and supporters of public sociology alike—have said very little about connecting the teaching of sociology to the promotion of public sociology. And among the few commentators who have discussed teaching, not a single one has devoted any attention to teaching below the undergraduate level. Besides Burawoy himself, just one of the contributors to a 2004 Social Forces symposium on public sociology even mentioned teaching—and only in footnotes. The journal Social Problems also published a symposium on public sociology in 2004. Like the papers in the Social Forces special issue, its essays, with two minor exceptions, had nothing to say about teaching.

One year after the Social Forces and Social Problems symposia appeared, The American Sociologist published a special double issue on public sociology, an ambitious project which was eventually turned into a book. Only Edna Bonacich devoted any space—a page and a half, to be exact—to the link between public sociology and the teaching of sociology. She admitted to "try[ing] to inject a 'Marxist' understanding" into her undergraduate course on Research Methods in Ethnic Studies. She did so by asking students to participate in group projects on "'lessening inequality' or 'countering racism'". Bonacich's essay, as well as her teaching style, clearly leaned towards outright activism rather than sociology. Nichols's edited volume on public sociology is one of three that have appeared over the past couple of years. Clawson collection offers essays by fifteen prominent sociologists—including two by Michael Burawoy himself—on public sociology. And a third collection of essays on public sociology, Blau and Smith's Public Sociologies Reader, appeared one year earlier. Among the three books, there is not a single mention of using high school classrooms as what Burawoy would call "laboratories" for public sociology. As the person who has spearheaded the campaign for public sociology, Burawoy has written relatively little about teaching. And, like previous authors' comments, his observations have applied only to college and university classrooms. Perhaps Burawoy's clearest statement of the relationship between teaching and public sociology is in his contribution to the Social Forces symposium, where he suggested, quite correctly, that "students are our first public". Borrowing the terms and themes found in Freire and, before him, in Gramsci, Burawoy differentiated "traditional" from "organic" public sociology, and discussed in one paragraph—of a 16-page article—how each results in a unique teaching style:

- In the traditional approach we treat [students] as empty vessels to be filled with knowledge. The lecturer stands above the lectured in a position of unquestioned authority—the possessor and disseminator of truth. Dialogue, if it takes place at all, does so behind the back of the lecturer. In the organic approach to teaching, students are treated not as tabula rasa but as carriers of accumulated experience, brought to the surface and turned into knowledge through dialogue. That experience may be cultivated from a student's own biography and augmented through specific engagements— the underlying presumption is that the teacher and taught have an organic relation, that the educator too must be educated.

Few sociologists are likely to disagree with Burawoy's "organic" conceptualization of teaching. As attractive as it might be, however, it says very little about precisely how it qualifies as public sociology. It is undoubtedly true that our students are our first public. But undergraduates are not necessarily the largest or most important, and they are certainly not the only, potential public audience for sociology. In fact, more than one-third of a million students each year take a sociology course at some point in high school. These students constitute a very large and very public captive audience. Practicing and aspiring public sociologists alike should begin examining the kinds of sociology that high school students have been and are exposed to—as well as thinking about the kinds they should be exposed to. What follows is an outline of the past, present, and probable future presentations of Burawoy's four "faces" of sociology in high school courses. I suggest that none of the four has ever been prevalent in high schools, and only one—professional sociology—is at all likely to become more common. But even establishing that type, I argue, will take a lot of work. The fact of the matter is that without a deep and solid foundation of professional sociology, the other three types cannot and will not develop at the secondary level. This is a lesson, incidentally, that carries over into each subfield of sociology, insofar as its practitioners are interested in connecting their particular area of expertise to public sociology.

**Presenting Sociology's Four "Faces" to High School Students**

Regardless of one's own orientation to sociology, every self-reflective sociologist would do well to devote serious attention to Burawoy's fourfold conceptualization of the field; that is, professional, critical, policy, and public sociologies. Though one may quarrel with Burawoy's definition of each type, and with the relationships he observes among them, it is difficult to deny that his model neatly categorizes an increasingly messy field. It also provides a useful framework for thinking about the ways in which sociology has been taught in high schools—a subject that has been largely ignored by sociologists for more than 95 years.

***Professional Sociology***

Burawoy suggests that the role of professional sociology is to “suppl[y] true and tested methods, accumulated bodies of knowledge, orienting questions, and conceptual frameworks” for the practice of both policy and public sociology; indeed, it provides the legitimacy of and expertise for both types. Professional sociology, in short, consists not only of currently popular sociological theories and research methods, but of the historical debates and questions that led to their development. For the purposes of this chapter, it is helpful to think of professional sociology as composed of the theoretical frameworks, methodological approaches, and historical perspectives that dominate American sociology. “Professional sociologist” is, for most academic sociologists, our “master identity.” Importantly, as Burawoy correctly points out, “there can be neither policy nor public sociology without a professional sociology.” It is the seed from which the other three types grow.

Obviously, then, if sociologists hope to develop a more public—or, for that matter, critical or policy—brand of sociology in high school courses, it is necessary, first, for professional sociology to have a strong presence. Unfortunately, the little existing research indicates that professional sociology—that is, academic sociology: formal theory, research methods, and a history of the discipline’s development—has been a relative rarity in high school sociology courses over the past 95 years. Marlene Weber carried out a survey of high school sociology courses and teachers in Wisconsin. She found that more than half of the courses focused solely on social problems or on a combination of social problems and sociology. Very little academic sociology—in other words, professional sociology—was being taught at the time. In a follow-up to Weber’s research, Kraft again surveyed Wisconsin high school sociology teachers. He found that only about one-fourth of the courses stressed “sociology as a scientific discipline”; the other 73 percent emphasized social problems, practical interpersonal relationships, or some other approach.

Like Weber, Kraft was forced to the conclusion that, at least in Wisconsin’s high schools, there was a “lack of emphasis on the scientific nature of sociology” in favour of a superficial focus on current social problems. Though it would be nearly two decades before the publication of the next study of the content of high school sociology courses, it was apparent that nothing had changed. Dennick-Brecht, Lashbrook, and De Cesare found that a large majority of high school sociology teachers in Pennsylvania, New York, and Connecticut, respectively, used a standard introductory textbook—the overwhelming majority of which were college-level texts. Though a large percentage of teachers in each state ostensibly relied on a textbook as their primary teaching tool, at least some in Connecticut used it “as a reference more than a bible,” as one teacher told me during an interview. In other words, some teachers use texts solely to introduce students to the terminology and definitions that supposedly characterize sociology. To provide the "meat" of the course, teachers utilized readings other than a textbook. Lashbrook and DeCesare both reported that an overwhelming percentage of the teachers in their respective samples incorporated national and local newspaper as well as newsmagazine articles. These, of course, focus almost exclusively on current events, and are notoriously deficient in the extent of the sociological analyses they offer. There is more. Research methods, which is generally considered one of the two pillars of professional sociology, was included in fewer than threefourths of the courses I studied in Connecticut; the second pillar of professional sociology, theoretical perspectives, was offered in fewer than half of the courses.

These topics were included in 82.4 percent and 43.7 percent of New York high school sociology courses, respectively. The most recent results clearly indicate that professional sociology is being deemphasized in favour of topics that teachers perceive to be more interesting and relevant to students: culture, race, gender, deviance, and the like. One final point has to do with teachers' conceptions of sociology as a field of study. In short, they do not take it all that seriously. When pressed during interviews, teachers are typically much more concrete and articulate about sociology's perceived weaknesses than they are about its strengths. Here is how one teacher responded to my asking whether he thought high school students were capable of understanding sociology: "Why wouldn't they be able to understand sociology? It's not rocket science!". The perception that sociology is easier and "softer" than most other courses in the curriculum is a fairly common one among high school sociology teachers, and is likely due primarily to their lack of training in sociology.

Based on the conclusions drawn by recent researchers, and on past teachers' published course descriptions, it is clear that professional sociology, as Burawoy describes it, is not, and rarely has been, an important feature of high school sociology courses. It is crucially important for more professional sociology to be incorporated into high school sociology courses. In its absence, a high school course in the field becomes little more than a meandering discussion of current events. In its absence, high school students will leave the course with the impression that sociology lacks any theoretical or methodological rigour. In its absence, and most importantly for the purposes of this chapter, Burawoy's other three types of sociology cannot develop. Consider the following, as just one illustration of the vast difference between the usual high school sociology course and one that is based on professional sociology. Many high school teachers include a section on education in their sociology courses. As it stands, it typically includes an unfocused—and unsociological—group discussion of the pros and cons of the controversial topic du jour: school vouchers, bussing, the No Child Left Behind Act, and the like. Injected with a dose of professional sociology, it would become something far more useful, interesting, and sociological. The teacher might lead students through an examination of the social functions of schooling by framing the in-class discussions with fundamentally sociological questions: Who is involved in the process of schooling, and what roles do they play, what statuses do they occupy, and how are they structurally and organizationally connected to one another? How are districts, schools, curricula, and classrooms organized, and why do they look as they do? In what ways do teachers and school environments socialize children into acceptable behaviour, attitudes, and morals? How does the education system produce and perpetuate inequality based on sex, race, and social class? What are the causes of stratification both within and across schools and school districts? Fruitful sociological questions abound, and would be both more interesting and more relevant to high school students than stale, cliché-ridden debates about this or that contentious topic in education.

***Critical Sociology***

Lest professional sociology get too carried away by its self-assumed virtues, critical sociology lies in wait, ready to "examine [its] foundations— both the explicit and the implicit, both normative and descriptive". Critical sociologists, in short, keep professional sociologists intellectually honest. Burawoy cites Robert Lynd, C. Wright Mills, Alvin Gouldner, and Alfred Lee as exemplars of the former. Certainly, critical sociology has, almost from the field's beginnings in this country, played a crucial role in its development. But critical sociology is virtually non-existent at the high school level. I am not aware of a single study that has documented the importance to high school teachers of a Lynd, a Mills, a Gouldner, or a Lee. It is true that a few teachers include an excerpt from The Sociological Imagination in their courses. It is also true that some aim for Mills's primary goal of enabling students to recognize the relationship between biography and history. But these teachers are very few, and even they do not seem to fully appreciate or understand the criticisms leveled by sociology's best-known intellectual pugilist. In fact, judging by the available data, many teachers have never heard of these figures, let alone thought about how to incorporate either them or their sharp criticisms into their sociology courses.

Critical sociology, as Burawoy has argued, is intimately tied to professional sociology. Since professional sociology is difficult to find in high school sociology courses, it should come as no surprise that a critical appraisal of it is also rare. Critical sociology can only grow where professional sociology is deeply rooted. If and when professional sociology begins to stretch its roots into the fertile ground offered by high schools, teachers must also be prepared to offer students a critical assessment of the field. As they should be in undergraduate courses, Mills, Gouldner, Lynd, and the like should become familiar figures in high school courses. Students should be exposed to the epistemological, theoretical, and methodological debates that currently characterize the field. For without exposure to Mills, Gouldner, and Lynd, and the ideas they offered to students of the field, high school students who take a course in sociology will depart with the badly mistaken belief that sociology is, and always has been, an incontestable body of truth, facts, and knowledge. They will be ignorant of the debates about sociology's status as a science, the "proper" role of sociology in society, sociologists' ongoing struggle with objectivity, the relationship between sociology and democracy, and the place of values in sociological research—among other things. Sociology, of course, is a fluid and contestable body of knowledge, and it should be taught to high school students as such.

### *Policy Sociology*

Policy sociology, according to Burawoy, is "sociology in the service of a goal defined by a client"; its purpose is typically to "provide solutions to problems that are presented to us, or to legitimate solutions that have already been reached". Policy sociology, in other words, may be thought of as the practical side of professional sociology. Given high school teachers' well-documented lack of training in sociological theory and methods, it is perhaps not surprising that very little policy sociology is practiced in high school sociology courses. Some, however, is, though necessarily in rudimentary form. A few teachers I interviewed reported conducting a schoolwide survey of students' attitudes and beliefs, and subsequently presenting the results to administrators in the hope of changing school policies. After the infamous shooting at Colorado's Columbine High School in 1999, for instance, one Connecticut teacher guided her sociology students through the development and administration of a questionnaire about bullying. The students presented their results at a schoolwide assembly that also included a panel discussion about how to use the study's results to inform new policies about bullying and violence.

A second example of policy sociology was a study of academic cheating. One teacher I interviewed had her sociology class carry out a survey of nearly 200 of their peers' attitudes towards cheating. One questionnaire item asked: "If you saw someone cheating, would you report it?" Of the 182 students who responded, nearly 97 percent said "no." This result, along with some commentary on it from the teacher of the course, were ultimately incorporated into an academic journal article. Inspiring and interesting as these few examples are, they are rare. Policy sociology, like critical sociology, is difficult to find in high school classrooms. This is unfortunate, since policy-related activities can teach students at least three important lessons: that sociologists study important and timely issues, that sociology is applicable to the "real world," and that sociological research can and does inform policy makers' decisions. These are lessons that are too often lacking in introductory sociology courses—at both the high school and college levels. Nevertheless, projects in policy sociology obviously should not be undertaken by teachers who have not been trained in sociological theory and methods. At best, any empirical results they and their students obtain would likely be meaningless; at worst, they could become dangerous if they fell or were placed into the wrong hands.

### *Public Sociology*

Given what has been said so far, what are the prospects of using high school classrooms to develop a more public "face" for sociology? Not promising. From my perspective, individuals who seek to teach or practice public sociology—teachers, sociologists, or whoever else—first need a clear and comprehensive understanding of professional sociology. Very few teachers, as I and others have pointed out, have been adequately trained in professional sociology, and even fewer have any familiarity with critical or policy sociology. They are, quite simply, teachers, first and foremost. And they are teachers of a variety of disciplines including, typically, history and psychology in addition to sociology. It is telling that I have not met one high school sociology teacher who thinks of him or herself primarily as a sociologist. Still, and despite the rarity of it at the moment, public sociology can be found here and there in high school classrooms.

Consider again the earlier example of the panel discussion of bullying that was prompted by one sociology teacher's class project. It is, I think, a prime example of public sociology. In this case, the dialogue was between the student researchers, the panel, and the larger student body. The goals were to raise awareness about bullying and violence in school, to stimulate a wider discussion of it, and to produce a meaningful and positive change in school policy. It is not difficult to imagine other topics that would lend themselves to a similar public conversation in schools: drug use, techniques for becoming a successful student, gangs, peer pressure, and the like. If sociology courses were taught by teachers who had been trained in professional sociology, conversations about these and other subjects would be the norm, rather than the exception. The public sociologist's mind boggles at what could be, if only we were to devote serious attention to increasing sociology teachers' exposure to professional sociology.

**Paths Forward**

But all is not lost. Critical, policy, and public sociologies can, I believe, be incorporated into high school classrooms. But there is a lot of work to be done before that can happen. Quite simply, but most importantly, high school sociology teachers must be better trained in professional sociology. There must be a sustained effort on the part of the ASA both in lobbying for legislation aimed at increasing the required coursework in sociology for high school teachers of the discipline, and in working to professionalize high school sociology teachers into the discipline. This is quite an old argument, but it is worth repeating. The simple fact is that the ASA is the largest and most powerful sociological organization in the country; if there is to be change in the high school sociology classroom, it must be actively supported, if not initiated, by the association. The ASA might take a cue from another national body, the National Council for the Social Studies. Several of the teachers I surveyed have benefited from this organization. One said: "I went to the NCSS conference in Washington, DC during November [2001]. One meeting was for sociology teachers. Four teachers showed up plus the V[ice] P[resident] from the American Sociology [sic] Association. A high school teachers [sic] network of some sort seemed to be a good goal among those who attended. This would be centered on the internet." Only a national organization—like the ASA or the NCSS—

has the resources necessary for managing a national network of high school sociology teachers, even if it were only to be housed on the Internet. A more general and consistent emphasis placed by the ASA upon teaching might draw some high school sociology teachers into the discipline's professional activities. As long as high school sociology teachers continue to view sociologists solely as researchers, rather than as teachers like themselves, they will not be motivated to participate in the discipline's professional activities. It is therefore another one of the tasks of the ASA to change the misperceptions of sociology and sociologists held by most high school sociology teachers. The association might consider, for example, offering free trial subscriptions of its journal Teaching Sociology, rather than Contexts, to high school teachers. One of the virtues of the former is that it makes clear the fact that all sociologists are not just researchers; most are teachers as well, and many study ways in which to improve the teaching of sociology.

Unfortunately, these recommendations are the latest in a series of recent— and largely unheeded—suggestions made by sociologists for how the association might increase teachers' involvement and improve the quality of their courses. Thus far, the ASA has rested content with trying to reform the courses themselves, rather than with trying to improve teachers' training. In the early 1990s, for example, the ASA created a Task Force on Content Standards for the High School Course in Sociology. It marked the first time since the New Social Studies movement of the 1960s and early 1970s that an ASA-sponsored body devoted itself to sociology in high school. The Task Force published "The High School Course in Sociology: Objectives and Standards" in 1993. The group's position on sociology was made clear in the first sentence: "Sociology is the scientific study of human behaviour."

The authors of the "Objectives and Standards" went on to describe sociology as an "empirical science" and a "social science". The fourteen major sections of the proposed course were:

- Sociological Perspectives,
- Sociological Methods of Enquiry,
- Culture,
- Social Structure,
- Socialization,
- The Self and Social Interaction,
- Deviance, Crime, and Social Control,
- Groups and Organizations,
- Social Institutions,
- Stratification,
- Racial and Ethnic Relations,
- Gender and Age Inequality,
- Demography and Urbanization, and
- Social Change.

To readers who have seen any of the dozens of introductory college textbooks currently on the market, this list will look familiar—the topics correspond quite nicely to the typical textbook's chapters. Each of the sections of the semester-long course had at least nine objectives that "the learner will be able to do" after being exposed to that section. There were a total of 167 course objectives. An important point is that only one of the six task force members, Robert Greene, was a high school teacher; he was also the only teacher of the ten "contributors and collaborators in the formation of standards." It is unclear when the task force was dissolved, and it is unknown how much of an influence, if any, its statement of content and objectives actually had. Nevertheless, its goal was clear: to introduce more professional sociology into high school sociology courses.

Ten years after establishing the Task Force on Content Standards, the ASA appointed a Task Force on the Advanced Placement Course for Sociology in High School in 2001. Of the fourteen original members, it was again only Robert Greene who was a high school teacher. Before disbanding in 2005, the task force worked with representatives from both the American Political Science Association and the American Association of Geographers in an effort to set forth curriculum standards for the AP course in sociology. Such standards are a necessary precursor to the implementation of an AP exam. In 2002, the Task Force completed a first draft of its proposed AP sociology curriculum. Because the curriculum represented the first official ASA statement about course content and objectives to appear in ten years, and because it embodies the ASA's current vision for high school sociology, the proposed AP curriculum deserves careful examination. Here is the course outline:

- The Sociological Perspective,
- Research Methods,
- Culture,
- Socialization,
- Social Organization,
- Social Inequalities,
- Deviance and Conformity,
- Social Institutions, and
- Social Change.

Readers will notice, first, the similarity to "textbook" sociology. In comparing this list to the list created by the 1993 task force, little change is evident. Perhaps most obvious is the fact that this outline is much shorter than the 1993 outline, even though it is for a year-long rather than a semester-long course. However, most of the topics that appeared on the earlier list were subsumed under one of these nine more general headings. Second, this is a course in academic, or professional, sociology. According to the draft of the course outline, the first topic, "The Sociological Perspective," introduces "sociology as a discipline that is both basic science and applied science." More of the same follows in the second section on "Sociological Theory and Research Methods":

"This section of the course introduces students to the dynamic interplay between theory and the logic of the scientific method in sociology.... They will recognize that sociology is a science." The ASA's approach to high school sociology courses has been consistent since the 1960s: teach academic sociology. But the ASA has missed a crucial point: teachers have never been trained to teach academic sociology. Teachers' lack of formal training in, and professional exposure to, the discipline is the primary reason why they have taken a current events approach. Most high school teachers never learned much, if anything, about sociological theory, research methods, or sociology's scientific aspirations. It seems unreasonable, therefore, to expect them to teach professional sociology.

On the other hand, anyone who reads the newspaper or newsmagazines, or who watches television, is familiar with social problems and current events. It seems that teachers, by focusing on social problems, are teaching what they are most familiar with. They will likely continue to do so. Even if they are trained in professional sociology, many teachers will probably continue to teach current events instead of sociology anyway. There are at least two major reasons for this. The first is purely pragmatic: Teachers need to consider what will be of interest and seem relevant to high school students, especially in the context of an elective course that could very quickly disappear from the curriculum if students do not enroll in it. Put bluntly, students are more interested in social problems that they are already familiar with than in sociological theory or research methods. Of the subjects covered in high school, which are more relevant and interesting to students than current events? Probably not many—and certainly not the principles of a vaguely defined scientific discipline. There is no reason to think that adolescent students' interests will change any time soon.

A second reason that teachers are unlikely to introduce professional sociology into their classrooms, even if they are familiar with it, is structural: they teach in the context of a social studies curriculum. Citizenship education and the teaching of democratic values have been the dual goals of the American social studies curriculum since it was first developed immediately following the Civil War. This point is often overlooked in the scholarship on teaching sociology, even though early sociologists were aware of it. Edward Hayes, for example, observed more than eighty years ago the "insistent and wide-spread demand among public school authorities for important changes in the curriculum of the public schools designed to provide a more adequate preparation for citizenship."

Put bluntly, teachers have little incentive to tinker with such a long-standing curricular objective, and little power to do so; indeed, they may be sanctioned by administrators, colleagues, parents, and students for attempting to do so. Given the position that high school sociology teachers are in, it is not surprising that the ASA's course-based efforts to inject more professional sociology into high schools have largely failed. We must begin to focus on the backgrounds, training, and experiences of teachers themselves if we hope to change the nature of high school sociology courses. The ASA has long ignored this point, and evidently continues to do so. An alternative avenue for increasing teachers' professionalization runs through our various regional and state associations. I agree with Rienerth that most regional sociological associations need to take a more active role in changing high school sociology for the better. It is somewhat disheartening that this suggestion was first put forth by Harold Meyer before World War II, and it is still being echoed today. Regional and state associations might follow the examples of the Wisconsin Sociological Association and the North Central Sociological Association, which regularly offer workshops for high school teachers at their annual meetings. It might be the case that high school teachers would be more willing to attend the smaller, and often more personal, annual meetings of the regional and state sociological associations than the large-scale annual ASA meetings. This presents an opportunity, especially for state sociological associations, to provide a "home" for high school sociology teachers.

The smaller associations might also be able to communicate more effectively with local high school teachers than the ASA. For these reasons alone, regional and state sociological associations should encourage high school teachers to participate more consistently in the activities they sponsor. Local and regional associations should also continue lobbying their respective legislatures to impose minimum educational requirements on high school sociology teachers. Even without the support of the ASA, regional and state organizations could make a significant impact on legislators, local boards of education, and school administrators. After all, as Rienerth et al. point out, educational policy is made at the state and local levels. A big enough impact at these levels may even provoke the ASA into more sustained action on behalf of high school sociology and teacher training. In other words, rather than waiting for change to trickle down from the top of the organizational hierarchy, regional and state sociological associations should attempt to introduce change from below.

The discipline's organizations should and must take the lead in improving relations between their members and high school teachers. But it is up to sociologists themselves to do the "dirty work." If sociologists want to see a different type of sociology offered in high schools, they must model it, not simply write and talk about it. For when sociologists write and talk about anything, it is usually to one another. Former ASA representatives Carla Howery and Felice Levine have each provided a list of the many ways in which sociologists can assist in the teaching of high school sociology, and in the process, contribute to the professionalization of teachers. Among them are to deliver guest lectures in high schools; prepare a videotape about sociology; help with the school's career day; visit annual meetings of state councils on the social studies to talk to teachers; become active in local school board activities and serve on advisory panels for curriculum materials, guest speakers, and selection of films; look at teacher training programmes in schools of education and how sociologists can contribute to strengthening them. It is important that sociologists continue these efforts and initiate new ones. Perhaps it should not be just any sociologists, at least at first. Jeff Lashbrook and I have argued that the discipline's "charismatic leaders" should take the lead in bringing about the kinds of changes sociologists wish to see in the high school classroom. This suggestion is not as outlandish as it might seem. As early as the 1920s, prominent sociologists like Robert Park, Emory Bogardus, and Charles Cooley were taking public positions on sociology's unique contribution to the high school social studies curriculum. Others, during the 1960s, publicly debated the merits of high school courses in sociology. Still others served in important roles in the nationwide Sociological Resources for the Secondary Schools project during the 1960s and early 1970s. Today's most famous sociologists should follow the lead of their best-known predecessors, and make the professionalization of high school sociology a top priority.

Doing so would likely convince other sociologists of the topic's importance. There is also the possibility that forming partnerships between university sociology departments and high school social studies departments might encourage some sociology students—both undergraduates and graduates—to pursue a teaching career in a secondary school. Other researchers have also recognized the importance of encouraging sociology majors to teach in high school. Irving Horowitz once went so far as to call for "the activation of some of the very best sociologists in [high school sociology] programmes." Through the forging of partnerships between secondary and post-secondary institutions, academic sociologists with an interest in teaching could, for instance, teach one fewer class at the college level and assist high school instructors in the teaching of their courses during a given semester.

On the bright side, several university sociology departments have begun reaching out to their local high schools: Appalachian State University, Fairfield University, UCLA, Temple University, and the University of Rhode Island have been at the forefront of this cooperative effort. The important point in all of the suggestions offered above is that there must be a more sustained commitment on the parts of both the ASA and its regional affiliates, as well as academic departments of sociology, to increase the presence of professional sociology in high school sociology courses. As I argued above, it is only after professional sociology becomes a normal feature of the courses that we can hope to develop any of Burawoy's other three types.

## TEACHING AND PUBLIC SOCIOLOGY

Public sociology is centrally related to the teaching of sociology, given that it is defined by Burawoy as reflexive knowledge aimed at extraacademic audiences. Students taking sociology for the first time are part of larger extra-academic audiences. Many who become interested in sociology wonder what they might do with it. Burawoy's typology of sociological labour greatly enlarges the spectrum of possibilities for students happy to see that there are a number of career pathways they could follow in sociology, across the quadrants of Burawoy's fourfold table encompassing professional, critical, policy, and public sociologies. This chapter first describes Burawoy's four types of sociology and how they relate to the teaching of sociology. It then considers some relationships, tensions, and synergies between parts of the typology, how each might influence the others, and the implications of such influences for teaching and learning. In his 2005 presidential address, Burawoy clearly situates professional sociology as the foundation for everything else, because professional sociology provides the scientific knowledge base. Like other fields, sociology needs an institutional base, with clear training and career paths. Graduate education is not financially sustainable on its own in most departments, so the fiscal viability of sociology in research universities depends on undergraduate majors, minors, and service courses. Moreover, since 70 to 80 percent of advanced degree holders in sociology work in academia, the majority of career paths involve teaching.

Critical sociology examines “the foundations—both the explicit and the implicit, both normative and descriptive—of the research programmes of professional sociology.... and it attempts to make professional sociology aware of its biases, silences, promoting new research programmes built on alternative foundations. Critical sociology is the conscience of professional sociology just as public sociology is the conscience of policy sociology”. Critical sociology does more than simply challenge or criticize existing arrangements and practices, although it certainly does that. It also scrutinizes what we know and how we know it. It asks what is missing from a particular formulation of a question or problem, whose perspectives are omitted, what other perspectives might be possible in a given situation, and what are the hidden assumptions in a statement? It includes issues of major concern to professional sociologists, such as sampling and representativeness, measurement and measurement error, design, and inferences about causal relationships, and considers the difference between explanation and prediction. It anticipates unintended consequences from planned social action and looks for them. One of the major contributions of sociological research has been to bring out the unintended consequences of planned social changes, as in Karen’s discussion of the No Child Left Behind legislation.

Because of the likelihood of such unanticipated consequences, both professional and critical sociology encourage the building of meaningful feedback loops into social policies. “Policy sociology is sociology in the service of a goal defined by a client.” Its purpose “is to provide solutions to problems that are presented to us, or to legitimate solutions that have already been reached”. Policy sociology is concerned with the application of sociological principles, insights, and empirical research in everyday life, for the amelioration of social problems, usually in the realm of government. Policy sociology can also provide strategic research sites for understanding better how things work in the social world.

Public sociology, the fourth cell in Burawoy’s typology, “brings sociology into a conversation with publics”. It is one thing to recognize public sociology, but “another thing to make it an integral part of our discipline”. In contrast to policy sociology, public sociology “strikes up a dialogic relation between sociologist and public in which the agenda of each is brought to the table, in which each adjusts to the other”. “The approaches of public and policy sociology are neither mutually exclusive nor even antagonistic". "Public sociology is often an avenue for the marginalized, locked out of the policy arena and ostracized in the academy". It may well be about efforts to change "official" practices. As such, it is often rooted in non-governmental organizations and may even be distinct from any institutional setting. For some examples, see Willie, Ridini, and Willard 2008.

**Analyzing Teaching in Terms of the Four Types**

How do these four types illuminate the teaching of sociology, in terms of cognitive practices and potential pitfalls? Teaching sociology is a form of public sociology because teachers are bringing sociology into dialogue with new publics. But they may also have professional audiences and peers if they publish about teaching. We consider below what some sociologists say about their teaching in each quadrant and how they involve students.

***Professional Sociology***

Professional sociology is related to where teaching occurs, what is taught, and to norms of accountability. Although teaching generally occurs within the academy, the home space of professional sociology, its arena is broadening to include high schools, community colleges, four-year colleges, comprehensive regional universities, as well as research universities. Teaching and learning may also occur in labour unions or communities, or through digital media including public television, web learning, and podcasts. Teaching always involves decisions about what to teach. Do we teach primarily or only "professional" sociology, or do we also teach public, policy, and critical sociology? In what proportions? Studies of teachers, syllabi, and an ASA task force have all identified the importance of teaching key theoretical and empirical sociological knowledge and scientific norms. Another study of peer-designated leaders in professional sociology revealed that the most important understanding they wanted students to gain was a sense of what is distinctive about a sociological perspective on social phenomena, compared to psychological, economic, political, anthropological, or linguistic perspectives.

They see this as the power of social factors—contexts, relationships, structures, patterns, institutions, values, attitudes, and practices—for explaining social behaviour over and beyond individual personalities, temperaments, and traits. Leaders also wanted students to understand the difference between opinions, on the one hand, and views based on systematic principles and empirical research that observe publicly discussed standards. They want students to understand that "science is a method not a subject matter" and want them to be exposed to research. Indeed, the leaders interviewed placed somewhat greater stress on research and getting students involved in research than did sociology teachers more generally. Professional sociology includes numerous research programmes dealing with inequality, whether based on social class, race, gender, sexual orientation, age, disability, or some other characteristic, and leaders saw the centrality of inequality in power and resources as another key understanding they wanted students to gain. Moreover, they often stressed the importance of incorporating data on these issues into courses. Another important understanding concerned the concept of the social construction of reality, including things that economists posit as "natural laws" or deities, such as the "market" or market forces. This perspective asks students to constantly question definitions that appear to be rooted in nature, such as gender, race, and deviance, and explicate their socially constructed features. Professional sociology affects more than the place and content of teaching. It also offers standards for professional practice and accountability. A major example is the way "Professional sociology is accountable to peer review".

One of the big debates in teaching currently is how to make it more accountable to peers. There are greater calls for professional practices in teaching, not only in sociology but in many academic subjects. Since teaching is often done away from the scrutiny of fellow professionals and since the principle "publics" for teaching are students who can assess only certain aspects, students may not be in the best position to assess the degree to which a course or a programme represents various issues, approaches, or knowledge in a field. Also, while students may be able to say whether or not they like particular methods of teaching, they may not always be able to assess the effectiveness of various methods, at least in the short term. Interviews with peer-recognized leaders in the field of sociology reveal that some feel a tension between what might be seen as a professional responsibility for teaching introductory sociology students something about the field of sociology, its history, major theories, and how it differs from other social and behavioural sciences, and, on the other hand, teaching students about society and/or social problems, which students might find more immediately engaging and useful. Some articulated this as a tension between seeing introductory courses as gateways and background for the major, in contrast to seeing them as a chance to recruit new students into sociology.

Many sociology electives serve dual purposes as gateways to a major and as "service" courses for students who will never be majors. Such service enrollments are often important for the fiscal accounting increasingly imposed on academic departments. Moreover, even among majors, usually only a few are planning to study sociology in graduate school. Nevertheless, the "professional sociologist" in many teachers seems to think that even if few students go to graduate school, they should be prepared for the theory, methods, and substance they will encounter. This dilemma raises the question of who is our public in teaching sociology? To whom are we accountable? Our answers shape the learning goals and content of courses and curricula. The pathology of professional sociology can become evident in teaching if instructors are so embedded within the subject that they are not able to explain clearly why a problem is important or how it connects to issues beyond the walls of academic sociology. When instructors can help learners connect the concepts, theories, and methods of sociology to their everyday lives, students are better able to understand and remember what they are learning. Probably all of us have had the experience of taking a course in a subject we thought would be fascinating, and then finding that the instructor spent all his/her time dissecting minute debates between scholars in the field, without ever connecting to larger issues. Professional sociology bases its legitimacy on adherence to scientific norms but that may be an inadequate basis for its legitimacy in teaching.

***Critical Sociology***

In relation to reflexive critical sociology, teaching faces additional dilemmas, including concern for sociology's identity and public image, a tension between moral and scientific passions and the possibility that exposing ideological biases will limit its influence on publics. Leaders placed critical thinking very high on their list of what students should understand, although they were not too explicit about what they meant by it and how it relates to critical sociology. Some of what the leaders mentioned includes the use of comparative and international examples and analyses particularly at the societal level, to show how things vary. Others mentioned the importance of students learning to question scientific knowledge and methods. However, if we use critical sociology to challenge the foundational basis of sociological knowledge, students may not take sociology seriously or believe the empirical findings that are reasonably reliable and valid. Critical sociology suggests that the legitimacy of sociology may be undermined if it fails to offer moral visions that go beyond scientific norms. However, if we venture too far into normative truths, we make it difficult for students to identify the difference between normative and empirical statements. If we offer a moral vision, they may reject it and simply choose an alternative moral vision. It may seem easier, more comfortable, and more defensible for many of us to say that learning sociology is like becoming familiar with the culture and mores of another society, and that we want them to learn this new way of thinking.

On the other hand, if we neglect the moral visions of sociologists and the motivation to improve the world that brought many into the field, we risk giving students a desiccated version of sociology when they could have a juicy full meal. Reflecting on teaching, Burawoy notes that it is "as if graduate school is organized to winnow away at the moral commitments that inspired the interest in sociology in the first place". He seems to be suggesting that if there were some space for students' moral commitments to exist within sociology, fewer students might be lost along the path to a Ph.D., with the further implication that sociology would be less likely to lose those with moral commitments. There are times, however, as Brint noted, when there is a conflict between what we learn and what we want to believe, leading to tension between our moral and scientific passions. Neither Burawoy nor Brint discusses how to deal with such tensions. While often seen as primarily concerned with critiquing existing policies, formulations, or practices, critical sociology can also offer positive alternatives, according to some scholars. Building on Sorokin's thinking, Jeffries suggests considering notions of the good, something that is very consistent with Burawoy's view that critical sociology "supply moral visions" to both the profession of sociology and society. The "assessment of values is fundamental in evaluating what problems should be studied... and in justifying their relative importance". Sorokin suggests that sociological analysis could utilize "ideas from religious traditions pertaining to topics such as human nature, the characteristics of goodness and of perfection, the ends of human existence, and moral and ethical precepts", since they are especially relevant to the sociocultural level and to individual personality. As an example, Sorokin notes the importance of the sociological problem of solidarity and antagonism when he writes, had we the knowledge "to increase the familistic and eliminate the antagonistic from interpersonal and intergroup relationships" we might have been able to reduce or eliminate war, crime, coercion, compulsion, and misery. Such a statement seems to assume that crime, coercion, compulsion, and misery do not occur in families, an assumption that is seriously challenged by numerous conflict and feminist researchers studying the family. Similarly, when Sorokin suggested that "love is the heart and soul of ethical goodness itself and of all great religions" is he failing to consider how often love is subverted or corrupted in many religious institutions and families, and how difficult it has been to realize in human practices? Nevertheless, starting with values can lead to a focus on "the goodness of individuals, defined in terms of virtue and benevolent love... [and on] social solidarity, the manifestation of goodness in interaction."

This raises two general theoretical and research problems: "how culture and society influence individual goodness, and how individual goodness influences culture and society". Sorokin's and Jeffries's points raise the question of whether there is agreement on "the good," but I think they might be pleased to see that question being debated. I wonder what they would think of Barrington Moore Jr.'s comment in Reflections on the Causes of Human Misery and upon Certain Proposals to Eliminate Them on how people might be more likely to agree on their conceptions of misery, but might disagree more in their conceptions of happiness. We might also wonder if happiness is equivalent to "the good." How do we reconcile conditional, contingent, and contextual scientific thinking with normative and moral thinking? Can normative thinking get beyond either/or binaries and into more contextualized and nuanced ways of responding to the social world? Is it possible to suspend judgement while we try to find out what is actually happening?

What about the human potential for evil as well as for good? How do societies and individuals protect against that? Who is involved in the definitions of good and evil, and their implementation in practice? To what degree should sociology be involved in such philosophical questions? If constructive critical sociology discussed these and similar questions it might help instructors bring notions of the good into their classrooms. The pathology of critical sociology can certainly surface in teaching if classrooms or departments become dogmatic in their moral or normative beliefs and practices. A monochromatic climate will stifle enquiry and debate, make students who don't share the "party line" feel unwelcome and drive them away. An additional drawback is that such an atmosphere, lacking sufficient challenges, is unlikely to sharpen and strengthen the quality of arguments and evidence offered to support the dominant position. I remember a student who was very outspoken in an introduction to sociology class when we were discussing the American Dream and social mobility. Not only did he believe that the system was open to anyone who worked hard, he was also convinced that there were no longer differences between men and women in terms of their occupations and earnings. In his group research project he investigated that topic and was very surprised to find that while younger men and women were becoming more similar with respect to occupations, their earnings were far from equal. He and the other students also became more sensitive to how men and women were distributed by particular occupations within the broader category of "professional and managerial workers." The data clearly did not confirm his worldview. Perhaps because they found the data themselves and because he was free to articulate an alternative hypothesis initially, he was convinced by what they found.

***Policy Sociology***

While professional and critical sociology usually exist within the academy, policy sociology and public sociology are less common there. One of the unexpected findings from the interviews with professional sociology leaders was the value a number placed on "improving the world."

They wanted their own sociological work to do this and they wanted to imbue students with the idea that sociology could and should be used to improve the world. Policy issues, for example, regarding the death penalty or legal options for abortion or stem cell research, may help students learn how to use empirical data to analyze possible costs and consequences of various social policies, as well as become better at identifying competing value claims, some of which cannot be adjudicated by empirical evidence. Policy sociology often legitimates itself in terms of whether the policies advocated are effective in accomplishing a stated goal. One major issue is who articulates that goal?

Is it a particular client, who may have a specific perspective on the issue and one that may differ from those of other participants in the situation? Because policy sociologists may be relatively more dependent on the funding received from clients than academic sociologists are, they may be less able to resist restrictions on how studies are framed, who is included, or open publication of results. This is the only form of policy sociology articulated by Burawoy.

However, some sociologists have initiated policy sociology by clearly identifying problems and presenting specific solutions to them, as Jeffries notes about the work of Sorokin, Yablonsky, and Jacobson. It is noteworthy that Sorokin held an academic post at Harvard, Yablonsky holds an academic position in the California State College System, and Jacobson worked in government and currently directs the Vera Institute for Justice, suggesting the importance of a relatively secure position for providing the relative autonomy and insulation from client pressures needed to initiate independent policy proposals.

***Public Sociology***

In one sense, all teaching is public sociology in that it is talking about sociology to non-sociologists. Instructors have considerable freedom to decide what and how to teach in many of their courses. To what degree do they acquaint students with all four of the quadrants of sociology identified by Burawoy? Decisions about content have been supplemented by efforts to provide students with opportunities to learn more about public sociology by doing it in some form. Since public sociology depends for legitimacy on its relevance, its risk of pathology comes from judging issues by their current trendiness, whether by virtue of being in the news or on afternoon talk shows. There, the quest for novelty may trump social significance, for example by ignoring how widespread or enduring a particular phenomenon is. There is the further risk that public sociology "can be held hostage to outside forces... [or] pander to and flatter its publics... thereby compromising professional and critical commitments". Besides the independent contributions each of the four types of practice may make individually, they may also interact with each other in interesting ways.

**Relationships, Tensions, and Synergies within the Typology**

Professional sociology can offer experience in using peer review as a way of setting standards and assessing quality that would be useful for teaching, public sociology, and policy sociology. For example, the rigorous standard of theoretically grounded and empirically validated knowledge raises the bar for assessing teaching effectiveness, potentially enhancing teaching and learning. For several years in this decade, the Carnegie Foundation for the Advancement of Teaching ran summer workshops for faculty from a variety of fields who struggled to ascertain whether their approaches to teaching were helping students to learn, and why or why not. Both professional and policy sociologists have a battery of quantitative and qualitative research methods to study such questions. The development of the Scholarship of Teaching and Learning in sociology and other fields is one effort to make teaching more professional, in several senses. It focuses on making teaching public at least within the community of scholars so that it can be critiqued, improved, and ultimately validated by peer review.

Making teaching public includes publishing thoughtful, annotated course syllabi online that explicate the rationale for organizing and constructing a course in a particular way, indicating what worked well and did not work as well, and suggesting ways of improving teaching in the future. Rather than being fixed, such a syllabus becomes a living, reflexive work in progress. Another direction is to encourage people to conduct systematic research on teaching and learning and to publish it in peer-reviewed journals. While some sociologists have achieved fruitful synergies by using professional sociological concepts and perspectives to inform enquiries into teaching and learning, SoTL in sociology could even more vigorously develop or apply sociological theories to teaching and learning. Many studies in Teaching Sociology focus on individual teaching, and sometimes learning, without always contextualizing that learning. Others have consciously discussed ways of bringing current professional sociological research into teaching. It cannot only be because SoTL is an interdisciplinary field that sociological theory has not been used more, because another very interdisciplinary field, namely social gerontology, uses sociological theories. SoTL might incorporate more professional sociology by including issues in the political economy of education such as reductions in federal and state spending for higher education, growing state expenditures on prisons and incarceration compared to education, and how the rising costs of higher education are being passed on to families and students; by analyzing the absorption of corporate models and practices into higher education; or by studying social movements within higher education aimed at including new curricula in women's studies, Black Studies, Asian-American Studies, or Queer Studies.

Such developments could be related to learning goals, instructional practices, or student learning. Sociologists investigate colleges and universities as organizations, how environments and university managements affect new organizational contexts and forms, and how diversity in higher education affects social relationships. In my view, the scholarship of teaching and learning badly needs to stay connected with professional sociology and to engage in theoretical and methodological dialogues with it. When it does that, perhaps the total absence of citations to articles in Teaching Sociology in the American Sociological Review, noted by Purvin and Kain will change. Some might include within the purview of critical sociology such educational works as Paulo Freire's Pedagogy of the Oppressed; Lisa Delpit's "The Silenced Dialogue: Power and Pedagogy in Educating Other People's Children"; bell hooks's Teaching to Transgress; and Berenice Malka Fisher's No Angel in the Classroom: Teaching through Feminist Discourse.

These are powerful, emotionally, and morally moving works. How can they inspire new research about teaching and learning or about professional sociology? They might be brought into fruitful dialogue with insights from professional sociology, such as the social psychological research of Cohen and Lotan and Skvoretz, Webster, and Whitmeyer on how status and power differentials among individuals affect interactions in group tasks.

***How Might Each Form of Practice Influence the Others in Teaching***

If critical, policy, and public sociology were brought into the teaching of professional sociology, how might it change? If critical sociology were a stronger presence in professional sociology graduate study, would it change what was presented to students, even as examples of professional sociology? This is an interesting question that could be empirically studied if departments with strong programmes in critical sociology could be found and compared with departments with weak or non-existent critical sociology programmes. Are there differences in how professional sociology is taught in the two types of settings? The question of how a strong presence of critical sociology affects students' levels of interest is more difficult to address because of possible confounding from the self-selection of students into programmes with particular characteristics. However, one could investigate the question of what types of students are attracted to which kinds of programmes. Even without such empirical studies, some mix of professional sociology, with its relatively orthodox "normal" science paradigms, and critical sociology that steps out of those intellectual boxes to examine the epistemological and normative assumptions of professional sociology could very well benefit a new generation of thinkers who might make major innovations in our field and beyond.

Acquainting students with professional sociology, critical sociology, policy, and public sociology may be somewhat analogous to learning different types of dance steps, for example, the Waltz and Foxtrot which are more prescriptive and formed compared to the Monkey or other contemporary dance forms which are more open-ended. Each of these dances is relevant for different purposes and in different contexts. Would our students be better prepared if they understood and appreciated a broader range of sociologies, like the cultural omnivores identified by Peterson and Kern? Most students need to see a broader spectrum of the social world, especially those who have grown up in communities relatively segregated by colour, class, or ethnicity. To the degree that sociology students can get out of the academy to do some form of policy or public sociology, they will bring more diverse social experiences and observations to their study of professional sociology. Some instructors make a point of not only telling their students about the existence of policy and public sociology, but provide opportunities to do them.

*For example, one of the peer-recognized leaders described the following project he had his students do:*

- We did statewide surveys in Pennsylvania with a sample of 2000 to 3000 people. Each student in the class had to interview 35–40 people, although some did as many as 75 or 80. Some students went with me to Harrisburg to present the findings to the state government. When Governor Ridge was there, he generally liked the surveys. We did questions on issues the legislature was debating. We worked with the Pennsylvania Education Association, a state environmental group, or other state agencies to get enough support to pay the telephone bill and pay for the report. The deal we made was that we would release all our findings; they could not restrict or suppress any of them. We were doing it as a public service. We got about a 30% response rate which was helped by the students doing the calling who told people they had to do the interviews for the course. They did some pre-election surveys that predicted the winners of the national elections very closely. They never missed in their predictions. That gained credibility for their surveys.

By doing this, students were able to learn a great deal about the concrete, pragmatic, and potentially effective nature of one type of policy sociology, namely opinion research on current issues. They dealt with clients and learned something about how instrumental knowledge generated by sociological methods could enter the public arena and be viewed as legitimate because it was accurate. Another professor had students in a research methods course do a qualitative and quantitative study about student satisfaction on campus, based on a random sample of students.

Together the entire class of about 25 students wrote the final report, which included tables and qualitative interview results. The report was sent to the department chair and undergraduate dean on the campus. While technically in that case the dean was not the "client," the students did feel that they were doing something real that might have an impact on college policy and practice in the future. They also learned a lot about how to do surveys and interviews. The way these projects were structured avoided the potential pathology of policy sociology, namely of being "captured by clients who impose strict contractual obligations on their funding". Other professors have students do public sociology.

Several leaders encourage students to work with one or more local grassroots social movements. While not the same as public sociology, many sociology programmes have instituted service learning as either a requirement or an option within the major. Service learning gets students out of the classroom into a variety of public settings, where they can begin to talk with and hopefully understand the perspectives of a wider range of people than they have previously met. Others, for example, Edna Bonacich involve students in multi-faceted projects that have clear research and action implications. She says she tries to "inject a 'Marxist' understanding into research and discussions." By this she means considering such questions as: "What are the underlying power relations? What are the major economic interests? How does the capitalist system shape social policy? How are people and communities, especially racialized groups, hurt by these policies? What community organizations have formed to combat them, and how can we help them?".

Within classrooms there are many opportunities for students to practice developing communicative knowledge and consensus building. Many instructors have used collaborative and cooperative research projects with the express goal of helping students to understand they can learn from the knowledge of others and to see how knowledge is built by moving towards consensus about values, warranted methods of obtaining information, and provisional understandings. In my own teaching, when I ask students to point out what they don't understand in a reading, I learn from them.

For example, I wrote in a scholarly paper that several research studies have found that when you control for seven socioeconomic factors, the black-white gap in test scores disappears. When questioned by my students I learned the importance of adding "statistically control for" to indicate that there was not yet a situation that I know of in the United States where the gap has disappeared, since the control factors are all correlated with race. Dialogue with more broadly defined publics may also be built into certain assignments. For example, Eleen Baumann and Richard Mitchell had students conduct research on the amount of violence on TV. Besides writing up the results of their research, the students were required to write a letter to the editor of a local newspaper and to a television network president reporting on the results of their research and indicating how they felt about those results.

Graduate courses on writing sociology and the American Sociological Association's recently inaugurated award for "Excellence in the Reporting of Social Issues" also illustrate efforts to improve the communication of professional sociology's insights. How might professional, policy, and public sociology influence the teaching of critical sociology? Professional sociology might ask sociology of knowledge research questions about critical sociology, such as under what social conditions is critical sociology more or less likely, more or less isolated, or more or less likely to fall prey to pathologies? Public and policy sociology might press to include examples from their work in the teaching of critical sociology, to illustrate the kinds of critical concerns that could be raised. Each of these three forms of practice might ask of critical sociology, "what difference does it make for what we do?"

For example, do values change the research questions we ask, how we conduct research, or teaching practices? How do we keep our values from biasing how we do research? Professional, critical, and public sociology might also interrogate policy sociology. Are policies or policy makers taking too narrow a view of the issues at hand? Are there relevant theories that might illuminate the phenomenon? Can theoretical questions be addressed by a particular policy study?

Are the research methods flawed or biased? Are key constituencies being ignored? Will a given policy benefit some publics more than others? When policy is taught, perspectives from other forms of practice need to be included. For example, when military sociologists help the U.S. Army accomplish military goals, professional sociology might ask what new knowledge could be generated here? Critical and public sociologists might question whether sociological knowledge should be used to help one group in a conflict harm members of another group.

Thus, instruction could include raising a broader set of theoretical and ethical questions, as well as policy concerns. Professional, critical, and policy practices can influence public sociology. If we take the case of teaching as a form of public sociology, we see that considerable thought has gone into bringing more professional sociology into teaching, so that students can know more about sociological research. Examples beyond those discussed above include the experimental use of issues of Contexts in teaching and the recent publication of a Contexts reader for teaching. Critical perspectives and practices might seek to elevate the value placed on teaching and the place of values in teaching. A policy perspective might call for more research on the effectiveness of various methods of teaching, and might suggest a variety of methods for assessing efficacy.

***How Might Critical, Policy, and Public Sociology Influence the Teaching of Professional Sociology***

Professional sociology might ask critical sociology to bring their discussions about the good to the formulation of existing or future research questions and to the shaping of course content. Policy sociologists might press professional sociologists about the policy implications of their work, and ask that they bring such concerns into their teaching, particularly if they are teaching in a law, business, education, public policy, medical, or other professional school. Does bringing all four forms of sociological practice into the teaching of professional sociology require curricular change? Some of the incorporations described occur within existing classes, whether in research methods, social problems, or topics courses. Others involve additional curricular components such as internships or service learning. Should there be formal courses or programmes in critical sociology, policy sociology, or public sociology? In the 2007 ASA Guide to Graduate Departments of Sociology, only 11 departments indicate a specialty in Policy Analysis/Public Policy and one of these is in criminology only. Some universities, for example, Columbia, have organized extracurricular seminars that bring together academics, policy makers, and community practitioners to discuss issues of common concern. One way to begin approaching the issue of curricula might be to collect syllabi, exercises, and other teaching materials being used in existing courses and programmes in critical, policy, and public sociology, and get them published by the Teaching Resources Center of the American Sociological Association. In the early years of sociology, course syllabi were published in the American Journal of Sociology in an effort to share curricular content. Before it is likely that there would be significant curricular changes in graduate departments of sociology, I think it would be important to have a critical mass of existing courses, scholarly materials, and teaching resources, as well as visible career paths for students pursuing such specialties.

## PSYCHOLOGY APPLIED TO TEACHING AND LEARNING PSYCHOLOGY AT THE UNIVERSITY LEVEL

This chapter is a summary of current research from cognitive and educational psychology highlighting important aspects of good teaching in psychology, particularly by lecture, since this seems to be the most common method used internationally. It is written by Steve Newstead, who gained his first degree at the University of Oxford and subsequently received a PhD from the University of Nottingham. His PhD research was on the psychology of language and thinking, an area in which he maintains a strong interest to this day. However, he has also established an international reputation in the applications of psychology to teaching and learning in higher education, and has published seminal articles on student cheating, biases in the assessment of students, and the reliability of the marking process. This work has led to invitations to give keynote addresses in a number of different countries, including Russia and Malta. He is currently a Professor of psychology and Dean at the University of Plymouth, where he has worked for more than 30 years. In 1999 he received the British Psychological Society's Award for Distinguished Contributions to the Teaching of Psychology. He was President of the British Psychological Society in 1995-96, and has also served the Society in a number of other roles, including chair of the Special Group of Teachers in Psychology, chair of the Membership and Qualifications Board, and Centenary Vice-President in 2001 when the Society celebrated its 100th anniversary. He recently chaired the group which drew up the national benchmarks for psychology teaching in the UK, and has also chaired a group set up by the Economic and Social Research Council to draw up guidelines for the research training of psychology postgraduates. He chaired a pan-European Task Force which looked at psychology training across Europe, which led to a number of publications, including an overview of psychology training which appeared in *European Psychologist.*

**THE APPLIED PSYCHOLOGY OF TEACHING**

There is little excuse for teaching psychology badly. Psychology teachers are in a privileged position since teaching is in many ways just a branch of applied psychology and hence psychologists should be able to apply the principles and knowledge they possess to the advantage of their students. This does not mean, of course, that psychologists always do this. As Gale put it: 'The typical psychology degree on offer today falls very short of the ideal. A major problem, or cause of the problem, is that psychologists have not used psychology itself to design the degree.' In this chapter I will try to summarise some of the main areas of psychology that are relevant to teaching. My aim will be to indicate how psychological knowledge is relevant and how it might be applied to the teaching situation.

The review will not be comprehensive, but hopefully there will be sufficient coverage to both assist teachers in their practice and also to inspire them to examine further their own psychological knowledge in order to seek out other implications and applications of psychological knowledge. While teaching and learning can take many forms, the most prevalent form of contact between students is in the formal classes that are held, and of these much the most common format is the lecture. As Ramsden says: "lecturing remains the pre-eminent method of teaching in most subjects." In a more recent review, Bligh starts his book, provocatively titled 'What's the use of lectures?' with these words: 'In the United States lecturing is the most common method when teaching adults. And so it is all over the world.'

I am sure that few would disagree with these claims. Hence in this chapter the focus is on the effectiveness of lectures, and brings in research on individual differences, memory, learning, and evaluation, as well as more educationally-oriented research which investigates lectures directly. In a later chapter I examine other areas of psychology which have implications for the way in which we assess students. The focus in this chapter on lecturing is intentional since this is one of the few near-universals in psychology teaching; I know of no country or institution where lecturing does not play a major part in the education of psychology students.

**WHAT IS THE PURPOSE OF LECTURES?**

Although lectures are in such widespread use, there is not necessarily agreement as to what they are designed to achieve. Are they intended to transmit facts? To inspire students to read round a topic? To persuade them of the validity of a certain point of view? To encourage critical thinking? Or something other than these? There are those who argue that the use of lectures is based more on habit and false assumptions than on any real insight into what their purpose is. Gibbs wrote a booklet entitled “Twenty terrible reasons for lecturing”, in which he summarised some of the false assumptions that are often made. These include the assumption that lectures are the only way to ensure that the ground is covered, that they are the best way to get facts across, that they get students to think, and that they ensure that students have a proper set of notes. Gibbs challenges each of these claims, and suggests that the real reasons why we lecture are rather different, being based more on ignorance of the facts and of the alternative methods available, on unwillingness (or lack of time) to put in the additional work to explore and introduce other methods, and on a variety of other factors including conservatism and resistance to change in both teachers and institutions. Hence for some the new conventional wisdom is against lectures. However, a more sober assessment comes from a widely used textbook in this area - Beard and Hartley’s Teaching and Learning in Higher Education. They say:

- One aspect of learning that is rarely mentioned is its efficiency.... It is possible to teach large audiences within one building.... Lecturing is still an economical method.

One way of ascertaining the purpose of lectures is to ask the lecturers what they are trying to achieve. This provides a variety of different answers and a clear indication that different lecturers are trying to achieve different things. In a widely-cited study, Trigwell and Prosser described five different approaches to lecturing derived from interviews with lecturers, ranging from strongly teacher-focused (aimed at information transmission) to strongly student-focused (aimed at bringing about conceptual change in students). The approach taken depends in part on the context in which the teaching takes place (institutional expectation, discipline taught) but also on the person doing the teaching. In other words, lecturers have different conceptions of the purposes of teaching. A further complication here is that there is evidence that there may be a mismatch between what lecturers say and what they actually do. Murray and Macdonald found that lecturers who believed in student-focused teaching actually taught in ways which were more indicative of information transmission. Other evidence suggests that this disjunction between beliefs and actions represents a compromise between teachers’ beliefs and the institutional and social context in which the teaching is delivered. Thus we have some advocating the reduced usage, if not abandonment, of lectures on the grounds that they do not achieve the aims that lecturers have for them and that those aims are often misinformed. There are others who suggest that lectures are an efficient method and implying that, given the large numbers of students that need to be taught, their use should be maintained and even increased. Who is right? This of course is where research comes in. It is an empirical question as to whether lectures achieve the aims they are claimed to have, and one which psychologists are well fitted to address.

## HOW WELL DO LECTURES ACHIEVE THEIR AIMS?

Given that lectures are both widely used and controversial, it might be expected that there would be a mass of research investigating how effective they are at achieving their aims. It is indeed the case that there have been numerous studies, but these are not always of high quality and certainly use a wide variety of different methods. Bligh's now classic text on lecturing provides a convenient review of the studies. However, it is noticeable that much of this research was conducted some time ago and there may be a problem in relying on Bligh's review since things have moved on since much of this research was done. Higher education in many countries has changed almost out of recognition in recent years, with an increased emphasis on the need for high level qualifications and the assumption that the 'knowledge-based economy' is the key to future prosperity. In many countries, this has led to a massive expansion of the higher education system. In addition, the availability of new technology has had a major impact on lecturing, not just on the lectures themselves but also on ways of supplementing lecture material, for example through the use of web-based material. Bligh considered four main purposes of lectures. Firstly, their usefulness as a means of transmitting information; secondly, their use in the promotion of independent thought and critical thinking; thirdly their effectiveness in bringing about changes in attitudes; and fourthly their use in the teaching of behavioural skills. With respect to the first of these, Bligh found, in his meta-analysis, that lectures were as effective as other methods at transmitting information but not more so. In other words, some studies showed that lectures were better (*e.g.,* as demonstrated in a multiple-choice questionnaire), others showed lectures to be worse, others showed no difference. This is a slightly simplified version of Bligh's table:

**Table. An Owniew of Studies Favctuing Lecture vs. Other Methods of Teaching**

| Alternative method investigated | Number of studies favouring lectures | Number of studies favouring alternative method | Number of studies with no difference |
|---|---|---|---|
| Programmed learning | 8 | 20 | 17 |
| Discussion | 22 | 18 | 54 |
| Reading and | | | |

| | | | |
|---|---|---|---|
| independent study | 9 | 10 | 21 |
| Project | 3 | 6 | 6 |
| Other | 20 | 27 | 57 |

As can be seen from Table, the majority of studies showed no significant difference between lectures and other methods of teaching. The only possible exception (and even this is not clear cut) is that certain types of programmed learning, specifically one called Personalised System of Instruction, did in some studies prove more effective than lectures in getting information across. With respect to the second aim, Bligh concluded that lectures were not more effective than other methods at promoting critical thinking and analysis. Relatively few studies have been done, but those that have been conducted have shown that other methods (primarily discussion groups) are more effective. One problem in this area, of course, is that it is not easy to measure thinking skills. There are standardised tests but these are of dubious validity. Most studies have used specially designed tests of the ability to solve problems or to apply the principles learned; these have the merit of being relevant to what the lecturer is trying to do but the de-merit that we have no information as to how well they measure general thinking skills. With regard to the third aim, Bligh's review of the literature on lectures as a way of changing students' attitudes suggests that lectures are ineffective at doing this. Lectures are actually worse than other methods (principally group discussion) at increasing students' interest in the subject, helping students acquire the values associated with the subject, or promoting changes in students' personality. The fourth use of lectures that Bligh considers is in teaching behavioural skills. Lectures are relatively ineffective and other methods (such as practising the skill) are better. Thus Bligh came to the simple (and intuitively sensible) conclusion that lectures are a reasonably effective means of communicating information, but less effective at encouraging critical thinking, changing attitudes and acquiring skills. This conclusion seems inherently plausible and few would contest it.

## HOW CAN LECTURES BE IMPROVED?

Recent work has tended to focus not on the general effectiveness of lectures but on ways of improving their effectiveness. One focus has been on the use of active learning in classes. Active learning techniques involve a wide variety of possibilities. They may involve buzz groups, where the lecture group is broken down into smaller discussion groups to tackle a specific problem or issue; or short problem solving sessions, for example based on a case study; or discussions between the lecturer and the students; or short talks by students; in fact any activity in which students are actively involved rather than passive recipients of information. Gibbs, Habeshaw and Habeshaw produced a booklet entitled '53 interesting things to do in your lectures', and most of these involve the students in some sort of group or individual activity. While many of these might be fun and might fill up the time in an entertaining way, the important question is whether they actually improve the effectiveness of lectures. Some traditionalists might argue that these activities serve to distract students' attention way from the more important material and might lead to less comprehensive coverage of the topic. Benjamin is one example of work in this area. She describes the ways in which she has used active learning techniques in her own very large lectures, and has even managed to personalise classes in groups of several hundred. In their feedback students were generally positive about these classes, but what is lacking is any thorough critical evaluation of the effectiveness of the techniques. Another fairly typical example of the kind of research conducted — and of the associated problems – is the study by Birchmeier and Bakker. They delivered two associated classes in social psychology, one of which involved very little actual lecturing but instead held discussion groups, mostly led by students and on topics of the students' own choosing. The other class was the control group which was a standard lecture with some (though limited) discussion. Students in the active learning group felt that the material they had covered was more valuable and indicated that they were more motivated by the course. Such findings are not untypical since students often report that they find active learning programmes more engaging and enjoyable. But what is lacking, as in the Benjamin study, is evidence that they have actually learned more or that they do any better in their courses. Of course, student engagement and satisfaction is a good thing in itself, but it is important that these should not be the only criteria against which educational innovations are judged.

## WHAT ARE THE EFFECTS OF CLASS SIZE?

One recurring theme in research on lectures is the effect of class size. As size increases, so do classes become more and more impersonal, and one might expect that students' interest and involvement in the classes would diminish. Furthermore, lecturers almost always say that they prefer and are more effective in smaller classes. Again, what does the empirical evidence show? The picture is a little variable, but in general the evidence indicates that, beyond a certain size, it has relatively little effect. Jenkins reviewed the (mostly American) literature, and concluded that as class size increased up to about 20 students performance is affected by increasing size, but that further increases after that seem to make no difference. Indeed, on some measures, such as student feedback, there is even evidence that larger classes are preferred.

On the basis of this evidence it is difficult to claim that large numbers in lecture classes are necessarily a bad thing. However, Gibbs, Lucas and Simonite came to a slightly different conclusion in their research on the modular degree course at Oxford Brookes University in the UK. On this programme students have a very wide choice of modules which leads to the students being placed in classes of very different size. The authors were able to study how performance varied across modules as a function of the number of students in the module. There was a very clear relationship between size and performance. In other words, the smaller the classes, the better the student performance. How can these conflicting results be reconciled? Both studies agree that, with smaller sizes of classes, increased size does lead to less effective teaching. Over a certain size it may be that size makes little difference, and performance may be more attributable to the teacher than to class size. Jenkins' review was, as indicated earlier, based primarily on American studies, and it may be that the better lecturers are assigned to the larger classes, thus cancelling out any effect of class size; or it may be that lecturers make extra efforts with larger classes. But there is little evidence that larger classes in themselves are inevitably a source of poor lecturing.

## NOTE-TAKING IN LECTURES

Note-taking by students in lectures is an almost universal activity, but what exactly is the function of taking notes? Most students (and lecturers) would claim that notes help to focus attention, and serve as a memory and revision aid. But just how effective are they in this, or is note-taking just something that students do out of conformity – because everyone else does it? Kiewra makes a distinction between two main functions of note-taking: storage and encoding. The storage function emphasises that notes provide the basis for review which facilitates memory. The encoding function refers to the extent to which note-taking facilitates learning and comprehension even without review; it claims that note-taking helps students take in the information even if they never review their notes. Most studies of the effects of note-taking are in fact studies of both encoding and review: students have taken notes (encoding) and then reviewed them. Only recently have researchers looked at the effects of review of notes taken by other people, where the possible benefits of encoding have not been present. Hartley presents a review which suggests that whether it be from lectures, texts, films or whatever, recall is better if notes are taken. This is a summary of his findings:

**Table. Effects of Note-taking on Learning**

| | Helps teaming | No effect | Hinders Learning |
|---|---|---|---|
| NOTE TAKING | | | |
| Audio Pmentation | 21 | 15 | 2 |
| Text Presentation | 8 | 3 | 1 |
| REVIEWINGNOTES | | | |
| Audio Presentation | 13 | 4 | 0 |
| Text Presentation | 9 | 2 | 0 |

More recent studies have tended to confirm these findings, though not unequivocally. Most of this recent research on note-taking has been conducted in the USA by Kiewra and his colleagues. They have looked primarily at the effects of note-taking on recall, and most of their research has looked at multiple-choice type tests and real-life note-taking. Kiewra gives an excellent review, and many of the points I will make are derived from this summary. The general consensus is that note-taking on its own (*i.e.,* without review) does encourage more active learning and is better than not taking notes. If groups of students are given a lecture and required either to take notes or not take notes, the groups taking notes tend to do better. However, this is not the typical use to which notes are put: it is not usual to take notes and then do nothing with them. More normal would be to review the notes, sometimes recoding them into revision aids, sometimes using them directly as revision aids.

Most research done on review of notes has indicated that taking notes and reviewing them leads to better recall than just listening to a lecture or than taking notes and not reviewing them. This has led to another question being asked: is it better to use your own notes or will someone else's do? Kiewra, du Bois, Christina and McShane found that it made little difference - those who borrowed notes did as well as those who took their own notes! A similar result derives from studies of provided notes (*i.e.,* prepared by the lecturer). As long as these are sufficiently detailed, people will remember at least as much (and often more) from provided notes as from their own. The reason for this is probably that students' own notes are notoriously inaccurate, with both omissions and inaccuracies.

The lecturer's notes are presumably more accurate and serve to eliminate such errors. It does lead one to ask again whether lectures are on their own the best way of transmitting information. How effective would it be just to provide a summary of the points and not give the lecture at all? Matrix notes have in some studies been found to be better than outline notes or free-format notes. Matrix notes are ones where the student is given a complicated matrix covering the lecture and has to fill in the various elements as the lecture proceeds. Since these effectively force students to contrast elements of the lecture, it is not surprising that they led to better essays, but it is far from clear that they will always do so. Outline notes - being presented with an overall summary of the main points and having to fill in the details as the lecture progresses - were no better than free-format notes. Hence the research would seem to suggest that note-taking is useful in ensuring students actively engage with the material presented and helps them to pay attention to the lecture. And notes are also very useful in review and revision – though the notes can be prepared by someone else, they do not necessarily have to be the students' own notes!

**STUDENTS' PERSPECTIVES ON NOTE-TAKING**

A number of studies have looked at the kinds of notes that students make, and the results are sobering. Baker and Lombardi looked at the notes students took on a lecture which was an actual part of their course. The lecture was carefully prepared and the lecturer indicated what were the most important points being made in the lecture. The average number of these points actually recorded by the students in their notes was just over a quarter — though students did tend to accurately note down the very central points. Interestingly, students tended to note down accurately everything that was presented on an overhead. Baker and Lombardi also noted that there was a correlation between the presence of information in lecture notes and its subsequent recall in a test. Information that is noted down tends to be better remembered. A number of studies have looked at ways of improving notes. Moore looked at this by trying to cue students as to what they should note down. He held up red and green cards to indicate whether a topic was one that should be noted down or not. Perhaps not surprisingly, this act of theatre led students to recall the principal points better than those who had not had such cueing. A study by van Meter, Yokoi and Pressley looked at the students' perspectives on note-taking. The main findings to emerge from this qualitative study were:

1. Students are goal directed in note-taking in that the notes are thought to maintain attention, facilitate understanding, and serve as a study and revision aid.
2. Notes take different forms, in that some notes serve to select important information whereas others are aide-memoires. Some students tend to paraphrase information while others record what is said verbatim.
3. Note-taking is affected by contextual factors such as the pace of the lecture, the nature of the lecture outline, signalling of information by lecturer, and the organisation of the lecture
4. There are differences between students dependent on their familiarity with the material and their skill as note-takers.
5. Course demands affect note-taking; it makes a difference whether the lecture is covering facts or concepts, whether the material is easy or difficult, and what the course requirements are.
6. Notes have different fates: some are ignored, but most are re-read; some students re-write their notes; notes are often used in exam preparation; and some notes are made more personally meaningful to students.

Hence students try to be strategic in their note taking, varying their style of note-taking depending on the perceived purpose and context of the lecture. However, it is difficult to see how students can be completely successful in these strategic aims if their notes are as inaccurate as Baker and Lombardi found.

## WHAT USE DO STUDENTS MAKE OF NOTES?

Most of the available evidence suggests that students rely heavily on lecture notes in preparing essays, be these for exams or coursework. Entwistle and Entwistle conducted a retrospective study of students' approaches to their assessment using in-depth interviews with students (mainly psychology students) who had graduated a few months earlier. The authors categorised the students' approaches to revision into one of these categories:

1. Reproducing content from lecture notes without a clear structure
2. Reproducing content and logical framework from lecture notes
3. Using own structure for individual topics, mainly from lecture notes
4. Adjusting structures from strategic reading to meet exam requirements
5. Developing an individual conception of the discipline from wide reading

The students reported using the first of these only in their early days on the degree course. They later tended to use strategies to organise the information, such as developing schematic representations. Most of their methods were geared primarily towards remembering what they had learned in order to reproduce it in the exams. What is more, this was a deliberate strategy that they believed would lead to a good mark in the exam. Even students who had reported using deep approaches to understanding the material in the first place tended to use fairly superficial revision strategies, aimed only at committing the information to memory. Students seem to believe, rightly or wrongly, that regurgitating information in an exam is what will get them good marks, and they gear their revision strategies accordingly; and they believe that their lecture notes help them in this. Other studies have actually tried to investigate the source of information that is contained in essays.

The most significant study here is one by Norton and Hartley. We have already seen that note-taking seems to serve as a useful memory aid; what Norton and Hartley did was to study where the information reproduced in actual exam essays came from. For two successive years students in a subsidiary psychology class were given the same lecture on programmed and computer assisted learning.

After the lecture, students' lecture notes were collected in and photocopied before being returned to students.

The same exam question was set in both years of the study. Norton and Hartley then examined all the essays produced by these students and categorised every sentence as to whether the information in it was derived from the lecture notes, the lecture handout, the two main course texts or outside reading. One third of the students presented material in their essays which could be directly attributable to their own lecture notes; this was less than could be attributed to the course texts and outside reading, but more than could be traced back to the lecture handout. If the analysis is restricted to those students who actually attended the relevant lecture, then more than three quarters of them used their notes in writing their essay answers.

Norton and Hartley also observed a small positive correlation between the quality of notes taken and the overall mark awarded, and that women took more notes than men Kiewra and his colleagues have also carried out research on the use to which lecture notes are put in writing essays. In the study by Benton, Kiewra, Whitfill and Dennison participants watched a video-recording of a 19- minute lecture on creativity, following which they were asked to write a short essay. The researchers measured the quality of the essays by looking at the number of words, the number of idea units and measures of coherence and cohesion.

Students who took notes and had access to them during essay-writing produced longer, more coherent essays than non-note-takers (*i.e.,* students who just listened to the lecture). Students who took notes but did not have access to them during the writing of the essay performed no better than those not allowed to take notes. This rather lengthy review of note-taking suggests some fairly simple conclusions.

Firstly, students would be well advised to take notes. All the evidence suggests that those who take notes pay better attention to the material being presented and take it in better. Although taking notes in itself is useful, students should also review and refer back to their notes; those who do so seem to recall the material better and are more likely to use it effectively in exams. However, it is important to ensure that notes are accurate since there is rather disturbing evidence that note-taking can be decidedly unreliable. Students perhaps need training in how to take effective notes. Lecturers should seriously consider providing ready made notes for students, as long as these are ones that the students needs to think about and add to; matrix notes have been shown in some studies to be effective in doing this.

**REMEMBERING INFORMATION FROM LECTURES**

As we have seen, one of the main purposes of lecturing (sometimes it seems as if it is the only purpose) is to enable students to remember information. Psychologists have studied memory in considerable depth and can provide an insight into the best ways to ensure that memory is as effective as possible. Of especial relevance are those studies which have looked at what we remember from information presented in classes, how notes help or hinder in this, and how information is remembered in exams. An interesting study in this context is that by Nakamura, Graesser, Zimmerman and Riha who looked at memory for events in a lecture.

Some of the events were part of the lecture script (pointing to information on the blackboard, using eraser) others were not (wiping off glasses, bending a coffee stirrer). Non-scripted activities were better remembered. Why? Because we activate the script, and once activated we find it difficult to discriminate between events that are part of the script and events that have actually happened. Distinctive events, *i.e.,* those that are not part of the script, are much more likely to be accurately remembered. Perhaps disturbingly, such distinctive events are likely to be remembered better than the actual content of the lecture: students may be much more likely to remember a minor disruption in the lecture than what the lecturer said!

Such research illustrates clearly the role that schemas play in memory. If we are asked to remember something, *e.g.,* last week's lecture, our schemas will help. We know roughly what form a lecture will take, and can use this knowledge in re-constructing what actually happened, Indeed, even if we could not remember a thing (*e.g.,* someone who had not attended the lecture at all) it would still be possible to "remember" it by using our schemas.

Indeed, a lot of memory may be largely reconstructive rather than based on memory for what actually happened. Other research has looked at long-term memory for material presented in class. Some of this has investigated very long term memory over 30 years or more. Bahrick has looked at memory for the names and faces of people who you were at school with, memory for a second language learned at school and for mathematics.

Fairly typical is the study by Bahrick and Hall which looked at memory for mathematics (algebra and geometry). Those who had scored well in school exams or who had high SAT scores tended to remember better; but the loss of information was unrelated to these factors.

The main determinant of loss was the level to which mathematics had been studied. Those who took mathematics to college level seemed to retain the information almost perfectly for up to half a century. But those who gave up maths at high school seemed to forget most of what they had learned. As Bahrick and Hall say: "Those who have taken three or more college mathematics courses with the highest level above calculus show minimal losses during the course of more than 50 years even though they report no rehearsal activities during that period." Bahrick and Hall conclude that information which is learned well and understood conceptually and perhaps over-learned tends to be remembered indefinitely.

They propose the idea of a permastore; once information has been put in to this store it tends not to be forgotten. One explanation for this is in terms of schema theory; if the information has been incorporated into a conceptual schema, then it is resistant to decay; but if it has not, and is just another fact, then it is forgotten. If the aim of teaching is to ensure students remember the information (and this is surely one of the aims) this finding has implications for how we teach. Bahrick and Hall suggest that teaching might be changed to improve retention, for example by having cumulative reexaminations of material or by extending the time period over which information is taught.

Psychologists have investigated long term memory for material taught in psychology classes. Conway, Gardiner, Perfect, Anderson, and Cohen investigated memory for a psychology course - actually a cognitive psychology course run by the Open University in the UK. They studied graduates from the course up to ten years after they had graduated (longer term memory could not be studied because the course only started 10 years before the study). This course was in some ways ideal: it is a highly structured course, which changed little over the years.

Conway et al examined memory for five different areas, including familiarity with proper names and concepts; fact verification; concept grouping; cued recall; and research methods. A perhaps surprising finding to emerge was that recall declined considerably over the first 3 or 4 years, but after that did not decline further. This applied to both recall and recognition of names and concepts, and to concept grouping, though this latter declined less rapidly.

This coincides closely with Bahrick's results. Fact retrieval showed a different pattern with only a small decline overall, even over the first 3 or 4 years. Research methods showed no decline at all over time. This latter may be because the material was overlearned through being used in practical classes and projects as well as in the cognitive psychology course. It is encouraging that students remember some of the information they have learned in psychology courses for considerable periods of time. However, much is forgotten, especially information which has not been used regularly or overlearned in some way. If psychology lecturers want students to remember better, they need to ensure that the content is conceptually processed and continually used.

# PROSPECTIVE TEACHERS' PERSPECTIVES ON TEACHING AND SOCIAL JUSTICE

## INTRODUCTION

As lecturers in an undergraduate teacher education programme we aim to encourage students to engage critically in, and with, the worlds in which they live and work. We consider that for educators generally, and teachers in particular, such a critical capacity is vital. Broadly put, education is not simply imparting knowledge about things. It is an inherently political process that actively works for certain ends, or futures, over other possibilities. For us one such end is social justice where education is directed towards the attainment of a better, more open and humanly possible world. We agree with Ayres and his assertion that education and schooling are contested arenas of both hope and struggle:

- … hope for a better life and struggle over how to understand … and achieve that better life. … At that moment we realize that no teaching is or ever can be innocent –it must be situated in a cultural context, an historical flow, an economic condition. Teaching must be towards something; it must take a stand; it is either for or against; it must account for the specific within the universal.

This paper reports on a study into the beliefs of 535 first year prospective teachers. These students studied in two compulsory first year topics; a sociology of education topic entitled 'Key Educational Ideas' and a philosophy of education topic entitled 'Ways of Explaining Education'.

Data were gathered from two sources for this report. First, student reflexive journals that were part of their assessment in the sociology of education topic were collated and analysed, and second student responses to weekly propositions about education were collated and analysed. Roughly, these data give us both qualitative and quantitative aspects to understanding how this group of prospective teachers think about teaching and education at this time in their teaching careers.

In the first section of the paper we outline the theoretical idea of 'ideology'. 'Ideology' is the concept we use to describe the dispositions and attitudes that these students have to teaching and education. The notion of 'discourse' is also sometimes used, however we do not go to any great lengths to distinguish between the two concepts and their theoretical and political differences. Our primary aim is to describe the student's responses, and their naturalised ways of understanding and explaining key ideas in education. We then outline the study and go onto describe and make sense of the student responses in the second half of the paper.

## IDEOLOGY AND PROSPECTIVE TEACHERS IDEAS ABOUT TEACHING AND EDUCATION

As we mentioned at the beginning of this paper teaching and education are shaped by politics, or more precisely, by one's beliefs, values and attitudes to teaching, learning, students, schools and other key ideas within Education. These beliefs and values arise from someone's social background and cultural experiences within particular historical and cultural contexts to describe the student's responses, and their naturalised ways if understanding and explaining key ideas in education. Taken as those deep and often unquestioned assumptions about the world, ideologies are vital to understanding students' dispositions to teaching and education. While ideology is generally recognised within the social sciences as "one of the most equivocal and elusive concepts one can find", we take that our ideologies are the foundations from which practical engagements with the socio-political world emerge and are justified. In this sense, ideologies – or, more particularly, political ideologies – are not dispassionate theories but sets of collective beliefs that come to pass as the common-sense bases for sensible action. Ideologies are to be understood in a positive sense: as expressions of specific world views and certain collective interests. In short, ideologies are not irresistible forces but emergent features of specific historical and cultural conditions. Likewise 'discourse' refers to the sense of ideas and networks of practices that we take for granted and use habitually to make sense and to act in the world. The strength of this idea over ideology is that it gives us a sense of the relationship between the agent, or the individual and their ways of understanding themselves and their action in a global context. In this sense individual beliefs are part of the broad networks of ideas that inform elites and institution as much as less powerful peoples and groups throughout culture. Hence, as we look at the student responses we are both considering their local and individualised dispositions to teaching and education but also thinking to some extent about how they fit within, inform and are informed by broader cultural networks of meaning and practice.

## THE STUDY – BACKGROUND AND METHOD

This research is based upon data collected from students in two large, compulsory, first year prospective teacher topics undertaken as part of the Bachelor of Education at Flinders University in South Australia. The topic 'Key Educational Ideas' is an introductory sociology of education topic that attempts to generate awareness of the social and cultural aspects of the field of education, from the classroom to educational policy. The topic implicitly addresses issues of social justice; that is, questions of social and cultural difference are considered in relation to issues of inclusion, access and the distribution of knowledge and resources.

Social justice in this topic is considered in its breadth, with the aim being to help students begin to understand the contested character of education and just teaching practice. From this topic we collected and analysed materials taken from student reflexive journals that were part of their assessment. From 535 students 60 journals were analysed.

The cohort include students undertaking Bachelor of Education degrees in junior primary, middle school and secondary teaching and cover students studying arts and science in more general degrees. Of the journals collected there were 27 male and 33 female responses. Twenty-nine responses were collected from students studying junior primary teaching, 17 middle school and 14 secondary school responses. The topic 'Ways of Explaining Education' introduces students to a number of philosophical positions concerning the nature and purposes of education. It aims to develop students' skills in identifying the raft of assumptions that inform their standing views on what education is and ought to be about human nature, childhood, the individual and the collective, morals, and in critically reflecting on the relative merits of these through engagement with others holding to more or less different views. From this topic data were derived from student responses to weekly propositions about education arranged in terms of Likert scales. Both of these topics attempt to develop reflexivity, an awareness of asymmetries in the distribution of life-chances and power, and promote sensitivity to difference in prospective teachers.

The instruments from which our data are derived were components of their assessment for these topics. These did not seek out right or wrong answers to content driven questions but rather attempted to get students to position themselves and reflect upon this positioning on questions and propositions related to the students own sense of social justice as it related to education; for example on the issues of collective influence, the cultivation of the self, the good social order and deviations from this, individual and collective responsibility and their limits, equality and sameness and difference.

They thus also served as a useful source of insights into the as yet largely unquestioned opinions and thoughts about education and teaching held by these students. Here we are particularly interested in how the student through their responses mediates the social context and the individual, and in a more abstract manner the dynamic of structure and agency. These ideas are mediated through notions of equality, order, difference, culture and power and social justice at the sites of education, teaching, the self, and the student or childhood more broadly. Australia is a liberal democracy, and is dominated by the ideological traditions of Europe and North America. One of the key tenets of liberalism, which we are seeking to explore in this article, is the embodiment of individualism. The implications of individualism, as an ontology, demonstrate particular discourses of the individual, responsibility, action, social influence which are deployed through key ideas in education, and used as ways of explaining education and the role of the teacher. These ideas have implications for what we term broadly as 'social justice', that is the ways that prospective teachers see their place in addressing questions of disadvantage and privilege in the education system.

## PERSPECTIVES AND IDEOLOGIES: MAKING SENSE OF THE RESPONSES

What insights into the ideological predispositions of the 2006 cohort of first year pre-service teachers can give a deceptively fair appearance from our data? The first, very general impression that emerges from student self-identifications is that the greatest number of these people subscribe to what, in lay terms could be described as a 'nice liberalism'. What do we mean by this? Entries in the reflective journals support a distribution of particular ideologies. When presented with a range of educational ideologies - conservative, liberal, social-democratic, socialist – and asked which they identified with most closely, the predominance of students opted for a liberal position, quite often tinged with social-democratic and even, though more rarely and only lightly, socialist hues. The following quote is typical of the most common responses:

- I find myself agreeing with Liberal perspectives…….Each person is different and therefore will achieve different outcomes from education. By seeing everyone as individual, the main purpose of education is to better that person, rather than society……By helping an individual to work to his/her potential the outcome of class divisions in society may be reduced.

Here the individual is positioned front and centre of consideration, with the social only emerging subsequently, and secondarily as a derivative of individual action. We can recognise here a number of key characteristics of a standard liberal world view; the naturally unique and sovereign pre-social individual voluntarily entering into relations with other individuals in the course of realizing their potential, and the good social order forming spontaneously from the aggregation of individual interactions. When ideas taken from the social-democratic or socialist traditions are introduced these tend to be grafted onto this basic liberal rootstock, as evidenced in the use of the language of equal opportunity. Thus for example:

- I find myself agreeing with the social democratic ways of education…..all people should be given the same opportunities despite their birthright and the state should help in ensuring there is equal opportunity
- I find myself agreeing more with socialism, but with a lean towards liberalism…..equal opportunity, equal access and inclusivity……are all important but the socialist also takes the notion of 'equal worth' to mean 'equal power'….. It doesn't support power or privilege of some if it means the majority of others suffer.

We can note at this point that there were far fewer takers for the type of hard-core neo-classical economic liberalism, that which would place education in the marketplace with consumer choice presented as the truest expression of individual freedom. Nor for the type of social conservatism, with whom it often keeps company in contemporary debates, emphasising discipline, traditional values and a curriculum filled to the brim with 'real', standardised and testable knowledge. Hence the designation 'nice liberals': liberal in the emphasis on maximising individual autonomy, a capacity for self-assertion and the fulfilment of the native potentials of all. 'Nice' in the sense of an awareness of and concern about the ways society can and does frustrate and injure the capacity to individuate, and that it may do so more for some than others – and in the roseate view that through a bit of tinkering with our existing arrangements for living together these accidents or oversights might be overcome. Too often absent or underdeveloped, however, is a clear sense of the ways socio-cultural contexts condition and suggest prevailing interpretations about what it means to be an individual, and hence how one ought to go about the task of constituting oneself as an individual; or of the ways the manner and methods through which people respond to this task affects the structure and ambiance of the society we share. That is to say there is a common difficulty in grasping how the uneven distribution of material resources, power and respect, and as such significant differences in practical capacities to individuate, results from systemic contradictions rather than deficits in the individuals and groups most adversely effected by these. This in turn impacts upon where people draw the lines demarcating what they see as the outer limits of their legitimate responsibilities. This general impression should not come as a surprise. When asked, in their journals, to reflect upon the kinds of literacies and other forms of cultural capital that supported their transition through schooling many either identified explicitly as middle class, or they described features typical of a middle class habitus. It might be said that that ideal of the freestanding, selfrealising individual has always sat quite comfortably with or comes more naturally to those whose position in the social order allows them credibly to believe that in many important respects they are their own boss, the authors of their own destiny.

## THE ROLE OF EDUCATION A MEDIATOR OF RELATIONS BETWEEN THE INDIVIDUAL AND SOCIETY

From the journal responses we have understood our students as presenting with a largely naive form of liberalism, which demonstrates some conception of responsibility to others but without a strong sense of the social and cultural forces which shape responsibility, for both the individual and the social. The discourse of individualism here delimits their capacity for any sophisticated notion of the social. We further elaborate this understanding by considering responses to the educational propositions in the Ways of Explaining Education topic. Under this heading we paired subjects' scaled responses to propositions concerning the aims of education and the importance of justice relative to efficiency and productivity with journal reflections on the role of education in society and, more specifically on the relative weight that should be accorded individual fulfilment over the needs to secure the conditions for a viable workforce. The distributions of rank scaled responses are summarised in Figures.

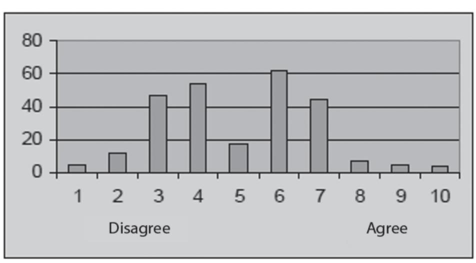


**Fig.** Education's main aim should be the transmission of community norms

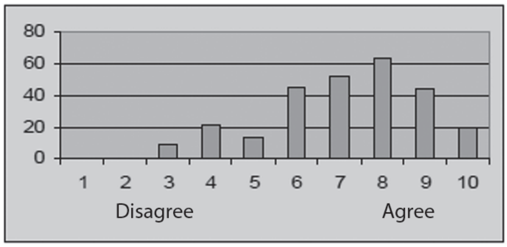


**Fig.** Education's main aim should be the fulfilment of the individual student

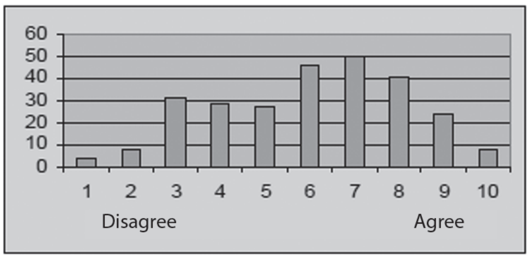


**Fig.** Justice is more important than efficiency, productivity and prosperity

In Figure, we see what appears to be a significant divide, though not extreme, over whether the main aim of education ought to be the transmission of community norms. Figure, however, would suggest that the type of norms many subjects supporting this proposition had in mind were inclined towards those that valued individual fulfilment. In the journals we find something similar taking place. Here there are some takers for what we might call a conventional even conservative view of education as a mechanism for the intergenerational maintenance of community norms:

- …we need teachers capable of passing on the knowledge of how society works and the values it wishes to instil in every person.

A more common response was to suggest that the route to maintaining a healthy society was through the development of individual potentials:

- …helping individuals to grow, develop and reach their potential to eventually become a contributing member of a changing world.

Moreover as indicated in Figure, individual fulfilment tended to be understood in terms much wider than those promoted by the more puristic forms of economic liberalism, where the good person is the rational calculating market actor, and the just society one organised around market ideals of free exchange, efficiency and productivity. Thus while a few did subscribe to this more limited view:

- To be successful or to get ahead in today's world you have to have money. The more money you have the more successful you are seen to be. A high level of education will give you a better chance of gaining a high paying job….the needs of society can at times totally eclipse individual aims leaving an individuals life goals unfulfilled.
- A far larger number tended to write in the language of self-actualisation;

- ….education is about liberating people so they can reach the potential of their choice….Schools should also educate a social conscience in their students which will inspire them to take on crucial jobs later on in life.

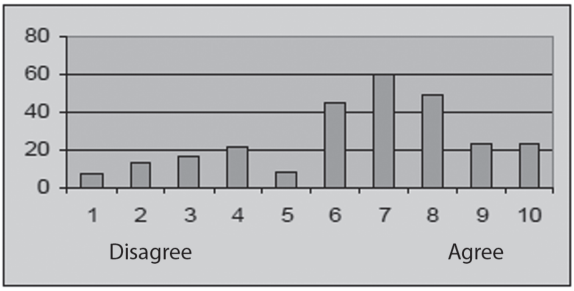


**Fig.** The enforcement of rules such as the wearing of uniform, showing respect for the teacher, is essentials in education

Here we find a repetition of the idea that if we provide conditions conducive to the development and fulfilment of individual potential then the prosperous economy, as with the good society, will take care of itself.

- ….people fulfilling their potential is the most important. I believe that everybody has gifts and talents for different things and that helping people discover their potential is vital. In saying that for economy to go around we need to maintain the workforce, but I believe that if everyone fulfils their potential with education there will in turn be a maintained workforce.

A few were able to take this further, including under 'the development of individual potentials':

- …preparing them to face an ever-changing world, and to develop in them the attributes to be a discerning individual that is able to critically analyse the global environment of which they are a part.

## ORDER, NORMS AND VALUES

We can take these observations further and press more explicitly towards some of the social justice issues noted above through looking at subjects' responses to propositions concerning the importance of school rules, the grounds for these, patriotism and citizenship. These responses are summarised in Figures.

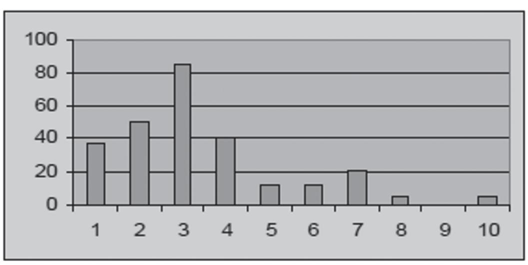


**Fig.** School rules are arbitrary conventions that have no moral value

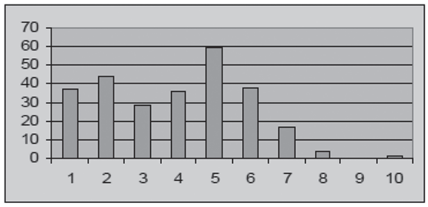


**Fig.** Patriotism is good. But no politics

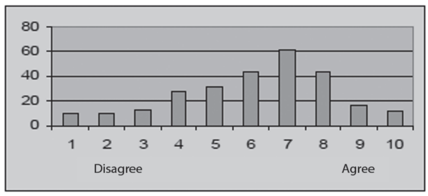


**Fig.** We should be forming a society of citizens commited to higher values than themselves

These data can be narratively summarised in the following simple points. Rules and their enforcement are crucial ingredients in effective schooling. These rules ought to be rooted in some core principles, involve something more than just order for the sake of order. The core principles seen to be at stake here bear little resemblance to those popularly associated with the figure of the patriot, one who prizes most highly the unchanging survival of a particular cultural form of life. The core principles seen to be at stake here do have something to do with the figure of the citizen, the member of a culturally diverse public-political community willing and capable of participating in deliberation over and decision about how best to organise collective living arrangements. What support for this interpretation can be found within the journals? Here we looked at responses to questions about the cultural dimensions of education and the importance of a cultural understanding for teaching practice, as well as those asking respondents to consider what constitutes the so-called ‘good’ student. The type of narrative sequence that can be read from these responses both complements that just outlined and moves it forward in several interesting ways. First we find recognition that order is a necessary or at least unavoidable fact of socio-cultural life, that schools are no exception and that growing up in and into society, a function schools are tasked with overseeing, involves in large part an acceptance of the need for rules.

- Education is a way of passing on information and our way of life onto others…the transfer of culture from one person to another…which results in them becoming culturally the same as you……Information needs to be passed down the generations in order to maintain and continue to develop a functional world. A culture is defined by the behaviours that are acceptable within a particular society. I don’t think good kids are necessarily more mature, just more aware of the behaviour desired of them.

This is often followed closely by an observation that the task of presenting oneself as ‘good’ from the perspective of the prevailing order is far easier for some than for others, and that this depends in large part on the degree of fit between the culture of the school and the student's own cultural background.

- Obeying rules gets more important to you if they are your rules. Even though we are said to be a multicultural society, I think that if you are white things come to you a lot easier. People tend to see a person of another race as a troublemaker.

Around this point, however our narrative breaks in two different directions. One develops this account of differences in cultural fit in terms of a deficit view of the cultural background of the maladjusted or difficult student. What is not considered here is the possibility that schools, and the wider social order within which they are embedded and represent, might be unresponsive to cultural and other differences in life-situations, or that the attribution of dysfunctional qualities to difference might itself be symptomatic of a tendency to take the dominant culture as a sort of Archimedean point against which all departures are viewed, by degrees, as somehow pathological.

- Good kids fit into the establishment more easily than naughty or disruptive kids as they are different culturally. Good kids have support at home, their parents talk to them on how to behave and understand the rules of the school and the playground….. The disruptive and naughty kid comes from a home with little support or knowledge of the schooling process, their parents would have had a bad experience at school as well as resulting in the child having difficulty understanding the establishment and how to interact with teachers and students.

A second strand did attempt to place the problem of difference in cultural fit, and hence of experiences and outcomes of schooling, in the light of an historical cultural unresponsiveness and inflexibility on the part of teachers, schools and education systems.

- Australia's education was originally taught through the eyes of the white Australian. However recently as minorities have become increasingly recognised and their history which was overlooked is now being placed into many of the schools' curriculum. …the cultural factor in Australian education….has long been based predominantly on 'white' western Christian middle class values. Thus mainstream education in Australia may benefit some students more than others based on cultural values.

What is interesting here, and especially so in view of the following section, is the way these cultural differences are couched in a language wherein culture and the ideas of race and ethnicity are seldom out of each other's sight. Without delving into the mass of scholarship that gather around observations such as these we might surmise that cultural differences that make a difference may be more readily drawn and grasped when appended to ideas of race and ethnicity, those where the lines of division between 'our kind of people' and 'the others' may be represented in a tangible form; skin deep may be made to mean very deep. Other divisions that are routine features of scholarly accounts of social inequality, privilege and disadvantage and so forth, those for instance taking in the categories of class and gender, would seem to belong to the past having been put out of business by the corrective workings of equal opportunity law and policy. These are, of course, divisions that occur within the parameters of 'our kind of people', where there may be a strong temptation to view, given 'we' are seen to inhabit the same socio-cultural worlds, success or failure on individual terms.

## EQUALITY, SAMENESS AND DIFFERENCE, RESPONSIBILITY AND PUBLIC GOODS

Under this head we bring together materials that relate directly to specific concerns in the area of social justice and education, equality and inequality, sameness and difference, and those that explore perceptions of the extent and limits of responsibility as indicated in support or otherwise for education as a public good along with the grounds for this. Figure suggests that for our cohort there is a solid base of support for a well resourced and universally available education system.

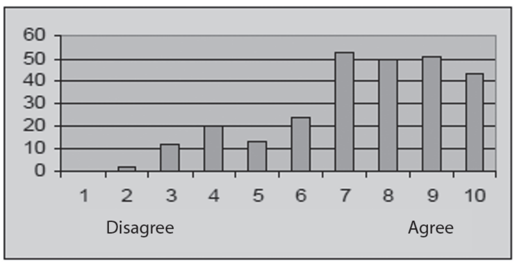


**Fig.** The state should be responsible for the provision of education to all kids (and university students too). But not a minimum education. A maximum education. No private schools. All public schools, but terrific schools

At the same time universal is not intended to mean uniform. There is also considerable support for the proposition that this education should be responsive to differences. Difference here is construed in terms consistent with a child-centred approach; that is in terms of individual potentials, needs and desires. We will look at other dimensions of difference below.

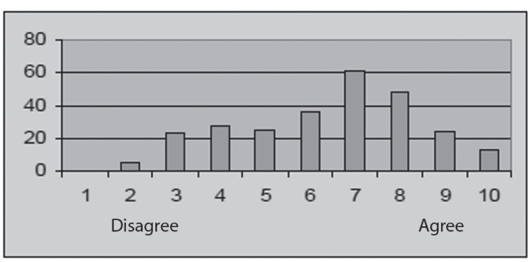


**Fig.** It is the State's duty to allow for the identification of each child's potential and the provision of appropriate education

Moreover this support would seem to be for education as a public good, one that serves all, not just those who have paid, or paid the most, for it. As Figure indicates there seems to be little confidence in letting the marketplace determine the type of education system that should obtain.

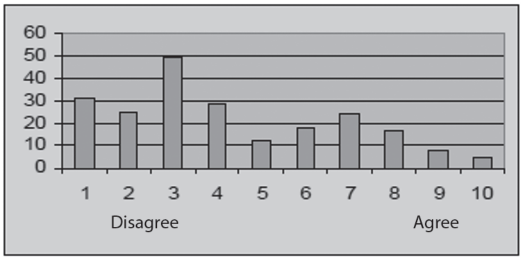


**Fig.** The state should provide for the same basic education for all, and if the family wants anything extra they can arrange for it and pay for it

Turning once more to the journal reflections we can elaborate further on these views. Here we have placed respondents' observations under sub-headings to assist in drawing out specific themes.

## CLASS INEQUALITIES

When pressed our respondents found little difficulty acknowledging the possibility that the existing distribution of material and cultural resources along with power can work to ease or retard the passage of students through schooling. We need to be mindful here that this may be as much an artefact of their having been exposed to the idea of class as a part of their programme of studies as it is a reflection of the schemes they routinely use to organise their perceptions and experiences of the world. Beyond a rather loose and vague reference to the 'middle class' or, less so, 'working class when asked to identify in such terms, there is little across the body of journal reflections to suggest that class occupies an important place in their systems of identification. For many it is the distinction between public and private schools that best allows them to grasp and describe the relation between class inequalities and schooling.

- Schools are separated into three categories…public schools, catholic schools and independent…..Most families on low income or multiple children have very limited choice about what schools they can send their children to…..Even between public schools, which are supposed to be equal as they are run by the same government, they are divided. This division is done by the suburbs they are situated in, how much the parents earn that send their children there…..Some also present their schools to be imposing to newcomers and give the opinion they are exclusive.

Others noted the effects of low family incomes on the ability to meet the financial demands of schooling while a few pursued this further into a discussion of cultural capital.

- I have seen school teachers set work that is impossible for disadvantaged and low income families to complete. Not all students have access to the internet at

> home so it would be inappropriate to set homework that requires that type of research. ….Monetary requirements in a school outside of traditional school fees are also out of reach for some students.

**MERITOCRATIC AND DEFICIT VIEWS**

This willingness to grapple with the idea of class needs, however, to be balanced against a strong meritocratic bias within respondents' accounts of the determinants of educational outcomes. When asked what they considered to be the main factors influencing success or failure at school most nominated hard work and talent, quite often with a glance in the direction of social circumstances though seeing these as being readily able to be offset through a bit of extra effort.

- ..a student's educational achievement is most likely affected by hard work and individual talent. All students start off with the same potential and with hard work and individual talent, students need to use both excellent results…. However…a student's circumstances within a community may mean that they may not have all the same opportunities as others. Despite this many students, in the past, have risen above the social standing to achieve top results.

On those occasions where a student's social circumstances were accorded a higher priority we found a recurrence of tendencies mentioned earlier; a deficit view of disadvantage and the treatment of cultural disparities between home and school in keeping with an ethnic or race conception of culture.

- Social circumstances may govern the way a student learns but it is not the sole factor of achievement or failure. A student with a hard background may have other things to consider…. …. before learning… (so that) …. schooling becomes a lower priority. On the other hand a student from a middle class or upper class family has more resources and opportunities to study and therefore achieve…. Those of lower class or different backgrounds may decide to work hard and achieve in school so they can create more opportunities for themselves and their families.
- …to achieve a high standard of work you need to push yourself and work hard. In saying that though many people are disadvantaged by their social circumstances and by this I mostly mean their parents, upbringing and life at home. Children may be from a wealthy background of highly educated parents who have pushed them to do well and taught them and helped them with their schooling…On the other hand children may come from a non-English speaking background, which already disadvantages them, or they may have parents that don't have much of an education and so haven't been pushed and helped with any education.

What is noticeably absent across nearly all responses to this question is any consideration of the possibility that socio-cultural advantages and disadvantages might be systematically related through the way society organises the production and distribution of material and symbolic goods. Once again this should not surprise. As already observed most students identified, when pressed, as middle class and saw their educational experiences enhanced by attending schools where the majority of their associates shared roughly the same circumstances and outlook as themselves. If ones primary reference group is made up of people just like you then it is not difficult to arrive at the conclusion that it is effort and native talent that separates the successes from the failures.

## POSITIVE DISCRIMINATION

Many respondents did nevertheless support some form of positive discrimination, though once again this support tended to be referenced against an idea of cultural difference as ethnic or racial differences.

- I can't see a good reason not to. Scholarships are generally given on merit, and for people from disadvantaged groups lacking financial resources can make all the difference. It's hard to stay afloat in someone else's culture, and it's hard to keep up when you're starting from behind.

Some took the view that if disadvantages had a significant cultural component then something more than just monetary assistance was required.

- I believe disadvantaged groups should receive assistance to give them greater opportunities to gain education. However scholarships are of no use if teaching practices and pedagogies do not meet their specific needs culturally, academically and socially.

Among the more interesting responses here took what is effectively a conservative position towards equality, namely equality as identical treatment.

- No more than any other citizen is entitled to. A balance needs to be maintained as some groups that society considers disadvantaged may be very offended if it was suggested that they were unable to cope and given handouts.

This is interesting in that by raising the issue of potential stigmatisation it opens out to a central issue confronting those working in the field of social justice within an individualised culture; that is, the way the term dependency, having been progressively denuded of any positive or even neutral connotations it might once have held, being now understood in entirely negative terms.

## WELFARE OR WORK

The final set of reflections in this study respond to the proposition, "Those who are able to work and refuse the opportunity should not expect society's support". This proposition quite brutally pushes tolerance for difference and an awareness of the high costs of staying different, collective as against individual responsibility, and even a commitment to individual fulfilment to their limits. In the majority of responses these limits were unequivocally declared.

- I think that people who choose not to work and who are able to should not expect society's support…. I don't think you can use cultural background as an excuse for not working if you are living in Australia.

- If there is a job there and they need a job, as unfortunate as the situation may be they should take it. For who is to say that if they take that job and work hard they may be able to rise up through the ranks..to a position they are more comfortable with.
- If too many people…start to refuse work our societies structure may not be able to hold.

Yet a small, but significant number opposed the proposition. For some it was a matter of striking a balance between competing values.

- I was raised to believe that you need to earn things in life, not just expect it …On the other hand I was brought up with the value that we need to help others as well…those who need it.

Others, often drawing anecdotally on their own experiences or those of familiar others, criticized the proposition for being too harsh and allowing stereotypes to eclipse empathy.

- I have heard stories from already working mothers being told to work more, to pregnant women being told to look for work right up until the baby drops.
- ….people seem to have a particular stereotype in mind…uneducated or poorly, no culture etc. This is not always the case…consider this case: a migrant, 50 years of age, OK English ability but not perfectly fluent…His job in his country of origin was a prestigious one, which relied on his university professor-level education, his superb talent in his native language….and his network of friends, peers and connections. In Australia, the only work that is guaranteed to him is a factory job, which pays less than the equivalent of his welfare benefits.

Perhaps the most compelling negative response worked between the lines of a commitment to fulfilling individual potential and the wellbeing of society as a whole.

- …putting a person in an underpaid, overworked, uninteresting job will not do much for society. Their self-esteem would suffer as well as mental health and confidence. A dissatisfied, apathetic dysfunctional workforce does not seem appealing….. The use of safety nets as in unemployment benefits is crucial not just to the health of the non-working person, but a safety net for society itself…For a person refusing to work may be lazy and should not expect support, but the inability to survive in society without resorting to crime is apparent. The costs to society in mental health care, increased crime rates, and flow down effects onto children are insurmountable.

# VOCATIONAL TEACHING AND LEARNING IN PRACTICE

## METHOD OF ANALYSIS

The approach to the analysis broadly drew on grounded theory. In grounded theory, the theory is generated from data, in the process of conducting the research. Key points in the written records of the observations and interviews were coded. The codes were then grouped into similar concepts and a framework for analysis was generated. Initially, three broad groupings emerged: teaching skills; teaching strategies; and underpinning teaching and learning theories and models. We then departed from grounded theory and referred to the literature review to establish a structure for synthesising and reporting the findings. The findings from the observations and interviews aligned closely with the four ways of thinking about teaching, illustrated in Figure 1 below, that was developed from the Improving the Quality of Education for All (IQEA) research project (Hopkins, 2007). This framework encompasses four different elements of effective teaching – Teaching Skills, Teaching Relationships, Teacher Reflection and Teaching Models. Importantly, it is only when these four elements are in synergy that they are able to support effective teaching. Creemers, who analysed the factors and variables in the teaching and learning process to identify those that could explain the differences in outcomes for comparable groups of learners, informs us that 'isolated components or effective elements of individual components do not result in strong effects on student achievement'.

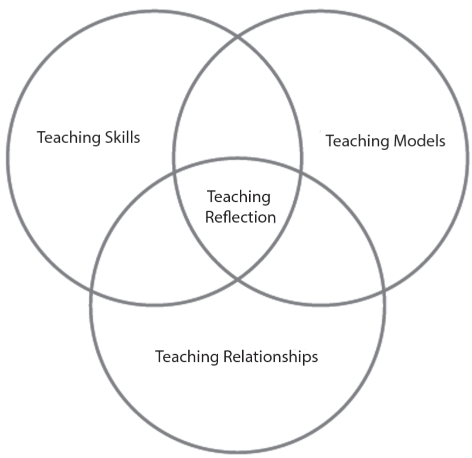


**Fig.** Four ways of thinking about teaching, (Hopkins, 2007)

These four components: teaching relationships, teaching models, teaching skills and teacher reflection, were adopted as the basis for analysis. We also drew on effect-size research 1 which identified consistently high correlations between learner achievement scores and classroom processes. From this stage, the process of analysis and the refinement of the framework for analysis has been an iterative one.

**DEFINITION OF TERMS**

Given the interchangeable use of terminology and the different meanings evident in the literature for these four concepts and the terms used to describe them, we have set out exactly what we mean in this report by each of these essential components and the relevance/importance of each to effective vocational teaching and learning.

Teaching relationships encompasses both the teachers' commitments to their learners and the relationships they develop with their learners. In the observations, teachers identified that their relationships with their learners was of critical importance to the effectiveness of their teaching and learning. 'Teaching relationships' also covers the range of roles that a teacher can take within a session and varies between 'high structure,' in which the teacher's role is dominant, directing the learning and 'low structure' in which learners take more control of the process of learning.

Teaching models are derived from theories about teaching and learning. Each model can be described as a structured sequence, which is designed to elicit a particular type of thinking or response, to achieve specific learning outcomes. The choice and use of the appropriate model (or combination of models) is influenced by the type of learning objective and nature of the learner as well as other factors such as the repertoire of teaching strategies available and skills of the teacher. A strong body of research and practice suggests that the quality of teaching and learning and learners' attainments can be enhanced by the use of specific models (DfES, 2004b, Hattie, 2009 and Marzano, 1998). We have defined 'teaching strategies' as the 'tools for teaching and learning' that teachers have available to them and 'teaching skills' as the ways in which teachers select and use the 'tools' at their disposal to achieve effective learning. Since we found these to be closely aligned, we have put them together for our analysis

The fourth component of 'teacher reflection' is a threefold process comprising direct experience, analysis of beliefs, values or knowledge about that experience, and consideration of the options which should lead to action as a result of the analysis (Whitton *et al,* 2004).

As the analysis progressed against the framework of these four components it became clear that there was one additional, distinctive feature that in part defines vocational learning and that is the context within which it takes place. Thus a new, fifth component was emerging to add to the framework. This is discussed further in the conclusions drawn from the analysis, where we present and discuss a new framework for effective vocational learning with five components. However, for clarity, we provide a definition of context at this point.

Teaching context covers a mixture of aspects and includes the nature of the vocational subject and the setting where teaching and learning takes place, including the specialist facilities and resources required for that vocational subject. It also includes the learning objectives and desired outcomes for a session plus specifications of the qualification. The nature of the learners, their level, and how they learn best including their learning styles, is also a part of the context.

A teacher's choice of teaching strategy or model to enable effective teaching and learning is affected by context in that, for example, it would be difficult to do 'role play' or whole class 'questioning' in a noisy workshop with confined space. To show how all components might work together in practice, we provide a worked example of a sequence of activities taken from an observed session

**PRESENTATION OF ANALYSIS**

In this section we now consider each of these components in turn and provide a selection of illustrative examples and quotations, drawn from the observations and interviews. *Please note that the examples have been selected to illustrate various points and should not be considered as exemplars to copy or necessarily as examples of outstanding practice*. It is also important to stress that in this chapter we are looking at practice from the perspective of each component so inevitably we will also make reference to other components. This is due to the holistic interrelationships of the components in practice.

In the next chapter, we draw our conclusions, and present our Framework for Developing Effective Vocational Teaching and Learning. We also provide a worked example drawn from an observed session that indicates how the Framework may be used to analyse the teaching and learning taking place or to illustrate practice. In presenting our analysis of findings we start with teaching relationships. It would be possible to start with any component and to present the analyses in any order but we have chosen to start with teaching relationships since the teacher/learner relationship is of such fundamental importance and was highlighted during the interviews as highly important in the delivery of effective teaching and learning.

## TEACHING RELATIONSHIPS

Teaching relationships refer to the relationships teachers develop with their learners as well as how learners relate to each other. The literature review within this report identified that in FE specifically, the tutor-learner relationships are identified as 'the most important link in the learning process', (TLRP, 2006). A meta-analysis of learner-centred teacher-learner relationships confirmed the importance. It reported that positive teacher-learner relationships are associated with optimal, holistic learning with above average mean correlations when compared with other educational innovations for cognitive and behavioural outcomes (Cornelius-White, 2007).

The way in which a teacher interacts with learners sets the scene for the subsequent learning to take place. As indicated earlier, within the research, teachers felt that their relationships with learners were of prime importance for the teaching and learning to be effective. The features of effective teacher relationships that were identified from the observations, included:

- Getting to know learners – knowing which learners need more attention
- Good rapport including listening
- High expectations
- Building trust
- Humour – appropriate, not sarcasm
- Relaxed atmosphere – relaxed learning with elements of fun
- Mutual respect – respect of other people's opinions
- Behaviour management – so that all of the group have the chance to learn.

Active learning, while carrying out assignments or projects for instance, gave many opportunities for teachers to build relationships with learners. The teacher's role during this activity can take various forms: demonstrator, organiser, coach, mentor, facilitator, reflector and even co-learner. A relationship of trust between the teacher and learners is likely to develop while working together and discussing aspects at various stages of the assignment, so that the teacher becomes an 'accomplice' in the learning process rather than the knowledge base. The following is a selection of examples taken from the observations and interviews.

Part of relationship building is the skill of the teacher in managing the behaviour of groups and individuals within a learning session. For example, a teacher might not allow learner discussion during the creation of plans, so that work is individual and then, might use pairings to discuss how each plan was created and the advantages and disadvantages of each plan. Therefore, managing behaviour is in part an individual teacher activity but also an organisational activity.

In one college (shown in the example below), within the study, they decided to change the culture of the college making behaviour one of the issues – behaviour of all staff and all learners. Over a period of time the culture changed so that relationships between teachers and teachers, teachers and learners and learners and learners became ones of mutual respect in which learning could flourish. Within each teaching model there is a 'social system', that is, the roles and relationships that learners and teachers take within each teaching model. For example, in some teaching models, the teacher is the source of information, the demonstrator or organiser. In these cases the teacher provides the structure and is in control as in a session within this study, where the teacher was organising the learners to play a game of dominos where one half of the domino was a question and the other half was an answer to a different question. The teacher needed to orchestrate the 'play' and the learners followed the instructions in the process, gaining knowledge from the game. In other teaching models, activity (and control) is divided more equally between the teachers and learners and the teacher acts as a facilitator, questioner or reflector as in the following example where a teacher is working with learners on an ICT assignment and facilitating their progress.

The final stage is where the learners take complete control and are learning independently. In this following example the class decides they now want to get on with the practical themselves and they don't need any more demonstration from the teacher. Importantly the teacher knows the group well enough to feel confident that the learners will be able to progress on their own. A useful way of considering the social system or describing the respective roles and responsibilities of teacher and learner has been provided by Fisher (2008) in what he describes as '*a gradual release of responsibility model*'. Four stages are described, moving from teacher directed or focused activity to independent work:

- Teacher focus – teachers are in control and might demonstrate or 'model' what is required from learners

- Guided instruction – teachers prompt, question, facilitate, or lead learners through tasks that increase their understanding of the content
- Collaborative learning - to consolidate their understanding of the content, learners need opportunities to problem solve, discuss, negotiate, and think with their peers. Collaborative learning opportunities ensure that learners practice and apply their learning while interacting with their peers
- Independent work – this is the overall goal. Independent learning provides learners with practice in applying information in new ways. In doing so, they synthesise information, transform ideas, and solidify their understanding.

Fisher points out that importantly, the gradual release of responsibility model is not linear. Learners move back and forth between each stage as they progress with their learning.

Teachers themselves decide within the teaching models that they are using, what actions or reactions to take to further the learning for individuals and groups. In some teaching models, the teacher tries to shape behaviour by rewarding certain learner activities and maintaining a neutral stance towards others. In other teaching models, such as those designed to develop creativity, the teacher tries to maintain a non-evaluative, equal stance so that the learner becomes self-directing. Principles of reaction help the teacher respond to what the learner does.

They can help teachers select the responses they will have in their interaction with the learners and provide them with guidelines by which they can better tune in to the learners and select model-appropriate responses to what the learners do (Ji-Ping and Collis, 1995).

**TEACHING RELATIONSHIPS – COMMENTS**

The more teachers know their learners, then the more able are they to ensure that each individual learner is learning in as effective a way as possible and that the group collectively is being managed in the best way for effective learning to take place. Behaviour management is made easier if teacher-learner relationships are well developed and trust is part of the culture of the group.

The literature review identified that teacher-learner relationships were the most important link in the learning process - a crucial part of the teaching and learning framework – and this was confirmed by the observations and interviews in this study. The extent of the teaching relationship development with a group of learners is likely to affect the choices and operation of the other components in the Framework, including the teaching model.

In some teaching models the teacher takes the lead and provides the structure for learning while in other teaching models there is a sharing of control between the teacher and the learners or ultimately, learner independence in learning. It is important that the teacher has understanding of individuals within the group to be able to make effective decisions on teaching strategies and teaching model choice.

# TEACHING MODELS

Teaching models are derived from theories about teaching and learning. Each model can be described as a structured sequence, which is designed to elicit a particular type of thinking or response, to achieve specific learning outcomes. The choice or use of the appropriate model (or combination of models) is influenced by the type of learning objective and nature of the learner as well as other factors such as teaching strategies and teaching skills. A strong body of research and practice suggests that the consistent use of specific models can make learning more effective (DfES, 2004b, Hattie, 2009 and Marzano, 1998).

In our analysis of the observations and interviews we found no evidence of the intentional, planned and systematic use of teaching models, as defined above. That is not to say that teachers' practice was uninformed by theory. In some cases teachers stated explicitly the aspects of teaching and learning theories that underpinned their practice – experiential learning and learning styles theories being predominant. Teachers referred to the importance of actively engaging learners and the observations and interviews showed that experiential, activity-based learning was the norm. Although there was no direct reference by teachers to any particular theory of experiential learning. When referring to learning styles, teachers frequently mentioned that their learners were for example, 'visual learners' or 'kinaesthetic learners' and explained how they had planned their teaching to take account of these factors. However, conceptualisations of teaching models did not, in the data available for analysis, appear to inform their practice. Teachers appeared to draw pragmatically from a very wide range of strategies, based on experience of what worked for them with their learners, in their contexts. Sessions were generally complex with a very large number of different activities within sessions that could combine the use of a number of different strategies. Teachers did not generally articulate a relationship between the strategies they chose and 'teaching models'. This presented us with a dilemma. Since there is evidence in the literature to suggest that learning could be enhanced by the consistent use of models (DfES, 2004b, Hattie, 2009 and Marzano, 1998), we needed to establish whether they could be applied to vocational learning. Given that teaching models are currently not prevalent or explicit in the vocational learning discourse, our task in analysing the data was to see if we could infer what might be described as teaching models from the observed and described practice. In analysing teaching models we have drawn on the DfES (2004b) guidance on teaching models and the taxonomy of models produced by Joyce et al (2008) as a framework for reference.

# 6
# CHAPTER

## Administration of Vocational Education

### ADMINISTRATOR'S PERCEPTION OF THE STRUCTURE OF BUSINESS AND INDUSTRY

Business administration is management of a business. It includes all aspects of overseeing and supervising business operations and related field which include Accounting, Finance and Marketing. The administration of a business includes the performance or management of business operations and decision making, as well as the efficient organization of people and other resources, to direct activities towards common goals and objectives. In general, administration refers to the broader management function, including the associated finance, personnel and MIS services.

In some analysis, management is viewed as a subset of administration, specifically associated with the technical and operational aspects of an organization, distinct from executive or strategic functions. Alternatively, administration can refer to the bureaucratic or operational performance of routine office tasks, usually internally oriented and reactive rather than proactive. Administrators, broadly speaking, engage in a common set of functions to meet the organization's goals. These "functions" of the administrator were described by Henri Fayol as "the five elements of administration". Sometimes *creating output*, which includes all of the processes that create the product that the business sells, is added as a sixth element.

A business administrator oversees a business and its operations. The job is to ensure that the business meets its goals and is properly organized and managed. The tasks a person in this position has are both wide and varied, and often include ensuring that the right staff members are hired and properly trained, making plans for the business' success, and monitoring daily operations. When organizational changes are necessary, a person in this position usually leads the way as well. In some cases, the person who he, starts or owns the business serves as its administrator, but this is not always the case, as sometimes a company hires an individual for the job.

When a person has the title of business administrator, they are essentially the manager of the company and its other managers. The person oversees those in managerial positions to ensure that they follow company policies and work towards the company's goals in the most efficient manner.

For example, they may work with the managers of the human resources, production, finance, accounting, and marketing departments to ensure that they function properly and are working inline with the company's goals and objectives. Additionally, they might interact with people outside the company, such as business partners and vendors.

## ACADEMIC DEGREES

### Bachelor of Business Administration

The Bachelor of Business Administration (BBA, B.B.A., B.Sc.) is a bachelor's degree in commerce and business administration. The degree is designed to give a broad knowledge of the functional aspects of a company and their interconnection, while also allowing for specialization in a particular area.

The degree also develops the student's practical, managerial and communication skills, and business decision-making capability. Many programmes incorporate training and practical experience, in the form of case projects, presentations, internships, industrial visits, and interaction with experts from industry.

### Master of Business Administration

The Master of Business Administration is a master's degree in business administration with a large focus on management. The MBA degree originated in the United States in the early 20th century when the country industrialized and companies sought scientific approaches to management.

The core courses in an MBA programme cover various areas of business such as accounting, finance, marketing, human resources, and operations in a manner most relevant to management analysis and strategy. Most programmes also include elective courses.

The MBA is a terminal professional degree (the DBA is the terminal research doctoral degree). Accreditation bodies specifically for MBA programmes ensure consistency and quality of education. Business schools in many countries offer programmes tailored to full-time, part-time, executive (abridged coursework typically occurring on nights and/or weekends), and distance learning students, many with concentrations.

**Doctor of Business Administration**

The Doctor of Business Administration (abbreviated DBA, D.B.A., DrBA, or Dr.B.A.) is a research doctorate awarded on the basis of advanced study and research in the field of business administration. The D.B.A. is a terminal degree in business administration, and is equivalent to the Ph.D in Business Administration.

**PhD in Management**

PhD in Management is the highest academic degree awarded in the study of management science. The degree was designed for those seeking academic research and teaching careers as faculty or professors in the study of management at business schools worldwide.

**Doctor of Management**

A newer form of a management doctorate is the Doctor of Management (D.M., D.Mgt. or DMan). It is a doctoral degree conferred upon an individual who is trained through advanced study and research in the applied science and practice of professional management. This doctorate has elements of both research and practice relative to social and managerial concerns within society and organizations.

# ACADEMIC ADMINISTRATION

Academic administration is a branch of university or college employees responsible for the maintenance and supervision of the institution and separate from the faculty or academics, although some personnel may have joint responsibilities. Some type of separate administrative structure exists at almost all academic institutions, as fewer and fewer schools are governed by employees who are also involved in academic or scholarly work. Many senior administrators are academics who have advanced degrees and no longer teach or conduct research.

**KEY RESPONSIBILITIES**

Key broad administrative responsibilities (and thus administrative units) in academic institutions include:

- Admissions
- Supervision of academic affairs such as hiring, promotion, tenure, and evaluation (with faculty input where appropriate);
- Maintenance of official records (typically supervised by a registrar in the US – In the UK, not all institutions have a Registrar, who would have varying responsibilities for non- academic matters depending on the organisation);
- Maintenance and audit of financial flows and records;
- Maintenance and construction of campus buildings (the *physical plant*);
- Maintenance of the campus grounds;

- Safety and security of people and property on the campus (often organized as an office of public safety or campus police);
- Maintenance and construction
- Supervision and support of campus computers and network (information technology).
- Fundraising from private individuals and foundations (“development” or “advancement”)
- Research administration (including grants and contract administration, and institutional compliance with federal and state regulations)
- Public affairs (including relations with the media, the community, and local, state, and federal governments)
- Student services such as disability services, career counselling and library staff.

## ADMINISTRATIVE TITLES

The chief executive, the administrative and educational head of a university, depending on tradition and location, may be termed the university president, the provost, the chancellor (the United States), the vice-chancellor (many Commonwealth countries), principal (Scotland and Canada), or rector (Europe, Russia, Asia and the Middle East).

An administrative executive in charge of a university department or of some schools, may be termed a dean or some variation, such as dean emeritus. The chief executive of academic establishments other than universities, may be termed headmaster or head teacher (schools), director (used to reflect various positions ranging from the head of an institution to the head of a programme), or principal, as used in primary education.

## BY COUNTRY

Academic administrations are structured in various ways at different institutions and in different countries.

### Australia

Full-time tertiary education administrators emerged as a distinct role in Australia from the mid-1970s, as institutions sought to deal with their increasing size and complexity, along with a broadening of their aspirations. As the professionalism of tertiary administrators has developed, there has been a corresponding push to recognise the uniqueness and validity of their role in the academic environment.

As of 2004, general staff comprised over half the employees at Australian universities. Around 65% of these are female. There has recently been a shift in the preferred nomenclature for non-academic staff at Australian universities, from “general staff” to “professional staff”. It has been argued that the changing in role of the professional staff has been due to the changing work that they are performing, as professional staff assist students with technology.

The overarching body for all staff working in administration and management in Australia is the Association for Tertiary Education Management.

## United Kingdom

### *Administrative Structures*

The structures for administration and management in higher education in the United Kingdom vary significantly between institutions. Any description of a general structure will therefore not apply to some or even many institutions, and therefore any general statement of structures may be misleading. Not all UK universities have the post of Registrar.

The Director of Finance may report to the Registrar or directly to the Vice-Chancellor, whilst other senior posts may or may not report to the Registrar. This next tier of senior positions might include Directors of Human Resources, Estates, and Corporate Affairs. The Academic Registrar is often included in this next tier. Their role is mostly to accomplish student-facing administrative processes such as admissions, student records, complaints, and graduation.

### *Professional Associations*

The overarching body for all staff working in administration and management in the UK is the Association of University Administrators.

## United States

### *Presidents and Chancellors*

In the United States, a college or university is typically supervised by a President or Chancellor who reports regularly to a Board of Trustees (made up of individuals from outside the institution) and who serves as Chief Executive Officer. Most large colleges and universities now use an administrative structure with a tier of vice presidents, among whom the Provost (or Vice President for Academic Affairs, or Academic Dean) serves as the chief academic officer.

### *Deans*

Deans may supervise various and more specific aspects of the institution, or may be CEOs of entire campuses. They may report directly to the president or chancellor. The division of responsibility among deans varies widely among institutions; some are chiefly responsible for clusters of academic fields (such as the humanities or natural sciences) or whole academic units (such as a graduate school or college), while others are responsible for non-academic but campus-wide concerns such as minority affairs. In some cases a provost supervises the institution's entire academic staff, occupying a position generally superior to any dean. In other instances the Dean of a College may be the equivalent to a Provost or Vice Chancellor or Vice President for Academic Affairs. Below deans in the administrative hierarchy are heads of individual academic departments and of individual administrative departments from groundskeeping to libraries to registrars of records. These heads (commonly styled “chairs” or “directors”) then supervise the faculty and staff of their individual departments.

***Departmental Chairs***

The Chair of a department is typically a tenured or at least tenure-track faculty member, supported by administrative staff.

***Administrative Staff***

All levels of the university have administrative staff, reporting to Chancellors, Presidents, Provosts, Deans, and Departmental Chairs. Each department is typically led by a Management Services Officer (MSO), the departmental head of administrative staff, who reports to the Chair of the department. Front Desk staff, Admissions staff, Graduate and Undergraduate Student Counselors, and other staff report to the MSO.

# LEADERSHIP DIMENSIONS OF ADMINISTRATION

Leadership is both a research area and a practical skill encompassing the ability of an individual or organization to “lead” or guide other individuals, teams, or entire organizations. The literature debates various viewpoints, contrasting Eastern and Western approaches to leadership, and also (within the West) US vs. European approaches. US academic environments define leadership as “a process of social influence in which a person can enlist the aid and support of others in the accomplishment of a common task”. Leadership seen from a European and non-academic perspective encompasses a view of a leader who can be moved not only by communitarian goals but also by the search for personal power. Leadership can be derived from a combination of several factors. Studies of leadership have produced theories involving traits, situational interaction, function, behaviour, power, vision and values, charisma, and intelligence, among others.

## HISTORICAL VIEWS

Aristocratic thinkers have postulated that leadership depends on one’s “blue blood” or genes. Monarchy takes an extreme view of the same idea, and may prop up its assertions against the claims of mere aristocrats by invoking divine sanction. Contrariwise, more democratically inclined theorists have pointed to examples of meritocratic leaders, such as the Napoleonic marshals profiting from careers open to talent.

In the autocratic/paternalistic strain of thought, traditionalists recall the role of leadership of the Roman *pater familias*. Feminist thinking, on the other hand, may object to such models as patriarchal and posit against them emotionally attuned, responsive, and consensual empathetic guidance, which is sometimes associated with matriarchies. Comparable to the Roman tradition, the views of Confucianism on "right living" relate very much to the ideal of the (male) scholar-leader and his benevolent rule, buttressed by a tradition of filial piety.

- Leadership is a matter of intelligence, trustworthiness, humaneness, courage, and discipline... Reliance on intelligence alone results in rebelliousness. Exercise of humaneness alone results in weakness. Fixation on trust results in folly. Dependence on the strength of courage results in violence. Excessive discipline and sternness in command result in cruelty. When one has all five virtues together, each appropriate to its function, then one can be a leader. — Sun Tzu
- Machiavelli's *The Prince*, written in the early 16th century, provided a manual for rulers ("princes" or "tyrants" in Machiavelli's terminology) to gain and keep power.

In the 19th century the elaboration of anarchist thought called the whole concept of leadership into question. (Note that the *Oxford English Dictionary* traces the word "leadership" in English only as far back as the 19th century.) One response to this denial of élitism came with Leninism, which demanded an élite group of disciplined cadres to act as the vanguard of a socialist revolution, bringing into existence the dictatorship of the proletariat.

Other historical views of leadership have addressed the seeming contrasts between secular and religious leadership. The doctrines of Caesaro-papism have recurred and had their detractors over several centuries. Christian thinking on leadership has often emphasized stewardship of divinely provided resources—human and material—and their deployment in accordance with a Divine plan. Compare servant leadership. For a more general take on leadership in politics, compare the concept of the statesperson.

## THEORIES

### Early Western History

The search for the characteristics or traits of leaders has continued for centuries. Philosophical writings from Plato's *Republic* to Plutarch's *Lives* have explored the question "What qualities distinguish an individual as a leader?" Underlying this search was the early recognition of the importance of leadership and the assumption that leadership is rooted in the characteristics that certain individuals possess. This idea that leadership is based on individual attributes is known as the "trait theory of leadership".

A number of works in the 19th century – when the traditional authority of monarchs, lords and bishops had begun to wane – explored the trait theory at length: note especially the writings of Thomas Carlyle and of Francis Galton, whose works have prompted decades of research. In *Heroes and Hero Worship* (1841), Carlyle identified the talents, skills, and physical characteristics of men who rose to power. Galton's *Hereditary Genius* (1869) examined leadership qualities in the families of powerful men. After showing that the numbers of eminent relatives dropped off when his focus moved from first-degree to second-degree relatives, Galton concluded that leadership was inherited. In other words, leaders were born, not developed. Both of these notable works lent great initial support for the notion that leadership is rooted in characteristics of a leader.

Cecil Rhodes (1853–1902) believed that public-spirited leadership could be nurtured by identifying young people with "moral force of character and instincts to lead", and educating them in contexts (such as the collegiate environment of the University of Oxford) which further developed such characteristics. International networks of such leaders could help to promote international understanding and help "render war impossible". This vision of leadership underlay the creation of the Rhodes Scholarships, which have helped to shape notions of leadership since their creation in 1903.

**Rise of Alternative Theories**

In the late 1940s and early 1950s, a series of qualitative reviews of these studies (*e.g.,* Bird, 1940; Stogdill, 1948; Mann, 1959) prompted researchers to take a drastically different view of the driving forces behind leadership. In reviewing the extant literature, Stogdill and Mann found that while some traits were common across a number of studies, the overall evidence suggested that persons who are leaders in one situation may not necessarily be leaders in other situations. Subsequently, leadership was no longer characterized as an enduring individual trait, as situational approaches posited that individuals can be effective in certain situations, but not others. The focus then shifted away from traits of leaders to an investigation of the leader behaviours that were effective. This approach dominated much of the leadership theory and research for the next few decades

**Reemergence of Trait Theory**

New methods and measurements were developed after these influential reviews that would ultimately reestablish trait theory as a viable approach to the study of leadership. For example, improvements in researchers' use of the round robin research design methodology allowed researchers to see that individuals can and do emerge as leaders across a variety of situations and tasks. Additionally, during the 1980s statistical advances allowed researchers to conduct meta-analyses, in which they could quantitatively analyze and summarize the findings from a wide array of studies. This advent allowed trait theorists to create a comprehensive picture of previous leadership research rather than rely on the qualitative reviews of the past.

*Equipped with new methods, leadership researchers revealed the following:*

- Individuals can and do emerge as leaders across a variety of situations and tasks.
- Significant relationships exist between leadership emergence and such individual traits as:
  - Intelligence

- – Adjustment
- – Extraversion
- – Conscientiousness
- – Openness to experience
- – General self-efficacy.

While the trait theory of leadership has certainly regained popularity, its reemergence has not been accompanied by a corresponding increase in sophisticated conceptual frameworks.

*Specifically, Zaccaro (2007) noted that trait theories still:*

- Focus on a small set of individual attributes such as "The Big Five" personality traits, to the neglect of cognitive abilities, motives, values, social skills, expertise, and problem-solving skills.
- Fail to consider patterns or integrations of multiple attributes.
- Do not distinguish between the leadership attributes that are generally not malleable over time and those that are shaped by, and bound to, situational influences.
- Do not consider how stable leader attributes account for the behavioural diversity necessary for effective leadership.

**Attribute Pattern Approach**

Considering the criticisms of the trait theory outlined above, several researchers have begun to adopt a different perspective of leader individual differences—the leader attribute pattern approach. In contrast to the traditional approach, the leader attribute pattern approach is based on theorists' arguments that the influence of individual characteristics on outcomes is best understood by considering the person as an integrated totality rather than a summation of individual variables. In other words, the leader attribute pattern approach argues that integrated constellations or combinations of individual differences may explain substantial variance in both leader emergence and leader effectiveness beyond that explained by single attributes, or by additive combinations of multiple attributes..

**Behavioural and Style Theories**

In response to the early criticisms of the trait approach, theorists began to research leadership as a set of behaviours, evaluating the behaviour of successful leaders, determining a behaviour taxonomy, and identifying broad leadership styles. David McClelland, for example, posited that leadership takes a strong personality with a well-developed positive ego. To lead, self-confidence and high self-esteem are useful, perhaps even essential.

Kurt Lewin, Ronald Lipitt, and Ralph White developed in 1939 the seminal work on the influence of leadership styles and performance. The researchers evaluated the performance of groups of eleven-year-old boys under different types of work climate.

In each, the leader exercised his influence regarding the type of group decision making, praise and criticism (feedback), and the management of the group tasks (project management) according to three styles: authoritarian, democratic, and laissez-faire.

In 1945, Ohio State University conducted a study which investigated observable behaviours portrayed by effective leaders, They would then identify if these particular behaviours reflective in leadership effectiveness. They were able to narrow their findings to two identifiable distinctions The first dimension was identified as “Initiating Structure”, which described how a leader clearly and accurately communicates with their followers, defines goals, and determine how tasks are performed. These are considered “task oriented” behaviours The second dimension is “Consideration”, which indicates the leader’s ability to build an interpersonal relationship with their followers, to establish a form of mutual trust. These are considered “social oriented” behaviours.

The Michigan State Studies, which were conducted in the 1950’s, made further investigations and findings that positively correlated behaviours and leadership effectiveness. Although they similar findings as the Ohio State studies, they did contribute an additional behaviour identified in leaders. This was participative behaviour; allowing the followers to participate in group decision making and encouraged subordinate input. Another term used to describe this is “Servant Leadership”, which entails the leader to reject a more controlling type of leadership and allow more personal interaction between themselves and their subordinates.

The managerial grid model is also based on a behavioural theory. The model was developed by Robert Blake and Jane Mouton in 1964 and suggests five different leadership styles, based on the leaders’ concern for people and their concern for goal achievement.

***Positive Reinforcement***

B. F. Skinner is the father of behaviour modification and developed the concept of positive reinforcement. Positive reinforcement occurs when a positive stimulus is presented in response to a behaviour, increasing the likelihood of that behaviour in the future. The following is an example of how positive reinforcement can be used in a business setting. Assume praise is a positive reinforcer for a particular employee. This employee does not show up to work on time every day. The manager of this employee decides to praise the employee for showing up on time every day the employee actually shows up to work on time. As a result, the employee comes to work on time more often because the employee likes to be praised. In this example, praise (the stimulus) is a positive reinforcer for this employee because the employee arrives at work on time (the behaviour) more frequently after being praised for showing up to work on time.

The use of positive reinforcement is a successful and growing technique used by leaders to motivate and attain desired behaviours from subordinates. Organizations such as Frito-Lay, 3M, Goodrich, Michigan Bell, and Emery Air Freight have all used reinforcement to increase productivity. Empirical research covering the last 20 years suggests that reinforcement theory has a 17 percent increase in performance. Additionally, many reinforcement techniques such as the use of praise are inexpensive, providing higher performance for lower costs.

**Situational and Contingency Theories**

Situational theory also appeared as a reaction to the trait theory of leadership. Social scientists argued that history was more than the result of intervention of great men as Carlyle suggested. Herbert Spencer (1884) (and Karl Marx) said that the times produce the person and not the other way around. This theory assumes that different situations call for different characteristics; according to this group of theories, no single optimal psychographic profile of a leader exists. According to the theory, “what an individual actually does when acting as a leader is in large part dependent upon characteristics of the situation in which he functions.”

Some theorists started to synthesize the trait and situational approaches. Building upon the research of Lewin et al., academics began to normalize the descriptive models of leadership climates, defining three leadership styles and identifying which situations each style works better in. The authoritarian leadership style, for example, is approved in periods of crisis but fails to win the “hearts and minds” of followers in day-to-day management; the democratic leadership style is more adequate in situations that require consensus building; finally, the laissez-faire leadership style is appreciated for the degree of freedom it provides, but as the leaders do not “take charge”, they can be perceived as a failure in protracted or thorny organizational problems. Thus, theorists defined the style of leadership as contingent to the situation, which is sometimes classified as contingency theory. Four contingency leadership theories appear more prominently in recent years: Fiedler contingency model, Vroom-Yetton decision model, the path-goal theory, and the Hersey-Blanchard situational theory.

The Fiedler contingency model bases the leader’s effectiveness on what Fred Fiedler called *situational contingency*. This results from the interaction of leadership style and situational favourability (later called *situational control*). The theory defined two types of leader: those who tend to accomplish the task by developing good relationships with the group (relationship-oriented), and those who have as their prime concern carrying out the task itself (task-oriented). According to Fiedler, there is no ideal leader. Both task-oriented and relationship-oriented leaders can be effective if their leadership orientation fits the situation. When there is a good leader-member relation, a highly structured task, and high leader position power, the situation is considered a “favourable situation”. Fiedler found that task-oriented leaders are more effective in extremely favourable or unfavourable situations, whereas relationship-oriented leaders perform best in situations with intermediate favourability.

Victor Vroom, in collaboration with Phillip Yetton (1973) and later with Arthur Jago (1988), developed a taxonomy for describing leadership situations, which was used in a normative decision model where leadership styles were connected to situational variables, defining which approach was more suitable to which situation. This approach was novel because it supported the idea that the same manager could rely on different group decision making approaches depending on the attributes of each situation. This model was later referred to as situational contingency theory.

The path-goal theory of leadership was developed by Robert House (1971) and was based on the expectancy theory of Victor Vroom. According to House, the essence of the theory is "the meta proposition that leaders, to be effective, engage in behaviours that complement subordinates' environments and abilities in a manner that compensates for deficiencies and is instrumental to subordinate satisfaction and individual and work unit performance". The theory identifies four leader behaviours, *achievement-oriented*, *directive*, *participative*, and *supportive*, that are contingent to the environment factors and follower characteristics. In contrast to the Fiedler contingency model, the path-goal model states that the four leadership behaviours are fluid, and that leaders can adopt any of the four depending on what the situation demands. The path-goal model can be classified both as a contingency theory, as it depends on the circumstances, and as a transactional leadership theory, as the theory emphasizes the reciprocity behaviour between the leader and the followers.

The Situational Leadership® Model proposed by Hersey suggests four leadership-styles and four levels of follower-development. For effectiveness, the model posits that the leadership-style must match the appropriate level of follower-development. In this model, leadership behaviour becomes a function not only of the characteristics of the leader, but of the characteristics of followers as well.

**Functional Theory**

Functional leadership theory (Hackman and Walton, 1986; McGrath, 1962; Adair, 1988; Kouzes and Posner, 1995) is a particularly useful theory for addressing specific leader behaviours expected to contribute to organizational or unit effectiveness. This theory argues that the leader's main job is to see that whatever is necessary to group needs is taken care of; thus, a leader can be said to have done their job well when they have contributed to group effectiveness and cohesion (Fleishman et al., 1991; Hackman and Wageman, 2005; Hackman and Walton, 1986). While functional leadership theory has most often been applied to team leadership (Zaccaro, Rittman, and Marks, 2001), it has also been effectively applied to broader organizational leadership as well (Zaccaro, 2001). In summarizing literature on functional leadership observed five broad functions a leader performs when promoting organization's effectiveness. These functions include environmental monitoring, organizing subordinate activities, teaching and coaching subordinates, motivating others, and intervening actively in the group's work.

A variety of leadership behaviours are expected to facilitate these functions. In initial work identifying leader behaviour, Fleishman (1953) observed that subordinates perceived their supervisors' behaviour in terms of two broad categories referred to as consideration and initiating structure. Consideration includes behaviour involved in fostering effective relationships. Examples of such behaviour would include showing concern for a subordinate or acting in a supportive manner towards others. Initiating structure involves the actions of the leader focused specifically on task accomplishment. This could include role clarification, setting performance standards, and holding subordinates accountable to those standards.

**Integrated Psychological Theory**

The Integrated Psychological theory of leadership is an attempt to integrate the strengths of the older theories (*i.e.,* traits, behavioural/styles, situational and functional) while addressing their limitations, largely by introducing a new element – the need for leaders to develop their leadership presence, attitude towards others and behavioural flexibility by practicing psychological mastery. It also offers a foundation for leaders wanting to apply the philosophies of servant leadership and authentic leadership.

Integrated Psychological theory began to attract attention after the publication of James Scouller's Three Levels of Leadership model (2011). Scouller argued that the older theories offer only limited assistance in developing a person's ability to lead effectively.

*He pointed out, for example, that:*

- Traits theories, which tend to reinforce the idea that leaders are born not made, might help us select leaders, but they are less useful for developing leaders.
- An ideal style (*e.g.,* Blake and Mouton's team style) would not suit all circumstances.
- Most of the situational/contingency and functional theories assume that leaders can change their behaviour to meet differing circumstances or widen their behavioural range at will, when in practice many find it hard to do so because of unconscious beliefs, fears or ingrained habits. Thus, he argued, leaders need to work on their inner psychology.
- None of the old theories successfully address the challenge of developing "leadership presence"; that certain "something" in leaders that commands attention, inspires people, wins their trust and makes followers want to work with them.

Scouller proposed the Three Levels of Leadership model, which was later categorized as an "Integrated Psychological" theory on the Businessballs education web site. In essence, his model aims to summarize what leaders have to do, not only to bring leadership to their group or organization, but also to develop themselves technically and psychologically as leaders.

*The three levels in his model are Public, Private and Personal leadership:*

- The first two – public and private leadership – are "outer" or behavioural levels. These are the behaviours that address what Scouller called "the four dimensions of leadership". These dimensions are:
  - A shared, motivating group purpose;

- Action, progress and results;
- Collective unity or team spirit;
- Individual selection and motivation.

Public leadership focuses on the 34 behaviours involved in influencing two or more people simultaneously. Private leadership covers the 14 behaviours needed to influence individuals one to one.

- The third – personal leadership – is an “inner” level and concerns a person’s growth towards greater leadership presence, knowhow and skill. Working on one’s personal leadership has three aspects:
  - Technical knowhow and skill
  - Developing the right attitude towards other people – which is the basis of servant leadership
  - Psychological self-mastery – the foundation for authentic leadership.

Scouller argued that self-mastery is the key to growing one’s leadership presence, building trusting relationships with followers and dissolving one’s limiting beliefs and habits, thereby enabling behavioural flexibility as circumstances change, while staying connected to one’s core values (that is, while remaining authentic). To support leaders’ development, he introduced a new model of the human psyche and outlined the principles and techniques of self-mastery, which include the practice of mindfulness meditation.

**Transactional and Transformational Theories**

Bernard Bass and colleagues developed the idea of two different types of leadership, transactional that involves exchange of labour for rewards and transformational which is based on concern for employees, intellectual stimulation, and providing a group vision.

The transactional leader (Burns, 1978) is given power to perform certain tasks and reward or punish for the team’s performance. It gives the opportunity to the manager to lead the group and the group agrees to follow his lead to accomplish a predetermined goal in exchange for something else. Power is given to the leader to evaluate, correct, and train subordinates when productivity is not up to the desired level, and reward effectiveness when expected outcome is reached.

**Leader–member Exchange Theory**

This LMX theory addresses a specific aspect of the leadership process is the leader–member exchange (LMX) theory, which evolved from an earlier theory called the vertical dyad linkage (VDL) model. Both of these models focus on the interaction between leaders and individual followers. Similar to the transactional approach, this interaction is viewed as a fair exchange whereby the leader provides certain benefits such as task guidance, advice, support, and/or significant rewards and the followers reciprocate by giving the leader respect, cooperation, commitment to the task and good performance. However, LMX recognizes that leaders and individual followers will vary in the type of exchange that develops between them. LMX theorizes that the type of exchanges between the leader and specific followers can lead to the creation of *in-groups* and *out-groups*. In-group members are said to have *high-quality exchanges* with the leader, while out-group members have *low-quality exchanges* with the leader.

***In-group Members***

In-group members are perceived by the leader as being more experienced, competent, and willing to assume responsibility than other followers. The leader begins to rely on these individuals to help with especially challenging tasks. If the follower responds well, the leader rewards him/her with extra coaching, favourable job assignments, and developmental experiences. If the follower shows high commitment and effort followed by additional rewards, both parties develop mutual trust, influence, and support of one another. Research shows the in-group members usually receive higher performance evaluations from the leader, higher satisfaction, and faster promotions than out-group members. In-group members are also likely to build stronger bonds with their leaders by sharing the same social backgrounds and interests.

***Out-group Members***

Out-group members often receive less time and more distant exchanges than their in-group counterparts. With out-group members, leaders expect no more than adequate job performance, good attendance, reasonable respect, and adherence to the job description in exchange for a fair wage and standard benefits. The leader spends less time with out-group members, they have fewer developmental experiences, and the leader tends to emphasize his/her formal authority to obtain compliance to leader requests. Research shows that out-group members are less satisfied with their job and organization, receive lower performance evaluations from the leader, see their leader as less fair, and are more likely to file grievances or leave the organization.

**Emotions**

Leadership can be perceived as a particularly emotion-laden process, with emotions entwined with the social influence process. In an organization, the leader's mood has some effects on his/her group. These effects can be described in three levels:

1. The mood of individual group members. Group members with leaders in a positive mood experience more positive mood than do group members with leaders in a negative mood. The leaders transmit their moods to other group members through the mechanism of emotional contagion. Mood contagion may be one of the psychological mechanisms by which charismatic leaders influence followers.
2. The affective tone of the group. Group affective tone represents the consistent or homogeneous affective reactions within a group. Group affective tone is an aggregate of the moods of the individual members of the group and refers to mood at the group level of analysis. Groups with leaders in a positive mood

have a more positive affective tone than do groups with leaders in a negative mood.

3. Group processes like coordination, effort expenditure, and task strategy. Public expressions of mood impact how group members think and act. When people experience and express mood, they send signals to others. Leaders signal their goals, intentions, and attitudes through their expressions of moods. For example, expressions of positive moods by leaders signal that leaders deem progress towards goals to be good. The group members respond to those signals cognitively and behaviourally in ways that are reflected in the group processes.

In research about client service, it was found that expressions of positive mood by the leader improve the performance of the group, although in other sectors there were other findings.

Beyond the leader's mood, her/his behaviour is a source for employee positive and negative emotions at work. The leader creates situations and events that lead to emotional response. Certain leader behaviours displayed during interactions with their employees are the sources of these affective events. Leaders shape workplace affective events. Examples – feedback giving, allocating tasks, resource distribution. Since employee behaviour and productivity are directly affected by their emotional states, it is imperative to consider employee emotional responses to organizational leaders. Emotional intelligence, the ability to understand and manage moods and emotions in the self and others, contributes to effective leadership within organizations.

### Neo-emergent Theory

The neo-emergent leadership theory (from the Oxford school of leadership) sees leadership as created through the emergence of information by the leader or other stakeholders, not through the true actions of the leader himself. In other words, the reproduction of information or stories form the basis of the perception of leadership by the majority. It is well known that the naval hero Lord Nelson often wrote his own versions of battles he was involved in, so that when he arrived home in England he would receive a true hero's welcome. In modern society, the press, blogs and other sources report their own views of leaders, which may be based on reality, but may also be based on a political command, a payment, or an inherent interest of the author, media, or leader. Therefore, one can argue that the perception of all leaders is created and in fact does not reflect their true leadership qualities at all.

## LEADERSHIP EMERGENCE

Many personality characteristics were found to be reliably associated with leadership emergence. The list include, but is not limited to following (list organized in alphabetical order): assertiveness, authenticity, Big Five personality factors, birth order, character strengths, dominance, emotional intelligence, gender identity, intelligence, narcissism, self-efficacy for leadership, self-monitoring and social motivation. Leadership emergence is the idea that people born with specific characteristics become leaders, and those without these characteristics do not become leaders. People like Mahatma Gandhi, Abraham Lincoln, and Nelson Mandela all share traits that an average person does not. This includes people who choose to participate in leadership roles, as opposed to those who do not. Research indicates that up to 30% of leader emergence has a genetic basis. There is no current research indicating that there is a “leadership gene”, instead we inherit certain traits that might influence our decision to seek leadership. Both anecdotal, and empirical evidence support a stable relationship between specific traits and leadership behaviour. Using a large international sample researchers found that there are three factors that motivate leaders; affective identity (enjoyment of leading), non-calculative (leading earns reinforcement), and social-normative (sense of obligation).

**Assertiveness**

The relationship between assertiveness and leadership emergence is curvilinear; individuals who are either low in assertiveness or very high in assertiveness are less likely to be identified as leaders.

**Authenticity**

Individuals who are more aware of their personality qualities, including their values and beliefs, and are less biased when processing self-relevant information, are more likely to be accepted as leaders.

**Big Five Personality Factors**

Those who emerge as leaders tend to be more (order in strength of relationship with leadership emergence): extroverted, conscientious, emotionally stable, and open to experience, although these tendencies are stronger in laboratory studies of leaderless groups. Agreeableness, the last factor of the Big Five personality traits, does not seem to play any meaningful role in leadership emergence

**Birth Order**

Those born first in their families and only children are hypothesized to be more driven to seek leadership and control in social settings. Middle-born children tend to accept follower roles in groups, and later-borns are thought to be rebellious and creative

**Character Strengths**

Those seeking leadership positions in a military organization had elevated scores on a number of indicators of strength of character, including honesty, hope, bravery, industry, and teamwork.

**Dominance**

Individuals with dominant personalities – they describe themselves as high in the desire to control their environment and influence other people, and are likely to express their opinions in a forceful way – are more likely to act as leaders in small-group situations.

**Emotional Intelligence**

Individuals with high emotional intelligence have increased ability to understand and relate to people. They have skills in communicating and decoding emotions and they deal with others wisely and effectively. Such people communicate their ideas in more robust ways, are better able to read the politics of a situation, are less likely to lose control of their emotions, are less likely to be inappropriately angry or critical, and in consequence are more likely to emerge as leaders.

**Gender Identity**

Masculine individuals are more likely to emerge as leaders than are feminine individuals. This trend is expected to change in the modern era as in more developed countries we have seen how women have begun to rise to leadership position in the society as they were given equal rights compared to men.

**Intelligence**

Individuals with higher intelligence exhibit superior judgement, higher verbal skills (both written and oral), quicker learning and acquisition of knowledge, and are more likely to emerge as leaders. Correlation between IQ and leadership emergence was found to be between.25 and.30. However, groups generally prefer leaders that do not exceed intelligence prowess of average member by a wide margin, as they fear that high intelligence may be translated to differences in communication, trust, interests and values

**Narcissism**

Individuals who take on leadership roles in turbulent situations, such as groups facing a threat or ones in which status is determined by intense competition among rivals within the group, tend to be narcissistic: arrogant, self-absorbed, hostile, and very self-confident.

**Self-efficacy for Leadership**

Confidence in one’s ability to lead is associated with increases in willingness to accept a leadership role and success in that role.

**Self-monitoring**

High self-monitors are more likely to emerge as the leader of a group than are low self-monitors, since they are more concerned with status-enhancement and are more likely to adapt their actions to fit the demands of the situation

### Social Motivation

Individuals who are both success-oriented and affiliation-oriented, as assessed by projective measures, are more active in group problem-solving settings and are more likely to be elected to positions of leadership in such groups

## LEADERSHIP STYLES

A leadership style is a leader's style of providing direction, implementing plans, and motivating people. It is the result of the philosophy, personality, and experience of the leader. Rhetoric specialists have also developed models for understanding leadership (Robert Hariman, *Political Style*, Philippe-Joseph Salazar, *L'Hyperpolitique. Technologies politiques De La Domination*).

Different situations call for different leadership styles. In an emergency when there is little time to converge on an agreement and where a designated authority has significantly more experience or expertise than the rest of the team, an autocratic leadership style may be most effective; however, in a highly motivated and aligned team with a homogeneous level of expertise, a more democratic or Laissez-faire style may be more effective.

The style adopted should be the one that most effectively achieves the objectives of the group while balancing the interests of its individual members. A field in which leadership style has gained strong attention is that of military science, recently expressing a holistic and integrated view of leadership, including how a leader's physical presence determines how others perceive that leader. The factors of physical presence are military bearing, physical fitness, confidence, and resilience. The leader's intellectual capacity helps to conceptualize solutions and acquire knowledge to do the job. A leader's conceptual abilities apply agility, judgement, innovation, interpersonal tact, and domain knowledge. Domain knowledge for leaders encompasses tactical and technical knowledge as well as cultural and geopolitical awareness.

### Autocratic or Authoritarian

Under the autocratic leadership style, all decision-making powers are centralized in the leader, as with dictators.

Autocratic leaders do not entertain any suggestions or initiatives from subordinates. The autocratic management has been successful as it provides strong motivation to the manager. It permits quick decision-making, as only one person decides for the whole group and keeps each decision to him/herself until he/she feels it needs to be shared with the rest of the group.

### Participative or Democratic

The democratic leadership style consists of the leader sharing the decision-making abilities with group members by promoting the interests of the group members and by practicing social equality. This has also been called shared leadership.

**Laissez-faire or Free-rein**

In Laissez-faire or free-rein leadership, decision-making is passed on to the sub-ordinates. The sub-ordinates are given complete right and power to make decisions to establish goals and work out the problems or hurdles.

**Task-oriented and Relationship-oriented**

Task-oriented leadership is a style in which the leader is focused on the tasks that need to be performed in order to meet a certain production goal. Task-oriented leaders are generally more concerned with producing a step-by-step solution for given problem or goal, strictly making sure these deadlines are met, results and reaching target outcomes.

Relationship-oriented leadership is a contrasting style in which the leader is more focused on the relationships amongst the group and is generally more concerned with the overall well-being and satisfaction of group members. Relationship-oriented leaders emphasize communication within the group, show trust and confidence in group members, and show appreciation for work done.

Task-oriented leaders are typically less concerned with the idea of catering to group members, and more concerned with acquiring a certain solution to meet a production goal. For this reason, they typically are able to make sure that deadlines are met, yet their group members' well-being may suffer. Relationship-oriented leaders are focused on developing the team and the relationships in it. The positives to having this kind of environment are that team members are more motivated and have support. However, the emphasis on relations as opposed to getting a job done might make productivity suffer.

**LEADERSHIP DIFFERENCES AFFECTED BY SEX**

Another factor that covaries with leadership style is whether the person is male or female. When men and women come together in groups, they tend to adopt different leadership styles. Men generally assume an agentic leadership style. They are task-oriented, active, decision focused, independent and goal oriented. Women, on the other hand, are generally more communal when they assume a leadership position; they strive to be helpful towards others, warm in relation to others, understanding, and mindful of others' feelings. In general, when women are asked to describe themselves to others in newly formed groups, they emphasize their open, fair, responsible, and pleasant communal qualities. They give advice, offer assurances, and manage conflicts in an attempt to maintain positive relationships among group members. Women connect more positively to group members by smiling, maintaining eye contact and respond tactfully to others' comments. Men, conversely, describe themselves as influential, powerful and proficient at the task that needs to be done. They tend to place more focus on initiating structure within the group, setting standards and objectives, identifying roles, defining responsibilities and standard operating procedures, proposing solutions to problems, monitoring compliance with procedures, and finally, emphasizing the need for productivity and efficiency in the work that needs to be done. As leaders, men are primarily task-oriented, but women tend to be both task- and relationship-oriented. However, it is important to note that these sex differences are only tendencies, and do not manifest themselves within men and women across all groups and situations.

## PERFORMANCE

In the past, some researchers have argued that the actual influence of leaders on organizational outcomes is overrated and romanticized as a result of biased attributions about leaders (Meindl and Ehrlich, 1987). Despite these assertions, however, it is largely recognized and accepted by practitioners and researchers that leadership is important, and research supports the notion that leaders do contribute to key organizational outcomes (Day and Lord, 1988; Kaiser, Hogan, and Craig, 2008). To facilitate successful performance it is important to understand and accurately measure leadership performance.

Job performance generally refers to behaviour that is expected to contribute to organizational success (Campbell, 1990). Campbell identified a number of specific types of performance dimensions; leadership was one of the dimensions that he identified. There is no consistent, overall definition of leadership performance (Yukl, 2006). Many distinct conceptualizations are often lumped together under the umbrella of leadership performance, including outcomes such as leader effectiveness, leader advancement, and leader emergence (Kaiser et al., 2008). For instance, leadership performance may be used to refer to the career success of the individual leader, performance of the group or organization, or even leader emergence. Each of these measures can be considered conceptually distinct. While these aspects may be related, they are different outcomes and their inclusion should depend on the applied or research focus.

A toxic leader is someone who has responsibility over a group of people or an organization, and who abuses the leader–follower relationship by leaving the group or organization in a worse-off condition than when he/she joined it.

## TRAITS

Most theories in the 20th century argued that great leaders were born, not made. Current studies have indicated that leadership is much more complex and cannot be boiled down to a few key traits of an individual. Years of observation and study have indicated that one such trait or a set of traits does not make an extraordinary leader. What scholars have been able to arrive at is that leadership traits of an individual do not change from situation to situation; such traits include intelligence, assertiveness, or physical attractiveness.

However, each key trait may be applied to situations differently, depending on the circumstances. The following summarizes the main leadership traits found in research by Jon P. Howell, business professor at New Mexico State University and author of the book *Snapshots of Great Leadership*.

Determination and drive include traits such as initiative, energy, assertiveness, perseverance and sometimes dominance. People with these traits often tend to wholeheartedly pursue their goals, work long hours, are ambitious, and often are very competitive with others.

Cognitive capacity includes intelligence, analytical and verbal ability, behavioural flexibility, and good judgement. Individuals with these traits are able to formulate solutions to difficult problems, work well under stress or deadlines, adapt to changing situations, and create well-thought-out plans for the future. Howell provides examples of Steve Jobs and Abraham Lincoln as encompassing the traits of determination and drive as well as possessing cognitive capacity, demonstrated by their ability to adapt to their continuously changing environments.

Self-confidence encompasses the traits of high self-esteem, assertiveness, emotional stability, and self-assurance. Individuals who are self-confident do not doubt themselves or their abilities and decisions; they also have the ability to project this self-confidence onto others, building their trust and commitment.

Integrity is demonstrated in individuals who are truthful, trustworthy, principled, consistent, dependable, loyal, and not deceptive. Leaders with integrity often share these values with their followers, as this trait is mainly an ethics issue. It is often said that these leaders keep their word and are honest and open with their cohorts. Sociability describes individuals who are friendly, extroverted, tactful, flexible, and interpersonally competent.

Such a trait enables leaders to be accepted well by the public, use diplomatic measures to solve issues, as well as hold the ability to adapt their social persona to the situation at hand. According to Howell, Mother Teresa is an exceptional example who embodies integrity, assertiveness, and social abilities in her diplomatic dealings with the leaders of the world.

Few great leaders encompass all of the traits listed above, but many have the ability to apply a number of them to succeed as front-runners of their organization or situation.

## ONTOLOGICAL-PHENOMENOLOGICAL MODEL

One of the more recent definitions of leadership comes from Werner Erhard, Michael C. Jensen, Steve Zaffron, and Kari Granger who describe leadership as “an exercise in language that results in the realization of a future that wasn’t going to happen anyway, which future fulfills (or contributes to fulfilling) the concerns of the relevant parties...”. This definition ensures that leadership is talking about the future and includes the fundamental concerns of the relevant parties. This differs from relating to the relevant parties as “followers” and calling up an image of a single leader with others following. Rather, a future that fulfills on the fundamental concerns of the relevant parties indicates the future that wasn’t going to happen is not the “idea of the leader”, but rather is what emerges from digging deep to find the underlying concerns of those who are impacted by the leadership.

## CONTEXTS

### Organizations

An organization that is established as an instrument or means for achieving defined objectives has been referred to as a *formal organization*. Its design specifies how goals are subdivided and reflected in subdivisions of the organization. Divisions, departments, sections, positions, jobs, and tasks make up this work structure. Thus, the formal organization is expected to behave impersonally in regard to relationships with clients or with its members. According to Weber's definition, entry and subsequent advancement is by merit or seniority. Employees receive a salary and enjoy a degree of tenure that safeguards them from the arbitrary influence of superiors or of powerful clients. The higher one's position in the hierarchy, the greater one's presumed expertise in adjudicating problems that may arise in the course of the work carried out at lower levels of the organization. It is this bureaucratic structure that forms the basis for the appointment of heads or chiefs of administrative subdivisions in the organization and endows them with the authority attached to their position.

In contrast to the appointed head or chief of an administrative unit, a leader emerges within the context of the *informal organization* that underlies the formal structure. The informal organization expresses the personal objectives and goals of the individual membership. Their objectives and goals may or may not coincide with those of the formal organization. The informal organization represents an extension of the social structures that generally characterize human life — the spontaneous emergence of groups and organizations as ends in themselves.

In prehistoric times, humanity was preoccupied with personal security, maintenance, protection, and survival. Now humanity spends a major portion of waking hours working for organizations. The need to identify with a community that provides security, protection, maintenance, and a feeling of belonging has continued unchanged from prehistoric times. This need is met by the informal organization and its emergent, or unofficial, leaders.

Leaders emerge from within the structure of the informal organization. Their personal qualities, the demands of the situation, or a combination of these and other factors attract followers who accept their leadership within one or several overlay structures. Instead of the authority of position held by an appointed head or chief, the emergent leader wields influence or power. Influence is the ability of a person to gain co-operation from others by means of persuasion or control over rewards. Power is a stronger form of influence because it reflects a person's ability to enforce action through the control of a means of punishment.

A leader is a person who influences a group of people towards a specific result. It is not dependent on title or formal authority. (Elevos, paraphrased from Leaders, Bennis, and Leadership Presence, Halpern and Lubar.) Ogbonnia (2007) defines an effective leader "as an individual with the capacity to consistently succeed in a given condition and be viewed as meeting the expectations of an organization or society." Leaders are recognized by their capacity for caring for others, clear communication, and a commitment to persist. An individual who is appointed to a managerial position has the right to command and enforce obedience by virtue of the authority of their position. However, she or he must possess adequate personal attributes to match this authority, because authority is only potentially available to him/her. In the absence of sufficient personal competence, a manager may be confronted by an emergent leader who can challenge her/his role in the organization and reduce it to that of a figurehead. However, only authority of position has the backing of formal sanctions. It follows that whoever wields personal influence and power can legitimize this only by gaining a formal position in the hierarchy, with commensurate authority. Leadership can be defined as one's ability to get others to willingly follow. Every organization needs leaders at every level.

**Management**

Over the years the philosophical terminology of "management" and "leadership" have, in the organizational context, been used both as synonyms and with clearly differentiated meanings. Debate is fairly common about whether the use of these terms should be restricted, and generally reflects an awareness of the distinction made by Burns (1978) between "transactional" leadership (characterized by *e.g.,* emphasis on procedures, contingent reward, management by exception) and "transformational" leadership (characterized by *e.g.,* charisma, personal relationships, creativity).

**Group**

In contrast to individual leadership, some organizations have adopted group leadership. In this so-called shared leadership, more than one person provides direction to the group as a whole. It is furthermore characterized by shared responsibility, cooperation and mutual influence among the team members. Some organizations have taken this approach in hopes of increasing creativity, reducing costs, or downsizing. Others may see the traditional leadership of a boss as costing too much in team performance. In some situations, the team members best able to handle any given phase of the project become the temporary leaders. Additionally, as each team member has the opportunity to experience the elevated level of empowerment, it energizes staff and feeds the cycle of success.

Leaders who demonstrate persistence, tenacity, determination, and synergistic communication skills will bring out the same qualities in their groups. Good leaders use their own inner mentors to energize their team and organizations and lead a team to achieve success.

*According to the National School Boards Association (USA):*

These Group Leaderships or Leadership Teams have specific characteristics:

*Characteristics of a Team:*

- There must be an awareness of unity on the part of all its members.

- There must be interpersonal relationship. Members must have a chance to contribute, and learn from and work with others.
- The members must have the ability to act together towards a common goal.

*Ten characteristics of well-functioning teams:*

- *Purpose:* Members proudly share a sense of why the team exists and are invested in accomplishing its mission and goals.
- *Priorities:* Members know what needs to be done next, by whom, and by when to achieve team goals.
- *Roles:* Members know their roles in getting tasks done and when to allow a more skillful member to do a certain task.
- *Decisions:* Authority and decision-making lines are clearly understood.
- *Conflict:* Conflict is dealt with openly and is considered important to decision-making and personal growth.
- *Personal traits:* Members feel their unique personalities are appreciated and well utilized.
- *Norms:* Group norms for working together are set and seen as standards for every one in the groups.
- *Effectiveness:* Members find team meetings efficient and productive and look forward to this time together.
- *Success:* Members know clearly when the team has met with success and share in this equally and proudly.
- *Training:* Opportunities for feedback and updating skills are provided and taken advantage of by team members.

**Self-leadership**

Self-leadership is a process that occurs within an individual, rather than an external act. It is an expression of who we are as people.

**Primates**

Mark van Vugt and Anjana Ahuja in *Naturally Selected: The Evolutionary Science of Leadership* present evidence of leadership in non-human animals, from ants and bees to baboons and chimpanzees. They suggest that leadership has a long evolutionary history and that the same mechanisms underpinning leadership in humans can be found in other social species, too. Richard Wrangham and Dale Peterson, in *Demonic Males: Apes and the Origins of Human Violence*, present evidence that only humans and chimpanzees, among all the animals living on Earth, share a similar tendency for a cluster of behaviours: violence, territoriality, and competition for uniting behind the one chief male of the land. This position is contentious. Many animals beyond apes are territorial, compete, exhibit violence, and have a social structure controlled by a dominant male (lions, wolves, etc.), suggesting Wrangham and Peterson's evidence is not empirical. However, we must examine other species as well, including elephants (which are matriarchal and follow an alpha female), meerkats (who are likewise matriarchal), and many others.

By comparison, bonobos, the second-closest species-relatives of humans, do *not* unite behind the chief male of the land. The bonobos show deference to an alpha or top-ranking female that, with the support of her coalition of other females, can prove as strong as the strongest male. Thus, if leadership amounts to getting the greatest number of followers, then among the bonobos, a female almost always exerts the strongest and most effective leadership. However, not all scientists agree on the allegedly peaceful nature of the bonobo or its reputation as a “hippie chimp”.

## MYTHS

Leadership, although largely talked about, has been described as one of the least understood concepts across all cultures and civilizations. Over the years, many researchers have stressed the prevalence of this misunderstanding, stating that the existence of several flawed assumptions, or myths, concerning leadership often interferes with individuals’ conception of what leadership is all about (Gardner, 1965; Bennis, 1975).

### Leadership is Innate

According to some, leadership is determined by distinctive dispositional characteristics present at birth (*e.g.,* extraversion; intelligence; ingenuity). However, according to Forsyth (2009) there is evidence to show that leadership also develops through hard work and careful observation. Thus, effective leadership can result from nature (*i.e.,* innate talents) as well as nurture (*i.e.,* acquired skills).

### Leadership is Possessing Power Over Others

Although leadership is certainly a form of power, it is not demarcated by power *over* people – rather, it is a power *with* people that exists as a reciprocal relationship between a leader and his/her followers (Forsyth, 2009). Despite popular belief, the use of manipulation, coercion, and domination to influence others is not a requirement for leadership. In actuality, individuals who seek group consent and strive to act in the best interests of others can also become effective leaders (*e.g.,* class president; court judge).

### Leaders are Positively Influential

The validity of the assertion that groups flourish when guided by effective leaders can be illustrated using several examples. For instance, according to Baumeister et al. (1988), the bystander effect (failure to respond or offer assistance) that tends to develop within groups faced with an emergency is significantly reduced in groups guided by a leader. Moreover, it has been documented that group performance, creativity, and efficiency all tend to climb in businesses with designated managers or CEOs. However, the difference leaders make is not always positive in nature. Leaders sometimes focus on fulfilling their own agendas at the expense of others, including his/her own followers (*e.g.,* Pol Pot; Josef Stalin). Leaders who focus on personal gain by employing stringent and manipulative leadership styles often make a difference, but usually do so through negative means.

**Leaders Entirely Control Group Outcomes**

In Western cultures it is generally assumed that group leaders make *all* the difference when it comes to group influence and overall goal-attainment. Although common, this romanticized view of leadership (*i.e.,* the tendency to overestimate the degree of control leaders have over their groups and their groups' outcomes) ignores the existence of many other factors that influence group dynamics. For example, group cohesion, communication patterns among members, individual personality traits, group context, the nature or orientation of the work, as well as behavioural norms and established standards influence group functionality in varying capacities. For this reason, it is unwarranted to assume that all leaders are in complete control of their groups' achievements.

**All Groups have a Designated Leader**

Despite preconceived notions, not all groups need have a designated leader. Groups that are primarily composed of women, are limited in size, are free from stressful decision-making, or only exist for a short period of time (*e.g.,* student work groups; pub quiz/trivia teams) often undergo a diffusion of responsibility, where leadership tasks and roles are shared amongst members (Schmid Mast, 2002; Berdahl and Anderson, 2007; Guastello, 2007).

**Group Members Resist Leaders**

Although research has indicated that group members' dependence on group leaders can lead to reduced self-reliance and overall group strength, most people actually prefer to be led than to be without a leader (Berkowitz, 1953). This "need for a leader" becomes especially strong in troubled groups that are experiencing some sort of conflict. Group members tend to be more contented and productive when they have a leader to guide them. Although individuals filling leadership roles can be a direct source of resentment for followers, most people appreciate the contributions that leaders make to their groups and consequently welcome the guidance of a leader (Stewart and Manz, 1995).

**ACTION-ORIENTED ENVIRONMENTS**

In most cases, these teams are tasked to operate in remote and changeable environments with limited support or backup (action environments). Leadership of people in these environments requires a different set of skills to that of front line management. These leaders must effectively operate remotely and negotiate the needs of the individual, team, and task within a changeable environment. This has been termed action oriented leadership. Some examples of demonstrations of action oriented leadership include extinguishing a rural fire, locating a missing person, leading a team on an outdoor expedition, or rescuing a person from a potentially hazardous environment.

Other examples include modern technology deployments of small/medium-sized IT teams into client plant sites. Leadership of these teams requires hands on experience and a lead-by-example attitude to empower team members to make well thought out and concise decisions independent of executive management and/or home base decision makers. Zachary Hansen was an early adopter of Scrum/Kanban branch development methodologies during the mid 90's to alleviate the dependency that field teams had on trunk based development. This method of just-in-time action oriented development and deployment allowed remote plant sites to deploy up-to-date software patches frequently and without dependency on core team deployment schedules satisfying the clients need to rapidly patch production environment bugs as needed.

## CRITICAL THOUGHT

Carlyle's 1840 "Great Man theory", which emphasized the role of leading individuals, met opposition in the 19th and 20th centuries. Karl Popper noted in 1945 that leaders can mislead and make mistakes - he warns against deferring to "great men".

Noam Chomsky and others have subjected the concept of leadership to critical thinking and have provided an analysis that asserts that people abrogate their responsibility to think and will actions for themselves. While the conventional view of leadership may satisfy people who "want to be told what to do", these critics say that one should question why they are being subjected to a will or intellect other than their own if the leader is not a subject-matter expert (SME).

Concepts such as autogestion, employeeship, and common civic virtue, etc., challenge the fundamentally anti-democratic nature of the leadership principle by stressing individual responsibility and/or group authority in the workplace and elsewhere and by focusing on the skills and attitudes that a person needs in general rather than separating out "leadership" as the basis of a special class of individuals.

Similarly, various historical calamities (such as World War II) can be attributed to a misplaced reliance on the principle of leadership as exhibited in dictatorship.

The idea of leaderism paints leadership and its excesses in a negative light.

## EXECUTIVES

Executives are energetic, outgoing, and competitive. They can be visionary, hard-working, and decisive. However, managers need to be aware of unsuccessful executives who once showed management potential but who are later dismissed or retired early. They typically fail because of personality factors rather than job performances.

*Terms fallacies in their thinking are:*

- *Unrealistic optimism fallacy*: Believing they are so smart that they can do whatever they want

- *Egocentrism fallacy*: Believing they are the only ones who matter, that the people who work for them don't count
- *Omniscience fallacy*: Believing they know everything and seeing no limits to their knowledge
- *Omnipotence fallacy*: Believing they are all powerful and therefore entitled to do what they want
- *Invulnerability fallacy*: Believing they can get away with doing what they want because they are too clever to get caught; even if they are caught, believing they will go unpunished because of their importance.

## EDUCATIONAL LEADERSHIP

School leadership is the process of enlisting and guiding the talents and energies of teachers, pupils, and parents towards achieving common educational aims. This term is often used synonymously with educational leadership in the United States and has supplanted *educational management* in the United Kingdom. Several universities in the United States offer graduate degrees in educational leadership. Certain obstacles of educational leadership can be overcome. A self-assessment technique can help examine equity and justice that affects student diversity, especially with selection of candidates.

### History

The term *school leadership* came into currency in the late 20th century for several reasons. Demands were made on schools for higher levels of pupil achievement, and schools were expected to improve and reform. These expectations were accompanied by calls for accountability at the school level. Maintenance of the *status quo* was no longer considered acceptable. *Administration* and *management* are terms that connote stability through the exercise of control and supervision. The concept of leadership was favoured because it conveys dynamism and pro-activity. The principal or school head is commonly thought to be the school leader; however, school leadership may include other persons, such as members of a formal leadership team and other persons who contribute towards the aims of the school.

While *school leadership* or *educational leadership* have become popular as replacements for *educational administration* in recent years, *leadership* arguably presents only a partial picture of the work of school, division or district, and ministerial or state education agency personnel, not to mention the areas of research explored by university faculty in departments concerned with the operations of schools and educational institutions. For this reason, there may be grounds to question the merits of the term as a catch-all for the field. Rather, the etiology of its use may be found in more generally and con-temporarily experienced neo-liberal social and economic governance models, especially in the United States and the United Kingdom. On this view, the term is understood as having been borrowed from business.

In the United States, the superintendency, or role of the chief school administrator, has undergone many changes since the creation of the position—which is often attributed to the Buffalo Common Council that approved a superintendent on June 9, 1837. If history serves us correctly, the superintendency is about 170 years old with four major role changes from the early 19th century through the first half of the 20th century and into the early years of the 21st century. Initially, the superintendent's main function was clerical in nature and focused on assisting the board of education with day-to-day details of running the school. At the turn of the 20th century, states began to develop common curriculum for public schools with superintendents fulfilling the role of teacher-scholar or master educator who had added an emphasis on curricular and instructional matters to school operations. In the early 20th century, the Industrial Revolution affected the superintendent's role by shifting the emphasis to expert manager with efficiency in handling non-instructional tasks such as budget, facility, and transportation. The release of *A Nation at Risk* in 1983 directly impacted public school accountability and, ultimately, the superintendency. The early 1980s initiated the change that has continued through today with the superintendent viewed as chief executive officer, including the roles of professional adviser to the board, leader of reforms, manager of resources and communicator to the public.

**Graduate Studies**

The term "educational leadership" is also used to describe programmes beyond schools. Leaders in community colleges, proprietary colleges, community-based programmes, and universities are also *educational leaders*.

Some United States university graduate masters and doctoral programmes are organized with higher education and adult education programmes as a part of an educational leadership department. In these cases, the entire department is charged with educating *educational leaders* with specific specialization areas such as university leadership, community college leadership, and community-based leadership (as well as school leadership). Masters of education are offered at a number of universities around the United States in traditional and online formats including the University of Texas at El Paso, University of Massachusetts, Saint Mary's University of Minnesota, Pepperdine University, Capella University, Northcentral University, and the University of Scranton. Some United States graduate programmes with a tradition of graduate education in these areas of specialization have separate departments for them. The area of higher education may include areas such as student affairs leadership, academic affairs leadership, community college leadership, community college and university teaching, vocational, adult education and university administration, and educational wings of non-governmental organizations.

In Europe, similar degrees exist at the University of Bath and Apsley Business School - London, where the focus is on the management systems of education, especially as British schools move away from state funding to semi-autonomous Free Schools and

Academies. In fact in these schools, the focus is on traditional MBA disciplines, such as HR, Change Management and Finance. The so-called “Academisation” of British education is highly contentious and political issue with many headteachers resisting moves to what they see as forced privatization. In mainland Europe, Educational Leadership is not taught formally, with senior educationalists having come through academic pathways, not administration.

**Literature, Research and Policy**

Educational leadership draws upon interdisciplinary literature, generally, but ideally distinguishes itself through its focus on pedagogy, epistemology and human development. In contemporary practice it borrows from political science and business. Debate within the field relates to this tension. Numerous educational leadership theories and perspectives have been presented and explored, such as: (a) instructional leadership; (b) distributed leadership; (c) transformational leadership; (d) social justice leadership; and (e) Teacher leadership. Researchers have explored how different practices and actions impact student achievement, teacher job satisfaction, or other elements related to school improvement. Moreover, researchers continue to investigate the methodology and quality of principal preparation programmes.

A number of publications and foundations are devoted to studying the particular requirements of leadership in these settings, and educational leadership is taught as an academic discipline at a number of universities.

Several countries now have explicit policies on school leadership, including policies and budgets for the training and development of school leaders.

In the USA, formal “curriculum audits” are becoming common, in which educational leaders and trained auditors evaluate school leadership and the alignment of curriculum with goals and objectives. Curriculum audits and curriculum mapping were developed by Fenwick W. English in the late 1970s. The educational leaders and auditors who conduct the audits are certified by Phi Delta Kappa. Research shows how educational leadership influences student learning.

## TEACHER LEADERSHIP

Teacher leadership is a term used in K-12 schools for classroom educators who simultaneously take on administrative roles outside of their classrooms to assist in functions of the larger school system.

Teacher leadership tasks may include but are not limited to: managing teaching, learning, and resource allocation. Teachers who engage in leadership roles are generally experienced and respected in their field which can both empower them and increase collaboration among peers.

In these types of school environments, teachers are able to make decisions based on the work they do directly with students. When a school system places the decision-making on the teachers, the action is happening one level closer to the people who are most closely impacted by the decisions (generally the students and the teachers), rather than two or more levels above at the principal, superintendent, or school board level.

The extent to which teacher leaders adopt additional roles varies in degree and description:

- Administration leadership (traditional school leadership/educational leadership): Administrative staff carries out the majority of the leadership duties.
- Teacher networks (professional learning community/professional community/networked improvement communities/community of practice/distributed leadership):
- All teachers collectively take on decision-making roles about curriculum and school climate. This practice is facilitated by and supported by an administrative leader.
- Teacher leaders (instructional leadership/instructional coaches): Some teachers take on individual leadership roles that directly impact educational practices under the leadership of a school administrator.
- Teacher co-ops (teacher-powered schools/teacher-led schools/worker cooperative/professional partnerships/teacherpreneurs): All teachers collectively take on leadership and administrative tasks that would traditionally be done by a principal or administrative team.

**Research Supportive of the Philosophy**

The National Education Association (NEA) (2011) describes teacher leaders as, "experienced professionals who have earned the respect of their students and colleagues and have gained a set of skills that enable them to work effectively and collaboratively with colleagues. They work closely with principals who have been trained to develop and implement effective mechanisms of support for teachers and teacher leaders."

Teacher leaders are teachers who, "want to remain closely connected to the classroom and students, but are willing to assume new responsibilities that afford them leadership opportunities in or outside the classroom while still teaching full or part-time."

In his book, Kolderie (2014) cites Hubbs in describing the benefits of decentralizing the authority in schools: "You just can't beat a decentralized system. It gets closest to the level where the action really is. Education should have an advantage in moving into it, because your locations and your people are already physically dispersed."

Another potential benefit of decentralizing authority is as Spillane et al. describe: "The interactions among two or more leaders in carrying out a particular task may amount to more than the sum of those leaders' practice."

The NEA (2011) reported that, "Research indicates that in order to increase the likelihood that Gen "Y" teachers remain in the profession, they need opportunities to participate in decision making at the school and district level; a positive and supportive school culture which fosters teamwork and effective lines of communication; professional opportunities that include collaboration and technology; in-depth feedback and support from administrators and colleagues; time set aside for regular collaboration; and fair pay and a differentiated pay structure which includes rewarding outstanding performance, acquiring new knowledge and skills, and assuming new roles and responsibilities (Behrstock and Clifford, 2009)."

Kolderie (2014) emphasizes, "If teachers can control what matters for student success teachers will accept accountability for student success."

Teacherpowered.org is a resource for this kind of work, and they note that teachers, "feel increased passion for the job and have greater ability to make the dramatic changes in schools that they determine are needed to truly improve student learning and the teaching profession."

Lieberman (2000) cited Newmann and Wehlage (1995) in their search for, "an understanding of how schools developed the capacity to inspire student learning of high intellectual quality. They found that a self-conscious professional community was a salient characteristic of those schools most successful with students."

**Spectrum of Authority**

To frame the types of work in which Teacher leaders participate, it is important to look at the roles taken on by Educational leadership more broadly.

*Halverson, Kelley, and Shaw (2013) identify the following domains of school leadership:*

1. "Focus on Learning
2. Monitoring teaching and learning
3. Building nested learning communities
4. Acquiring and allocating resources
5. Establishing a safe and effective learning environment."

Styles of school leadership can be placed on a spectrum in which on one end the leadership is completely owned by the administration to the opposite end in which leadership in completely owned by the teachers.

***Teacher Networks***

The structure of teacher networks, or as Kruse defines them, professional communities, includes "collective responsibility, professional control, and flexible boundaries". Kruse continues to describe the day-to-day norms of a professional community, including: "reflective dialogue..., de-privatized, collective focus on student learning, collaboration, [and] shared norms and values." Lieberman (2000) explains that, "having a professional community differentiated those teachers who worked together to change the culture of their classrooms and their departments from those teachers who either tried new ideas in fragmented ways on their own or who blamed students for their inabilities to learn." Bryk/Gomez describe a type of teacher network, "a networked improvement community" as, "an intentionally formed social organization" whose "improvement goals impose specific demands on the rules and norms of participation."

***Teacher Leaders***

Danielson (2006) explains, “Many schools have instituted structures in which teachers assume formal leadership roles in the school, such as master teacher, department chair, team leader, helping teacher, or mentor.”

“Teachers in leadership roles work in collaboration with principals and other school administrators by facilitating improvements in instruction and promoting practices among their peers that can lead to improved student learning outcomes. By doing so, they support school leaders in encouraging innovation and creating cultures of success in school.

Teacher leadership can neither be effective nor successful without principal support, but neither can the principal maximize his or her effectiveness without harnessing the talents and expertise of teachers in leadership roles.”

“Teachers who serve in leadership roles may do so formally or informally. Rather than having positional authority, teachers become leaders in their schools by being respected by their peers, being continuous learners, being approachable, and using group skills and influence to improve the educational practice of their peers. They model effective practices, exercise their influence in formal and informal contexts, and support collaborative team structures within their schools.”

“The Teacher Leader Model Standards consist of seven domains describing the many dimensions of teacher leadership:

- *Domain I:* Fostering a Collaborative Culture to Support Educator Development and Student Learning
- *Domain II:* Accessing and Using Research to Improve Practice and Student Learning
- *Domain III:* Promoting Professional Learning for Continuous Improvement
- *Domain IV:* Facilitating Improvements in Instruction and Student Learning
- *Domain V:* Promoting the Use of Assessments and Data for School and District Improvement
- *Domain VI:* Improving Outreach and Collaboration with Families and Community Domain VII: Advocating for Student Learning and the Profession.”

One way in which teachers serve in these leadership or instructional coaching roles is to do instructional rounds in their school. Rounds consist of teachers spending time observing in one another’s classrooms and then meeting to discuss the findings. City et al. describe the goals of rounds to: “1.

Build skills of network members by coming to a common understanding of effective practice and how to support it. 2. Support instructional improvement at the host site (school or district) by sharing what the network learns and by building skills at the local level.”

Some additional forms of teacher or instructional leadership may include: problem-solving teams, peer mentoring, and coaching, which support of the work of the administration without replacing it.

***Teacher Co-ops***

The Teacher Powered Schools organization explains this teacher co-op or teacher-powered model as: “1. collaboratively designed and implemented by teachers. 2. teachers having collective autonomy to make the decisions influencing the success of a school, project, or professional endeavor.”

In these teacher co-ops or professional partnerships, Kolderie explains, “The authority pyramid inverts, and school moves to the dual-leader model typical of partnerships in most areas. (Think of the managing partner and the administrator of a law firm, or the chief of the medical staff and the executive of a hospital.).”

The structure of a teacher co-op could vary from, “...handling the whole school [which] involves the teachers in the administration and management of the school as well as in the learning.

Alternatively, teachers in a large secondary school—while continuing to be district employees—could form a partnership to handle the math department, the science department or the English department. They could then focus on learning; the principal would handle the school administration. A partnership of teachers might also organize to take responsibility for a programme serving several schools across a district; Montessori, for example.”

**Factors of Implementation**

Lieberman (2000) found that, “Networks that last, that hold their members, and continue to attract new teachers understand that they must account for the daily pressures of teaching, even as they seek to advance larger ideals.”

In order to be successful, teacher leadership should be implemented strategically.

*“Among the issues that will need to be considered are:*

1. Resources (new or reallocated) necessary to provide professional development to teachers and administrators,
2. Availability of time within schools to create a collaborative work environment,
3. Re-structuring of current teacher compensation systems,
4. The development of cost-efficient, equitable, and streamlined systems to recognize and reward teachers serving in leadership roles,
5. Assessing the effectiveness of various teacher leadership models, and
6. Replicating and scaling-up effective teacher leadership practices.”

Teacher leadership can be initiated from within the staff itself, however, it is important that the teachers have administrative support as well as peer-to-peer support to carry out their tasks. Halverson, Kelley, and Shaw (2013) reference: “Louis, Kruse and Bryk (1995) conclude that the most important task for school leaders is to create meaningful opportunities for teachers across the school to work together on pressing issues of common interest.”

Kruse (1993) outlines the conditions that support teacher leadership in the specific case of a professional community:

*"Structural conditions:*

- Time to meet
- Physical proximity
- Interdependent teaching roles
- Communication structures
- Teacher empowerment and school autonomy.

*Social and human resources:*

- Openness to improvement
- Trust and respect
- Cognitive skills base
- Supportive leadership
- Socialization of new members."

Schools may implement a teacher leadership model as a strategy to down-size and cut costs for the school. In most cases, distributing administrative among the teachers could reduce overall personnel costs.

# PLANNING DIMENSIONS OF ADMINISTRATION

Planning (also called forethought) is the process of thinking about and organizing the activities required to achieve a desired goal. It involves the creation and maintenance of a plan, such as psychological aspects that require conceptual skills. There are even a couple of tests to measure someone's capability of planning well. As such, planning is a fundamental property of intelligent behaviour.

Also, planning has a specific process and is necessary for multiple occupations (particularly in fields such as management, business, etc.). In each field there are different types of plans that help companies achieve efficiency and effectiveness. An important, albeit often ignored aspect of planning, is the relationship it holds to forecasting. Forecasting can be described as predicting what the future will look like, whereas planning predicts what the future should look like for multiple scenarios. Planning combines forecasting with preparation of scenarios and how to react to them. Planning is one of the most important project management and time management techniques. Planning is preparing a sequence of action steps to achieve some specific goal. If a person does it effectively, they can reduce much the necessary time and effort of achieving the goal. A plan is like a map. When following a plan, a person can see how much they have progressed towards their project goal and how far they are from their destination.

## PLANNING TOPICS

### Psychological Aspects

Planning is one of the executive functions of the brain, encompassing the neurological processes involved in the formulation, evaluation and selection of a sequence of thoughts and actions to achieve a desired goal. Various studies utilizing a combination of neuropsychological, neuropharmacological and functional neuroimaging approaches have suggested there is a positive relationship between impaired planning ability and damage to the frontal lobe.

A specific area within the mid-dorsolateral frontal cortex located in the frontal lobe has been implicated as playing an intrinsic role in both cognitive planning and associated executive traits such as working memory.

Disruption of the neural pathways, via various mechanisms such as traumatic brain injury, or the effects of neurodegenerative diseases between this area of the frontal cortex and the basal ganglia specifically the striatum (cortico-striatal pathway), may disrupt the processes required for normal planning function.

Individuals who were born Very Low Birth Weight (<1500 grams) and Extremely Low BirthWeight (ELBW) are at greater risk for various cognitive deficits including planning ability.

**Neuropsychological Tests**

There are a variety of neuropsychological tests which can be used to measure variance of planning ability between the subject and controls.

- Tower of Hanoi (TOH-R), a puzzle invented in 1883 by the French mathematician Édouard Lucas. There are different variations of the puzzle, the classic version consists of three rods and usually seven to nine discs of subsequently smaller size. Planning is a key component of the problem solving skills necessary to achieve the objective, which is to move the entire stack to another rod, obeying the following rules:
  - Only one disk may be moved at a time.
  - Each move consists of taking the upper disk from one of the rods and sliding it onto another rod, on top of the other disks that may already be present on that rod.
  - No disk may be placed on top of a smaller disk.
- Tower of London (TOL) is another test that was developed in 1992 (Shallice 1992) specifically to detect deficits in planning as may occur with damage to the frontal lobe. Test participants with damage to the left anterior frontal lobe demonstrated planning deficits (*i.e.,* greater number of moves required for solution).

In test participants with damage to the right anterior, and left or right posterior areas of the frontal lobes showed no impairment. The results implicating the left anterior frontal lobes involvement in solving the TOL were supported in concomitant neuroimaging studies which also showed a reduction in regional cerebral blood flow to the left pre-frontal lobe. For the number of moves, a significant negative correlation was observed for the left prefrontal area: *i.e.,* subjects that took more time planning their moves showed greater activation in the left prefrontal area.

**Planning in Public Policy**

Public policy planning includes environmental, land use, regional, urban and spatial planning. In many countries, the operation of a town and country planning system is often referred to as "planning" and the professionals which operate the system are known as "planners".

It is a conscious as well as sub-conscious activity. It is "an anticipatory decision making process" that helps in coping with complexities. It is deciding future course of action from amongst alternatives. It is a process that involves making and evaluating each set of interrelated decisions. It is selection of missions, objectives and "translation of knowledge into action." A planned performance brings better results compared to an unplanned one. A manager's job is planning, monitoring and controlling. Planning and goal setting are important traits of an organization. It is done at all levels of the organization. Planning includes the plan, the thought process, action, and implementation. Planning gives more power over the future. Planning is deciding in advance what to do, how to do it, when to do it, and who should do it. This bridges the gap from where the organization is to where it wants to be. The planning function involves establishing goals and arranging them in logical order. A well planned organization achieve faster goals than the ones that don't plan before implementation.

## PLANNING PROCESS

*Patrick Montana and Bruce Charnov outline a three-step result-oriented process for planning:*

1. Choosing a destination
2. Evaluating alternative routes, and
3. Deciding the specific course of your plan.

In organizations, planning is a management process, concerned with defining goals for company's future direction and determining on the missions and resources to achieve those targets. To meet the goals, managers may develop plans such as a business plan or a marketing plan. Planning always has a purpose. The purpose may be achievement of certain goals or targets.

Main characteristics of planning in organizations are:

Planning increases the efficiency of an organization. It reduces the risks involved in modern business activities. It utilizes with maximum efficiency the available time and resources. The concept of planning is to identify what the organization wants to do by using the four questions which are "where are we today in terms of our business or strategy planning? Where are we going? Where do we want to go? How are we going to get there?..." ‘’

## URBAN PLANNING EDUCATION

Urban planning education is a practice of teaching and learning urban theory, studies, and professional practices. The interaction between public officials, professional planners and the public involves a continuous education on planning process. Community members often serve on a city planning commission, council or board. As a result, education outreach is effectively an ongoing cycle.

Formal education is offered as an academic degree in urban, city or regional planning, and awarded as a bachelor's degree, master's degree, or doctorate.

Since planning programmes are usually small, they tend not to be housed in distinct "planning schools" but rather, as part of an architecture school, a design school, a geography department, or a public policy school, since these are cognate fields. Generally speaking, planning programmes in architecture schools focus primarily on physical planning and design, while those in policy schools tend to focus on policy and administration.

As urban planning is such a broad and interdisciplinary field, a typical planning degree programme emphasizes breadth over depth, with core coursework that provides background for all areas of planning. Core courses typically include coursework in history/theory of urban planning, urban design, statistics, land use/planning law, urban economics, and planning practice.

Many planning degree programmes also allow a student to "concentrate" in a specific area of interest within planning, such as land use, environmental planning, housing, community development, economic development, historic preservation, international development, urban design, transportation planning, or geographic information systems (GIS). Some programmes permit a student to concentrate in real estate, however, graduate real estate education has changed giving rise to specialized real estate programmes.

**Bachelor's Degree in Urban Planning**

Bachelor of Planning (B.Plan) is an undergraduate academic degree designed to train applicants in various aspects of designing, engineering, managing and resolving challenges related to urban human settlements. It is awarded for a course of study that lasts up to four years and contextual to modern challenges of urbanisation. It goes into the techniques and theories related to settlement design starting at the site planning level of a neighbourhood and moving up to the regional city planning context. Understanding relations between built forms and the citizens in cities and rural areas, and their implications on local environment, supporting utilities, transport networks, and physical infrastructure forms the core of the planning course. With an engineering orientation, the graduates emerging as urban planners are equipped with not only tools for rational comprehensive planning but also participatory and social development.

The degree may be awarded as a Bachelor of Arts in Geography with an emphasis in urban planning, *Bachelor of Arts in Urban Planning*, or Bachelor of Science in Urban and Regional Planning. The distinction reflects university policies, or some universities may have greater course offerings in urban planning, design, sociology, or a related degree.

**Master of Urban Planning**

The Master of Urban Planning (MUP) is a two-year academic/professional master's degree that qualifies graduates to work as urban planners. Some schools offer the degree as a Master of City Planning (MCP), Master of Community Planning, Master of Regional Planning (MRP), Master of Town Planning (MTP), Master of Planning (MPlan), Master of Environmental Planning (MEP) or in some combination of the aforementioned (*e.g.,* Master of Urban and Regional Planning), depending on the program's specific focus. Some schools offer a Master of Arts or Master of Science in planning. Regardless of the name, the degree remains generally the same.

A thesis, final project or capstone project is usually required to graduate. Additionally, an internship component is almost always mandatory due to the high value placed on work experience by prospective employers in the field.

Like most professional master's degree programmes, the MUP is a terminal degree. However, some graduates choose to continue on to doctoral studies in urban planning or cognate fields. The Ph.D. is a research degree, as opposed to the professional MUP, and thus focuses on training planners to engage in scholarly activity directed towards providing greater insight into the discipline and underlying issues related to urban development.

**Accreditation in North America**

Accreditation is a system for recognizing educational institutions and professional programmes affiliated with those institutions for a level of performance, integrity, and quality.

The Planning Accreditation Board is the sole accreditor of planning programmes in the United States. The Planning Accreditation Board (PAB) accredits graduate and undergraduate planning programmes in the United States and Canada.

As of January 1, 2016, PAB accredits 15 undergraduate programmes and 71 graduate programmes in 75 North America Universities.

- The Planning Accreditation Board (PAB) accredits university programmes in North America leading to bachelors and masters degrees in planning. The accreditation process is based on standards approved by the PAB and its sponsoring organizations: the American Planning Association (APA); the American Institute of Certified Planners (AICP) (the professional planners' institute within the American Planning Association); and the Association of Collegiate Schools of Planning (ACSP).

Graduation from a PAB accredited programme allows a graduate to sit for the American Institute of Certified Planners (AICP) Certification Exam earlier in their career than a student with a degree from a non-accredited programme or school.

Programmes that desire accreditation through the PAB must apply for candidacy status. The programme seeking candidacy must demonstrate that they meet the five preconditions to accreditation.

*The five preconditions are:*

- Programme graduation of at least 25 students in the degree.
- Program's parent institution must be accredited by an institutional accrediting body recognized by the Council for Higher Education Accreditation (CHEA).
- Formal title of programme and degree offered must include the term "Planning".
- Undergraduate programmes must offer 4 full-time years of study or equivalent, while graduate programmes must be 2 full-time years of study or equivalent.
- Program's primary goal is to educate students to become practicing planning professionals.

Once the preconditions have successfully been met by the programme, the programme must complete and submit a Self-Study Report. Through the Self-Study Report, the programme assesses their performance and compliance with PAB's accreditation standards. This report serves as the basis of review for the Planning Accreditation Board, along with a formal meeting with the Programme Administrator at the Board meeting.

If candidacy is awarded, the Planning Accreditation Board will send a three-member team to visit and formally review the programme during a semester. The three member team will meet with faculty, staff, students, and members of the local planning community.

The team will then submit a Site Visit Report to the Planning Accreditation Board. During the meeting of the Planning Accreditation Board, the board will review the Self-Study Report, Site Visit Report and other documentation and meet with the Programme Administrator.

At the conclusion of the meeting, the Board decides if the programme is awarded accreditation and the length of accreditation. Accreditation length is dependent on the extent the programme complies with requirements of the Planning Accreditation Board, with the maximum length awarded is 7 years. Programmes must re-apply for accreditation in the year their accreditation term expires.

**Accredited Planning Programmes**

For current information on PAB accredited programmes, please visit the Planning Accreditation Board web site.

***United States of America and Puerto Rico***

| School | City | State | Undergraduate | Since | Graduate | Since | Accreditation Through |
|---|---|---|---|---|---|---|---|
| Alabama A&M University | Normal | AL | Bachelor of Science in Urban Planning | 1986 | Master of Urban & Regional Planning | 1976 | December 31, 2018 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Arizona State University | Tempe | AZ | Bachelor of Science in Planning *[no longer accredited]* | 2002 | Master of Urban & Environmental Planning | 1992 | December 31, 2018 |
| University of Arizona | Tucson | AZ | *not offered* | | Master of Science in Planning | 1998 | December 31, 2017 |
| California Polytechnic State University | San Luis Obispo | CA | Bachelor of Science in City & Regional Planning | 1973 | Master of City & Regional Planning | 1993 | December 31, 2019 |
| California State Polytechnic University, Pomona | Pomona | CA | Bachelor of Science in Urban & Regional Planning | 1970 | Master of Urban & Regional Planning | 1972 | December 31, 2022 |
| San Jose State University | San Jose | CA | *not offered* | | Master of Urban Planning | 1972 | December 31, 2018 |
| University of California, Berkeley | Berkeley | CA | *not offered* | | Master of City Planning | 1960 | December 31, 2017 |
| University of California, Irvine | Irvine | CA | *not offered* | | Master of Urban & Regional Planning | 1998 | December 31, 2021 |
| University of California, Los Angeles | Los Angeles | CA | *not offered* | | Master of Urban & Regional Planning | 1971 | December 31, 2019 |
| University of Southern California | Los Angeles | CA | *offered but not accredited* | | Master of Planning | 1967 | December 31, 2021 |
| University of Colorado, Denver | Denver | CO | *not offered* | | Master of Urban & Regional Planning | 1975 | December 31, 2023 |
| Florida Atlantic University | Fort Lauderdale | FL | *offered but not accredited* | 2011 | Master of Urban & Regional Planning | 1995 | December 31, 2018 |
| Florida State University | Tallahassee | FL | *not offered* | | Master of Science in Planning | 1965 | December 31, 2017 |
| University of Florida | Gainesville | FL | *not offered* | | Master of Arts in Urban & Regional Planning | 1978 | December 31, 2019 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Georgia Institute of Technology | Atlanta | GA | *not offered* | | Master of City & Regional Planning | 1969 | December 31, 2019 |
| University of Hawaii | Honolulu | HI | *not offered* | | Master of Urban & Regional Planning | 1981 | December 31, 2020 |
| University of Illinois at Chicago | Chicago | IL | *offered but not accredited* | | Master of Urban Planning & Policy | 1979 | December 31, 2019 |
| University of Illinois at Urbana-Champaign | Champaign | IL | Bachelor of Arts in Urban Planning | 1953 | Master of Urban Planning | 1945 | December 31, 2021 |
| Ball State University | Muncie | IN | Bachelor of Urban Planning & Development | 1995 | Master of Urban & Regional Planning | 1993 | December 31, 2017 |
| Iowa State University | Ames | IA | Bachelor of Science in Community & Regional Planning | 1979 | Master of Community & Regional Planning | 1979 | December 31, 2017 |
| University of Iowa | Iowa City | IA | *not offered* | | Master of Urban & Regional Planning | 1970 | December 31, 2020 |
| Kansas State University | Manhattan | KS | *not offered* | | Master of Regional & Community Planning | 1961 | December 31, 2022 |
| University of Kansas | Lawrence | KS | *not offered* | | Master of Urban Planning | 1983 | December 31, 2017 |
| University of Louisville | Louisville | KY | *not offered* | | Master of Urban Planning | 2010 | December 31, 2016 |
| University of New Orleans | New Orleans | LA | *offered but not accredited* | | Master of Urban & Regional Planning | 1976 | December 31, 2018 |
| Morgan State University | Baltimore | MD | *not offered* | | Master of City & Regional Planning | 1973 | December 31, 2020 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| University of Maryland at College Park | College Park | MD | *not offered* | | Master of Community Planning | 1978 | December 31, 2020 |
| Harvard University | Cambridge | MA | *not offered* | | Master in Urban Planning | 1923–1984; 1997 | December 31, 2019 |
| Massachusetts Institute of Technology | Cambridge | MA | *offered but not accredited* | | Master in City Planning | 1932 | December 31, 2018 |
| Tufts University | Medford | MA | *not offered* | | Master of Arts in Urban, Environmental Policy & Planning | 2004 | December 31, 2018 |
| University of Massachusetts Amherst | Amherst | MA | *not offered* | | Master of Regional Planning | 1987 | December 31, 2019 |
| Eastern Michigan University | Ypsilanti | MI | Bachelor of Science or Arts in Urban & Regional Planning | 1998 | *offered but not accredited* | | December 31, 2020 |
| Michigan State University | East Lansing | MI | Bachelor of Science in Urban & Regional Planning | 1959 | Master in Urban & Regional Planning | 1959 | December 31, 2019 |
| University of Michigan | Ann Arbor | MI | *not offered* | | Master of Urban Planning | 1968 | December 31, 2022 |
| Wayne State University | Detroit | MI | *not offered* | | Master of Urban Planning | 1975–1985; 1997 | December 31, 2019 |
| University of Minnesota | Minneapolis | MN | *offered but not accredited* | | Master of Urban & Regional Planning | 1982 | December 31, 2020 |
| Jackson State University | Jackson | MS | *not offered* | | Master of Arts in Urban & Regional Planning | 2010 | December 31, 2020 |
| Missouri State University | Springfield | MO | Bachelor of Science in Planning | 2004 | *not offered* | | December 31, 2018 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| University of Nebraska-Lincoln | Lincoln | NE | *not offered* | | Master of Community & Regional Planning | 1978 | December 31, 2018 |
| Rutgers, The State University of New Jersey | New Brunswick | NJ | *offered but not accredited* | | Master of City & Regional Planning | 1968 | December 31, 2021 |
| University of New Mexico | Albuquerque | NM | *offered but not accredited* | | Master of Community & Regional Planning | 1987 | December 31, 2021 |
| Columbia University | New York | NY | *not offered* | | Master of Science in Urban Planning | 1945 | December 31, 2019 |
| Cornell University | Ithaca | NY | *offered but not accredited* | | Master of Regional Planning | 1959 | December 31, 2021 |
| Hunter College, City University of New York | New York | NY | *not offered* | | Master in Urban Planning | 1969 | December 31, 2017 |
| New York University | New York | NY | *not offered* | | Master of Urban Planning | 1961 | December 31, 2019 |
| Pratt Institute | Brooklyn | NY | *not offered* | | Master of Science in City & Regional Planning | 1962 | December 31, 2020 |
| University at Albany, State University of New York | Albany | NY | *offered but not accredited* | | Master of urban & Regional Planning | 2000 | December 31, 2023 |
| University at Buffalo, State University of New York | Buffalo | NY | *not offered* | | Master of Urban Planning | 1988 | December 31, 2021 |
| East Carolina University | Greenville | NC | Bachelor of Science in Urban & Regional Planning | 2003 | *not offered* | | December 31, 2019 |
| The University of North Carolina at Chapel Hill | Chapel Hill | NC | *not offered* | | Master of City & Regional Planning | | December 31, 2019 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Cleveland State University | Cleveland | OH | *not offered* | | Master of Urban Planning, Design & Development | 1998 | December 31, 2019 |
| Ohio State University | Columbus | OH | Bachelor of Science in City and Regional Planning** | | Master of City & Regional Planning | 1961 | December 31, 2018 |
| University of Cincinnati | Cincinnati | OH | Bachelor of Urban Planning | 1966 | Master of Community Planning | 1964 | December 31, 2019 |
| University of Oklahoma | Norman | OK | *not offered* | | Master of Regional & City Planning | 1957–1966; 1992 | December 31, 2016 |
| Portland State University | Portland | OR | *not offered* | | Master of Urban & Regional Planning | 1980 | December 31, 2020 |
| University of Oregon | Eugene | OR | *not offered* | | Master of Community & Regional Planning | 1970 | December 31, 2016 |
| Indiana University of Pennsylvania | Indiana | PA | Bachelor of Science in Regional Planning | | *not offered* | | December 31, 2016 |
| Temple University | Philadelphia | PA | Bachelor of Science in Community Development | | Master of Science in City & Regional Planning | 2002 | December 31, 2020 |
| University of Pennsylvania | Philadelphia | PA | *not offered* | | Master of City Planning | 1969 | December 31, 2019 |
| Clemson University | Clemson | SC | *not offered* | | Master of City & Regional Planning | 1972 | December 31, 2018 |
| University of Memphis | Memphis | TN | *not offered* | | Master of City & Regional Planning | 1981 | December 31, 2020 |
| Texas A&M University | College Station | TX | *not offered* | | Master of Urban Planning | 1968 | December 31, 2018 |
| Texas Southern University | Houston | TX | *not offered* | | Master of Urban Planning & Environmental Policy | 2009 | December 31, 2018 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| The University of Texas at Arlington | Arlington | TX | *not offered* | | Master of City & Regional Planning | 1978 | December 31, 2018 |
| The University of Texas at Austin | Austin | TX | *not offered* | | Master of Science in Community & Regional Planning | 1969 | December 31, 2017 |
| University of Utah | Salt Lake City | UT | *offered but not accredited* | | Master of City & Metropolitan Planning | 2011 | December 31, 2021 |
| University of Virginia | Charlottesville | VA | Bachelor of Urban & Environmental Planning | 1963 | Master of Urban & Environmental Planning | 1968 | December 31, 2020 |
| Virginia Commonwealth University | Richmond | VA | *not offered* | | Master of Urban & Regional Planning | 1977 | December 31, 2021 |
| Virginia Polytechnic Institute & State University | Blacksburg | VA | *not offered* | | Master of Urban & Regional Planning | 1961 | December 31, 2019 |
| Eastern Washington University | Cheney | WA | Bachelor of Arts in Urban & Regional Planning | 1983 | Master of Urban & Regional Planning | 1983 | December 31, 2019 |
| University of Washington | Seattle | WA | *offered but not accredited* | | Master of Urban Planning | 1936 | December 31, 2020 |
| Western Washington University | Bellingham | WA | Bachelor of Arts in Urban Planning and Sustainable Development | | *not offered* | 2016 | December 31, 2020 |
| University of Wisconsin-Madison | Madison | WI | *not offered* | | Master of Science in Urban & Regional Planning | 1962 | December 31, 2016 |
| University of Wisconsin–Milwaukee | Milwaukee | WI | *not offered* | | Master of Urban Planning | 1974 | December 31, 2018 |
| Universidad de Puerto Rico | San Juan | Puerto Rico | *Offered but Not Accredited* | 1977 | Maestría en Planificación Urbana y | 1997 | December 31, 2017 |

## *Canada*

| School | City | Province/ Territory | Undergraduate | Since | Graduate | Since | Accreditation Through |
|---|---|---|---|---|---|---|---|
| Université de Montréal | Montréal | Quebec | Baccalauréat Spécialisé en Urbanisme | 1982 | Maîtrise en Urbanisme | 1965 | December 31, 2013 |
| University of British Columbia | Vancouver | British Columbia | *Not Offered* | | Master of Arts in Planning, Master of Science in Planning, and Master of Community and Regional Planning | 1970 | December 31, 2017 |
| University of Guelph | Guelph | Ontario | *Not Offered* | | Master of Science in Rural Planning & Development, and Master of Planning (MPlan) | | |
| McGill University | Montreal | Quebec | *Not Offered* | | Master of Urban Planning | | |
| Université du Québec à Montréal | Montreal | Quebec | Baccalauréat ès sciences spécialisé en urbanisme | | *Master in Urban planning* | | |
| Queen's University | Kingston | Ontario | *Not Offered* | | Master of Urban & Regional Planning | | |
| Ryerson University | Toronto | Ontario | Bachelor of Urban and Regional Planning, Post Degree Bachelor of Urban and Regional Planning, and Post Diploma Bachelor of Urban and Regional Planning | | Master of Urban & Regional Planning | | |
| University of Saskatchewan | Saskatoon | Saskatchewan | Bachelor of Arts in Regional and Urban Planning | | *Not Offered* | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Simon Fraser University | Burnaby | British Columbia | *Not Offered* | | Master of Resource Management (Planning) | | |
| University of Toronto | Toronto | Ontario | *Not Offered* | | Master of Science in Planning | | |
| University of Waterloo | Waterloo | Ontario | Bachelor of Environmental Studies in Planning, and Bachelor of Environmental Studies in Co-Op Planning | | Master of Arts in Planning, Master of Environmental Studies in Planning, and Master of Planning (MPlan) | | |
| York University | Toronto | Ontario | *Not Offered* | | Master of Environmental Studies in Planning | | |
| University of Northern British Columbia | Prince George | British Columbia | Bachelor of Planning | | *Not Offered* | | |
| University of Manitoba | Winnipeg | Manitoba | *Not Offered* | | Master of City Planning | | |
| Université Laval | Quebec City | Quebec | *Not Offered* | | Maîtrise en aménagement du territoire et développement régional | | |
| Dalhousie University | Halifax | Nova Scotia | Bachelor of Community Design with majors in; Environmental Planning, Urban Design and Planning, and Sustainability | | Master of Planning | | |
| University of Calgary | Calgary | Alberta | *Not Offered* | | Master of Planning | | |
| Vancouver Island University | Nanaimo | British Columbia | *Not Offered* | | **Master of Community Planning | | |
| University of Alberta | Edmonton | Alberta | Bachelor of Arts in Planning, and Bachelor of Science in Planning | | Master of Science in Planning | | |

## ACCREDITATION IN THE REPUBLIC OF INDIA

Though Planning as a Profession is not a recognized profession under Indian law, the profession was started long back with School of Planning and Architecture in 1941 as a Department of Architecture of Delhi College of Engineering now the Delhi Technological University. It was later integrated with the School of Town and Country Planning which was established in 1955 by the Government of India to provide facilities for rural, urban and regional planning. On integration, the School was renamed as school of planning and architecture in 1959. Today it is one of the premier schools of pursuing planning studies at bachelor, masters and post doctorate levels. The Institute of Town Planners, India set up on the lines of the [Royal Town Planning Institute, London] is the body representing planning professionals in India. A small group formed itself into an Indian Board of Town Planners which after three years of continuous work formed the Institute of Town Planners, India. The Institute which was established in July 1951, Today, has a membership of over 2,800, apart from a sizable number of student members, many of whom have qualified Associateship Examination (AITP) conducted by ITPI. Institutes under ITPI offers a 4-year undergraduate degree in Planning.

| School | Location | Undergraduate | Graduate |
|---|---|---|---|
| Visvesvaraya National Institute of Technology, Nagpur | Nagpur, Maharashtra | Not Offered | Master of Technology in Urban Planning |
| School of Planning & Architecture, Delhi (SPA-D)<br><br>(Autonomous institution established by Ministry of Human Resource Development (India)) | New Delhi | Bachelor of Planning | Master of Planning |
| School of Planning and Architecture, Vijayawada (SPA-V)<br><br>(Autonomous institution established by Ministry of Human Resource Development (India)) | Vijayawada, AP | Bachelor of Planning | Master of Planning |
| School of Planning and Architecture, Bhopal (SPA-B)<br><br>(Autonomous institution established by Ministry of Human Resource Development (India)) | Bhopal, MP | Bachelor of Planning | Master of Planning |
| School of Planning, Bhaikaka Centre for Human Settlements, APIED | Vallabh Vidhyanagar, Anand, Gujarat | not offered | Master of Urban Planning |
| Maulana Azad National Institute of Technology, Bhopal | Bhopal, MP | Bachelor of Planning | Master of Urban Development & Planning |
| Sardar Vallabhbhai National Institute of Technology, Surat | Surat, Gujarat | *not offered* | Master of Technology in Urban Planning |

| | | | |
|---|---|---|---|
| Indian Institute of Technology Kharagpur | Kharagpur, WB | *not offered* | Master of Planning |
| Indian Institute of Technology Roorkee | Roorkee, UK | *not offered* | Master of Planning |
| Pune Institute of Engineering & Technology | Pune, MH | *not offered* | Master of Planning |
| College of Engineering, Pune | Shivaji Nagar, Pune, MH | B.Tech (Planning) | Master of Urban & Regional Planning |
| Centre for Environmental Planning and Technology | Ahmadabad, GJ | Bachelor of Planning | Master of Planning |
| Guru Nanak Dev University | Amritsar, PB | Bachelor of Technology (Planning) | Master of Urban Planning |
| Anna University | Chennai, TN | *not offered* | Master of Planning |
| University of Mysore | Mysore, KN | *not offered* | Master of Urban & Regional Planning |
| Bengal Engineering and Science University | Howrah, WB | *not offered* | Master of Urban & Regional Planning |
| Malaviya National Institute of Technology, | Jaipur (Rajasthan) | not offered | Master of Planning |
| College of Engineering, | Thiruvananthapuram (Kerala) | not offered | M. Plan (Housing) |
| Birla Institute of Technology | Ranchi (MESRA), Jharkhand | not offered | Master of Urban Planning (Town Planning) |
| Jawaharlal Nehru Technology University | Hyderabad, AP | Bachelor of Planning | Master of Planning |

## Rankings

### *United States of America*

While there is no official ranking of the graduate programmes for planning, planning-community site Planetizen and college guide Best Colleges publish periodic lists of the Top US planning graduate programmes, and the AICP provides a listing of currently- and formerly-accredited programmes sorted by proportion of recent graduates passing its certification exam. The Best Colleges and Planetizen top-ranked schools aren't all at the top of the exam pass-rate rankings, but the two guides broadly agree on which schools are best.

| University (in alphabetical order) | Best Colleges Rank (2013) | Planetizen Rank (2015) | Planetizen Rank (2012) | Planetizen Rank (2009) | Planetizen Rank (2006) | AICP Pass Rate |
|---|---|---|---|---|---|---|
| Cornell University Master of Regional Planning | 7 | 7 | 2 | 6 | 7 | 82% (58/71) |
| Georgia Institute of Technology Master of City and Regional Planning | NR | 5 | 8 | NR | NR | 84% (132/157) |
| Harvard University Master in Urban Planning | 4 | 10 | NR | 7 | 4 | 79% (44/56) |
| Massachusetts Institute of Technology Master in City Planning | 8 | 1 | 1 | 1 | 1 | 82% (65/75) |
| Rutgers, The State University of New Jersey Master of City and Regional Planning | 6 | 6 | 3 | 4 | 8 | 76% (102/135) |
| University of California, Berkeley Master of City Planning | 9 | 2 | 4 | 2 | 2 | 90% (62/69) |
| University of California, Los Angeles Master of Urban and Regional Planning | 2 | 4 | 9 | 8 | 6 | 84% (57/68) |
| University of Illinois at Urbana-Champaign Master of Urban Planning | 5 | 3 | 5 | 5 | 10 | 88% (75/85) |
| University of North Carolina, Chapel Hill Master of City and Regional Planning | 10 | 8 | 6 | 3 | 3 | 93% (95/102) |
| University of Pennsylvania Master of City Planning | 3 | NR | 10 | 10 | 5 | 89% (151/169) |
| University of Southern California Master of Planning | 1 | 9 | 7 | 9 | 9 | 69% (68/99) |

## INDIVIDUAL LEARNING PLAN

Individual Learning Plan or ILP is a user (student) specific programme or strategy of education or learning that takes into consideration the student's strengths and weaknesses. While normal classroom or distance education is based on the premise that all should get equal attention (a democratic principle), be exposed to same curriculum and evaluated on the same pattern ('One size fits all'), ILP presumes that the needs of individual students are different, and thus, must be differently addressed. Emphasis on the student's role in the learning experience has been shown in research to be crucial to a productive learning experience.

The Individual Learning Plan can also be used by an individual on their own or as part of a community of interest, a team or an organization to manage learning over the course of their life.

This is explored further in the article "Learning Plan."

Adopted by many institutes as a teaching methodology, ILP for a student is generated after interaction between the student and the teacher, and is based upon assessment made therein. Further, ILP:

- Incorporates long-term goals of the student
- Synthesizes with the larger educational framework
- Gives credence to the student's aspirations - cultural, artistic, social, or personal

Individual Learning Plans are mandatory to complete for all students in Alaska (2010), Arizona (2008), Colorado (2010), Connecticut (2012), Delaware (2007), Georgia (2009), Hawaii (2009), Idaho (2011), Indiana (2006), Iowa (2008), Kentucky (2002), Louisiana (1998), Maryland (2008), Massachusetts (2007), Michigan (2009), Minnesota (2013), Missouri (2006), Oregon (2011), Rhode Island (2011), South Carolina (2006), South Dakota (2010), Virginia (2013), Washington (2000), West Virginia (1996), and Wisconsin (2013) in order to graduate.

### Purpose

*The Individual Learning plan has many purposes, including:*

- Discovery of many careers, beginning in the sixth grade
- Career matching making services
- Developing education plans
- Creating, maintaining and changing resumes
- Setting personal goals and keeping these insight as school progresses
- Saving and reflecting on activity including community service, work experience, career planning activities, and extra curricular activities
- Exploring colleges and postsecondary opportunities that fit with desired career, and other life goals
- Collecting personal information including assessment results, advising activities demographic information and educational history
- Keeping track of all courses taken.

Also, the Individual learning plan is set to establish college and career readiness throughout middle school and high school. According to the Alliance for Excellent Education, the graduating classes were more prepared for college-level work (in all four content areas of Mathematics, Reading, English and Science) after students created/used and ILP. On the other hand, in this same research, it was determined that nearly 16,200 students did not graduate in 2012, a fact that equates to a $4.2 billion lifetime earning loss for that class of students. This could be due to the lack of College and Career Readiness throughout states that do not implement Individual Learning Plans. The lack of college and career readiness does not only affect the student themselves, but also communities as a group. The lack of these critical skills in high school students can deny the community jobs and business due to the unfulfilled need for qualified employees.

## Functions

### *Resume Builder*

Tools located within the ILP can help you store and organize information more successfully, with no question about career development, activities, and experiences. Once a student saves information about a chosen career, school or other objective, students can record their opinion, interest, and thoughts. Also, through Career Matchmaker services students can save and name many different matchmaker results, and save which they feel they could be more successful doing. With this comes a tool very common and frequently used, the Resume Builder.

This enables help on creating personalized, and more professional-looking resumes. Information saved to an individual's ILP is automatically shared with the Resume Builder.

### *Career Match Maker/Career Exploration*

The Career Exploration section offers a number of assessment tools to help you discover your skills, abilities, and learning preferences. With this it uses that information to identify suitable career options for the students using this tool. Career Cruising and Career Matchmaker can help a student better understand how interest and career choices go hand in hand, due to shared likes, skills, and interest to occupations in certain programmes.

Throughout the Career Cruising there are hundreds of occupations with profiles. These profiles include interviews with people within the career, salary information, job outlook, and locations. In the employee interviews, located in the profiles of each career, they ask key questions including about their typical workday, pros and cons, and advise for students interested in pursuing that career. There are a couple ways to search for careers, including by keyword, index, school, subjects, career cluster, or career selector given to you after using Career Matchmaker.

***College Search***

Individual Learning Plan provides detailed school profiles for thousands of 2 and 4 year colleges, career and technical schools across the country. You can either search for particular schools/ programmes, or use the School Selector to find schools that meet a variety of criteria. You can see which schools offer your chosen programme, cost of enrollment, and the type of the community.

**Teacher's Point of View**

The process of advising students is a shared responsibility that can have a significant impact on transition to postsecondary education. Teachers are required to lead, guide and direct students in setting and reaching goals. This can be done through ILP or by day-to-day contact with students. This leads to several opportunities for teachers to share college and career readiness dialogue that assists students in focusing on their futures. This also enables students and teachers to focus on one specific goal for on student, therefore the conversations are more centered around that individual student. When teachers utilize the resources in the Individual Learning Plan (ILP) during classroom instruction, students can see the relevance of the content as it relates to their plans for life after high school. This may lead students to become more motivated on reaching their goals. There are many benefits of encompassing the ILP into classroom instruction include:

- More obvious sight of direct connections between classes and future goals made for students
- More individual centered learning for each student for more successful engagement
- More motivated students and more willing to take the steps necessary to reach goals through focusing on the work and assignment, which will lead them to gain the skills and knowledge necessary for a career
- Teachers see a more direct correlation between the content they are teaching and the students' goals for college and/or the workforce

The role of Counselors, Teachers, and other staff who work with students is to guide, facilitate, and support in the process of developing their ILPs. Also, ensure that students are reflecting on their previous records in their ILP.

**UNESCO INTERNATIONAL INSTITUTE FOR EDUCATIONAL PLANNING**

The International Institute for Educational Planning (IIEP – UNESCO) is an arm of UNESCO created in 1963 in Paris, France. It develops the capacities of education actors to plan and manage their education systems through its programmes of training, technical assistance, policy research and knowledge sharing.

**Origins**

A Committee chaired by Sir Alexander Carr-Saunders with Guy Benveniste as Rapporteur met in June 1962 and recommended the creation of the Institute. The

UNESCO General Conference adopted their recommendations in the fall of 1962 and the French Government provided a building to house this new institution. Philip Hall Coombs who had been the first Assistant Secretary of State for Educational and Cultural Affairs in the Kennedy administration, was appointed as its first Director in 1963. The Institute, while an arm of UNESCO, was established as a semi independent institution with its own board headed by a chair appointed by the Director General of UNESCO. Originally the Institute was financed jointly by UNESCO and the World Bank (IBRD) since that second institution was just beginning to finance educational projects in developing countries. Help was also provided in those first years by the Ford Foundation. Later, it was integrated more closely with, and financed mainly by, UNESCO. The Institute organized its first major seminar in the spring of 1964 when 80 participants from Latin America attended meetings in Paris. The first publication from this encounter was issued in 1965.

**Mission**

As the only specialized organization with the mandate to support educational policy, planning and management, IIEP plays a unique role within the United Nations system. IIEP's mission is to strengthen the capacity of countries. It helps UNESCO's Member States to manage their educational system and to achieve the Education 2030 Agenda. The International Institute for Educational Planning offers training in educational planning and management, but also explores fields such as statistical tools for educational planning, strategies and policy options, projects, budgets, monitoring and evaluating educational quality and access. Its programmes are designed for planners, policy-makers and researchers. IIEP targets both institutions and individuals, and works in both the national and international arenas. IIEP's research projects identify new approaches that planners could adopt to improve equity, access and quality in the various educational sectors. Costs and financing, along with governance and management, are also important research fields at IIEP. IIEP's technical assistance projects offer on-site coaching to ministry planning departments, so that they can quickly become autonomous in the performance of their basic duties. By building the capacity of individuals and of local, regional and national institutions, IIEP's technical assistance enables countries to make the most of their own expertise and to minimize their use of outside consultants. For instance, IIEP has created tailored programmes to help governments in emergencies and fragile contexts, to maintain or rebuild their educational system.

**IIEP and Education Sector Planning**

IIEP supports ministries of education to plan and prepare their education sector plans through long-term technical involvement or more focused interventions. Making progress in education demands that countries have a clear vision of their priorities and how to achieve these. Many ministries therefore prepare strategic plans, which reflect this vision and help mobilize people and resources. Strategic planning guides educational development by giving a common vision and shared priorities. Educational planning is both visionary and pragmatic, engaging a wide range of actors in defining education's future and mobilizing resources to reach its goals. A wide range of ministries worked in partnership with IIEP to develop their plans. In some countries, the Institute supports the whole process of formulation, implementation, monitoring and evaluation of these plans; in others, it offers advice and assistance in specific areas as requested by the ministry. In all cases, the joint efforts is aimed at strengthening the capacity and autonomy of ministries and their staff. IIEP gives special attention to working with countries faced with the challenges of emergencies and reconstruction (sometimes referred to as 'fragile' contexts) which are farthest away from achieving the EFA and the MDG goals.

**LESSON PLAN**

A lesson plan is a teacher's detailed description of the course of instruction or "learning trajectory" for a lesson. A daily lesson plan is developed by a teacher to guide class learning. Details will vary depending on the preference of the teacher, subject being covered, and the needs of the students. There may be requirements mandated by the school system regarding the plan. A lesson plan is the teacher's guide for running a particular lesson, and it includes the goal (what the students are supposed to learn), how the goal will be reached (the method, procedure) and a way of measuring how well the goal was reached (test, worksheet, homework etc.).

**Development**

*While there are many formats for a lesson plan, most lesson plans contain some or all of these elements, typically in this order:*

- *Title* of the lesson
- *Time* required to complete the lesson
- List of required *materials*
- List of *objectives*, which may be *behavioural objectives* (what the student can *do* at lesson completion) or *knowledge objectives* (what the student *knows* at lesson completion)
- The *set* (or lead-in, or bridge-in) that focuses students on the lesson's skills or concepts—these include showing pictures or models, asking leading questions, or reviewing previous lessons
- An *instructional component* that describes the sequence of events that make up the lesson, including the teacher's instructional input and, where appropriate, guided practice by students to consolidate new skills and ideas
- *Independent practice* that allows students to extend skills or knowledge on their own
- A *summary*, where the teacher wraps up the discussion and answers questions
- An *evaluation* component, a test for mastery of the instructed skills or concepts—such as a set of questions to answer or a set of instructions to follow

- A *risk assessment* where the lesson's risks and the steps taken to minimize them are documented
- An *analysis* component the teacher uses to reflect on the lesson itself—such as what worked and what needs improving
- A *continuity* component reviews and reflects on content from the previous lesson.

Lesson Plan Phases There are eight lesson plan phases that are design to provide many many opportunities for teachers to recognize and correct students' misconceptions while extending understanding for future lessons. Phase 1: Introduction Phase 2: Foundation Phase 3: Brain Activation Phase 4: Body of New Information Phase 5: Clarification Phase 6: Practice and Review Phase 7: Independent Practice Phase 8: Closure

*Herbartian Approach: John Fedrick Herbert (1776-1841):*

- *Preparation/Instruction:* It pertains to preparing and motivating children to the lesson content by linking it to the previous knowledge of the student, by arousing curiosity of the children and by making an appeal to their senses. This prepares the child's mind to receive new knowledge. "To know where the pupils are and where they should try to be are the two essentials of good teaching." Lessons may be started in the following manner: a. Two or three interesting but relevant questions b. Showing a picture/s, a chart or a model c. A situation Statement of Aim: Announcement of the focus of the lesson in a clear, concise statement such as "Today, we shall study the..."
- *Presentation/Development:* The actual lesson commences here. This step should involve a good deal of activity on the part of the students. The teacher will take the aid of various devices, *e.g.,* questions, illustrations, explanation, expositions, demonstration and sensory aids, etc. Information and knowledge can be given, explained, revealed or suggested. The following principles should be kept in mind. a. Principle of selection and division: This subject matter should be divided into different sections. The teacher should also decide as to how much he is to tell and how much the pupils are to find out for themselves. b. Principle of successive sequence: The teacher should ensure that the succeeding as well as preceding knowledge is clear to the students. c. Principle of absorption and integration: In the end separation of the parts must be followed by their combination to promote understanding of the whole.
- *Association comparison:* It is always desirable that new ideas or knowledge be associated to daily life situations by citing suitable examples and by drawing comparisons with the related concepts. This step is important when we are establishing principles or generalizing definitions.
- *Generalizing:* This concepts is concerned with the systematizing of the knowledge learned. Comparison and contrast lead to generalization. An effort should be made to ensure that students draw the conclusions themselves. It should result in student's own thinking, reflection and experience.

- *Application:* It requires a good deal of mental activity to think and apply the principles learn to new situations. Knowledge, when it is put to use and verified, becomes clear and a part of the student's mental make-up.
- *Recapitulation:* Last step of the lesson plan, the teacher tries to ascertain whether the students have understood or grasped the subject matter or not. This is used for assessing/evaluating the effectiveness of the lesson by asking students questions on the contents of the lesson or by giving short objectives to test the student's level of understanding; for example, to label different parts on a diagram, etc.

***A Well-developed Lesson Plan***

A well-developed lesson plan reflects the interests and needs of students. It incorporates best practices for the educational field. The lesson plan correlates with the teacher's philosophy of education, which is what the teacher feels is the purpose of educating the students.

Secondary English programme lesson plans, for example, usually center around four topics. They are literary theme, elements of language and composition, literary history, and literary genre. A broad, thematic lesson plan is preferable, because it allows a teacher to create various research, writing, speaking, and reading assignments. It helps an instructor teach different literature genres and incorporate videotapes, films, and television programmes. Also, it facilitates teaching literature and English together. Similarly, history lesson plans focus on content (historical accuracy and background information), analytic thinking, scaffolding, and the practicality of lesson structure and meeting of educational goals. School requirements and a teacher's personal tastes, in that order, determine the exact requirements for a lesson plan.

*Unit plans* follow much the same format as a lesson plan, but cover an entire unit of work, which may span several days or weeks. Modern constructivist teaching styles may not require individual lesson plans. The unit plan may include specific objectives and timelines, but lesson plans can be more fluid as they adapt to student needs and learning styles.

Unit Planning is the proper selection of learning activities which presents a complete picture. Unit planning is a systematic arrangement of subject matter. Samford "A unit plan is one which involves a series of learning experiences that are linked to achieve the aims composed by methodology and contents". Dictionary of Education:"A unit is an organization of various activities, experiences and types of learning around a central problem or purpose developed cooperatively by a group of pupils under a teacher leadership involving planning, execution of plans and evaluation of results".

***Criteria of a Good Unit Plan***

- Needs, capabilities, interest of the learner should be considered.
- Prepared on the sound psychological knowledge of the learner.

- Provide a new learning experience; systematic but flexible.
- Sustain the attention of the learner till the end.
- Related to social and Physical environment of the learner.
- Development of learner's personality.

It is important to note that lesson planning is a thinking process, not the filling in of a lesson plan template. Lesson plan envisaged s a blue print, guide map for action, a comprehensive chart of classroom teaching-learning activities, an elastic but systematic approach for the teaching of concepts, skills and attitudes.

***Setting Objectives***

The first thing a teacher does is to create an objective, a statement of purpose for the whole lesson. An objective statement itself should answer what students will be able to do by the end of the lesson. Harry Wong states that, "Each [objective] must begin with a verb that states the action to be taken to show accomplishment. The most important word to use in an assignment is a verb, because verbs state how to demonstrate if accomplishment has taken place or not." The objective drives the whole lesson, it is the reason the lesson exists. Care is taken when creating the objective for each day's lesson, as it will determine the activities the students engage in. The teacher also ensures that lesson plan goals are compatible with the developmental level of the students. The teacher ensures as well that their student achievement expectations are reasonable.

***Types of Assignments***

The instructor must decide whether class assignments are whole-class, small groups, workshops, independent work, peer learning, or contractual:

- Whole-class—the teacher lectures to the class as a whole and has the class collectively participate in classroom discussions.
- Small groups—students work on assignments in groups of three or four.
- Workshops—students perform various tasks simultaneously. Workshop activities must be tailored to the lesson plan.
- Independent work—students complete assignments individually.
- Peer learning—students work together, face to face, so they can learn from one another.
- Contractual work—teacher and student establish an agreement that the student must perform a certain amount of work by a deadline.

These assignment categories (*e.g.,* peer learning, independent, small groups) can also be used to guide the instructor's choice of assessment measures that can provide information about student and class comprehension of the material. As discussed by Biggs (1999), there are additional questions an instructor can consider when choosing which type of assignment would provide the most benefit to students. These include:

- What level of learning do the students need to attain before choosing assignments with varying difficulty levels?

- What is the amount of time the instructor wants the students to use to complete the assignment?
- How much time and effort does the instructor have to provide student grading and feedback?
- What is the purpose of the assignment? (*e.g.,* to track student learning; to provide students with time to practice concepts; to practice incidental skills such as group process or independent research)
- How does the assignment fit with the rest of the lesson plan? Does the assignment test content knowledge or does it require application in a new context?
- Does the lesson plan fit a particular framework? For example, a Common Core Lesson Plan.

**BACKWARD DESIGN**

Backward Design is a method of designing educational curriculum by setting goals before choosing instructional methods and forms of assessment. Backward design of curriculum typically involves three stages:

1. Identify the results desired (big ideas and skills)
   – What should the students know, understand, and be able to do?
   – Consider the goals and curriculum expectations
   – Focus on the "big ideas" (principles, theories, concepts, point of views, or themes)
2. Determine acceptable levels of evidence that support that the desired results have occurred (culminating assessment tasks)
   – What will teachers accept as evidence that student understanding took place?
   – Consider culminating assessment tasks and a range of assessment methods (observations, tests, projects, etc.)
3. Design activities that will make desired results happen (learning events)
   – What knowledge and skills will students need to achieve the desired results?
   – Consider teaching methods, sequence of lessons, and resource materials

Backward design challenges "traditional" methods of curriculum planning. In traditional curriculum planning, a list of content that will be taught is created and/or selected. In backward design, the educator starts with goals, creates or plans out assessments and finally makes lesson plans. Supporters of backward design liken the process to using a "road map". In this case, the destination is chosen first and then the road map is used to plan the trip to the desired destination. In contrast, in traditional curriculum planning there is no formal destination identified before the journey begins.

The idea in backward design is to teach towards the "end point" or learning goals, which typically ensures that content taught remains focused and organized. This, in turn, aims at promoting better understanding of the content or processes to be learned for students. The educator is able to focus on addressing what the students need to learn, what data can be collected to show that the students have learned the desired outcomes (or learning standards) and how to ensure the students will learn. Although backward design is based on the same components of the ADDIE model, backward design is a condensed version of these components with far less flexibility.

**Curriculum Design, and Instructional Design**

Backward design is often used in conjunction with two other terms: curriculum design and instructional design.

Curriculum design is the act of designing or developing curricula for students. Curricula may differ from country to country and further still between provinces or states within a country. Curriculum is based on benchmark standards deemed important by the government. Typically, the time frame of attainment of these outcomes or standards is set by physical age.

Instructional design is a technology for the development of learning experiences and environments which promote the acquisition of specific knowledge and skill by students. In addition, instructional design models or theories may be thought of as frameworks for developing modules or lessons that increase and/or enhance the possibility of learning and encourage the engagement of learners so that they learn faster and gain deeper levels of understanding. There are numerous instructional design models available to instructors that hold significant importance when planning and implementing curriculum.

*Many of the models are quite similar in that they essentially all address the same four components in some form or another:*

- The learners;
- The learning objectives;
- The method of instruction; and
- Some form of assessment or evaluation.

Based around those components, the instructor then has the opportunity to choose the design model and stages that work best for them in their specific situation. This way they can achieve the appropriate learning outcomes or create a plan or strategy for improvement. As learners and instructors may vary, instructional design must be a good fit for both and therefore different models can have behavioural, cognitive or constructivist roots.

**History**

Ralph W. Tyler introduced the idea of "backward design" (without using this particular term) in 1949 when referring to a *statement of objectives*. A statement of objectives is used to indicate the kinds of changes in the student to be brought about so that instructional activities can be planned and developed in a way likely to attain these objectives.

The term "backward design" was introduced to curriculum design in 1998/9 by Jay McTighe and Grant Wiggins *(Understanding by Design)*. The somewhat idiosyncratic term is ultimately due to James S. Coleman, who in his *Foundations of Social Theory* (1990) used it to parallel the term "backward policing" which he coined for a policy which he found in the production process in Honda factories.

**Advantages**

*According to Doug Buehl (2000), advantages of backward design include:*

- Students are not as likely to become so lost in the factual detail of a unit that they miss the point of studying the original topic.
- Instruction looks towards global understandings and not just daily activities; daily lessons are constructed with a focus on what the overall "gain" from the unit is to be.
- Assessment is designed before lesson planning, so that instruction drives students towards exactly what they need to know.

**The Importance of Assessment**

The primary starting point for backward design is to become familiar with the standards/outcomes for the grade level and curriculum being taught. The second part of curriculum planning with backward design is finding appropriate assessments. It can be difficult for "traditional" educators to switch to this model because it is hard to conceptualize an assessment before deciding on lessons and instruction. The idea is that the assessments (formative and/or summative) should meet the initial goals identified.

Wiggins and McTighe (2008) also utilize the "WHERE" approach during the assessment stage of the process.

- W stands for students knowing *where they are heading*, *why they are heading there*, *what they know*, *where they might go wrong in the process*, and *what is required of them*.
- H stands for *hooking* the students on the topic of study.
- E stands for students *exploring* and *experiencing ideas* and being *equipped with the necessary understanding* to master the standard/outcome being taught.
- R stands for providing opportunities for students to *rehearse, revise*, and *refine* their work.
- E stands for student *evaluation*.

**Supporting Research Using Backward Design**

- Shumway and Barrett (2004) used the backward design model to strengthen pre-service teachers' attitudes towards teaching. The experience appears to have allowed the pre-service teachers to do exactly that after using both the backward design model and a modified backward design. These pre-service teachers became more excited about their teaching profession and became better prepared as student teachers through the backward design that they had experienced.

- In the article, *Essential Questions — Inclusive Answers* (C.M. Jorgenson, 1995), Souhegan High School followed the steps of a backwards design model to reach all levels of student ability and create a school that promoted full inclusion. They concluded that all involved had experienced a richer experience because of the implementation of the backward design model.
- In *An Integration of "Backwards Planning" Unit Design with the "Two Step" Lesson Planning Framework* (Jones et al., 2009), a framework for employing backward planning in designing individual lessons is provided. Educators are provided with an integrated framework and more importantly a case study of the backward lesson planning in action.
- In the article, *Backward Design* (Childre, Sands, and Pope, 2009), examples of backward design are shown improving learning at both the elementary and high school levels. The research targets the depth of understanding for all learners. The fact that much research avoids the inclusion of special needs students is noted. The traditional instructional approaches that fail to engage disabled students were not an issue when backward design was implemented. The backward design was found to provide meaning and relevance to all levels of students.

**Application**

*Here is a practical example of a 5th grade teacher developing a three-week unit on nutrition:*

- *Stage 1:* Identify desired results: Based on three curriculum expectations about nutrition (concepts about nutrition, elements of a balanced diet, and understanding eating patterns), the take-away message that the teacher wants his/her students to understand is "Students will use an understanding of the element of good nutrition to plan a balanced diet for themselves and others".
- *Stage 2:* Determine acceptable evidence: The teacher has created an authentic task in which students will design a 3-day meal plan for a camp that uses food pyramid guidelines. The goal is a tasty and nutritionally balanced menu.
- *Stage 3:* Plan learning experiences and instruction: The teacher first considers the knowledge and skills that students will need in order to complete the authentic assessment. Specifically, students will need to know about different food groups, human nutritional needs (carbohydrates, proteins, sugars, vitamins, minerals etc.), and about what foods provide these needs. They will need to know how to read nutrition labels. Resources will be a pamphlet from the UDSA on food groups, the health textbook, and a video "Nutrition for You". Teaching methods will include direct instruction, inductive methods, cooperative learning, and group activities.

**Criticisms**

*Although this approach is widely accepted, the following are criticisms of the backward design approach:*

- Difficulties in dealing with issues of validity and reliability
- Textbooks and content standards do not always explicitly highlight the key concepts that students should learn
- This approach provides teachers with little support about how to enhance understanding of their students and the way people learn
- Teachers may misunderstand and misinterpret what their students should learn and what the big ideas are
- Teacher effectiveness is measured more on the success of the students based on formulated assessments rather than ability to connect knowledge and skills to the needs and interests of students. Thus, lack of concern with social and cultural differences within the classroom
- This process promotes lesson design through deductive reasoning. It does not fit in well in a constructivist ontology where the multifaceted nature of each student warrants consideration in planning. Similarly, it leaves little room for improvisation.
- Teachers who know their curriculum and lesson trajectory that was led by Backwards Design may find that over adherence depletes their agility to focus on the learning experience and, with students or colleagues, induce new routes towards learning goals. A focus on principles of creative development such as contextual probing, improvisation, and juxtaposition may lead students to discover and know that which was unanticipated by the teacher or curriculum developers. In this, BD is incomplete or a potential recipe for student boredom.
- Desired results may fall short of student potential. This model assumes the relative level of students, yet students may have the capacity to go beyond desired results. In assuming an end goal, students are not empowered to reach for their own goals or to follow a process that may lead to results that surprise both the student and the teacher. For the master teacher, it would be worthwhile to move past fixed goals and establish processes and student choices that lead them to relevant yet indeterminate locations.

# ECONOMIC DIMENSIONS OF ADMINISTRATION

Economic development is the process by which a nation improves the economic, political, and social well-being of its people. The term has been used frequently by economists, politicians, and others in the 20th and 21st centuries. The concept, however, has been in existence in the West for centuries. *Modernization*, *Westernization*, and especially *Industrialization* are other terms people have used while discussing economic development. Economic development has a direct relationship with the environment and environmental issues.

Whereas economic development is a policy intervention endeavor with aims of economic and social well-being of people, economic growth is a phenomenon of market productivity and rise in GDP. Consequently, as economist Amartya Sen points out, “economic growth is one aspect of the process of economic development”.

## TYPE

The scope of economic development includes the process and policies by which a nation improves the economic, political, and social well-being of its people.

The University of Iowa’s Center for International Finance and Development states that:

‘Economic development’ is a term that economists, politicians, and others have used frequently in the 20th century. The concept, however, has been in existence in the West for centuries. Modernization, Westernisation, and especially Industrialisation are other terms people have used while discussing economic development. Economic development has a direct relationship with the environment.

Although nobody is certain when the concept originated, some people agree that development is closely bound up with the evolution of capitalism and the demise of feudalism. Mansell and When also state that economic development has been understood since the World War II to involve economic growth, namely the increases in per capita income, and (if currently absent) the attainment of a standard of living equivalent to that of industrialized countries.

Economic development can also be considered as a static theory that documents the state of an economy at a certain time. According to Schumpeter and Backhaus (2003), the changes in this equilibrium state to document in economic theory can only be caused by intervening factors coming from the outside.

## HISTORY

Economic development originated in the post-war period of reconstruction initiated by the United States. In 1949, during his inaugural speech, President Harry Truman identified the development of undeveloped areas as a priority for the west:

- “More than half the people of the world are living in conditions approaching misery. Their food is inadequate, they are victims of disease. Their economic life is primitive and stagnant. Their poverty is a handicap and a threat both to them and to more prosperous areas. For the first time in history humanity possesses the knowledge and the skill to relieve the suffering of these people... I believe that we should make available to peace-loving peoples the benefits of our store of technical knowledge in order to help them realize their aspirations for a better life… What we envisage is a programme of development based on the concepts of democratic fair dealing... Greater production is the key to prosperity and peace. And the key to greater production is a wider and more vigorous application of modem scientific and technical knowledge.”

There have been several major phases of development theory since 1945. From the 1940s to the 1960s the state played a large role in promoting industrialization in developing countries, following the idea of modernization theory. This period was followed by a brief period of basic needs development focusing on human capital development and redistribution in the 1970s. Neo-liberalism emerged in the 1980s pushing an agenda of free trade and removal of Import Substitution Industrialization policies.

In economics, the study of economic development was borne out of an extension to traditional economics that focused entirely on national product, or the aggregate output of goods and services. Economic development was concerned in the expansion of people's entitlements and their corresponding capabilities, morbidity, nourishment, literacy, education, and other socio-economic indicators. Borne out of the backdrop of Keynesian, advocating government intervention, and neo-classical economics, stressing reduced intervention, with rise of high-growth countries (Singapore, South Korea, Hong Kong) and planned governments (Argentina, Chile, Sudan, Uganda), economic development, more generally development economics, emerged amidst these mid-20th century theoretical interpretations of how economies prosper. Also, economist Albert O. Hirschman, a major contributor to development economics, asserted that economic development grew to concentrate on the poor regions of the world, primarily in Africa, Asia and Latin America yet on the outpouring of fundamental ideas and models.

It has also been argued, notably by Asian and European proponents of infrastructure-based development, that systematic, long-term government investments in transportation, housing, education, and health care are necessary to ensure sustainable economic growth in emerging countries.

**Growth and Development**

Economic growth deals with increase in the level of output, but economic development is related to increase in output coupled with improvement in social and political welfare of people within a country. Therefore, economic development encompasses both growth and welfare values.

Dependency theorists argue that poor countries have sometimes experienced economic growth with little or no economic development initiatives; for instance, in cases where they have functioned mainly as resource-providers to wealthy industrialized countries. There is an opposing argument, however, that growth causes development because some of the increase in income gets spent on human development such as education and health.

According to Ranis et al., economic growth and development is a two-way relationship. According to them, the first chain consists of economic growth benefiting human development, since economic growth is likely to lead families and individuals to use their heightened incomes to increase expenditures, which in turn furthers human development. At the same time, with the increased consumption and spending, health, education, and infrastructure systems grow and contribute to economic growth. In addition to increasing private incomes, economic growth also generate additional resources that can be used to improve social services (such as health care, safe drinking water, etc.). By generating additional resources for social services, unequal income distribution will be mitigated as such social services are distributed equally across each community, thereby benefiting each individual.

Concisely, the relationship between human development and economic development can be explained in three ways. First, increase in average income leads to improvement in health and nutrition (known as Capability Expansion through Economic Growth). Second, it is believed that social outcomes can only be improved by reducing income poverty (known as Capability Expansion through Poverty Reduction). Lastly, social outcomes can also be improved with essential services such as education, health care, and clean drinking water (known as Capability Expansion through Social Services). John Joseph Puthenkalam's research aims at the process of economic growth theories that lead to economic development.

After analyzing the existing capitalistic growth-development theoretical apparatus, he introduces the new model which integrates the variables of freedom, democracy and human rights into the existing models and argue that any future economic growth-development of any nation depends on this emerging model as we witness the third wave of unfolding demand for democracy in the Middle East. He develops the knowledge sector in growth theories with two new concepts of 'micro knowledge' and 'macro knowledge'. Micro knowledge is what an individual learns from school or from various existing knowledge and macro knowledge is the core philosophical thinking of a nation that all individuals inherently receive. How to combine both these knowledge would determine further growth that leads to economic development of developing nations.

Yet others believe that a number of basic building blocks need to be in place for growth and development to take place. For instance, some economists believe that a fundamental first step towards development and growth is to address property rights issues, otherwise only a small part of the economic sector will be able to participate in growth. That is, without inclusive property rights in the equation, the informal sector will remain outside the mainstream economy, excluded and without the same opportunities for study.

## ECONOMIC DEVELOPMENT GOALS

The development of a country has been associated with different concepts but generally encompasses economic growth through higher productivity, political systems that represent as accurately as possible the preferences of its citizens, the extension of rights to all social groups and the opportunities to get them and the proper functionality of institutions and organizations that are able to attend more technically and logistically complex tasks (*i.e.,* raise taxes and deliver public services). These processes describe the State's capabilities to manage its economy, polity, society and public administration.

Generally, the goals of economic development policies attempt to solve issues in these topics. With this in mind, economic development is typically associated with improvements in a variety of areas or indicators (such as literacy rates, life expectancy, and poverty rates), that may be causes of economic development rather than consequences of specific economic development programmes. For example, health and education improvements have been closely related to economic growth, but the causality with economic development may not be obvious. In any case, it is important to not expect that particular economic development programmes be able to fix many problems at once as that would be establishing unsurmountable goals for them that are highly unlikely they can achieve. Any development policy should set limited goals and a gradual approach to avoid falling victim to something Prittchet, Woolcock and Andrews call 'premature load bearing'.

Many times the economic development goals of specific countries cannot be reached because they lack the State's capabilities to do so. For example, if a nation has little capacity to carry out basic functions like security and policing or core service delivery it is unlikely that a programme that wants to foster a free-trade zone (special economic zones) or distribute vaccinations to vulnerable populations can accomplish their goals. This has been something overlooked by multiple international organizations, aid programmes and even participating governments who attempt to carry out 'best practices' from other places in a carbon-copy manner with little success. This isomorphic mimicry –adopting organizational forms that have been successful elsewhere but that only hide institutional dysfunction without solving it on the home country –can contribute to getting countries stuck in 'capability traps' where the country does not advance in its development goals. An example of this can be seen through some of the criticisms of foreign aid and its success rate at helping countries develop.

Beyond the incentive compatibility problems that can happen to foreign aid donations –that foreign aid granting countries continue to give it to countries with little results of economic growth but with corrupt leaders that are aligned with the granting countries' geopolitical interests and agenda –there are problems of fiscal fragility associated to receiving an important amount of government revenues through foreign aid. Governments that can raise a significant amount of revenue from this source are less accountable to their citizens (they are more autonomous) as they have less pressure to legitimately use those resources. Just as it has been documented for countries with an abundant supply of natural resources such as oil, countries whose government budget consists largely of foreign aid donations and not regular taxes are less likely to have incentives to develop effective public institutions. This in turn can undermine the country's efforts to develop.

## ECONOMIC DEVELOPMENT POLICIES

*In its broadest sense, policies of economic development encompass three major areas:*

- Governments undertaking to meet broad economic objectives such as price stability, high employment, and sustainable growth. Such efforts include monetary and fiscal policies, regulation of financial institutions, trade, and tax policies.

- Programmes that provide infrastructure and services such as highways, parks, affordable housing, crime prevention, and K–12 education.
- Job creation and retention through specific efforts in business finance, marketing, neighbourhood development, workforce development, small business development, business retention and expansion, technology transfer, and real estate development. This third category is a primary focus of economic development professionals.

One growing understanding in economic development is the promotion of regional clusters and a thriving metropolitan economy. In today's global landscape, location is vitally important and becomes a key in competitive advantage.

International trade and exchange rates are a key issue in economic development. Currencies are often either under-valued or over-valued, resulting in trade surpluses or deficits. Furthermore, the growth of globalization has linked economic development with trends on international trade and participation in Global Value Chains (GVCs) and international financial markets. The last financial crisis had a huge effect on economies in developing countries. Economist Jayati Ghosh states that it is necessary to make financial markets in developing countries more resilient by providing a variety of financial institutions. This could also add to financial security for small-scale producers.

**Organization**

Economic development has evolved into a professional industry of highly specialized practitioners. The practitioners have two key roles: one is to provide leadership in policy-making, and the other is to administer policy, programmes, and projects. Economic development practitioners generally work in public offices on the state, regional, or municipal level, or in public–private partnerships organizations that may be partially funded by local, regional, state, or federal tax money. These economic development organizations function as individual entities and in some cases as departments of local governments. Their role is to seek out new economic opportunities and retain their existing business wealth.

There are numerous other organizations whose primary function is not economic development that work in partnership with economic developers. They include the news media, foundations, utilities, schools, health care providers, faith-based organizations, and colleges, universities, and other education or research institutions.

**International Economic Development Council**

With more than 20,000 professional economic developers employed worldwide in this highly specialized industry, the International Economic Development Council (IEDC) headquartered in Washington, D.C. is a non-profit organization dedicated to helping economic developers do their job more effectively and raising the profile of the profession. With over 4,500 members across the US and internationally, serving exclusively the economic development community, IEDC membership represents the entire range of the profession ranging from regional, state, local, rural, urban, and international economic development organizations, as well as chambers of commerce, technology development agencies, utility companies, educational institutions, consultants and redevelopment authorities. Many individual states also have associations comprising economic development professionals, who work closely with IEDC.

## DEVELOPMENT INDICATORS AND INDICES

There are various types of macroeconomic and sociocultural indicators or "metrics" used by economists and geographers to assess the relative economic advancement of a given region or nation. The World Bank's "World Development Indicators" are compiled annually from officially-recognized international sources and include national, regional and global estimates.

### GDP per Capita – Growing Development Population

GDP per capita is gross domestic product divided by mid year population. GDP is the sum of gross value added by all resident producers in the economy plus any product taxes and minus any subsidizes not included in the value of the products. It is calculated without making deductions for depreciation of fabricated assets or for depletion and degradation of natural resources.per capita income of India is increased.

### Modern Transportation

European development economists have argued that the existence of modern transportation networks- such as high-speed rail infrastructure constitutes a significant indicator of a country's economic advancement: this perspective is illustrated notably through the Basic Rail Transportation Infrastructure Index (known as BRTI Index) and related models such as the (Modified) Rail Transportation Infrastructure Index (RTI).

## COMMUNITY COMPETITION

One unintended consequence of economic development is the intense competition between communities, states, and nations for new economic development projects in today's globalized world. With the struggle to attract and retain business, competition is further intensified by the use of many variations of economic incentives to the potential business such as: tax incentives, investment capital, donated land, utility rate discounts, and many others. IEDC places significant attention on the various activities undertaken by economic development organizations to help them compete and sustain vibrant communities.

Additionally, the use of community profiling tools and database templates to measure community assets versus other communities is also an important aspect of economic development. Job creation, economic output, and increase in taxable basis are the most common measurement tools. When considering measurement, too much emphasis has been placed on economic developers for "not creating jobs". However, the reality is that economic developers do not typically create jobs, but facilitate the process for existing businesses and start-ups to do so. Therefore, the economic developer must make sure that there are sufficient economic development programmes in place to assist the businesses achieve their goals. Those types of programmes are usually policy-created and can be local, regional, statewide and national in nature.

## EDUCATION ECONOMICS

Education economics or the economics of education is the study of economic issues relating to education, including the demand for education, the financing and provision of education, and the comparative efficiency of various educational programmes and policies.

From early works on the relationship between schooling and labour market outcomes for individuals, the field of the economics of education has grown rapidly to cover virtually all areas with linkages to education.

### Education as an Investment

Economics distinguishes in addition to physical capital another form of capital that is no less critical as a means of production – human capital. With investments in human capital, such as education, three major economic effects can be expected:

- Increased expenses as the accumulation of human capital requires investments just as physical capital does,
- Increased productivity as people gain characteristics that enable them to produce more output and hence
- Return on investment in the form of higher incomes.

#### *Investment Costs*

Investments in human capital entail an investment cost, just as any investment does. Typically in European countries most education expenditure takes the form of government consumption, although some costs are also borne by individuals. These investments can be rather costly. EU governments spent between 3% and 8% of GDP on education in 2005, the average being 5%. However, measuring the spending this way alone greatly underestimates the costs because a more subtle form of costs is completely overlooked: the opportunity cost of forgone wages as students cannot work while they study. It has been estimated that the total costs, including opportunity costs, of education are as much as double the direct costs. Including opportunity costs investments in education can be estimated to have been around 10% of GDP in the EU countries in 2005. In comparison investments in physical capital were 20% of GDP. Thus the two are of similar magnitude.

***Returns on Investment***

Human capital in the form of education shares many characteristics with physical capital. Both require an investment to create and, once created, both have economic value. Physical capital earns a return because people are willing to pay to use a piece of physical capital in work as it allows them to produce more output. To measure the productive value of physical capital, we can simply measure how much of a return it commands in the market. In the case of human capital calculating returns is more complicated – after all, we cannot separate education from the person to see how much it rents for. To get around this problem, the returns to human capital are generally inferred from differences in wages among people with different levels of education. Hall and Jones have calculated from international data that on average that the returns on education are 13.4% per year for first four years of schooling (grades 1–4), 10.1% per year for the next four years (grades 5–8) and 6.8% for each year beyond eight years. Thus someone with 12 years of schooling can be expected to earn, on average, 1.134 × 1.101 × 1.068 = 3.161 times as much as someone with no schooling at all.

***Effects on Productivity***

Economy-wide, the effect of human capital on incomes has been estimated to be rather significant: 65% of wages paid in developed countries is payments to human capital and only 35% to raw labour. The higher productivity of well-educated workers is one of the factors that explain higher GDPs and, therefore, higher incomes in developed countries. A strong correlation between GDP and education is clearly visible among the countries of the world, as is shown by the upper left figure. It is less clear, however, how much of a high GDP is explained by education. After all, it is also possible that rich countries can simply afford more education.

To distinguish the part of GDP explained with education from other causes, Weil has calculated how much one would expect each country's GDP to be higher based on the data on average schooling. This was based on the above-mentioned calculations of Hall and Jones on the returns on education. GDPs predicted by Weil's calculations can be plotted against actual GDPs, as is done in the figure on the left, demonstrating that the variation in education explains some, but not all, of the variation in GDP.

Finally, the matter of externalities should be considered. Usually when speaking of externalities one thinks of the negative effects of economic activities that are not included in market prices, such as pollution. These are negative externalities. However, there are also positive externalities – that is, positive effects of which someone can benefit without having to pay for it. Education bears with it major positive externalities: giving one person more education raises not only his or her output but also the output of those around him or her. Educated workers can bring new technologies, methods and information to the consideration of others. They can teach things to others and act as an example. The positive externalities of education include the effects of personal networks and the roles educated workers play in them.

Positive externalities from human capital are one explanation for why governments are involved in education. If people were left on their own, they would not take into account the full social benefit of education – in other words the rise in the output and wages of others – so the amount they would choose to obtain would be lower than the social optimum.

## Demand for Education

### *Liberal Approaches*

The dominant model of the demand for education is based on human capital theory. The central idea is that undertaking education is investment in the acquisition of skills and knowledge which will increase earnings, or provide long-term benefits such as an appreciation of literature (sometimes referred to as cultural capital). An increase in human capital can follow technological progress as knowledgeable employees are in demand due to the need for their skills, whether it be in understanding the production process or in operating machines. Studies from 1958 attempted to calculate the returns from additional schooling (the percent increase in income acquired through an additional year of schooling). Later results attempted to allow for different returns across persons or by level of education.

Statistics have shown that countries with high enrollment/graduation rates have grown faster than countries without. The United States has been the world leader in educational advances, beginning with the high school movement (1910–1950). There also seems to be a correlation between gender differences in education with the level of growth; more development is observed in countries which have an equal distribution of the percentage of women versus men who graduated from high school. When looking at correlations in the data, education seems to generate economic growth; however, it could be that we have this causality relationship backwards. For example, if education is seen as a luxury good, it may be that richer households are seeking out educational attainment as a symbol of status, rather than the relationship of education leading to wealth.

Educational advance is not the only variable for economic growth, though, as it only explains about 14% of the average annual increase in labour productivity over the period 1915-2005. From lack of a more significant correlation between formal educational achievement and productivity growth, some economists see reason to believe that in today's world many skills and capabilities come by way of learning outside of traditional education, or outside of schooling altogether. An alternative model of the demand for education, commonly referred to as screening, is based on the economic theory of signalling. The central idea is that the successful completion of education is a signal of ability.

### *Marxist Critique*

Although Marx and Engels did not write widely about the social functions of education, their concepts and methods are theorized and criticized by the influence of

Marx as education being used in reproduction of capitalist societies. Marx and Engels approached scholarship as "revolutionary scholarship" where education should serve as a propaganda for the struggle of the working class. The classical Marxian paradigm sees education as serving the interest of capital and is seeking alternative modes of education that would prepare students and citizens for more progressive socialist mode of social organizations. Marx and Engels understood education and free time as essential to developing free individuals and creating many-sided human beings, thus for them education should become a more essential part of the life of people unlike capitalist society which is organized mainly around work and the production of commodities.

## FINANCING AND PROVISION

In most countries school education is predominantly financed and provided by governments. Public funding and provision also plays a major role in higher education. Although there is wide agreement on the principle that education, at least at school level, should be financed mainly by governments, there is considerable debate over the desirable extent of public provision of education.

Supporters of public education argue that universal public provision promotes equality of opportunity and social cohesion. Opponents of public provision advocate alternatives such as vouchers.

### Pre-primary Education Financing

Compared to other areas of basic education, globally comparable data on pre-primary education financing remain scarce. While much of existing non-formal and private programmes may not be fully accounted for, it can be deduced from the level of provision that pre-primary financing remains inadequate, especially when considered against expected benefits. Globally, pre-primary education accounts for the lowest proportion of the total public expenditure on education, in spite of the much-documented positive impact of quality early childhood care and education on later learning and other social outcomes.

## EDUCATION PRODUCTION FUNCTION

An *education production function* is an application of the economic concept of a production function to the field of education. It relates various inputs affecting a student's learning (schools, families, peers, neighbourhoods, etc.) to measured outputs including subsequent labour market success, college attendance, graduation rates, and, most frequently, standardized test scores.

The original study that eventually prompted interest in the idea of education production functions was by a sociologist, James S. Coleman. The Coleman Report, published in 1966, concluded that the marginal effect of various school inputs on student achievement was small compared to the impact of families and friends. Later work, by Eric A. Hanushek, Richard Murnane, and other economists introduced the structure of "production" to the consideration of student learning outcomes. Hanushek at al. (2008) reported a very high correlation between "adjusted growth rate" and "adjusted test scores". A large number of successive studies, increasingly involving economists, produced inconsistent results about the impact of school resources on student performance, leading to considerable controversy in policy discussions. The interpretation of the various studies has been very controversial, in part because the findings have directly influenced policy debates. Two separate lines of study have been particularly widely debated. The overall question of whether added funds to schools are likely to produce higher achievement (the "money doesn't matter" debate) has entered into legislative debates and court consideration of school finance systems. Additionally, policy discussions about class size reduction heightened academic study of the relationship of class size and achievement.

## PERSONNEL MANAGEMENT DIMENSIONS OF ADMINISTRATION

Human resource management in public administration concerns human resource management as it applies specifically to the field of public administration. It is considered to be an in-house structure that ensures unbiased treatment, ethical standards, and promotes a value-based system.

### FUNCTION

The function of human resources management is to provide the employees with the capability to manage: health care, record keeping, promotion and advancement, benefits, compensation, etc. The function, in terms of the employers benefit, is to create a management system to achieve long-term goals and plans. The management allows companies to study, target, and execute long-term employment goals. For any company to have an efficient ability to grow and advance human resource management is a key.

*Human resources are designed to manage the following:*

- *Employee Benefits:* Include various types of non-wage compensation provided to employees in addition to their normal wages or salaries.
- *Employee health care:* The identification of recognition of a disease by a physician/ physician's assistant/nurse practitioner.
- *Compensation:* Something, typically money, awarded to someone as a recompense for loss, injury, or suffering.
- *Annual, sick, and personal leave:* Excused (and generally unpaid) leave for unexpected (such as accident or sickness) or expected (anniversaries, birthdays, marriage) events important to the individual.
- *Sick banks:* A fund accumulated to pay off a corporate or public debt
- *Discipline:* The practice of training people to obey rules or a code of behaviour, using punishment to correct disobedience.

- *Records (tax information, personnel files, etc.):* Also known as records and information management or RIM, is the professional practice of managing the records of an organization throughout their life cycle, from the time they are created to their eventual disposal.
- Recruitment and employee retention strategies:refers to the ability of an organization to retain its employees
- *Salary and Wages Administrations:* Process of compensating an organization's employees in accordance with accepted policy and procedures.

**LEADING CONTRIBUTORS**

Historically, when a new president came into power, political leaders would appoint their supporters to political offices in thanks for the campaign assistance. This became known as the spoils system and became popular in the United States during the presidency of Andrew Jackson.

In his first address to Congress, Jackson defended the system; he believed that public offices should be rotated among supporters to help the nation achieve its ideals. Jackson maintained that to perform well in public office, did not require special intelligence or training and rotating the office would ensure that the government did not develop corrupt civil servants.

The system was viewed as a reward to supporters of the party and a way to build a stronger government. During the first 18 months of Jackson's presidency he replaced fewer than 1,000 of the 10,000 civil servants due to politics, and fewer than 20 percent of officeholders were removed. Many of the men Jackson appointed to offices came from backgrounds of wealth and high social status. The system continued after Jackson's presidency and opposition against the system began to grow. During the presidency of Ulysses S. Grant corruption and inefficiency began to reach staggering proportion. This led to a larger outcry against the system and helped bring about change in 1883.

George H. Pendleton: Senator from Ohio sponsored the Civil Service Reform Act in 1883, which sought to implement a merit-based programme in the federal government. Its principal tenets include:

- Hiring employees by merit
- Receiving pay according to position, not personal characteristics
- Protection from political interference and dismissal via regime changes
- Government workers have an obligation to accountability (transparency)

Chester Barnard: taught an organization was the cooperation of human activity and to survive an organization needed to have efficiency and effectiveness. His definition of effectiveness: being able to accomplish the goals that were set and efficiency – if the goals are reached by the individuals of the organization then cooperation among them will continue.

Paul C. Light: discusses the Shadow Government and how it is used to make the Federal government appear smaller, even as the Federal government grows. The Shadow

Government is made up of those entities that produce goods or services for the government under contracts, grants, or mandates.

Volcker Commission: also known as the National Commission on Public Service was established in 1989, to rebuild the federal civil service. The commission was established by the United States Chairman of the Federal Reserve Paul A. Volcker. The main concern of the commission was morale because it was beginning to fall as were recruitment and retention among civil service employees and would soon become a crisis. This possible crisis was believed to be hindering the ability of the government to function effectively as the demand on the government began to grow.

The commission identified three main threats:

- Public attitudes and political leadership: the public did not trust or respect the government and the leaders. This also included federal agencies.
- Internal management systems: the federal agencies were losing experienced personnel due to problems with the leadership in the federal agencies. Mid-level workers were leaving the departments and entry level recruits were rethinking the commitments they made to the government.

The commission made some recommendations to address the problems. These included:

- Strengthening the relationship between presidential appointees and career civil servants by building a spirit of partnership between the two.
- Reducing the number of presidential appointees so that there is more room at the top for civil servants.
- Providing competitive pay to aid in recruiting and retaining excellent people while demanding competitive performance of them.

## PERTINENT LEGISLATING

- *Pendleton Civil Service Reform Act 1883:* Designed to end to the spoils system and provide federal government jobs based on merit and be selected through competitive exams. The act also made firing and demoting employees for political reasons unlawful. It also made requiring employees to give political service or contributions unlawful. The act also established the Civil Service Commission to enforce these rules. It was used to reinforce the efficiency of government.
- *Civil Service Reform Act of 1978:* Encompassed a wide variety of reforms including the creation of the Office of Personnel Management (OPM), the Merit Systems Protection Board (MSPB), The Federal Labour Relations Authority, and abolished the Civil Service Commission. The act seeks greater accountability of federal employees for their performance. The act also provides protection for "whistleblowers" and employees calling attention to any government malpractices.
- *Hatch Act of 1939:* Was passed into legislation to prohibit federal government

employees from participating in certain political activities both on and off duty. The employee could not support or oppose a political party, partisan political group, or a candidate for a partisan political party.

- In 1993, Congress passed legislation that amended the act as it applies to federal employees. Under the amendment most federal employees are now able to take part in political management and political campaigns. The act also applies to local and state employees who are employed with programmes financed by loan or grants from the government or a federal agency. If the employee works for a research or educational institutions supported by a state, the employee is not under the restrictions of the act. The government employees that are covered by the new amendment are in executive agencies or in positions in the U.S. Postal Service and Postal Rate Commission.
- *Classification Act of 1949:* Established the classification standards programme, this law states that positions are to be classified based on the duties and responsibilities assigned and the qualifications required to do the work. The position classifications standards are built on the foundation of the grade levels.
- *Title VII from Civil Rights Act of 1964:* Founded the Equal Employment Opportunity Commission (EEOC) and forbade discrimination in hiring, firing, and compensation based on race, colour, religion, gender, or national origin. It is also unlawful for an employer to segregate, limit, or classify employees in any way that will deprive them of employment opportunities or affect their employment status. In addition, it is unlawful to discriminate on these five bases in an apprenticeship, training, or retraining programmes.
- *Enhancement of Health Department Capacity for Health Care- Associated Infection Prevention Through Recovery Act-Funded Programmes 2009:* This programme resulted in increases capacity for HAI surveillance and prevention across all 51 state and territorial health departments that received funding.

## HUMAN RESOURCE MANAGEMENT *FOCUS* IN EQUAL EMPLOYMENT OPPORTUNITY (EEO)

Equal Employment Opportunity is continually in the spotlight of human resource (HR) management even after over 40 years of progress. The number of EEO complaints and lawsuits remains significant, indicating that ongoing progress is needed to decrease employment discrimination. EEO issues in HR Management are so prevalent that it has become one of the biggest concerns for HR professionals.

While HR professionals agree that equal employment opportunities are a legitimate focus, there is considerable controversy over best way to achieve equality. One way is to use the “blind to differences” approach, which argues that differences among people should be ignored and everyone should be treated equally. The second common approach is affirmative action, through which employers are urged to employ people based on their race, age, gender, or national origin. The idea is to make up for historical discrimination by giving groups who have been affected enhanced opportunities for employment." The former approach emphasizes equal treatment regardless of individual differences; the latter emphasizes fairness based on individual circumstances.

Thus, it is important for HR professionals to understand Equal Employment Opportunity (EEO) discrimination process because of the significant complaints and lawsuits that will undoubtedly be encountered throughout HR Management. "This discussion is to enhance the reader's understanding of the EEO process; the parts in each section of this discussion track the EEO process as chronologically as possible. However, the goal of this discussion is not to provide an exhaustive study of complex legal subjects. Digest summaries and articles themselves do not have the force of law and the reader is advised to look to the actual decisions and other sources discussed for a more precise understanding of applicable EEO law."

**Initiating and Navigating the EEO Process**

***Part 1: The Pre-Complaint Process***

The EEO process is initiated when an individual contacts an EEO counselor concerning a suspected violation of one or more of the laws that the Equal Employment Opportunity Commission (EEOC) enforces. "The Commission's regulations, promulgated under applicable statutory law, can be found in relevant parts in Title 29 of the Code of Federal Regulations ("Labour"). The federal sector process itself is detailed in 29 C.F.R. Part 1614 (1999); and further amplified in Management Directive 110 (1999) (hereinafter, Maryland-110). MD-110 has often been referred to as the EEO counselor's "bible" for the wealth of information, appendices, and forms contained therein regarding the EEO process and is available online to the public at: http://www.eeoc.gov/federal/index.cfm"

Under the EEOC-enforced statutes currently in force, there are 8 bases of employment discrimination that may be alleged regarding an agency action, policy, or practice in the EEO process. "The U.S. Equal Employment Opportunity Commission (EEOC) is responsible for enforcing federal laws that make it illegal to discriminate against a job applicant or an employee because of the person's race, colour, religion, sex (including pregnancy), national origin, age (40 or older), disability or genetic information. It is also illegal to discriminate against a person because the person complained about discrimination, filed a charge of discrimination, or participated in an employment discrimination investigation or lawsuit."

"Most employers with at least 15 employees are covered by EEOC laws (20 employees in age discrimination cases). Most labour unions and employment agencies are also covered. The laws apply to all types of work situations, including hiring, firing, promotions, harassment, training, wages, and benefits."

Once an individual has filed a charge of discrimination, it is important that the individual must adhere to certain time frames and follow specified procedures in order to avoid dismissal of their complaints. For example, an EEO complaint may possibly be dismissed for failure to begin EEO counseling within 45 days of the suspected discriminatory incident or effective date of alleged discriminatory personnel action.

However, an "aggrieved person" or "counselee" must consult with an EEO counselor prior to filing a complaint in order to resolve the disputed matter informally. The counselor's role is to provide a solution of the alleged discrimination before the complaint is formally filed. After which, during the 30-day period the Counselor is to complete counseling, provide for the counselee (*i.e.,* the aggrieved person) a written list of the counselee's rights and responsibilities.

*These include the following:*

- The right to request a hearing or an immediate final decision after an investigation by the agency.
- The responsibility to exercise certain election rights (which will be specified later in this section).
- The right to file a civil action in federal court after filing with EEOC a notice of intent to sue under the ADEA instead of pursuing a complaint of age discrimination in the administrative EEO process.
- The need to be aware of administrative EEO and federal court time frames.
- An understanding that only the claims raised in pre-complaint counseling (or issues and claims like or related to issues or claims raised in pre-complaint counseling) may be alleged in a subsequent complaint filed with the agency.

"Such EEO contact must occur within 45 days of when the aggrieved person knew or should have known of the alleged discriminatory matter, or, in the case of a personnel action, within 45 days of the effective date of the personnel action. At the time of initiating EEO counseling and throughout the EEO process, the counselee is permitted to have a representative who may be, but is not required to be, an attorney. The counselor, who may be an agency employee and work either full-time in EEO or in a collateral duty role, is required to be neutral and favour neither the counselee nor the agency."

"Failure of the aggrieved person to raise a matter in counseling may result in subsequent dismissal of the formal EEO complaint. Through the counseling process claims are set forth and clarified, and the counselor conducts a limited enquiry (not an investigation) for the purpose of achieving resolution."

During the counselor's enquiry, the counselor may utilize certain procedures common to mediation but does not engage in actual mediation, even if that counselor is also a certified mediator. Throughout this counseling, or pre-complaint stage, the Counselor will also notify the counselee of pertinent legal choices that are available. Also, during this process the EEO counselor must inform the counselee that, where the agency offers ADR, the counselee must elect to choose between engaging in ADR or continuing informal counseling, but not both. Despite the choice of ADR or continuing the process of informal counseling, if resolution is not achieved, the counselee will have the opportunity to file a formal EEO complaint. "The ADR process in the pre-complaint phase is limited to a maximum of 90 days. However, the EEOC encourages the parties to engage in ADR to attempt to resolve their dispute at any subsequent time up to and including the appellate process."

Further, "Counselors must advise individuals of their duty to keep the agency and Commission informed of their current address and to serve copies of appeal papers on the agency. The notice [of the right to file a formal complaint within 15 days of the counselee's receipt of the notice] shall include notice of the right to file a class complaint. If the aggrieved person informs the Counselor that he or she wishes to file a class complaint, the counselor shall explain the class complaint procedures and the responsibilities of a class agent."

The Counselor must also notify the counselee of his or her right to remain anonymous until the grievance is officially filed, where and with whom the formal complaint is to be filed, "and of the complainant's duty to assure that the agency is informed immediately if the complainant retains counsel or a representative" In addition, "the Counselor shall not attempt in any way to hinder the aggrieved person from filing a complaint."

Before a formal complaint may go to federal court as a civil action to pursue the aggrieved individual's discrimination claims, the EEO administrative process reviews the claim. This is known as [exhaustion of administrative remedies]. "In complaints concerning Title VII, the Rehabilitation Act, and the ADEA—where the complainant chooses to go through the EEO process—the "exhaustion" requirement is satisfied after 180 days from the filing of the individual complaint or the class complaint if an appeal has not been filed and final action has not been taken by the agency." Equal Pay Act claims, on the other hand, must be filed within two years (or three years for willful violations) of the alleged discrimination, despite their standing in the administrative process.

There are exceptions to the above requirement. An EEO complaint filed under the ADEA may exempt a complaint from the above requirement. If the complaint gives the commission at least 30 days written notice of the intent to file an action, it may bypass the EEO process and go directly to a U.S. District Court and file a civil action naming the head of an allegedly discriminating agency. "A complainant who is asserting a claim under the EPA, however, may bypass the EEO administrative process completely and go directly to court. The filing of a civil action by the complainant will terminate the processing of an administrative complaint or appeal filed with the EEOC, and, therefore, the complainant should notify the agency and Commission when s/he has filed a civil action."

In addition to the claim processes discussed, a counselee may have to choose between continuing his or her claims in the negotiated grievance process or the EEO process. "When an aggrieved person is employed by an agency that is subject to 5 U.S.C. 7121 (d), and is covered by a collective bargaining agreement that permits claims of discrimination to be raised in a negotiated grievance procedure, that employee must elect to proceed either through the EEO process or the negotiated grievance procedure, but not both." However, a complainant should be aware that, if he or she chooses to pursue a negotiated grievance before filing an EEO complaint, the time limitations in the EEO process will not be extended unless the agency consents to an extension in writing.

Another important election that an EEO Counselor must inform the counselee of is mixed cases. Essentially, a mixed case is a claim of discrimination that is appealable to the Merit Systems Protection Board (MSPB). Regulations related to mixed cases can be found at 29 C.F.R. § 1614.302. To determine if MSPB may have jurisdiction there are two important questions that must be answered. First, does the employee have standing to appear before the MSPB? "For example, a probationary employee does not have standing to go to MSPB on an EEO-based claim. Employees of certain agencies, *e.g.,* the FBI, CIA, TVA, the U.S. Postal Service, and certain non-appropriated fund activities (such as the Army and Air Force Exchange) do not have standing. Those employees may, however, pursue their claims through the regular EEO process with their agency." Secondly, does the claim occur from an action appealable to MSPB? Commonly, the more severe the personnel action at hand, the more likely it will be appealable to MSPB, *e.g.,* removal, suspension for more than 14 days, and reduction in grade.

In short, an aggrieved individual can file a mixed case complaint with the agency or a mixed case appeal with the MSPB but not both at the same time. The aggrieved person must choose one or the other. In the initial case of a mixed case complaint being filed, the complaint proceeds through the EEO process as with any EEO complaint, with these exceptions:

- There is no right to a hearing before an EEOC administrative judge (AJ) after an investigation.
- The investigation is limited to 120 days (not 180).
- The agency must issue a final agency decision (FAD) within 45 days following the investigation.
- If dissatisfied with the FAD, complainant must appeal the FAD, within 30 days of receipt of the FAD, to the MSPB (not to the EEOC).

If the aggrieved individual chooses to file a mixed case "appeal" instead of a mixed case "complaint", then this individual may request a hearing before an MSPB administrative judge (AJ) but not an EEOC administrative judge (AJ). Afterwards, if the aggrieved individual is dissatisfied with the MSPB's verdict on his or her claims of discrimination under the statutes the EEOC enforces, they may choose to file a petition with the EEOC from the MSPB decision.

***Part 2: Filing the Individual Complaint***

Once the counseling is over and if there has been no resolution to the claim or claims, the EEO counselor must provide the counselee with a Notice of Final Interview and the Right to File a Formal Complaint with the appropriate agency official. The counselee then is required to file the formal complaint within a time period of 15 days once the Notice of Final Interview has been received. This complaint must be signed by either the complainant or the individual's attorney. The complaint is also required to contain a phone number and address where the complainant or his or her attorney can be reached. The complainant is responsible for proceeding with the complaint with or without a designated representative.

The formal complaint must contain a precise statement that identifies the aggrieved individual and the agency and the actions or practices that form the basis of the complaint. The agency in turn, must provide the complainant with written acknowledgement of the complaint and the date of filing.

*The agency's acknowledgement letter will include the following information:*

- The address of the EEOC office where a request for a hearing is to be sent.
- The right to appeal the final action on or dismissal of a complaint.
- The requirement that the agency conduct an impartial and appropriate investigation within 180 days of the filing of the complaint unless the parties agree in writing to extend the time period.

***Part 3: Amending and Consolidating Complaints***

A complainant may amend a pending complaint to add claims that are related or similar to those raised in the pending complaint, prior to the agency's mailing of the notice required by 29 C.F.R. § 1614.108 (f) at the conclusion of the investigation. If the complainant needs to add or amend a new incident of alleged discrimination during the processing of an EEO complaint, the complainant will be instructed by the investigator or other EEO staff person to submit a letter to the agency's EEO Director or Complaints Manager at that time. The letter submitted must describe the new incident or amendments added by the complainant. Once the EEO officials review this request and determine the proper handling of the amendment they will decide if new EEO counseling is required. No new counseling is required when:

- Additional evidence is offered in support of the existing claim, but does not raise a new claim.
- The incident raises a new claim that is like or related to the claims (s) raised in the pending complaint.

"Additional evidence becomes part of the investigation of the pending claim and the complainant is so notified. The complaint must be amended where a new claim is like or related to the claim (s) raised in the pending complaint, and the EEO official must notify both the complainant and the investigator, in writing, acknowledging receipt of the amendment and the date it was filed. The EEO official will also instruct the investigator to investigate the new claim. New counseling will be required if the new claim is not like or related to the claim (s) in the pending complaint. The new claim will be the subject of a separate complaint and be subject to all of the regulatory case processing requirements."

***Part 4: The Investigation***

EEO investigations are covered in 29 C.F.R. § 1614.108 and the instructions are contained in the Commission's Management Directives. An efficient investigation is one that is conducted impartially with and contains an appropriate factual record. A correct factual record is one that allows a reasonable fact finder to draw conclusions as to whether discrimination occurred. "All agency employees, including the complainant, are required to cooperate with the investigation and "witness testimony is given under oath or affirmation and without a promise that the agency will keep the testimony or information provided confidential."

During this process the investigators must thoroughly investigate the complaint and are authorized to administer oaths and require witness testimony and documentation. An investigator cannot make or suggest findings of discrimination and must be free of conflicts or the appearance of conflicts of interest throughout the investigation of complaints. The evidence gathered by the investigator should only be relevant to the case in order to determine whether discrimination has occurred and if so, create the "appropriate remedy." Generally, investigations should be completed within 180 days of the filing of the individual complaint, unless the parties agree in writing to extend the period an additional 90 days.

At the end of the investigation, the agency must present the complainant with a copy of the complaint file, including the report of investigation, and the notice of the right to request either an immediate final decision from the agency or a hearing before an EEOC AJ. A complainant also may request an AJ hearing after 180 days from the filing of the complaint even if the investigation has not been completed. The complainant must receive a copy of the complaint file, plus the report of investigation (ROI), and a copy of the hearing transcript if a hearing was held.

***An Overview of the EEO Process: Conclusion***

This discussion has provided the detailed EEO process with regard to the processing of individual EEO complaints of discrimination, in accordance with 29 C.F.R. Part 1614. "The principles reflected in those procedures are also intended to guide the processing of class complaints of discrimination under 29 C.F.R. § 1614.204." The overall purpose of this discussion is to enhance the reader's understanding of the EEO process; the parts in each section of this discussion track the EEO process as chronologically as possible. However, the goal of this discussion is not to provide an exhaustive study of complex legal subjects. Digest summaries and articles themselves do not have the force of law and the reader is advised to look to the actual decisions and other sources discussed for a more precise understanding of applicable EEO law"

**HR STRUCTURE**

- *Federal Level:* The Federal classification system is not a pay plan, but is vital to the structure and administrations of employee compensation. The pay system is influenced by the grade level and by quality of performance, length of service, and recruitment and retention considerations. The law requires the Office of Personnel Management (OPM) to define Federal occupations, establish official position titles, and describe the grades of various levels of work. OPM approves and issues position classification standards that must be used by federal agencies to determine the title, series, and grade of positions. The classification standards help assure that the Federal personnel management programme runs soundly because agencies are now becoming more decentralized and now have more authority to classify positions. Agencies are required to classify positions according to the criteria and the guidance that OPM has issued. The official titles that are published in classification standards have to be used for personnel, budget, and fiscal purposes. The occupations in federal agencies may change over time, but the duties, responsibilities, and qualifications remain the same so careful application of appropriate classification of the standards needs to be related to the kind of work for the position. When classifying a position the first decision to be made is the pay system. There is the General Schedule (GS) and the Federal Wage System (FWS), which covers trade, craft, or laboring experience.

- General Schedule Covers positions from grades GS”1 to GS”15 and consists of twenty two occupational groups and is divided into five categories:
- Professional – Requires knowledge either acquired through training or education equivalent to a bachelor’s degree or higher. It also requires the exercise of judgement, discretion and personal responsibility. Examples can be attorneys, medical officers, and biologists. Usually a person who is in the field of HR, and has gone through the education required, stays in the field for long-term career goals. People of this category are seen in the upper management of HR departments.
- Administrative – Requires the exercise of analytical ability, judgement, discretion, and personal responsibility, and the application of a substantial body of knowledge of principles, concepts, and practices applicable to one or more fields of administration or management. These positions do not require specialized education, but do require skills usually gained while attaining a college level education. Examples can be budget analysts and general supply specialists. These positions will most likely be filed by career employees that act in a managerial function.
- Technical – Requires extensive practical knowledge, gained through experience and specific training and these occupations may involve substantial elements of the work of the professional or administrative field. Technical employees usually carry out tasks, methods, procedures, and computations that are laid out either in published or oral instructions. Depending upon the level of difficulty of work, these procedures often require a high degree of technical skill, care, and

precision. Examples of the technical category would be forestry technician, accounting technician, and pharmacy technicians.

- Clerical – Involves work in support of office, business, or fiscal operations. Typically involves general office or programme support duties such as preparing, receiving, reviewing, and verifying documents; processing transactions; maintaining office records; locating and compiling data or information from files. Examples can be secretaries, data transcribers, and mail clerks.
- Other – There are some occupations in the General Schedule which do not clearly fit into one of the groups. Some firefighter and various law enforcement agencies have specialized positions that manage HR duties within the organization.

## CORPORATE EDUCATION

Corporate Education refers to a system of professional development activities provided to educate employees. It may consist of formal university or college training or informal training provided by non-collegiate institutions. The simplest form of corporate education may be training programmes designed "in-house" for an organization that may wish to train their employees on specific aspects of their job processes or responsibilities. More formal relationships may further exist where corporate training is provided to employees through contracts or relationships with educational institutions who may award credit, either at the institution or through a system of CEUs (Continuing Education Units).

Many institutions or trainers offering corporate education will provide certificates or diplomas verifying the attendance of the employee. Some employers use corporate and continuing education as part of a holistic human resources effort to determine the performance of the employee and as part of their review systems.

Increasingly organisations appear to be using corporate education and training as an incentive to retain managers and key employees within their organisation. This win-win arrangement creates better educated managers for the organisation and provides the employees with a more marketable portfolio of skills and, in many cases, recognised qualifications.

### The Difference Between Corporate Education and Corporate Training

Most organisations tend to think of corporate education as corporate training. Corporate training programmes are often competency based and related to the essential training employees need to operate certain equipment or perform certain tasks in a competent, safe and effective manner. The outcome of a corporate training programme is a participant who is either able to operate a piece of equipment or perform a specific task in an effective manner according to pre-determined training criteria.

The primary role of corporate training is to ensure an employee has the knowledge and skills to undertake a specific operation to enable an organisation can continue to operate. Fundamentally, corporate training is centred on knowledge transfer, with an instructor teaching or demonstrating a particular function and the student learning and demonstrating they can apply what they have learnt to a particular operation.

Corporate education, however, adds another dimension and depth to training by involving learners as participants in generating new knowledge that assists an organisation to develop and evolve, rather than maintain the status quo. Corporate education focuses on developing the capability of an organisation to be able to do things and, in particular, the right things in order to be a sustainable and successful organisation.

Corporate education involves a facilitator, rather than an instructor or trainer, to engage participants and encourage them to think about the what, how and why of what they are doing and to challenge their current paradigms. Corporate education is centred on introducing learning techniques to stimulate employees to think about what their organisation does, where it is heading, potential new opportunities for the organisation and new and better ways of doing things. While the role of corporate training is to develop the operational competency of individuals, the purpose of corporate education is to promote the development of capability of both an individual and their organisation.

Increasingly organisations appear to be using corporate education as an incentive to retain managers and key employees within their organisation. This win-win arrangement creates better educated managers and employees for the organisation and gives individual employees a more marketable portfolio of skills and, in many cases, recognised qualifications.

## TRAINING AND DEVELOPMENT

Human resource management regards training and development as a function concerned with organizational activity aimed at bettering the job performance of individuals and groups in organizational settings. Training and development can be described as "an educational process which involves the sharpening of skills, concepts, changing of attitude and gaining more knowledge to enhance the performance of employees". The field has gone by several names, including "Human Resource Development", "Human Capital Development" and "Learning and Development".

### History

The name of the discipline has been debated, with the Chartered Institute of Personnel and Development in 2000 arguing that "human resource development" is too evocative of the master-slave relationship between employer and employee for those who refer to their employees as "partners" or "associates" to feel comfortable with. Eventually, the CIPD settled upon "learning and development", although that was itself not free from problems, "learning" being an over-general and ambiguous name, and most organizations referring to it as "training and development".

**Practice**

*Training and development encompasses three main activities: training, education, and development:*

- *Training:* This activity is both focused upon, and evaluated against, the job that an individual currently holds.
- *Education:* This activity focuses upon the jobs that an individual may potentially hold in the future, and is evaluated against those jobs.
- *Development:* This activity focuses upon the activities that the organization employing the individual, or that the individual is part of, may partake in the future, and is almost impossible to evaluate.

The "stakeholders" in training and development are categorized into several classes. The sponsors of training and development are senior managers. The clients of training and development are business planners. Line managers are responsible for coaching, resources, and performance. The participants are those who actually undergo the processes. The facilitators are Human Resource Management staff. And the providers are specialists in the field. Each of these groups has its own agenda and motivations, which sometimes conflict with the agendas and motivations of the others.

The conflicts that are the best part of career consequences are those that take place between employees and their bosses. The number one reason people leave their jobs is conflict with their bosses. And yet, as author, workplace relationship authority, and executive coach, Dr. John Hoover points out, "Tempting as it is, nobody ever enhanced his or her career by making the boss look stupid." Training an employee to get along well with authority and with people who entertain diverse points of view is one of the best guarantees of long-term success. Talent, knowledge, and skill alone won't compensate for a sour relationship with a superior, peer, or customer.

Many training and development approaches available for organisations are proposed including: on-the-job training, mentoring, apprenticeship, simulation, web-based learning, instructor-led classroom training, programmed self-instruction, case studies/ role playing, systematic job rotations and transfers. etc.

Typical roles in the field include executive and supervisory/management development, new-employee orientation, professional-skills training, technical/job training, customer-service training, sales-and-marketing training, and health-and-safety training. Job titles may include vice-president of organizational effectiveness, training manager or director, management development specialist, blended-learning designer, training-needs analyst, chief learning officer, and individual career-development advisor.

Talent development is the process of changing an organization, its employees, its stakeholders, and groups of people within it, using planned and unplanned learning, in order to achieve and maintain a competitive advantage for the organization. Rothwell notes that the name may well be a term in search of a meaning, like so much in management, and suggests that it be thought of as selective attention paid to the top 10% of employees, either by potential or performance.

While talent development is reserved for the top management it is becoming increasingly clear that career development is necessary for the retention of any employee, no matter what their level in the company. Research has shown that some type of career path is necessary for job satisfaction and hence job retention. Perhaps organizations need to include this area in their overview of employee satisfaction.

The term talent development is becoming increasingly popular in several organizations, as companies are now moving from the traditional term training and development. Talent development encompasses a variety of components such as training, career development, career management, and organizational development, and training and development. It is expected that during the 21st century more companies will begin to use more integrated terms such as talent development.

**Benefits**

Training is crucial for organizational development and its success which is indeed fruitful to both employers and employees of an organization.

*Here are some important benefits of training and development:*

- Increased productivity
- Less supervision
- Job satisfaction
- Skills Development.

# Bibliography


A.R. Rather: *Theory and Principles of Education*, Discovery Publishing House, 2004.

Adult Education: *H D Kamat and A David*, Crescent Pub, 2007.

Albert Bandura: *Social learning theory*, Prentice Hall, 2007.

Alesina, A. & Dollar, D.: *Who gives foreign aid to who and why*? Journal of Economic Growth, 2000.

Al-Samarrai, S: *Achieving education for all: How much does money matter*? Journal of International Development, 2006.

Altschuler, D: *How patronage politics undermines parental participation and accountability: community-managed schools in Honduras and Guatemala*. Comparative Education Review, 2013

Alvarez-Valdivia, I., Chavez, L., Schneider, B.K., & Roberts, J.*: Parental involvement and the academic achievement and social functioning of Cuban school children*. School Psychology International, (2012, November.)

Andrews, Ralph L: *Post-Confederation Developments in Newfoundland Education 1949-1975*. St. John's: Harry Cuff Publications Limited, 1995.

Arksey, H., & Knight, P. T.: *Interviewing for social scientists: An introductory resources with examples,* Sage Publications, 1999.

Baecker, R.: *Readings in groupware and computer-supported co-operative work*. Lawrence Erlbaum Associates, Morgan Kaufman Publishers, Inc, 1992

Barker D: *Seven new ways to learn*, Byte, 1995.

Barker P: *Electronic course delivery, virtual universities and lifelong learning*, Educational Technology Review, 1999

Barker P: *Tools to support electronic lectures*, Educational Technology Review, 1997

Barrera-Osorio, F., Fasih, T., Patrinos, H.A., & Santibánez, L: *Decentralized decision-making in schools: the theory and evidence on school-based management*. Washington, DC: World Bank, 2009

Bates A: *Technology, Open Learning and Distance Education*, London/New York: Routledge, 1995

Becher, T.: *Academic Tribes and Territories: intellectual enquiry and the cultures of the disciplines*. Buckingham: SRHE/Open University Press, 1989

Beckett M: *Foresight Consultation on the next round of the Foresight Programme*, Office of Science and Technology, Department of Trade and Industry, 1998

Beetham H: *A Review of Learning and Teaching Innovation in UK HE*, CTISS Publications, 1998

Beetham H: *An alternative perspective on CPD*, Educational Developments, 2000

Beetham, H.: *Developing practice through networks and shared resources*, Educational Developments, 2001

Beetham, H.: *How Do Representations of Our Practice Enable Change to Happen*? Educational Developments, 2001.

Beetham, H.: *Learning Technology Scoping Study: Lessons for Educational Developers*. Educational developments, 2000.

Bell J : *Doing Your Research Project a guide for First-Time Researchers in Education and Social Science* (2nd edn), Milton-Keynes: Open University Press, 1997

Benavot, A., & Resnik, J. : Lessons from the past: A comparative socio-historical analysis of primary and secondary education, MIT Press, 2007

Bennett CK: *Reconceptualising staff development for technology diffusion*, Journal of Information Technology for Teacher Education, 1994

Bennett, S., Priest, A., and Macpherson, C.: *Learning about online learning: An approach to staff development for university teachers*. Australian Journal of Educational Technology, 1999

Billett, S.: *Workplaces, communities and pedagogy: an activity theory view*. In Lea, M. & Nicoll, K. (Eds), Distributed Learning: Social and Cultural Approaches to Practice, 2002.

Boud, D.: *Developing Student Autonomy in Learning*. Kogan Page: London, 1988

Bowman, M.J.: *An integrated framework for analysis of the spread of schooling in less developed countries*, 1994

Brahler CJ: *Developing on-line learning materials for higher education:* an overview of current issues, Educational Technology and Society, 1999.

Braun, D: *The rich get richer: the rise of income inequality in the United States and the world. Chicago*: Nelson-Hall, 1997

Bray, M: *Community partnerships in education: dimensions, variations and implications*. Paris: UNESCO, 2001

Brennan J: *What kind of university? International perspectives on knowledge, participation and governance*, Buckingham: SRHE and Open University Press, 1999

Bressler S: *Exploring the Electronic Marketplace*, VHS: Hewlett-Packard, 1997

Bridges, D.: *Transferable Skills in Higher Education*. Norwich: University of East Anglia, 1994

Briggs, C. L: *Learning how to ask: A sociolinguistic appraisal of the role of the interview in social science research,* Cambridge University Press. 2006.

Brown D, Burg JJ and Dominick JL: *A Strategic Plan for Ubiquitous Laptop Computing, Communications of the ACM* , 1998

Brown, A. L. : *Design experiments: Theoretical and methodological challenges in creating complex interventions in classroom settings*. Journal of the Learning Sciences, 2(2), 1992.

Brown, L.: *Introduction and acknowledgements. In L. Brown (ed.) Maintaining university primary education: Lessons from Commonwealth Africa* (pp. 1-8). London: Commonwealth Secretariat, 2009

Buffett, Rev. F.M.: *The Story of the Church in Newfoundland. Toronto: General Board of Religious Education*, 2002.

Burke, Louis. : "*Some Irish Contributors and Contributions to Newfoundland Education in the Last Century*." M.Litt. Thesis, University of Dublin, 1995.

Butler-Kisber, L.: *Qualitative inquiry: Thematic, narrative and arts-informed perspectives,* Sage Publications., 2010.

Caelli, K., Ray, L., & Miller, J. 'Clear as mud': *Toward greater clarity in generic qualitative research*, International Journal of Qualitative Methods, 2003.

Canadian Press. "*Churches lose bid to overturn change to Term* 17." The Telegram (29 February 2000.

Castells M: *The Informational City, Oxford*: Blackwell, 1993

Castells M: *The Rise of the Network Society*, London: Blackwell, 1996

Cheema, G. & Rondinelli, D. (Eds.): *Decentralization and development: Policy implications for developing countries*. London: Sage, 2003

Chesla, C. A.Translational research: *Essential contributions from interpretive nursing science*. Research in Nursing & Health, 2008.

Chirban, J. T.: *Interviewing in depth: The interactive-relational approach*. Thousand Oaks: Sage Publications, 1996.

Chris Abbott: *ICT- Changing Education*, Psychology Press, 2001.

Chris Jones: *Evaluating a collaborative online learning environment* , Active Learning, 1999

Clark, A. (Ed.): *Situational analysis: Grounded theory after the postmodern turn (1st ed.)* Sage Publications, 2005.

Clemens, M.: *The long walk to school: International education goals in historical perspective*. Working Paper Number 37. Washington, DC: Center for Global Development, 2004

Clément, Dominique: "*Searching for Rights in the Age of Activism*: The Newfoundland-Labrador Human Rights Association, 1968-1982.

Colclough, C.,& Al-Samarrai, S.: *Achieving schooling for all: Budgetary expenditures on education in Sub-Saharan Africa and South Asia*. IDS Working Paper 77. Brighton: University of Sussex Institute for Development Studies, 2000

Collins, M: *Computer-Mediated Communication and the Online Classroom* (New Jersey), 1994

Conole G and Oliver M: *A Pedagogical Framework for Embedding C&IT into the Curriculum*, ALT-Journal, 1997

Corrales, J.:*Political obstacles to expanding and improving schooling in developing countries.* In J. Cohen, D. Bloom, & M. Malin (eds.) educating all children: A global agenda. Cambridge, MA: MIT Press, 2007

Creswell, J. W. : *Qualitative inquiry and research design: Choosing among five traditions*. Thousand Oaks: Sage Publications, 1998.

Creswell, J. W.. *Educational research: Planning, conducting, and evaluating quantitative and qualitative research (2nd ed ed.)*. Upper Saddle River: Merrill, 2005.

Creswell, J. W.: *Research design: Qualitative, quantitative, and mixed methods approaches (3rd ed ed.)*. Thousand Oaks: Sage Publications, 2009.

Daniel, J.: *Mega-Universities and Knowledge Media: Technology Strategies for Higher Education.* London: Kogan Page, 2001

Davenport TH and Prusak L: *Working Knowledge: How Organizations Manage What They Know, Boston*, MA: Harvard Business School Press, 1998

Davenport TH: *Information Ecology*, Oxford: Oxford University Press, 1997

David Buckingham: *Media Education: Literacy, Learning and Contemporary Culture*, John Wiley & Sons, 2013.

David E. Kern: *Curriculum Development for Medical Education: A Six-step Approach*, JHU Press, 1998.

Davies, L., Harber, C., & Dzimadzi, C: *Educational decentralization in Malawi: A study of process*. Compare: A Journal of Comparative and International Education, 2003.

# Index

## A



## B



## C

## D



## E



## F

# H



# I



# L



# M



# N



# O



# P

## Q



## R



## S

# T



# U



# V